C8000V IPsec Testbed

Test evidence — every command run on every device

Run started	08 September 2026, 20:53 UTC
Duration	25.2 minutes
Result	234 of 234 tests passed
Suites	25
Device commands recorded	1627
Topology	hub-and-spoke: R1 hub, R2 and R3 spokes, one Linux host per router
Encryption	IKEv2 + static VTI, ESP-AES-256 / SHA256, one tunnel per spoke
Routing	eBGP inside the tunnels; AS 65001 hub, AS 65002 / 65003 spokes

Suite results

Suite	Tests	Passed	Failed	Commands	Time
01 Platform	5	5	0	10	1s
02 Configuration	5	5	0	14	2s
03 Wan	6	6	0	19	1s
04 Dmvpn	13	13	0	37	25s
05 Capture State	12	12	0	79	15s
06 Wire Encryption	3	3	0	27	34s
07 Ipsec Negative	3	3	0	63	124s
08 Management Api	6	6	0	7	3s
09 Bgp	26	26	0	623	302s
10 Host To Host	10	10	0	47	79s
11 Throughput	3	3	0	10	81s
12 Nat	10	10	0	51	79s
13 Nat Pool	13	13	0	50	91s
14 Ntp	12	12	0	39	3s
15 Syslog	10	10	0	21	47s
16 Snmpv3	13	13	0	87	17s

17 Snmp Polling	14	14	0	76	22s
18 Mtu	8	8	0	17	21s
19 Oob	9	9	0	69	36s
20 Tacacs	12	12	0	35	27s
21 Vty Acl	9	9	0	22	1s
22 Banner	6	6	0	6	21s
23 Destructive	7	7	0	178	271s
24 Terraform	9	9	0	20	10s
25 Terraform Nac	10	10	0	20	15s

Each test below lists the commands it issued, in order, with the device that ran them and the output it returned. Router commands are marked in teal, Linux host commands in amber. Long output is trimmed to the first 14 lines; complete captures are in `results/`.

01 Platform

1.01 PASS Both Routers Report Catalyst 8000V Software (0.2s, 2 commands)

Setup (suite) Open All Routers

Acceptance criteria

- output matches **Cisco IOS[-]XE Software, Version \d+\.\d+** (checked 2×)
- output contains **C8000V** (checked 2×)

How it was checked

R1 show version

```
Cisco IOS XE Software, Version 17.15.06
Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-UNIVERSALK9-M), Version 17.15.6, RELEASE
SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
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GPL code under the terms of GPL Version 2.0.  For more details, see the
documentation or "License Notice" file accompanying the IOS-XE software,
... 52 more lines
```

R2 show version

```
Cisco IOS XE Software, Version 17.15.06
Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-UNIVERSALK9-M), Version 17.15.6, RELEASE
SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
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documentation or "License Notice" file accompanying the IOS-XE software,
... 52 more lines
```

Teardown (suite) Close All Routers

1.02 PASS Hostnames Match The Day0 Bootstrap Configs (0.2s, 2 commands)

Setup (suite) Open All Routers

Acceptance criteria

- output contains **${expected}** (checked 2×)

How it was checked

R1 show running-config | include ^hostname

```
hostname c8000v-r1
```

R2 show running-config | include ^hostname

```
hostname c8000v-r2
```

Teardown (suite) Close All Routers

1.03 PASS Management Interfaces Are Up With DHCP Addresses (0.1s, 2 commands)

Setup (suite) Open All Routers

Acceptance criteria

- output matches **GigabitEthernet1\s+10\.0\.2\.\d+\s+YES\s+DHCP\s+up\s+up** (checked 2×)

How it was checked

```
R1 show ip interface brief GigabitEthernet1
  Interface              IP-Address      OK? Method Status                Protocol
  GigabitEthernet1       10.0.2.15       YES DHCP   up                    up
R2 show ip interface brief GigabitEthernet1
  Interface              IP-Address      OK? Method Status                Protocol
  GigabitEthernet1       10.0.2.15       YES DHCP   up                    up
```

Teardown (suite) Close All Routers

1.04 PASS Crypto Feature License Is Active (0.1s, 2 commands)

Without a feature license the crypto CLI does not exist at all, so every IPsec test downstream depends on this.

Setup (suite) Open All Routers

Acceptance criteria

- output matches **License Level:\s*\S+** (checked 2×)

How it was checked

```
R1 show version | include License Level
  License Level: network-premier
  Addon License Level: dna-premier
R2 show version | include License Level
  License Level: network-premier
  Addon License Level: dna-premier
```

Teardown (suite) Close All Routers

1.05 PASS Management Plane Services Are Enabled (0.2s, 2 commands)

Setup (suite) Open All Routers

Acceptance criteria

- output contains **netconf-yang** (checked 2×)
- output contains **restconf** (checked 2×)

How it was checked

```
R1 show running-config | include ^netconf-yang|^restconf
  netconf-yang
  restconf
R2 show running-config | include ^netconf-yang|^restconf
  netconf-yang
  restconf
```

Teardown (suite) Close All Routers

02 Configuration

2.01 PASS Create A Loopback Interface (0.1s, 6 commands)

Setup (suite) Open All Routers

Acceptance criteria

- output does not contain **% Invalid input**
- output does not contain **% Incomplete command**
- output does not contain **% Ambiguous command**

How it was checked

```
R1 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R1 interface Loopback100
  ${output} =
R1 description ROBOT-MANAGED
  ${output} =
R1 ip address 192.0.2.1 255.255.255.0
  ${output} =
R1 no shutdown
  ${output} =
R1 end
  ${output} =
```

Teardown (this test) Run Keyword If Test Failed (Log, ${TEST_NAME} failed -- see the console on port 5001)

2.02 PASS Loopback Appears In The Interface Table (0.0s, 1 commands)

Setup (suite) Open All Routers

Acceptance criteria

- output matches **${LOOPBACK}\s+${LOOPBACK_IP}\s+YES\s+manual\s+up\s+up**

How it was checked

```
R1 show ip interface brief Loopback100
  Interface              IP-Address      OK? Method Status                Protocol
  Loopback100            192.0.2.1       YES manual up                    up
```

Teardown (this test) Run Keyword If Test Failed (Log, ${TEST_NAME} failed -- see the console on port 5001)

2.03 PASS Loopback Is Reachable From The Device Itself (2.1s, 2 commands)

Setup (suite) Open All Routers

Acceptance criteria

- output contains **Success rate is 100 percent**

How it was checked

```
R1 ping 192.0.2.1 repeat 2
  Type escape sequence to abort.
  Sending 2, 100-byte ICMP Echos to 192.0.2.1, timeout is 2 seconds:
  .!
  Success rate is 50 percent (1/2), round-trip min/avg/max = 2/2/2 ms
R1 ping 192.0.2.1 repeat 5
  Type escape sequence to abort.
  Sending 5, 100-byte ICMP Echos to 192.0.2.1, timeout is 2 seconds:
  !!!!!
  Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
```

Teardown (this test) Run Keyword If Test Failed (Log, ${TEST_NAME} failed -- see the console on port 5001)

2.04 PASS Loopback Is Present In The Running Config (0.0s, 1 commands)

Setup (suite) Open All Routers

Acceptance criteria

- output contains **description ROBOT-MANAGED**
- output contains **ip address ${LOOPBACK_IP} ${LOOPBACK_MASK}**

How it was checked

R1 show running-config interface Loopback100

```
Building configuration...

Current configuration : 92 bytes
!
interface Loopback100
 description ROBOT-MANAGED
 ip address 192.0.2.1 255.255.255.0
end
```

Teardown (this test) Run Keyword If Test Failed (Log, ${TEST_NAME} failed -- see the console on port 5001)

2.05 PASS Removing The Loopback Cleans It Up (0.1s, 4 commands)

Setup (suite) Open All Routers

Acceptance criteria

- output does not contain **${LOOPBACK_IP}**

How it was checked

R1 configure terminal

```
Enter configuration commands, one per line.  End with CNTL/Z.
```

R1 no interface Loopback100

```
${output} =
```

R1 end

```
${output} =
```

R1 show ip interface brief Loopback100

```
^
% Invalid input detected at '^' marker.
```

Teardown (this test) Run Keyword If Test Failed (Log, ${TEST_NAME} failed -- see the console on port 5001)

03 Wan

3.01 PASS Every Router Is On The Shared WAN Segment (0.1s, 3 commands)

Setup (suite) Open All Routers

Acceptance criteria

- output matches **GigabitEthernet2\s+${WAN_IP}[${r}]\s+YES\s+\S+\s+up\s+up** (checked 3×)

How it was checked

```
R1 show ip interface brief | include GigabitEthernet2
  GigabitEthernet2       100.64.0.1      YES NVRAM  up                    up
R2 show ip interface brief | include GigabitEthernet2
  GigabitEthernet2       100.64.0.2      YES NVRAM  up                    up
R3 show ip interface brief | include GigabitEthernet2
  GigabitEthernet2       100.64.0.3      YES NVRAM  up                    up
```

Teardown (suite) Close All Connections

3.02 PASS Every Router Can Reach Every Other Router On It (0.2s, 6 commands)

Including spoke to spoke, which is the precondition for a direct tunnel and the property the VTI lab deliberately did not have. Tested at the underlay, before any tunnel exists.

Setup (suite) Open All Routers

Acceptance criteria

- output contains **Success rate is 100 percent** (checked 6×)

How it was checked

```
R2 ping 100.64.0.2 repeat 5
  Type escape sequence to abort.
  Sending 5, 100-byte ICMP Echos to 100.64.0.2, timeout is 2 seconds:
  !!!!!
  Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
R3 ping 100.64.0.3 repeat 5
  Type escape sequence to abort.
  Sending 5, 100-byte ICMP Echos to 100.64.0.3, timeout is 2 seconds:
  !!!!!
  Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
R1 ping 100.64.0.1 repeat 5
  Type escape sequence to abort.
  Sending 5, 100-byte ICMP Echos to 100.64.0.1, timeout is 2 seconds:
  !!!!!
  Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
R3 ping 100.64.0.3 repeat 5
  Type escape sequence to abort.
  Sending 5, 100-byte ICMP Echos to 100.64.0.3, timeout is 2 seconds:
  !!!!!
  Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
R1 ping 100.64.0.1 repeat 5
  Type escape sequence to abort.
  Sending 5, 100-byte ICMP Echos to 100.64.0.1, timeout is 2 seconds:
  !!!!!
  Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
R2 ping 100.64.0.2 repeat 5
  Type escape sequence to abort.
  Sending 5, 100-byte ICMP Echos to 100.64.0.2, timeout is 2 seconds:
  !!!!!
  Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
```

Teardown (suite) Close All Connections

3.03 PASS The Spokes Reach Each Other Directly, Not Through The Hub (0.0s, 1 commands)

One hop at the underlay. If this became two the spokes would still have connectivity, and DMVPN would still form tunnels -- but every "direct" spoke-to-spoke path would quietly be transiting the hub anyway.

Setup (suite) Open All Routers

Acceptance criteria

- output contains **${R3_WAN_IP}**
- output does not contain **${R1_WAN_IP}**

  the spokes reach each other via the hub at the underlay:\n${trace}

How it was checked

R2 traceroute 100.64.0.3 probe 1 timeout 1 numeric

```
Type escape sequence to abort.
Tracing the route to 100.64.0.3
VRF info: (vrf in name/id, vrf out name/id)
  1 100.64.0.3 2 msec
```

Teardown (suite) Close All Connections

3.04 PASS Each Router Learns The Others On The Segment (0.1s, 3 commands)

A shared segment, so every router should have resolved every other -- unlike a point-to-point link with exactly one neighbour.

Setup (suite) Open All Routers

Acceptance criteria

- output contains **${WAN_IP}[${peer}]** (checked 9×)

How it was checked

R1 show ip arp GigabitEthernet2

```
Protocol  Address          Age (min)  Hardware Addr   Type   Interface
Internet  100.64.0.1              -   5254.00c8.0102  ARPA   GigabitEthernet2
Internet  100.64.0.2            154   5254.00c8.0202  ARPA   GigabitEthernet2
Internet  100.64.0.3            159   5254.00c8.0302  ARPA   GigabitEthernet2
```

R2 show ip arp GigabitEthernet2

```
Protocol  Address          Age (min)  Hardware Addr   Type   Interface
Internet  100.64.0.2              -   5254.00c8.0202  ARPA   GigabitEthernet2
Internet  100.64.0.1            154   5254.00c8.0102  ARPA   GigabitEthernet2
Internet  100.64.0.3            104   5254.00c8.0302  ARPA   GigabitEthernet2
```

R3 show ip arp GigabitEthernet2

```
Protocol  Address          Age (min)  Hardware Addr   Type   Interface
Internet  100.64.0.3              -   5254.00c8.0302  ARPA   GigabitEthernet2
Internet  100.64.0.1            159   5254.00c8.0102  ARPA   GigabitEthernet2
Internet  100.64.0.2            104   5254.00c8.0202  ARPA   GigabitEthernet2
```

Teardown (suite) Close All Connections

3.05 PASS The Underlay Carries No Site Prefixes (0.1s, 3 commands)

The WAN is a transport, not a routing domain. A site prefix appearing here would mean traffic could reach a LAN without ever entering the tunnel -- unencrypted, and invisible to every test that watches the overlay.

Setup (suite) Open All Routers

Acceptance criteria

- output does not contain **${LAN_NET}[${peer}]** (checked 9×)

  would put site traffic on the underlay instead of the tunnel
- output contains **${WAN_NET}** (checked 3×)

How it was checked

R1 show ip route | include GigabitEthernet2

```
C        100.64.0.0/24 is directly connected, GigabitEthernet2
L        100.64.0.1/32 is directly connected, GigabitEthernet2
```

R2 show ip route | include GigabitEthernet2

```
C        100.64.0.0/24 is directly connected, GigabitEthernet2
L        100.64.0.2/32 is directly connected, GigabitEthernet2
```

R3 show ip route | include GigabitEthernet2

```
C        100.64.0.0/24 is directly connected, GigabitEthernet2
L        100.64.0.3/32 is directly connected, GigabitEthernet2
```

Teardown (suite) Close All Connections

3.06 PASS No Routing Protocol Runs On The WAN Segment (0.1s, 3 commands)

BGP peers over the tunnel, so the underlay should carry no adjacency at all. One here would be a second, unprotected path for the control plane.

Setup (suite) Open All Routers

Acceptance criteria

- output does not contain **${WAN_NET.rsplit('.', 2)[0]}.** (checked 3×)

How it was checked

R1 show bgp ipv4 unicast summary | begin Neighbor

```
Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
172.20.0.2      4        65002    3439    3446       34    0    0 09:05:04        3
172.20.0.3      4        65003    3448    3453       34    0    0 09:06:06        3
```

R2 show bgp ipv4 unicast summary | begin Neighbor

```
Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
172.20.0.1      4        65001    3446    3439       46    0    0 09:05:04        6
```

R3 show bgp ipv4 unicast summary | begin Neighbor

```
Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
172.20.0.1      4        65001    3453    3448      490    0    0 09:06:06        6
```

Teardown (suite) Close All Connections

04 Dmvpn

4.01 PASS Every Router Runs One Multipoint Tunnel With No Destination (0.2s, 4 commands)

The structural difference from a static VTI. A configured destination would make this point-to-point again and there would be nothing for NHRP to resolve.

Setup (suite) Open All Routers

Acceptance criteria

- output contains **tunnel mode gre multipoint** (checked 3×)
- output does not contain **tunnel destination** (checked 3×)
- output contains **tunnel source GigabitEthernet2** (checked 3×)
- output contains **ip address ${DMVPN_IP}[${r}]** (checked 3×)
- output contains **Number of lines which match regexp = 1**

the hub has more than one tunnel interface, which is the VTI shape

How it was checked

R1 show running-config interface Tunnel0

```
Building configuration...

Current configuration : 399 bytes
!
interface Tunnel0
 description DMVPN hub
 ip address 172.20.0.1 255.255.255.0
 no ip redirects
 ip nat outside
 ip nhrp authentication LabNhrp
 ip nhrp network-id 100
 ip nhrp holdtime 300
 ip nhrp redirect
 ip tcp adjust-mss 1360
... 6 more lines
```

R2 show running-config interface Tunnel0

```
Building configuration...

Current configuration : 433 bytes
!
interface Tunnel0
 description DMVPN spoke
 ip address 172.20.0.2 255.255.255.0
 no ip redirects
 ip nat outside
 ip nhrp authentication LabNhrp
 ip nhrp network-id 100
 ip nhrp holdtime 300
 ip nhrp nhs 172.20.0.1 nbma 100.64.0.1 multicast
 ip tcp adjust-mss 1360
... 6 more lines
```

R3 show running-config interface Tunnel0

```
Building configuration...

Current configuration : 433 bytes
!
interface Tunnel0
 description DMVPN spoke
 ip address 172.20.0.3 255.255.255.0
 no ip redirects
 ip nat outside
 ip nhrp authentication LabNhrp
 ip nhrp network-id 100
 ip nhrp holdtime 300
 ip nhrp nhs 172.20.0.1 nbma 100.64.0.1 multicast
 ip tcp adjust-mss 1360
... 6 more lines
```

R1 show ip interface brief | count Tunnel

```
Number of lines which match regexp = 1
```

Teardown (suite) Restore Nhrp, Close All Connections

4.02 PASS The Tunnel Is Protected By IPsec On Every Router (0.1s, 3 commands)

Setup (suite) Open All Routers

Acceptance criteria

- output contains **tunnel protection ipsec profile** (checked 3×)

How it was checked

R1 show running-config interface Tunnel0

```
Building configuration...

Current configuration : 399 bytes
!
interface Tunnel0
 description DMVPN hub
 ip address 172.20.0.1 255.255.255.0
 no ip redirects
 ip nat outside
 ip nhrp authentication LabNhrp
 ip nhrp network-id 100
 ip nhrp holdtime 300
 ip nhrp redirect
 ip tcp adjust-mss 1360
... 6 more lines
```

R2 show running-config interface Tunnel0

```
Building configuration...

Current configuration : 433 bytes
!
interface Tunnel0
 description DMVPN spoke
 ip address 172.20.0.2 255.255.255.0
 no ip redirects
 ip nat outside
 ip nhrp authentication LabNhrp
 ip nhrp network-id 100
 ip nhrp holdtime 300
 ip nhrp nhs 172.20.0.1 nbma 100.64.0.1 multicast
 ip tcp adjust-mss 1360
... 6 more lines
```

R3 show running-config interface Tunnel0

```
Building configuration...

Current configuration : 433 bytes
!
interface Tunnel0
 description DMVPN spoke
 ip address 172.20.0.3 255.255.255.0
 no ip redirects
 ip nat outside
 ip nhrp authentication LabNhrp
 ip nhrp network-id 100
 ip nhrp holdtime 300
 ip nhrp nhs 172.20.0.1 nbma 100.64.0.1 multicast
 ip tcp adjust-mss 1360
... 6 more lines
```

Teardown (suite) Restore Nhrp, Close All Connections

4.03 PASS NHRP Shares A Network ID And Authentication Everywhere (0.1s, 3 commands)

Both must match across the cloud; a mismatch in either leaves registrations silently unanswered.

Setup (suite) Open All Routers

Acceptance criteria

- output contains **ip nhrp network-id ${NHRP_NETWORK_ID}** (checked 3×)
- output contains **ip nhrp authentication ${NHRP_AUTH}** (checked 3×)

How it was checked

R1 show running-config interface Tunnel0

```
Building configuration...

Current configuration : 399 bytes
!
```

```
interface Tunnel0
 description DMVPN hub
 ip address 172.20.0.1 255.255.255.0
 no ip redirects
 ip nat outside
 ip nhrp authentication LabNhrp
 ip nhrp network-id 100
 ip nhrp holdtime 300
 ip nhrp redirect
 ip tcp adjust-mss 1360
... 6 more lines
```

R2 show running-config interface Tunnel0

```
Building configuration...

Current configuration : 433 bytes
!
interface Tunnel0
 description DMVPN spoke
 ip address 172.20.0.2 255.255.255.0
 no ip redirects
 ip nat outside
 ip nhrp authentication LabNhrp
 ip nhrp network-id 100
 ip nhrp holdtime 300
 ip nhrp nhs 172.20.0.1 nbma 100.64.0.1 multicast
 ip tcp adjust-mss 1360
... 6 more lines
```

R3 show running-config interface Tunnel0

```
Building configuration...

Current configuration : 433 bytes
!
interface Tunnel0
 description DMVPN spoke
 ip address 172.20.0.3 255.255.255.0
 no ip redirects
 ip nat outside
 ip nhrp authentication LabNhrp
 ip nhrp network-id 100
 ip nhrp holdtime 300
 ip nhrp nhs 172.20.0.1 nbma 100.64.0.1 multicast
 ip tcp adjust-mss 1360
... 6 more lines
```

Teardown (suite) Restore Nhrp, Close All Connections

4.04 PASS The Hub Is The Next Hop Server And The Spokes Point At It (0.1s, 4 commands)

Setup (suite) Open All Routers

Acceptance criteria

- output contains **ip nhrp nhs ${R1_DMVPN_IP} nbma ${R1_WAN_IP}** (checked 2×)
- output does not contain **ip nhrp nhs**

  the hub points at a next hop server, so it is behaving as a spoke
- output matches **${WAN_IP}[${r}]\s+${DMVPN_IP}[${r}]\s+UP** (checked 2×)
- **${dynamic}** has exactly **2** entries

  the hub does not show two dynamically registered peers

How it was checked

R2 show running-config interface Tunnel0

```
Building configuration...

Current configuration : 433 bytes
!
interface Tunnel0
 description DMVPN spoke
 ip address 172.20.0.2 255.255.255.0
 no ip redirects
 ip nat outside
 ip nhrp authentication LabNhrp
 ip nhrp network-id 100
 ip nhrp holdtime 300
 ip nhrp nhs 172.20.0.1 nbma 100.64.0.1 multicast
 ip tcp adjust-mss 1360
... 6 more lines
```

R3 show running-config interface Tunnel0

```
Building configuration...

Current configuration : 433 bytes
!
interface Tunnel0
 description DMVPN spoke
 ip address 172.20.0.3 255.255.255.0
 no ip redirects
 ip nat outside
 ip nhrp authentication LabNhrp
 ip nhrp network-id 100
 ip nhrp holdtime 300
 ip nhrp nhs 172.20.0.1 nbma 100.64.0.1 multicast
 ip tcp adjust-mss 1360
... 6 more lines
```

R1 show running-config interface Tunnel0

```
Building configuration...

Current configuration : 399 bytes
!
interface Tunnel0
 description DMVPN hub
 ip address 172.20.0.1 255.255.255.0
 no ip redirects
 ip nat outside
 ip nhrp authentication LabNhrp
 ip nhrp network-id 100
 ip nhrp holdtime 300
 ip nhrp redirect
 ip tcp adjust-mss 1360
... 6 more lines
```

R1 show dmvpn

```
Legend: Attrb --> S - Static, D - Dynamic, I - Incomplete
        N - NATed, L - Local, X - No Socket
        T1 - Route Installed, T2 - Nexthop-override, B - BGP
        C - CTS Capable, I2 - Temporary
        # Ent --> Number of NHRP entries with same NBMA peer
        NHS Status: E --> Expecting Replies, R --> Responding, W --> Waiting
        UpDn Time --> Up or Down Time for a Tunnel
==========================================================================

Interface: Tunnel0, IPv4 NHRP Details
Type:Hub, NHRP Peers:2,

 # Ent  Peer NBMA Addr Peer Tunnel Add State  UpDn Tm Attrb
 ----- --------------- --------------- ----- -------- -----
... 2 more lines
```

Teardown (suite) Restore Nhrp, Close All Connections

4.05 PASS This Is Phase 3, Not Phase 1 Or 2 (0.1s, 3 commands)

Redirect on the hub and shortcut on the spokes are what make the overlay reorganise itself. Without them the tests below about direct paths cannot pass, and the design is a different one.

Setup (suite) Open All Routers

Acceptance criteria

- output contains **ip nhrp redirect**

  the hub sends no NHRP redirects
- output contains **ip nhrp nhs** (checked 2×)

How it was checked

R1 show running-config interface Tunnel0

```
Building configuration...

Current configuration : 399 bytes
!
interface Tunnel0
 description DMVPN hub
 ip address 172.20.0.1 255.255.255.0
 no ip redirects
 ip nat outside
 ip nhrp authentication LabNhrp
```

```
  ip nhrp network-id 100
  ip nhrp holdtime 300
  ip nhrp redirect
  ip tcp adjust-mss 1360
 ... 6 more lines
```

R2 show running-config interface Tunnel0

```
 Building configuration...

 Current configuration : 433 bytes
 !
 interface Tunnel0
  description DMVPN spoke
  ip address 172.20.0.2 255.255.255.0
  no ip redirects
  ip nat outside
  ip nhrp authentication LabNhrp
  ip nhrp network-id 100
  ip nhrp holdtime 300
  ip nhrp nhs 172.20.0.1 nbma 100.64.0.1 multicast
  ip tcp adjust-mss 1360
 ... 6 more lines
```

R3 show running-config interface Tunnel0

```
 Building configuration...

 Current configuration : 433 bytes
 !
 interface Tunnel0
  description DMVPN spoke
  ip address 172.20.0.3 255.255.255.0
  no ip redirects
  ip nat outside
  ip nhrp authentication LabNhrp
  ip nhrp network-id 100
  ip nhrp holdtime 300
  ip nhrp nhs 172.20.0.1 nbma 100.64.0.1 multicast
  ip tcp adjust-mss 1360
 ... 6 more lines
```

Teardown (suite) Restore Nhrp, Close All Connections

4.06 PASS Both Spokes Register With The Hub (0.0s, 1 commands)

Registration is dynamic: the hub is configured with no spoke addresses at all and learns them as they arrive.

Setup (suite) Open All Routers

Acceptance criteria

- output matches **${WAN_IP}[${r}]\s+${DMVPN_IP}[${r}]\s+UP** (checked 2×)
- **${dynamic}** has exactly **2** entries

  the hub does not show two dynamically registered peers

How it was checked

R1 show dmvpn

```
 Legend: Attrb --> S - Static, D - Dynamic, I - Incomplete
         N - NATed, L - Local, X - No Socket
         T1 - Route Installed, T2 - Nexthop-override, B - BGP
         C - CTS Capable, I2 - Temporary
         # Ent --> Number of NHRP entries with same NBMA peer
         NHS Status: E --> Expecting Replies, R --> Responding, W --> Waiting
         UpDn Time --> Up or Down Time for a Tunnel
 ==========================================================================

 Interface: Tunnel0, IPv4 NHRP Details
 Type:Hub, NHRP Peers:2,

  # Ent  Peer NBMA Addr Peer Tunnel Add State  UpDn Tm Attrb
  ----- --------------- --------------- ----- -------- -----
 ... 2 more lines
```

Teardown (suite) Restore Nhrp, Close All Connections

4.07 PASS The Hub Maps Each Spoke's Tunnel Address To Its NBMA Address (0.0s, 1 commands)

Setup (suite) Open All Routers

Acceptance criteria

- output matches **(?s)${DMVPN_IP}[${r}]/32.*?${WAN_IP}[${r}]** (checked 2×)
  the hub has no NBMA mapping for ${r}
- output contains **registered**

How it was checked

R1 show ip nhrp
```
172.20.0.2/32 via 172.20.0.2
   Tunnel0 created 09:01:34, expire 00:04:50
   Type: dynamic, Flags: registered nhop bfd
   NBMA address: 100.64.0.2
172.20.0.3/32 via 172.20.0.3
   Tunnel0 created 09:01:34, expire 00:04:39
   Type: dynamic, Flags: registered nhop bfd
   NBMA address: 100.64.0.3
```

Teardown (suite) Restore Nhrp, Close All Connections

4.08 PASS Every DMVPN Peering Is Encrypted (0.1s, 2 commands)

One IPsec profile protects the multipoint tunnel, so each peering formed over it -- including ones the configuration never named -- is protected by construction.

Setup (suite) Open All Routers

Acceptance criteria

- output contains **${R1_WAN_IP}** (checked 2×)
- output matches **${R1_WAN_IP}\s+Tu0\s+.*UA** (checked 2×)

How it was checked

R2 show crypto session brief
```
Status: A- Active, U - Up, D - Down, I - Idle, S - Standby, N - Negotiating
        K - No IKE
ivrf = (none)
Peer            I/F          Username        Group/Phase1_id          Uptime   Status
100.64.0.1      Tu0                          100.64.0.1               09:01:34 UA
```

R3 show crypto session brief
```
Status: A- Active, U - Up, D - Down, I - Idle, S - Standby, N - Negotiating
        K - No IKE
ivrf = (none)
Peer            I/F          Username        Group/Phase1_id          Uptime   Status
100.64.0.1      Tu0                          100.64.0.1               09:01:35 UA
```

Teardown (suite) Restore Nhrp, Close All Connections

4.09 PASS A Spoke Learns The Far Spoke As The BGP Next Hop (0.0s, 2 commands)

The routing half of Phase 3, and the reason "next-hop-unchanged" is configured on the hub. If the hub rewrote the next hop to itself, every spoke-to-spoke prefix would resolve to the hub, NHRP would never be asked to find the far spoke, and no shortcut could form.

Setup (suite) Open All Routers

Acceptance criteria

- output contains **${R3_DMVPN_IP}**
  R2 does not see R3's tunnel address as the next hop for R3's prefix
- output contains **${R2_DMVPN_IP}**
  R3 does not see R2's tunnel address as the next hop for R2's prefix

How it was checked

R2 show bgp ipv4 unicast 10.20.3.1
```
BGP routing table entry for 10.20.3.1/32, version 43
Paths: (1 available, best #1, table default)
  Not advertised to any peer
  Refresh Epoch 1
  65001 65003
    172.20.0.3 from 172.20.0.1 (172.16.1.1)
```

```
        Origin IGP, localpref 100, valid, external, best
        Community: 65003:1
        rx pathid: 0, tx pathid: 0x0
        Updated on Sep 8 2026 15:48:47 UTC
```

R3 show bgp ipv4 unicast 10.20.2.1

```
  BGP routing table entry for 10.20.2.1/32, version 488
  Paths: (1 available, best #1, table default)
    Not advertised to any peer
    Refresh Epoch 1
    65001 65002
      172.20.0.2 from 172.20.0.1 (172.16.1.1)
        Origin IGP, localpref 100, valid, external, best
        Community: 65002:1
        rx pathid: 0, tx pathid: 0x0
        Updated on Sep 8 2026 15:49:16 UTC
```

Teardown (suite) Restore Nhrp, Close All Connections

4.10 PASS Spoke To Spoke Traffic Works Before Any Shortcut Exists (12.1s, 4 commands)

The fallback path. With the NHRP cache cleared there is no direct entry, so the first packets go through the hub -- which is what makes the shortcut an optimisation rather than a prerequisite for connectivity.

Setup (suite) Open All Routers

Acceptance criteria

- output does not contain **${WAN_IP}[${peer}]**
- output contains **Success rate is**
- **${loss}[0] > 0** holds

How it was checked

R2 clear ip nhrp

```
  ${output} =
```

R3 clear ip nhrp

```
  ${output} =
```

R2 show ip nhrp 172.20.0.3

```
  ${output} =
```

R2 ping 10.20.3.1 source Loopback1 repeat 30

```
  Type escape sequence to abort.
  Sending 30, 100-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:
  Packet sent with a source address of 10.20.2.1
  .!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
  Success rate is 96 percent (29/30), round-trip min/avg/max = 1/1/2 ms
```

Teardown (suite) Restore Nhrp, Close All Connections

4.11 PASS A Shortcut Forms On Demand (12.2s, 6 commands)

The defining behaviour. Cleared first so the entry cannot be left over from an earlier test: it has to be created by this traffic, here.

Setup (suite) Open All Routers

Acceptance criteria

- output does not contain **${WAN_IP}[${peer}]**
- output contains **${WAN_IP}[${peer}]** (checked 2×)

How it was checked

R2 clear ip nhrp

```
  ${output} =
```

R3 clear ip nhrp

```
  ${output} =
```

R2 show ip nhrp 172.20.0.3

```
  ${output} =
```

R2 ping 10.20.3.1 source Loopback1 repeat 20

```
  Type escape sequence to abort.
  Sending 20, 100-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:
  Packet sent with a source address of 10.20.2.1
  .!!!!!!!!!!!!!!!!!!!
  Success rate is 95 percent (19/20), round-trip min/avg/max = 1/1/2 ms
```

R2 show ip nhrp 172.20.0.3
```
172.20.0.3/32 via 172.20.0.3
   Tunnel0 created 00:00:02, expire 00:04:58
   Type: dynamic, Flags: router nhop bfd
   NBMA address: 100.64.0.3
```
R3 show ip nhrp 172.20.0.2
```
172.20.0.2/32 via 172.20.0.2
   Tunnel0 created 00:00:02, expire 00:04:57
   Type: dynamic, Flags: router implicit nhop bfd
   NBMA address: 100.64.0.2
```
Teardown (suite) Restore Nhrp, Close All Connections

4.12 PASS Traffic Then Leaves The Hub Out Of The Path (0.1s, 2 commands)

The point of the shortcut, and the strongest statement this suite makes: with the direct path installed, a trace from one spoke to the other reaches it in a single hop instead of transiting the hub.

Setup (suite) Open All Routers

Acceptance criteria

- output contains **${WAN_IP}[${peer}]**
- output contains **${R3_DMVPN_IP}**
- output does not contain **${R1_DMVPN_IP}**

  the trace still passes through the hub, so no shortcut is in use:\n${trace}
- **${hops}[-1]** equals **1**

  the far spoke is ${hops}[-1] hops away, expected 1

How it was checked

R2 show ip nhrp 172.20.0.3
```
172.20.0.3/32 via 172.20.0.3
   Tunnel0 created 00:00:02, expire 00:04:57
   Type: dynamic, Flags: router nhop bfd
   NBMA address: 100.64.0.3
```
R2 traceroute 10.20.3.1 source Loopback1 probe 1 timeout 1 numeric
```
Type escape sequence to abort.
Tracing the route to 10.20.3.1
VRF info: (vrf in name/id, vrf out name/id)
  1 172.20.0.3 1 msec
```
Teardown (suite) Restore Nhrp, Close All Connections

4.13 PASS The Shortcut Gets Its Own IPsec Session (0.0s, 2 commands)

A direct path between two routers that were never configured with each other's addresses, protected all the same, because the profile is bound to the tunnel rather than to a peer.

Setup (suite) Open All Routers

Acceptance criteria

- output contains **${WAN_IP}[${peer}]**
- output contains **${R3_WAN_IP}**

  R2 has no IPsec session with R3's NBMA address:\n${sess}
- **${count}** has exactly **2** entries

  R2 holds ${count} tunnel sessions, expected one to the hub and one to R3

How it was checked

R2 show ip nhrp 172.20.0.3
```
172.20.0.3/32 via 172.20.0.3
   Tunnel0 created 00:00:02, expire 00:04:57
   Type: dynamic, Flags: router nhop bfd
   NBMA address: 100.64.0.3
```
R2 show crypto session brief
```
Status: A- Active, U - Up, D - Down, I - Idle, S - Standby, N - Negotiating
        K - No IKE
ivrf = (none)
Peer            I/F          Username        Group/Phase1_id          Uptime   Status
100.64.0.3      Tu0                          100.64.0.3               00:00:02 UA
```

```
100.64.0.1      Tu0                       100.64.0.1          09:01:58 UA
```

Teardown (suite) Restore Nhrp, Close All Connections

05 Capture State

5.01 PASS Capture Running Config From Every Router (12.6s, 3 commands)

Setup (suite) Open All Routers, Open All Hosts, Open Nms

Acceptance criteria

- output contains **hostname** (checked 3×)
- output matches **(?m)^end\s*$** (checked 3×)

How it was checked

R1 show running-config

```
Building configuration...

Current configuration : 21292 bytes
!
! Last configuration change at 00:53:51 UTC Wed Sep 9 2026 by lab
!
version 17.15
service timestamps debug datetime msec
service timestamps log datetime msec show-timezone
service sequence-numbers
platform qfp utilization monitor load 80
platform sslvpn use-pd
platform console serial
!
... 624 more lines
```

R2 show running-config

```
Building configuration...

Current configuration : 19824 bytes
!
! Last configuration change at 00:53:31 UTC Wed Sep 9 2026 by lab
!
version 17.15
service timestamps debug datetime msec
service timestamps log datetime msec show-timezone
service sequence-numbers
platform qfp utilization monitor load 80
platform sslvpn use-pd
platform console serial
!
... 579 more lines
```

R3 show running-config

```
Building configuration...

Current configuration : 19832 bytes
!
! Last configuration change at 00:53:31 UTC Wed Sep 9 2026 by lab
!
version 17.15
service timestamps debug datetime msec
service timestamps log datetime msec show-timezone
service sequence-numbers
platform qfp utilization monitor load 80
platform sslvpn use-pd
platform console serial
!
... 579 more lines
```

Teardown (suite) Close Nms, Close All Connections

5.02 PASS Capture Routing Table From Every Router (0.2s, 3 commands)

Setup (suite) Open All Routers, Open All Hosts, Open Nms

Acceptance criteria

- output contains **Gateway of last resort** (checked 3×)
- output contains **Codes:** (checked 3×)

How it was checked

R1 show ip route

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
       n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       H - NHRP, G - NHRP registered, g - NHRP registration summary
       o - ODR, P - periodic downloaded static route, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR
       & - replicated local route overrides by connected

Gateway of last resort is 10.0.2.2 to network 0.0.0.0
... 24 more lines
```

R2 show ip route

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
       n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       H - NHRP, G - NHRP registered, g - NHRP registration summary
       o - ODR, P - periodic downloaded static route, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR
       & - replicated local route overrides by connected

Gateway of last resort is 10.0.2.2 to network 0.0.0.0
... 24 more lines
```

R3 show ip route

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
       n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       H - NHRP, G - NHRP registered, g - NHRP registration summary
       o - ODR, P - periodic downloaded static route, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR
       & - replicated local route overrides by connected

Gateway of last resort is 10.0.2.2 to network 0.0.0.0
... 24 more lines
```

Teardown (suite) Close Nms, Close All Connections

5.03 PASS Capture BGP Table And Neighbours From Every Router (0.2s, 6 commands)

Setup (suite) Open All Routers, Open All Hosts, Open Nms

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked

R1 show bgp ipv4 unicast

```
BGP table version is 34, local router ID is 172.16.1.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
              t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *>   10.20.1.1/32     0.0.0.0                  0         32768 i
 *>   10.20.2.1/32     172.20.0.2               0             0 65002 i
 *>   10.20.3.1/32     172.20.0.3               0             0 65003 i
 *>   10.30.1.0/24     0.0.0.0                  0         32768 i
 *>   10.30.2.0/24     172.20.0.2               0             0 65002 i
... 4 more lines
```

R1 show bgp ipv4 unicast summary

```
BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 34, main routing table version 34
9 network entries using 2232 bytes of memory
```

```
9 path entries using 1224 bytes of memory
7/7 BGP path/bestpath attribute entries using 2072 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
6 BGP community entries using 144 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 5720 total bytes of memory
BGP activity 303/294 prefixes, 324/315 paths, scan interval 60 secs
22 networks peaked at 02:27:59 Sep 8 2026 UTC (22:26:44.598 ago)

Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
... 2 more lines
```

R2 show bgp ipv4 unicast

```
BGP table version is 46, local router ID is 172.16.2.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
              t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *>   10.20.1.1/32     172.20.0.1               0             0 65001 i
 *>   10.20.2.1/32     0.0.0.0                  0         32768 i
 *>   10.20.3.1/32     172.20.0.3                             0 65001 65003 i
 *>   10.30.1.0/24     172.20.0.1               0             0 65001 i
 *>   10.30.2.0/24     0.0.0.0                  0         32768 i
... 4 more lines
```

R2 show bgp ipv4 unicast summary

```
BGP router identifier 172.16.2.1, local AS number 65002
BGP table version is 46, main routing table version 46
9 network entries using 2232 bytes of memory
9 path entries using 1224 bytes of memory
7/7 BGP path/bestpath attribute entries using 2072 bytes of memory
2 BGP AS-PATH entries using 64 bytes of memory
6 BGP community entries using 144 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 5736 total bytes of memory
BGP activity 333/324 prefixes, 348/339 paths, scan interval 60 secs
31 networks peaked at 02:28:40 Sep 8 2026 UTC (22:26:03.670 ago)

Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
... 1 more lines
```

R3 show bgp ipv4 unicast

```
BGP table version is 490, local router ID is 172.16.3.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
              t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *>   10.20.1.1/32     172.20.0.1               0             0 65001 i
 *>   10.20.2.1/32     172.20.0.2                             0 65001 65002 i
 *>   10.20.3.1/32     0.0.0.0                  0         32768 i
 *>   10.30.1.0/24     172.20.0.1               0             0 65001 i
 *>   10.30.2.0/24     172.20.0.2                             0 65001 65002 i
... 4 more lines
```

R3 show bgp ipv4 unicast summary

```
BGP router identifier 172.16.3.1, local AS number 65003
BGP table version is 490, main routing table version 490
9 network entries using 2232 bytes of memory
9 path entries using 1224 bytes of memory
7/7 BGP path/bestpath attribute entries using 2072 bytes of memory
2 BGP AS-PATH entries using 64 bytes of memory
6 BGP community entries using 144 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 5736 total bytes of memory
BGP activity 240/231 prefixes, 249/240 paths, scan interval 60 secs
9 networks peaked at 01:35:13 Sep 8 2026 UTC (23:19:30.752 ago)

Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
... 1 more lines
```

Teardown (suite) Close Nms, Close All Connections

5.04 PASS Capture Crypto State From Every Router (0.3s, 6 commands)

Setup (suite) Open All Routers, Open All Hosts, Open Nms

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked

R1 show crypto ipsec sa

```
interface: Tunnel0
    Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
   remote ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
   current_peer 100.64.0.2 port 500
     PERMIT, flags={origin_is_acl,}
    #pkts encaps: 13824, #pkts encrypt: 13824, #pkts digest: 13824
    #pkts decaps: 13835, #pkts decrypt: 13835, #pkts verify: 13835
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0
... 78 more lines
```

R1 show crypto ikev2 sa

```
IPv4 Crypto IKEv2  SA

Tunnel-id Local                 Remote                fvrf/ivrf            Status
1         100.64.0.1/500        100.64.0.3/500        none/none            READY
      Encr: AES-CBC, keysize: 256, PRF: SHA256, Hash: SHA256, DH Grp:14, Auth sign: PSK, Auth verify: PSK
      Life/Active Time: 86400/32545 sec
      CE id: 1154, Session-id: 149
      Local spi: 552EA031EC5AD516       Remote spi: DF541840554A4190

Tunnel-id Local                 Remote                fvrf/ivrf            Status
2         100.64.0.1/500        100.64.0.2/500        none/none            READY
      Encr: AES-CBC, keysize: 256, PRF: SHA256, Hash: SHA256, DH Grp:14, Auth sign: PSK, Auth verify: PSK
      Life/Active Time: 86400/32544 sec
      CE id: 1155, Session-id: 150
... 3 more lines
```

R2 show crypto ipsec sa

```
interface: Tunnel0
    Crypto map tag: Tunnel0-head-0, local addr 100.64.0.2

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
   remote ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
   current_peer 100.64.0.1 port 500
     PERMIT, flags={origin_is_acl,}
    #pkts encaps: 192424, #pkts encrypt: 192424, #pkts digest: 192424
    #pkts decaps: 192397, #pkts decrypt: 192397, #pkts verify: 192397
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0
... 33 more lines
```

R2 show crypto ikev2 sa

```
IPv4 Crypto IKEv2  SA

Tunnel-id Local                 Remote                fvrf/ivrf            Status
1         100.64.0.2/500        100.64.0.1/500        none/none            READY
      Encr: AES-CBC, keysize: 256, PRF: SHA256, Hash: SHA256, DH Grp:14, Auth sign: PSK, Auth verify: PSK
      Life/Active Time: 86400/32545 sec
      CE id: 1156, Session-id: 141
      Local spi: E99229818F7EF5EB       Remote spi: FA434BF0F1020BEE

 IPv6 Crypto IKEv2  SA
```

R3 show crypto ipsec sa

```
interface: Tunnel0
    Crypto map tag: Tunnel0-head-0, local addr 100.64.0.3

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
   remote ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
   current_peer 100.64.0.1 port 500
     PERMIT, flags={origin_is_acl,}
    #pkts encaps: 192390, #pkts encrypt: 192390, #pkts digest: 192390
    #pkts decaps: 192128, #pkts decrypt: 192128, #pkts verify: 192128
    #pkts compressed: 0, #pkts decompressed: 0
```

```
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0
... 33 more lines
```

R3 show crypto ikev2 sa

```
IPv4 Crypto IKEv2  SA

Tunnel-id Local                 Remote                fvrf/ivrf            Status
1         100.64.0.3/500        100.64.0.1/500        none/none            READY
      Encr: AES-CBC, keysize: 256, PRF: SHA256, Hash: SHA256, DH Grp:14, Auth sign: PSK, Auth verify: PSK
      Life/Active Time: 86400/32546 sec
      CE id: 1150, Session-id: 145
      Local spi: DF541840554A4190       Remote spi: 552EA031EC5AD516

 IPv6 Crypto IKEv2  SA
```

Teardown (suite) Close Nms, Close All Connections

5.05 PASS Capture NAT State From Every Router (0.5s, 6 commands)

Setup (suite) Open All Routers, Open All Hosts, Open Nms

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked

R1 show ip nat translations

```
Pro  Inside global         Inside local          Outside local         Outside global
---  10.30.1.128           192.168.10.88         ---                   ---
---  10.30.1.129           192.168.10.67         ---                   ---
---  10.30.1.113           192.168.10.72         ---                   ---
---  10.30.1.114           192.168.10.80         ---                   ---
---  10.30.1.116           192.168.10.87         ---                   ---
---  10.30.1.119           192.168.10.85         ---                   ---
---  10.30.1.123           192.168.10.71         ---                   ---
---  10.30.1.120           192.168.10.93         ---                   ---
---  10.30.1.126           192.168.10.75         ---                   ---
---  10.30.1.125           192.168.10.84         ---                   ---
---  10.30.1.101           192.168.10.91         ---                   ---
---  10.30.1.102           192.168.10.78         ---                   ---
---  10.30.1.111           192.168.10.70         ---                   ---
... 28 more lines
```

R1 show ip nat statistics

```
Total active translations: 40 (10 static, 30 dynamic; 0 extended)
Outside interfaces:
  Tunnel0
Inside interfaces:
  GigabitEthernet3
Hits: 457  Misses: 123
 Reserved port setting disabled provisioned no
 Dynamic overload mapping configured: 0
Expired translations: 123
Dynamic mappings:
-- Inside Source
[Id: 2] access-list NAT-POOL-SRC pool SITE-POOL refcount 30
 pool SITE-POOL: id 2, netmask 255.255.255.0
       start 10.30.1.100 end 10.30.1.129
... 9 more lines
```

R2 show ip nat translations

```
Pro  Inside global         Inside local          Outside local         Outside global
---  10.30.2.128           192.168.20.80         ---                   ---
---  10.30.2.129           192.168.20.75         ---                   ---
---  10.30.2.106           192.168.20.81         ---                   ---
---  10.30.2.105           192.168.20.74         ---                   ---
---  10.30.2.111           192.168.20.89         ---                   ---
---  10.30.2.108           192.168.20.82         ---                   ---
---  10.30.2.101           192.168.20.78         ---                   ---
---  10.30.2.102           192.168.20.90         ---                   ---
---  10.30.2.126           192.168.20.77         ---                   ---
---  10.30.2.125           192.168.20.79         ---                   ---
---  10.30.2.123           192.168.20.87         ---                   ---
---  10.30.2.120           192.168.20.68         ---                   ---
---  10.30.2.116           192.168.20.93         ---                   ---
... 28 more lines
```

R2 show ip nat statistics

```
Total active translations: 40 (10 static, 30 dynamic; 0 extended)
Outside interfaces:
```

```
    Tunnel0
  Inside interfaces:
    GigabitEthernet3
  Hits: 337  Misses: 83
   Reserved port setting disabled provisioned no
   Dynamic overload mapping configured: 0
  Expired translations: 83
  Dynamic mappings:
  -- Inside Source
  [Id: 2] access-list NAT-POOL-SRC pool SITE-POOL refcount 30
   pool SITE-POOL: id 2, netmask 255.255.255.0
          start 10.30.2.100 end 10.30.2.129
  ... 9 more lines
```

R3 show ip nat translations

```
  Pro  Inside global         Inside local          Outside local         Outside global
  ---  10.30.3.104           192.168.30.93         ---                   ---
  ---  10.30.3.107           192.168.30.89         ---                   ---
  ---  10.30.3.109           192.168.30.92         ---                   ---
  ---  10.30.3.110           192.168.30.91         ---                   ---
  ---  10.30.3.103           192.168.30.86         ---                   ---
  ---  10.30.3.100           192.168.30.90         ---                   ---
  ---  10.30.3.124           192.168.30.84         ---                   ---
  ---  10.30.3.127           192.168.30.82         ---                   ---
  ---  10.30.3.121           192.168.30.85         ---                   ---
  ---  10.30.3.122           192.168.30.78         ---                   ---
  ---  10.30.3.118           192.168.30.76         ---                   ---
  ---  10.30.3.117           192.168.30.80         ---                   ---
  ---  10.30.3.115           192.168.30.73         ---                   ---
  ... 28 more lines
```

R3 show ip nat statistics

```
  Total active translations: 40 (10 static, 30 dynamic; 0 extended)
  Outside interfaces:
    Tunnel0
  Inside interfaces:
    GigabitEthernet3
  Hits: 120  Misses: 40
   Reserved port setting disabled provisioned no
   Dynamic overload mapping configured: 0
  Expired translations: 40
  Dynamic mappings:
  -- Inside Source
  [Id: 2] access-list NAT-POOL-SRC pool SITE-POOL refcount 30
   pool SITE-POOL: id 2, netmask 255.255.255.0
          start 10.30.3.100 end 10.30.3.129
  ... 9 more lines
```

Teardown (suite) Close Nms, Close All Connections

5.06 PASS Capture NTP State From Every Router (0.1s, 6 commands)

Setup (suite) Open All Routers, Open All Hosts, Open Nms

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked

R1 show ntp status

```
  Clock is synchronized, stratum 3, reference is 192.168.99.10
  nominal freq is 250.0000 Hz, actual freq is 250.0020 Hz, precision is 2**10
  ntp uptime is 8400400 (1/100 of seconds), resolution is 4000
  reference time is EE4B201F.E45A1F20 (00:00:31.892 UTC Wed Sep 9 2026)
  clock offset is -11.9011 msec, root delay is 52.69 msec
  root dispersion is 62.22 msec, peer dispersion is 1.05 msec
  loopfilter state is 'CTRL' (Normal Controlled Loop), drift is -0.000008126 s/s
  system poll interval is 1024, last update was 3253 sec ago.
```

R1 show ntp associations

```
  address         ref clock       st   when   poll reach  delay  offset   disp
  *~192.168.99.10   192.207.55.254   2    630   1024   377  0.080 -11.901  1.050
   * sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

R2 show ntp status

```
  Clock is synchronized, stratum 3, reference is 192.168.99.10
  nominal freq is 250.0000 Hz, actual freq is 250.0022 Hz, precision is 2**10
  ntp uptime is 8400400 (1/100 of seconds), resolution is 4000
  reference time is EE4B0A01.03D70A48 (22:26:09.015 UTC Tue Sep 8 2026)
  clock offset is -6.7728 msec, root delay is 51.81 msec
  root dispersion is 138.91 msec, peer dispersion is 1.09 msec
  loopfilter state is 'CTRL' (Normal Controlled Loop), drift is -0.000008817 s/s
```

```
  system poll interval is 1024, last update was 8915 sec ago.
```

R2 show ntp associations

```
  address         ref clock       st   when   poll reach  delay  offset   disp
  *~192.168.99.10   192.207.55.254   2   1041   1024   377  0.065  -6.772  1.090
   * sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

R3 show ntp status

```
  Clock is synchronized, stratum 3, reference is 192.168.99.10
  nominal freq is 250.0000 Hz, actual freq is 250.0020 Hz, precision is 2**10
  ntp uptime is 8400400 (1/100 of seconds), resolution is 4000
  reference time is EE4B21E7.4A3D7170 (00:08:07.290 UTC Wed Sep 9 2026)
  clock offset is -11.8631 msec, root delay is 52.69 msec
  root dispersion is 61.55 msec, peer dispersion is 3.87 msec
  loopfilter state is 'CTRL' (Normal Controlled Loop), drift is -0.000008070 s/s
  system poll interval is 256, last update was 2797 sec ago.
```

R3 show ntp associations

```
  address         ref clock       st   when   poll reach  delay  offset   disp
  *~192.168.99.10   192.207.55.254   2    164    256   377  0.897 -11.863  3.875
   * sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

Teardown (suite) Close Nms, Close All Connections

5.07 PASS Capture Logging State From Every Router (0.2s, 3 commands)

Setup (suite) Open All Routers, Open All Hosts, Open Nms

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked

R1 show logging | include Trap logging|Logging to|Syslog logging

```
  Syslog logging: enabled (0 messages dropped, 3 messages rate-limited, 0 flushes, 0 overruns, xml disabled,
  filtering disabled)
      Trap logging: level informational, 2520 message lines logged
          Logging to 192.168.99.10  (MGMT) (udp port 514, audit disabled,
```

R2 show logging | include Trap logging|Logging to|Syslog logging

```
  Syslog logging: enabled (0 messages dropped, 3 messages rate-limited, 0 flushes, 0 overruns, xml disabled,
  filtering disabled)
      Trap logging: level informational, 3115 message lines logged
          Logging to 192.168.99.10  (MGMT) (udp port 514, audit disabled,
  002785: .Sep  8 15:46:25.039 UTC: %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 192.168.99.10 port 514
  (MGMT) started - reconnection
  002788: .Sep  8 15:46:35.460 UTC: %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 192.168.99.10 port 514
  (MGMT) started - reconnection
```

R3 show logging | include Trap logging|Logging to|Syslog logging

```
  Syslog logging: enabled (0 messages dropped, 3 messages rate-limited, 0 flushes, 0 overruns, xml disabled,
  filtering disabled)
      Trap logging: level informational, 2255 message lines logged
          Logging to 192.168.99.10  (MGMT) (udp port 514, audit disabled,
```

Teardown (suite) Close Nms, Close All Connections

5.08 PASS Capture SNMP State From Every Router (0.2s, 9 commands)

Setup (suite) Open All Routers, Open All Hosts, Open Nms

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked

R1 show snmp user

```
  User name: labmon
  Engine ID: 800000090300525400C80101
  storage-type: nonvolatile        active
  Authentication Protocol: SHA
  Privacy Protocol: AES128
  Group-name: LABGRP
```

R1 show snmp host

```
  Notification host: 192.168.99.10        udp-port: 1162  VRFName: MGMT   type: trap
  user: labmon    security model: v3 priv

  Notification host: 192.168.99.10        udp-port: 162   VRFName: MGMT   type: trap
  user: labmon    security model: v3 priv
```

R1 show snmp engineID

```
  Local SNMP engineID: 800000090300525400C80101
```

```
  Remote Engine ID          IP-addr    Port
```

R2 show snmp user

```
  User name: labmon
  Engine ID: 800000090300525400C80201
  storage-type: nonvolatile        active
  Authentication Protocol: SHA
  Privacy Protocol: AES128
  Group-name: LABGRP
```

R2 show snmp host

```
  Notification host: 192.168.99.10        udp-port: 1162  VRFName: MGMT   type: trap
  user: labmon    security model: v3 priv

  Notification host: 192.168.99.10        udp-port: 162   VRFName: MGMT   type: trap
  user: labmon    security model: v3 priv
```

R2 show snmp engineID

```
  Local SNMP engineID: 800000090300525400C80201
  Remote Engine ID          IP-addr    Port
```

R3 show snmp user

```
  User name: labmon
  Engine ID: 800000090300525400C80301
  storage-type: nonvolatile        active
  Authentication Protocol: SHA
  Privacy Protocol: AES128
  Group-name: LABGRP
```

R3 show snmp host

```
  Notification host: 192.168.99.10        udp-port: 1162  VRFName: MGMT   type: trap
  user: labmon    security model: v3 priv

  Notification host: 192.168.99.10        udp-port: 162   VRFName: MGMT   type: trap
  user: labmon    security model: v3 priv
```

R3 show snmp engineID

```
  Local SNMP engineID: 800000090300525400C80301
  Remote Engine ID          IP-addr    Port
```

Teardown (suite) Close Nms, Close All Connections

5.09 PASS Capture The NMS View Of Every Router (0.5s, 16 commands)

The same system group, but seen over SNMPv3 from the NMS rather than read off the console. Captured next to the CLI output so the two can be compared in the evidence PDF.

Setup (suite) Open All Routers, Open All Hosts, Open Nms

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked

NMS ip -br addr; ip route

```
  lo               UNKNOWN        127.0.0.1/8 ::1/128
  ens3             UP             10.0.2.15/24 metric 100 fec0::5054:ff:febb:101/64 fe80::5054:ff:febb:101/64
  ens4             UP             192.168.10.20/24 fe80::5054:ff:febb:102/64
  ens5             UP             192.168.99.10/24 fe80::5054:ff:febb:103/64
  default via 10.0.2.2 dev ens3 proto dhcp src 10.0.2.15 metric 100
  10.0.2.0/24 dev ens3 proto kernel scope link src 10.0.2.15 metric 100
  10.0.2.2 dev ens3 proto dhcp scope link src 10.0.2.15 metric 100
  10.0.2.3 dev ens3 proto dhcp scope link src 10.0.2.15 metric 100
  10.30.0.0/16 via 192.168.10.1 dev ens4
  192.168.10.0/24 dev ens4 proto kernel scope link src 192.168.10.20
  192.168.20.0/24 via 192.168.10.1 dev ens4
  192.168.30.0/24 via 192.168.10.1 dev ens4
  192.168.99.0/24 dev ens5 proto kernel scope link src 192.168.99.10
```

NMS ip -br addr; ip route

```
  lo               UNKNOWN        127.0.0.1/8 ::1/128
  ens3             UP             10.0.2.15/24 metric 100 fec0::5054:ff:febb:101/64 fe80::5054:ff:febb:101/64
  ens4             UP             192.168.10.20/24 fe80::5054:ff:febb:102/64
  ens5             UP             192.168.99.10/24 fe80::5054:ff:febb:103/64
  default via 10.0.2.2 dev ens3 proto dhcp src 10.0.2.15 metric 100
  10.0.2.0/24 dev ens3 proto kernel scope link src 10.0.2.15 metric 100
  10.0.2.2 dev ens3 proto dhcp scope link src 10.0.2.15 metric 100
  10.0.2.3 dev ens3 proto dhcp scope link src 10.0.2.15 metric 100
  10.30.0.0/16 via 192.168.10.1 dev ens4
  192.168.10.0/24 dev ens4 proto kernel scope link src 192.168.10.20
  192.168.20.0/24 via 192.168.10.1 dev ens4
  192.168.30.0/24 via 192.168.10.1 dev ens4
  192.168.99.0/24 dev ens5 proto kernel scope link src 192.168.99.10
```

NMS snmpget --version 2>&1 | head -1

```
NET-SNMP version: 5.9.4.pre2
```

NMS snmpget --version 2>&1 | head -1

```
NET-SNMP version: 5.9.4.pre2
```

NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 192.168.10.1 1.3.6.1.2.1.1

```
.1.3.6.1.2.1.1.1.0 = STRING: "Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-
UNIVERSALK9-M), Version 17.15.6, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2026 by Cisco Systems, Inc.
Compiled Wed 22-Jul-26 03:39 b"
.1.3.6.1.2.1.1.2.0 = OID: .1.3.6.1.4.1.9.1.3004
.1.3.6.1.2.1.1.3.0 = Timeticks: (8401422) 23:20:14.22
.1.3.6.1.2.1.1.4.0 = ""
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r1.lab.local"
.1.3.6.1.2.1.1.6.0 = ""
.1.3.6.1.2.1.1.7.0 = INTEGER: 78
.1.3.6.1.2.1.1.8.0 = Timeticks: (0) 0:00:00.00
```

NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 192.168.10.1 1.3.6.1.2.1.1

```
.1.3.6.1.2.1.1.1.0 = STRING: "Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-
UNIVERSALK9-M), Version 17.15.6, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2026 by Cisco Systems, Inc.
Compiled Wed 22-Jul-26 03:39 b"
.1.3.6.1.2.1.1.2.0 = OID: .1.3.6.1.4.1.9.1.3004
.1.3.6.1.2.1.1.3.0 = Timeticks: (8401422) 23:20:14.22
.1.3.6.1.2.1.1.4.0 = ""
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r1.lab.local"
.1.3.6.1.2.1.1.6.0 = ""
.1.3.6.1.2.1.1.7.0 = INTEGER: 78
.1.3.6.1.2.1.1.8.0 = Timeticks: (0) 0:00:00.00
```

NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 192.168.10.1 1.3.6.1.2.1.2.2.1.2

```
.1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
.1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
.1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
.1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
.1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"
.1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"
.1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"
.1.3.6.1.2.1.2.2.1.2.8 = STRING: "Loopback1"
.1.3.6.1.2.1.2.2.1.2.9 = STRING: "Tunnel0"
```

NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 192.168.10.1 1.3.6.1.2.1.2.2.1.2

```
.1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
.1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
.1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
.1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
.1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"
.1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"
.1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"
.1.3.6.1.2.1.2.2.1.2.8 = STRING: "Loopback1"
.1.3.6.1.2.1.2.2.1.2.9 = STRING: "Tunnel0"
```

NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 192.168.20.1 1.3.6.1.2.1.1

```
.1.3.6.1.2.1.1.1.0 = STRING: "Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-
UNIVERSALK9-M), Version 17.15.6, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2026 by Cisco Systems, Inc.
Compiled Wed 22-Jul-26 03:39 b"
.1.3.6.1.2.1.1.2.0 = OID: .1.3.6.1.4.1.9.1.3004
.1.3.6.1.2.1.1.3.0 = Timeticks: (8401461) 23:20:14.61
.1.3.6.1.2.1.1.4.0 = ""
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
.1.3.6.1.2.1.1.6.0 = ""
.1.3.6.1.2.1.1.7.0 = INTEGER: 78
.1.3.6.1.2.1.1.8.0 = Timeticks: (0) 0:00:00.00
```

NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 192.168.20.1 1.3.6.1.2.1.1

```
.1.3.6.1.2.1.1.1.0 = STRING: "Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-
UNIVERSALK9-M), Version 17.15.6, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2026 by Cisco Systems, Inc.
Compiled Wed 22-Jul-26 03:39 b"
.1.3.6.1.2.1.1.2.0 = OID: .1.3.6.1.4.1.9.1.3004
.1.3.6.1.2.1.1.3.0 = Timeticks: (8401461) 23:20:14.61
```

```
  .1.3.6.1.2.1.1.4.0 = ""
  .1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
  .1.3.6.1.2.1.1.6.0 = ""
  .1.3.6.1.2.1.1.7.0 = INTEGER: 78
  .1.3.6.1.2.1.1.8.0 = Timeticks: (0) 0:00:00.00
NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 192.168.20.1
1.3.6.1.2.1.2.2.1.2
  .1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
  .1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
  .1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
  .1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
  .1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"
  .1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"
  .1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"
  .1.3.6.1.2.1.2.2.1.2.8 = STRING: "Loopback1"
  .1.3.6.1.2.1.2.2.1.2.9 = STRING: "Tunnel0"
NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 192.168.20.1
1.3.6.1.2.1.2.2.1.2
  .1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
  .1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
  .1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
  .1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
  .1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"
  .1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"
  .1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"
  .1.3.6.1.2.1.2.2.1.2.8 = STRING: "Loopback1"
  .1.3.6.1.2.1.2.2.1.2.9 = STRING: "Tunnel0"
NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 192.168.30.1
1.3.6.1.2.1.1
  .1.3.6.1.2.1.1.1.0 = STRING: "Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-
  UNIVERSALK9-M), Version 17.15.6, RELEASE SOFTWARE (fc2)
  Technical Support: http://www.cisco.com/techsupport
  Copyright (c) 1986-2026 by Cisco Systems, Inc.
  Compiled Wed 22-Jul-26 03:39 b"
  .1.3.6.1.2.1.1.2.0 = OID: .1.3.6.1.4.1.9.1.3004
  .1.3.6.1.2.1.1.3.0 = Timeticks: (8401439) 23:20:14.39
  .1.3.6.1.2.1.1.4.0 = ""
  .1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r3.lab.local"
  .1.3.6.1.2.1.1.6.0 = ""
  .1.3.6.1.2.1.1.7.0 = INTEGER: 78
  .1.3.6.1.2.1.1.8.0 = Timeticks: (0) 0:00:00.00
NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 192.168.30.1
1.3.6.1.2.1.1
  .1.3.6.1.2.1.1.1.0 = STRING: "Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-
  UNIVERSALK9-M), Version 17.15.6, RELEASE SOFTWARE (fc2)
  Technical Support: http://www.cisco.com/techsupport
  Copyright (c) 1986-2026 by Cisco Systems, Inc.
  Compiled Wed 22-Jul-26 03:39 b"
  .1.3.6.1.2.1.1.2.0 = OID: .1.3.6.1.4.1.9.1.3004
  .1.3.6.1.2.1.1.3.0 = Timeticks: (8401439) 23:20:14.39
  .1.3.6.1.2.1.1.4.0 = ""
  .1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r3.lab.local"
  .1.3.6.1.2.1.1.6.0 = ""
  .1.3.6.1.2.1.1.7.0 = INTEGER: 78
  .1.3.6.1.2.1.1.8.0 = Timeticks: (0) 0:00:00.00
NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 192.168.30.1
1.3.6.1.2.1.2.2.1.2
  .1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
  .1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
  .1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
  .1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
  .1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"
  .1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"
  .1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"
  .1.3.6.1.2.1.2.2.1.2.8 = STRING: "Loopback1"
  .1.3.6.1.2.1.2.2.1.2.9 = STRING: "Tunnel0"
NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 192.168.30.1
1.3.6.1.2.1.2.2.1.2
  .1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
  .1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
  .1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
  .1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
  .1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"
  .1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"
  .1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"
  .1.3.6.1.2.1.2.2.1.2.8 = STRING: "Loopback1"
  .1.3.6.1.2.1.2.2.1.2.9 = STRING: "Tunnel0"
```

Teardown (suite) Close Nms, Close All Connections

5.10 PASS Capture Collector State From The NMS (0.3s, 12 commands)

What the collectors actually hold: the tail of each router's syslog file and of snmptrapd's decrypted log. This is the management-plane evidence that used to come off h1.

Setup (suite) Open All Routers, Open All Hosts, Open Nms

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked

```
NMS systemctl is-active rsyslog snmptrapd lab-trapcap
  active
  active
  active
NMS systemctl is-active rsyslog snmptrapd lab-trapcap
  active
  active
  active
NMS sudo ls -l /var/log/lab
  total 10600
  -rw-r--r-- 1 syslog syslog  527673 Sep  9 00:54 192.168.99.1.log
  -rw-r--r-- 1 syslog syslog  653218 Sep  9 00:54 192.168.99.2.log
  -rw-r--r-- 1 syslog syslog  455348 Sep  9 00:54 192.168.99.3.log
  -rw-r--r-- 1 root   root   2824382 Sep  9 00:54 snmptrapd.log
  -rw-r--r-- 1 root   root   6362155 Sep  9 00:54 traps-raw.log
NMS sudo ls -l /var/log/lab
  total 10600
  -rw-r--r-- 1 syslog syslog  527673 Sep  9 00:54 192.168.99.1.log
  -rw-r--r-- 1 syslog syslog  653218 Sep  9 00:54 192.168.99.2.log
  -rw-r--r-- 1 syslog syslog  455348 Sep  9 00:54 192.168.99.3.log
  -rw-r--r-- 1 root   root   2824382 Sep  9 00:54 snmptrapd.log
  -rw-r--r-- 1 root   root   6362155 Sep  9 00:54 traps-raw.log
NMS sudo tail -40 /var/log/lab/192.168.10.1.log
  tail: cannot open '/var/log/lab/192.168.10.1.log' for reading: No such file or directory
NMS sudo tail -40 /var/log/lab/192.168.10.1.log
  tail: cannot open '/var/log/lab/192.168.10.1.log' for reading: No such file or directory
NMS sudo tail -40 /var/log/lab/192.168.20.1.log
  tail: cannot open '/var/log/lab/192.168.20.1.log' for reading: No such file or directory
NMS sudo tail -40 /var/log/lab/192.168.20.1.log
  tail: cannot open '/var/log/lab/192.168.20.1.log' for reading: No such file or directory
NMS sudo tail -40 /var/log/lab/192.168.30.1.log
  tail: cannot open '/var/log/lab/192.168.30.1.log' for reading: No such file or directory
NMS sudo tail -40 /var/log/lab/192.168.30.1.log
  tail: cannot open '/var/log/lab/192.168.30.1.log' for reading: No such file or directory
NMS sudo tail -40 /var/log/lab/snmptrapd.log
  2026-09-09 00:54:16 <UNKNOWN> [UDP: [192.168.99.3]:50504->[192.168.99.10]:162]:
  iso.3.6.1.2.1.1.3.0 0:23:19:44.43       iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.171.2.0.7
  iso.3.6.1.4.1.9.9.171.1.3.2.1.9.148 3600        iso.3.6.1.4.1.9.9.171.1.3.2.1.8.148 4608000
  iso.3.6.1.4.1.9.9.171.1.3.3.1.3.148.1 1 iso.3.6.1.4.1.9.9.171.1.3.3.1.4.148.1 "64 40 00 03 "
  iso.3.6.1.4.1.9.9.171.1.3.3.1.5.148.1 "64 40 00 03 "    iso.3.6.1.4.1.9.9.171.1.3.3.1.9.148.1 1
  iso.3.6.1.4.1.9.9.171.1.3.3.1.10.148.1 "64 40 00 02 "   iso.3.6.1.4.1.9.9.171.1.3.3.1.11.148.1 "64 40 00 02
  "   iso.3.6.1.4.1.9.9.171.1.3.3.1.6.148.1 47        iso.3.6.1.4.1.9.9.171.1.3.3.1.7.148.1 0
  2026-09-09 00:54:16 <UNKNOWN> [UDP: [192.168.99.2]:51414->[192.168.99.10]:162]:
  iso.3.6.1.2.1.1.3.0 0:23:19:44.78       iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.171.2.0.7
  iso.3.6.1.4.1.9.9.171.1.3.2.1.9.150 3600        iso.3.6.1.4.1.9.9.171.1.3.2.1.8.150 4608000
  iso.3.6.1.4.1.9.9.171.1.3.3.1.3.150.1 1 iso.3.6.1.4.1.9.9.171.1.3.3.1.4.150.1 "64 40 00 02 "
  iso.3.6.1.4.1.9.9.171.1.3.3.1.5.150.1 "64 40 00 02 "    iso.3.6.1.4.1.9.9.171.1.3.3.1.9.150.1 1
  iso.3.6.1.4.1.9.9.171.1.3.3.1.10.150.1 "64 40 00 03 "   iso.3.6.1.4.1.9.9.171.1.3.3.1.11.150.1 "64 40 00 03
  "   iso.3.6.1.4.1.9.9.171.1.3.3.1.6.150.1 47        iso.3.6.1.4.1.9.9.171.1.3.3.1.7.150.1 0
  2026-09-09 00:54:16 <UNKNOWN> [UDP: [192.168.99.3]:50504->[192.168.99.10]:162]:
  iso.3.6.1.2.1.1.3.0 0:23:19:45.19       iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.10.137.0.1
  iso.3.6.1.4.1.9.10.137.1.2.1.8.1708097 0        iso.3.6.1.4.1.9.10.137.1.2.1.8.1708097 0
  iso.3.6.1.2.1.2.2.1.1.9 9
  2026-09-09 00:54:16 <UNKNOWN> [UDP: [192.168.99.2]:51414->[192.168.99.10]:162]:
  iso.3.6.1.2.1.1.3.0 0:23:19:45.55       iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.10.137.0.1
  iso.3.6.1.4.1.9.10.137.1.2.1.8.1708044 0        iso.3.6.1.4.1.9.10.137.1.2.1.8.1708044 0
  iso.3.6.1.2.1.2.2.1.1.9 9
  2026-09-09 00:54:17 <UNKNOWN> [UDP: [192.168.99.2]:51414->[192.168.99.10]:162]:
  iso.3.6.1.2.1.1.3.0 0:23:19:46.94       iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.10.62.2.0.4
  iso.3.6.1.4.1.9.10.62.1.2.3.1.1.2.14.84.117.110.110.101.108.48.45.104.101.97.100.45.48 3
  2026-09-09 00:54:17 <UNKNOWN> [UDP: [192.168.99.3]:50504->[192.168.99.10]:162]:
```

```
iso.3.6.1.2.1.1.3.0 0:23:19:46.59       iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.171.2.0.8
iso.3.6.1.4.1.9.9.171.1.3.2.1.10.148 216        iso.3.6.1.4.1.9.9.171.1.4.3.1.1.12.147 216
iso.3.6.1.4.1.9.9.171.1.4.3.1.1.10.147 3600     iso.3.6.1.4.1.9.9.171.1.4.3.1.1.9.147 4608000
iso.3.6.1.4.1.9.9.171.1.4.3.1.1.2.147 1 iso.3.6.1.4.1.9.9.171.1.4.3.2.1.5.147 1
iso.3.6.1.4.1.9.9.171.1.4.3.2.1.11.147 1        iso.3.6.1.4.1.9.9.171.1.4.3.2.1.6.147 "64 40 00 03 "
iso.3.6.1.4.1.9.9.171.1.4.3.2.1.7.147 "64 40 00 03 "    iso.3.6.1.4.1.9.9.171.1.4.3.2.1.12.147 "64 40 00 02
"   iso.3.6.1.4.1.9.9.171.1.4.3.2.1.13.147 "64 40 00 02 "   iso.3.6.1.4.1.9.9.171.1.4.3.2.1.8.147 47
iso.3.6.1.4.1.9.9.171.1.4.3.2.1.9.147 0
2026-09-09 00:54:18 <UNKNOWN> [UDP: [192.168.99.2]:51414->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:23:19:46.94       iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.171.2.0.8
iso.3.6.1.4.1.9.9.171.1.3.2.1.10.150 215        iso.3.6.1.4.1.9.9.171.1.4.3.1.1.12.149 215
iso.3.6.1.4.1.9.9.171.1.4.3.1.1.10.149 3600     iso.3.6.1.4.1.9.9.171.1.4.3.1.1.9.149 4608000
iso.3.6.1.4.1.9.9.171.1.4.3.1.1.2.149 1 iso.3.6.1.4.1.9.9.171.1.4.3.2.1.5.149 1
iso.3.6.1.4.1.9.9.171.1.4.3.2.1.11.149 1        iso.3.6.1.4.1.9.9.171.1.4.3.2.1.6.149 "64 40 00 02 "
iso.3.6.1.4.1.9.9.171.1.4.3.2.1.7.149 "64 40 00 02 "    iso.3.6.1.4.1.9.9.171.1.4.3.2.1.12.149 "64 40 00 03
"   iso.3.6.1.4.1.9.9.171.1.4.3.2.1.13.149 "64 40 00 03 "   iso.3.6.1.4.1.9.9.171.1.4.3.2.1.8.149 47
iso.3.6.1.4.1.9.9.171.1.4.3.2.1.9.149 0
... 26 more lines
```

NMS **sudo tail -40 /var/log/lab/snmptrapd.log**

```
2026-09-09 00:54:16 <UNKNOWN> [UDP: [192.168.99.3]:50504->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:23:19:44.43       iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.171.2.0.7
iso.3.6.1.4.1.9.9.171.1.3.2.1.9.148 3600        iso.3.6.1.4.1.9.9.171.1.3.2.1.8.148 4608000
iso.3.6.1.4.1.9.9.171.1.3.3.1.3.148.1 1 iso.3.6.1.4.1.9.9.171.1.3.3.1.4.148.1 "64 40 00 03 "
iso.3.6.1.4.1.9.9.171.1.3.3.1.5.148.1 "64 40 00 03 "    iso.3.6.1.4.1.9.9.171.1.3.3.1.9.148.1 1
iso.3.6.1.4.1.9.9.171.1.3.3.1.10.148.1 "64 40 00 02 "   iso.3.6.1.4.1.9.9.171.1.3.3.1.11.148.1 "64 40 00 02
"   iso.3.6.1.4.1.9.9.171.1.3.3.1.6.148.1 47        iso.3.6.1.4.1.9.9.171.1.3.3.1.7.148.1 0
2026-09-09 00:54:16 <UNKNOWN> [UDP: [192.168.99.2]:51414->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:23:19:44.78       iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.171.2.0.7
iso.3.6.1.4.1.9.9.171.1.3.2.1.9.150 3600        iso.3.6.1.4.1.9.9.171.1.3.2.1.8.150 4608000
iso.3.6.1.4.1.9.9.171.1.3.3.1.3.150.1 1 iso.3.6.1.4.1.9.9.171.1.3.3.1.4.150.1 "64 40 00 02 "
iso.3.6.1.4.1.9.9.171.1.3.3.1.5.150.1 "64 40 00 02 "    iso.3.6.1.4.1.9.9.171.1.3.3.1.9.150.1 1
iso.3.6.1.4.1.9.9.171.1.3.3.1.10.150.1 "64 40 00 03 "   iso.3.6.1.4.1.9.9.171.1.3.3.1.11.150.1 "64 40 00 03
"   iso.3.6.1.4.1.9.9.171.1.3.3.1.6.150.1 47        iso.3.6.1.4.1.9.9.171.1.3.3.1.7.150.1 0
2026-09-09 00:54:16 <UNKNOWN> [UDP: [192.168.99.3]:50504->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:23:19:45.19       iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.10.137.0.1
iso.3.6.1.4.1.9.10.137.1.2.1.8.1708097 0        iso.3.6.1.4.1.9.10.137.1.2.1.8.1708097 0
iso.3.6.1.2.1.2.2.1.1.9 9
2026-09-09 00:54:16 <UNKNOWN> [UDP: [192.168.99.2]:51414->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:23:19:45.55       iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.10.137.0.1
iso.3.6.1.4.1.9.10.137.1.2.1.8.1708044 0        iso.3.6.1.4.1.9.10.137.1.2.1.8.1708044 0
iso.3.6.1.2.1.2.2.1.1.9 9
2026-09-09 00:54:17 <UNKNOWN> [UDP: [192.168.99.2]:51414->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:23:19:46.94       iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.10.62.2.0.4
iso.3.6.1.4.1.9.10.62.1.2.3.1.1.2.14.84.117.110.110.101.108.48.45.104.101.97.100.45.48 3
2026-09-09 00:54:17 <UNKNOWN> [UDP: [192.168.99.3]:50504->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:23:19:46.59       iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.171.2.0.8
iso.3.6.1.4.1.9.9.171.1.3.2.1.10.148 216        iso.3.6.1.4.1.9.9.171.1.4.3.1.1.12.147 216
iso.3.6.1.4.1.9.9.171.1.4.3.1.1.10.147 3600     iso.3.6.1.4.1.9.9.171.1.4.3.1.1.9.147 4608000
iso.3.6.1.4.1.9.9.171.1.4.3.1.1.2.147 1 iso.3.6.1.4.1.9.9.171.1.4.3.2.1.5.147 1
iso.3.6.1.4.1.9.9.171.1.4.3.2.1.11.147 1        iso.3.6.1.4.1.9.9.171.1.4.3.2.1.6.147 "64 40 00 03 "
iso.3.6.1.4.1.9.9.171.1.4.3.2.1.7.147 "64 40 00 03 "    iso.3.6.1.4.1.9.9.171.1.4.3.2.1.12.147 "64 40 00 02
"   iso.3.6.1.4.1.9.9.171.1.4.3.2.1.13.147 "64 40 00 02 "   iso.3.6.1.4.1.9.9.171.1.4.3.2.1.8.147 47
iso.3.6.1.4.1.9.9.171.1.4.3.2.1.9.147 0
2026-09-09 00:54:18 <UNKNOWN> [UDP: [192.168.99.2]:51414->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:23:19:46.94       iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.171.2.0.8
iso.3.6.1.4.1.9.9.171.1.3.2.1.10.150 215        iso.3.6.1.4.1.9.9.171.1.4.3.1.1.12.149 215
iso.3.6.1.4.1.9.9.171.1.4.3.1.1.10.149 3600     iso.3.6.1.4.1.9.9.171.1.4.3.1.1.9.149 4608000
iso.3.6.1.4.1.9.9.171.1.4.3.1.1.2.149 1 iso.3.6.1.4.1.9.9.171.1.4.3.2.1.5.149 1
iso.3.6.1.4.1.9.9.171.1.4.3.2.1.11.149 1        iso.3.6.1.4.1.9.9.171.1.4.3.2.1.6.149 "64 40 00 02 "
iso.3.6.1.4.1.9.9.171.1.4.3.2.1.7.149 "64 40 00 02 "    iso.3.6.1.4.1.9.9.171.1.4.3.2.1.12.149 "64 40 00 03
"   iso.3.6.1.4.1.9.9.171.1.4.3.2.1.13.149 "64 40 00 03 "   iso.3.6.1.4.1.9.9.171.1.4.3.2.1.8.149 47
iso.3.6.1.4.1.9.9.171.1.4.3.2.1.9.149 0
... 26 more lines
```

Teardown (suite) Close Nms, Close All Connections

5.11 PASS Capture Config And Routing Table From Every Host (0.1s, 6 commands)

Setup (suite) Open All Routers, Open All Hosts, Open Nms

Acceptance criteria

- output contains **eth1** (checked 3×)
- output contains **default via** (checked 3×)

How it was checked

H1 **ip addr**

```
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue qlen 1000
```

```
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 52:54:00:aa:01:01 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
       valid_lft 45682sec preferred_lft 34882sec
    inet6 fec0::5054:ff:feaa:101/64 scope site dynamic noprefixroute flags 100
       valid_lft 86385sec preferred_lft 14385sec
    inet6 fe80::5054:ff:feaa:101/64 scope link
       valid_lft forever preferred_lft forever
... 84 more lines
```

H1 ip route

```
default via 10.0.2.2 dev eth0  src 10.0.2.15  metric 1002
10.0.2.0/24 dev eth0 scope link  src 10.0.2.15  metric 1002
10.30.0.0/16 via 192.168.10.1 dev eth1
169.254.0.0/16 dev eth1 scope link  src 169.254.229.55  metric 1003
192.168.10.0/24 dev eth1 scope link  src 192.168.10.10
192.168.20.0/24 via 192.168.10.1 dev eth1
192.168.30.0/24 via 192.168.10.1 dev eth1
```

H2 ip addr

```
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 52:54:00:aa:02:01 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
       valid_lft 45682sec preferred_lft 34882sec
    inet6 fec0::5054:ff:feaa:201/64 scope site dynamic noprefixroute flags 100
       valid_lft 86191sec preferred_lft 14191sec
    inet6 fe80::5054:ff:feaa:201/64 scope link
       valid_lft forever preferred_lft forever
... 84 more lines
```

H2 ip route

```
default via 10.0.2.2 dev eth0  src 10.0.2.15  metric 1002
10.0.2.0/24 dev eth0 scope link  src 10.0.2.15  metric 1002
10.30.0.0/16 via 192.168.20.1 dev eth1
169.254.0.0/16 dev eth1 scope link  src 169.254.199.41  metric 1003
192.168.10.0/24 via 192.168.20.1 dev eth1
192.168.20.0/24 dev eth1 scope link  src 192.168.20.10
192.168.30.0/24 via 192.168.20.1 dev eth1
```

H3 ip addr

```
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 52:54:00:aa:03:01 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
       valid_lft 45683sec preferred_lft 34883sec
    inet6 fec0::5054:ff:feaa:301/64 scope site dynamic noprefixroute flags 100
       valid_lft 86305sec preferred_lft 14305sec
    inet6 fe80::5054:ff:feaa:301/64 scope link
       valid_lft forever preferred_lft forever
... 84 more lines
```

H3 ip route

```
default via 10.0.2.2 dev eth0  src 10.0.2.15  metric 1002
10.0.2.0/24 dev eth0 scope link  src 10.0.2.15  metric 1002
10.30.0.0/16 via 192.168.30.1 dev eth1
169.254.0.0/16 dev eth1 scope link  src 169.254.207.242  metric 1003
192.168.10.0/24 via 192.168.30.1 dev eth1
192.168.20.0/24 via 192.168.30.1 dev eth1
192.168.30.0/24 dev eth1 scope link  src 192.168.30.10
```

Teardown (suite) Close Nms, Close All Connections

5.12 PASS Every Router Routing Table Holds All Three LANs (0.2s, 3 commands)

A content check on what was just captured: in a working hub-and-spoke every router reaches every LAN, the spokes reaching each other's through the hub.

Setup (suite) Open All Routers, Open All Hosts, Open Nms

Acceptance criteria

- output contains **${R1_LAN_NET}** (checked 3×)
- output contains **${R2_LAN_NET}** (checked 3×)
- output contains **${R3_LAN_NET}** (checked 3×)

How it was checked

R1 show ip route

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
       n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       H - NHRP, G - NHRP registered, g - NHRP registration summary
       o - ODR, P - periodic downloaded static route, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR
       & - replicated local route overrides by connected

Gateway of last resort is 10.0.2.2 to network 0.0.0.0
... 24 more lines
```

R2 show ip route

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
       n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       H - NHRP, G - NHRP registered, g - NHRP registration summary
       o - ODR, P - periodic downloaded static route, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR
       & - replicated local route overrides by connected

Gateway of last resort is 10.0.2.2 to network 0.0.0.0
... 24 more lines
```

R3 show ip route

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
       n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       H - NHRP, G - NHRP registered, g - NHRP registration summary
       o - ODR, P - periodic downloaded static route, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR
       & - replicated local route overrides by connected

Gateway of last resort is 10.0.2.2 to network 0.0.0.0
... 24 more lines
```

Teardown (suite) Close Nms, Close All Connections

06 Wire Encryption

6.01 PASS Protected Traffic Appears On The Link As ESP (15.6s, 9 commands)

Setup (suite) Open All Routers

Acceptance criteria

- output contains **ESP**
- output matches **${R1_WAN_IP}\s+->\s+${R2_WAN_IP}**

How it was checked

```
R1 no monitor capture ROBOTCAP
  Capture does not exist
R1 monitor capture ROBOTCAP interface GigabitEthernet2 both
  ${output} =
R1 monitor capture ROBOTCAP match ipv4 any any
  ${output} =
R1 monitor capture ROBOTCAP buffer size 5
  ${output} =
R1 monitor capture ROBOTCAP start
  Started capture point : ROBOTCAP
R1 ping 172.16.2.1 source Loopback0 repeat 5
  Type escape sequence to abort.
  Sending 5, 100-byte ICMP Echos to 172.16.2.1, timeout is 2 seconds:
  Packet sent with a source address of 172.16.1.1
  .....
  Success rate is 0 percent (0/5)
R1 monitor capture ROBOTCAP stop
  Stopped capture point : ROBOTCAP
R1 show monitor capture ROBOTCAP buffer brief
  ----------------------------------------------------------------------------
   #   size   timestamp     source             destination      dscp    protocol
   ----------------------------------------------------------------------------
     0  138    0.000000   100.64.0.2       -> 100.64.0.1       48 CS6  ESP
     1  138    0.000000   100.64.0.1       -> 100.64.0.2       48 CS6  ESP
     2  138    0.013000   100.64.0.1       -> 100.64.0.2       48 CS6  ESP
     3  170    0.046995   100.64.0.1       -> 100.64.0.3       48 CS6  ESP
     4  138    0.057996   100.64.0.1       -> 100.64.0.3       48 CS6  ESP
     5  138    0.057996   100.64.0.3       -> 100.64.0.1       48 CS6  ESP
     6  138    0.082989   100.64.0.1       -> 100.64.0.2       48 CS6  ESP
     7  138    0.083996   100.64.0.2       -> 100.64.0.1       48 CS6  ESP
     8  138    0.152992   100.64.0.2       -> 100.64.0.1       48 CS6  ESP
     9  154    0.168983   100.64.0.3       -> 100.64.0.1       48 CS6  ESP
    10  138    0.293036   100.64.0.3       -> 100.64.0.1       48 CS6  ESP
  ... 272 more lines
R1 no monitor capture ROBOTCAP
  ${output} =
```

Teardown (suite) Run Keyword And Ignore Error, Delete Capture, R1, Close All Routers

6.02 PASS Protected Endpoint Addresses Never Appear In Cleartext On The Link (15.7s, 9 commands)

The strongest statement this lab can make about encryption: traffic between the loopbacks crosses the wire without either loopback address being visible, and with no cleartext ICMP.

Setup (suite) Open All Routers

Acceptance criteria

- output does not contain **${R1_LOOPBACK}**
- output does not contain **${R2_LOOPBACK}**
- output does not contain **ICMP**

How it was checked

```
R1 no monitor capture ROBOTCAP
  Capture does not exist
R1 monitor capture ROBOTCAP interface GigabitEthernet2 both
```

```
${output} =
```

R1 **`monitor capture ROBOTCAP match ipv4 any any`**

```
${output} =
```

R1 **`monitor capture ROBOTCAP buffer size 5`**

```
${output} =
```

R1 **`monitor capture ROBOTCAP start`**

```
Started capture point : ROBOTCAP
```

R1 **`ping 172.16.2.1 source Loopback0 repeat 5`**

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.2.1, timeout is 2 seconds:
Packet sent with a source address of 172.16.1.1
.....
Success rate is 0 percent (0/5)
```

R1 **`monitor capture ROBOTCAP stop`**

```
Stopped capture point : ROBOTCAP
```

R1 **`show monitor capture ROBOTCAP buffer brief`**

```
----------------------------------------------------------------------------
 #   size   timestamp     source             destination      dscp    protocol
 ----------------------------------------------------------------------------
   0  138    0.000000   100.64.0.1       ->  100.64.0.3       48 CS6  ESP
   1  138    0.000000   100.64.0.3       ->  100.64.0.1       48 CS6  ESP
   2  138    0.020995   100.64.0.1       ->  100.64.0.2       48 CS6  ESP
   3  138    0.085002   100.64.0.2       ->  100.64.0.1       48 CS6  ESP
   4  138    0.085002   100.64.0.1       ->  100.64.0.2       48 CS6  ESP
   5  138    0.098002   100.64.0.1       ->  100.64.0.3       48 CS6  ESP
   6  138    0.119989   100.64.0.3       ->  100.64.0.1       48 CS6  ESP
   7  138    0.119989   100.64.0.1       ->  100.64.0.3       48 CS6  ESP
   8  138    0.126000   100.64.0.1       ->  100.64.0.2       48 CS6  ESP
   9  138    0.126000   100.64.0.2       ->  100.64.0.1       48 CS6  ESP
  10  138    0.209996   100.64.0.3       ->  100.64.0.1       48 CS6  ESP
... 275 more lines
```

R1 **`no monitor capture ROBOTCAP`**

```
${output} =
```

Teardown (suite) Run Keyword And Ignore Error, Delete Capture, R1, Close All Routers

6.03 PASS Unprotected Link Traffic Is Visible As Cleartext ICMP (2.4s, 9 commands)

Positive control for the capture itself. Without this, the absence asserted above could just mean the capture was broken.

Setup (suite) Open All Routers

Acceptance criteria

- output contains **ICMP**
- output matches **${R1_WAN_IP}\s+->\s+${R2_WAN_IP}\s+\S+\s+\S+\s+ICMP**

How it was checked

R1 **`no monitor capture ROBOTCAP`**

```
Capture does not exist
```

R1 **`monitor capture ROBOTCAP interface GigabitEthernet2 both`**

```
${output} =
```

R1 **`monitor capture ROBOTCAP match ipv4 any any`**

```
${output} =
```

R1 **`monitor capture ROBOTCAP buffer size 5`**

```
${output} =
```

R1 **`monitor capture ROBOTCAP start`**

```
Started capture point : ROBOTCAP
```

R1 **`ping 100.64.0.2 repeat 5`**

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 100.64.0.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
```

R1 **`monitor capture ROBOTCAP stop`**

```
Stopped capture point : ROBOTCAP
```

R1 **`show monitor capture ROBOTCAP buffer brief`**

```
----------------------------------------------------------------------------
 #   size   timestamp     source             destination      dscp    protocol
 ----------------------------------------------------------------------------
   0  114    0.000000   100.64.0.1       ->  100.64.0.2       0  BE   ICMP
   1  114    0.000000   100.64.0.2       ->  100.64.0.1       0  BE   ICMP
   2  114    0.000992   100.64.0.1       ->  100.64.0.2       0  BE   ICMP
```

```
   3  114    0.000992   100.64.0.2       ->  100.64.0.1       0  BE   ICMP
   4  114    0.001999   100.64.0.1       ->  100.64.0.2       0  BE   ICMP
   5  114    0.001999   100.64.0.2       ->  100.64.0.1       0  BE   ICMP
   6  114    0.002991   100.64.0.1       ->  100.64.0.2       0  BE   ICMP
   7  114    0.003998   100.64.0.2       ->  100.64.0.1       0  BE   ICMP
   8  114    0.003998   100.64.0.1       ->  100.64.0.2       0  BE   ICMP
   9  114    0.004989   100.64.0.2       ->  100.64.0.1       0  BE   ICMP
  10  138    0.167991   100.64.0.3       ->  100.64.0.1       48 CS6  ESP
... 48 more lines
```

R1 no monitor capture ROBOTCAP

```
${output} =
```

Teardown (suite) Run Keyword And Ignore Error, Delete Capture, R1, Close All Routers

07 Ipsec Negative

7.01 PASS IKEv2 Refuses To Establish With A Mismatched Pre-Shared Key (13.3s, 13 commands)

Proves authentication is actually enforced. Without this, a tunnel that accepted any peer would still pass suite 04.

Setup (suite) Open All Routers

Acceptance criteria

- output does not contain **Success rate is 100 percent**
- output does not contain **READY**

How it was checked

```
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 crypto ikev2 keyring LAB-KR
  ${output} =
R2 peer ANY
  ${output} =
R2 pre-shared-key WrongKeyOnPurpose999
  ${output} =
R2 end
  ${output} =
R1 clear crypto sa
  ${output} =
R1 clear crypto ikev2 sa
  ${output} =
R2 clear crypto sa
  ${output} =
R2 clear crypto ikev2 sa
  ${output} =
R3 clear crypto sa
  ${output} =
R3 clear crypto ikev2 sa
  ${output} =
R1 ping 172.16.2.1 source Loopback0 repeat 5
  Type escape sequence to abort.
  Sending 5, 100-byte ICMP Echos to 172.16.2.1, timeout is 2 seconds:
  Packet sent with a source address of 172.16.1.1
  .....
  Success rate is 0 percent (0/5)
R2 show crypto ikev2 sa
  ${output} =
```

Teardown (suite) Restore Baseline Crypto, Close All Routers

7.02 PASS Tunnel Recovers Once The Correct Key Is Restored (14.3s, 13 commands)

Also guarantees the previous test cannot leave the lab broken.

Setup (suite) Open All Routers

Acceptance criteria

- output contains **Success rate is 100 percent** (checked 2×)

How it was checked

```
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 crypto ikev2 keyring LAB-KR
  ${output} =
R2 peer ANY
  ${output} =
R2 pre-shared-key LabPreSharedKey123
  ${output} =
R2 end
  ${output} =
```

R1 **clear crypto sa**
```
${output} =
```
R1 **clear crypto ikev2 sa**
```
${output} =
```
R2 **clear crypto sa**
```
${output} =
```
R2 **clear crypto ikev2 sa**
```
${output} =
```
R3 **clear crypto sa**
```
${output} =
```
R3 **clear crypto ikev2 sa**
```
${output} =
```
R1 **ping 10.20.2.1 source Loopback1 repeat 5**
```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.20.2.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.1.1
...!!
Success rate is 40 percent (2/5), round-trip min/avg/max = 1/1/1 ms
```
R1 **ping 10.20.2.1 source Loopback1 repeat 5**
```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.20.2.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.1.1
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
```
Teardown (suite) Restore Baseline Crypto, Close All Routers

7.03 PASS IPsec Rekeys And Traffic Still Flows Afterwards (96.2s, 37 commands)

Drops the SA lifetime to the 120s platform minimum, waits for the outbound SPI to rotate, then re-checks connectivity. Rekey is where real tunnels break.

Setup (suite) Open All Routers

Acceptance criteria

- output contains **Success rate is 100 percent** (checked 3×)
- **${m}** is not empty (checked 10×)
- **${new}** differs from **${old}** (checked 9×)
  SPI ${new} has not rotated yet

How it was checked

R1 **configure terminal**
```
Enter configuration commands, one per line.  End with CNTL/Z.
```
R1 **crypto ipsec security-association lifetime seconds 120**
```
Warning! Lifetime value of 120 sec is lower than the recommended optimum value of 900 sec
```
R1 **end**
```
${output} =
```
R2 **configure terminal**
```
Enter configuration commands, one per line.  End with CNTL/Z.
```
R2 **crypto ipsec security-association lifetime seconds 120**
```
Warning! Lifetime value of 120 sec is lower than the recommended optimum value of 900 sec
```
R2 **end**
```
${output} =
```
R3 **configure terminal**
```
Enter configuration commands, one per line.  End with CNTL/Z.
```
R3 **crypto ipsec security-association lifetime seconds 120**
```
Warning! Lifetime value of 120 sec is lower than the recommended optimum value of 900 sec
```
R3 **end**
```
${output} =
```
R1 **clear crypto sa**
```
${output} =
```
R1 **clear crypto ikev2 sa**
```
${output} =
```
R2 **clear crypto sa**
```
${output} =
```
R2 **clear crypto ikev2 sa**
```
${output} =
```

R3 clear crypto sa
```
${output} =
```

R3 clear crypto ikev2 sa
```
${output} =
```

R1 ping 10.20.2.1 source Loopback1 repeat 5
```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.20.2.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.1.1
...!!
Success rate is 40 percent (2/5), round-trip min/avg/max = 1/1/1 ms
```

R1 ping 10.20.2.1 source Loopback1 repeat 5
```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.20.2.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.1.1
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
```

R1 show crypto ipsec sa
```
interface: Tunnel0
    Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
   remote ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
   current_peer 100.64.0.2 port 500
     PERMIT, flags={origin_is_acl,}
    #pkts encaps: 74, #pkts encrypt: 74, #pkts digest: 74
    #pkts decaps: 73, #pkts decrypt: 73, #pkts verify: 73
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0
... 94 more lines
```

R1 show crypto ipsec sa
```
interface: Tunnel0
    Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
   remote ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
   current_peer 100.64.0.2 port 500
     PERMIT, flags={origin_is_acl,}
    #pkts encaps: 74, #pkts encrypt: 74, #pkts digest: 74
    #pkts decaps: 73, #pkts decrypt: 73, #pkts verify: 73
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0
... 94 more lines
```

R1 show crypto ipsec sa
```
interface: Tunnel0
    Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
   remote ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
   current_peer 100.64.0.2 port 500
     PERMIT, flags={origin_is_acl,}
    #pkts encaps: 137, #pkts encrypt: 137, #pkts digest: 137
    #pkts decaps: 137, #pkts decrypt: 137, #pkts verify: 137
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0
... 78 more lines
```

R1 show crypto ipsec sa
```
interface: Tunnel0
    Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
   remote ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
   current_peer 100.64.0.2 port 500
     PERMIT, flags={origin_is_acl,}
    #pkts encaps: 197, #pkts encrypt: 197, #pkts digest: 197
    #pkts decaps: 197, #pkts decrypt: 197, #pkts verify: 197
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
```

```
     #pkts not decompressed: 0, #pkts decompress failed: 0
     #send errors 0, #recv errors 0
 ... 78 more lines
```

R1 show crypto ipsec sa

```
 interface: Tunnel0
     Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

    protected vrf: (none)
    local  ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
    remote ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
    current_peer 100.64.0.2 port 500
      PERMIT, flags={origin_is_acl,}
     #pkts encaps: 258, #pkts encrypt: 258, #pkts digest: 258
     #pkts decaps: 263, #pkts decrypt: 263, #pkts verify: 263
     #pkts compressed: 0, #pkts decompressed: 0
     #pkts not compressed: 0, #pkts compr. failed: 0
     #pkts not decompressed: 0, #pkts decompress failed: 0
     #send errors 0, #recv errors 0
 ... 78 more lines
```

R1 show crypto ipsec sa

```
 interface: Tunnel0
     Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

    protected vrf: (none)
    local  ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
    remote ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
    current_peer 100.64.0.2 port 500
      PERMIT, flags={origin_is_acl,}
     #pkts encaps: 317, #pkts encrypt: 317, #pkts digest: 317
     #pkts decaps: 322, #pkts decrypt: 322, #pkts verify: 322
     #pkts compressed: 0, #pkts decompressed: 0
     #pkts not compressed: 0, #pkts compr. failed: 0
     #pkts not decompressed: 0, #pkts decompress failed: 0
     #send errors 0, #recv errors 0
 ... 78 more lines
```

R1 show crypto ipsec sa

```
 interface: Tunnel0
     Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

    protected vrf: (none)
    local  ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
    remote ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
    current_peer 100.64.0.2 port 500
      PERMIT, flags={origin_is_acl,}
     #pkts encaps: 374, #pkts encrypt: 374, #pkts digest: 374
     #pkts decaps: 383, #pkts decrypt: 383, #pkts verify: 383
     #pkts compressed: 0, #pkts decompressed: 0
     #pkts not compressed: 0, #pkts compr. failed: 0
     #pkts not decompressed: 0, #pkts decompress failed: 0
     #send errors 0, #recv errors 0
 ... 78 more lines
```

R1 show crypto ipsec sa

```
 interface: Tunnel0
     Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

    protected vrf: (none)
    local  ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
    remote ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
    current_peer 100.64.0.2 port 500
      PERMIT, flags={origin_is_acl,}
     #pkts encaps: 443, #pkts encrypt: 443, #pkts digest: 443
     #pkts decaps: 442, #pkts decrypt: 442, #pkts verify: 442
     #pkts compressed: 0, #pkts decompressed: 0
     #pkts not compressed: 0, #pkts compr. failed: 0
     #pkts not decompressed: 0, #pkts decompress failed: 0
     #send errors 0, #recv errors 0
 ... 78 more lines
```

R1 show crypto ipsec sa

```
 interface: Tunnel0
     Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

    protected vrf: (none)
    local  ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
    remote ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
    current_peer 100.64.0.2 port 500
      PERMIT, flags={origin_is_acl,}
     #pkts encaps: 501, #pkts encrypt: 501, #pkts digest: 501
     #pkts decaps: 501, #pkts decrypt: 501, #pkts verify: 501
     #pkts compressed: 0, #pkts decompressed: 0
```

```
      #pkts not compressed: 0, #pkts compr. failed: 0
      #pkts not decompressed: 0, #pkts decompress failed: 0
      #send errors 0, #recv errors 0
  ... 78 more lines
```

R1 show crypto ipsec sa

```
  interface: Tunnel0
      Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

     protected vrf: (none)
     local  ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
     remote ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
     current_peer 100.64.0.2 port 500
       PERMIT, flags={origin_is_acl,}
      #pkts encaps: 559, #pkts encrypt: 559, #pkts digest: 559
      #pkts decaps: 557, #pkts decrypt: 557, #pkts verify: 557
      #pkts compressed: 0, #pkts decompressed: 0
      #pkts not compressed: 0, #pkts compr. failed: 0
      #pkts not decompressed: 0, #pkts decompress failed: 0
      #send errors 0, #recv errors 0
  ... 78 more lines
```

R1 ping 10.20.2.1 source Loopback1 repeat 5

```
  Type escape sequence to abort.
  Sending 5, 100-byte ICMP Echos to 10.20.2.1, timeout is 2 seconds:
  Packet sent with a source address of 10.20.1.1
  !!!!!
  Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
```

R1 configure terminal

```
  Enter configuration commands, one per line.  End with CNTL/Z.
```

R1 no crypto ipsec security-association lifetime seconds 120

```
  ${output} =
```

R1 end

```
  ${output} =
```

R2 configure terminal

```
  Enter configuration commands, one per line.  End with CNTL/Z.
```

R2 no crypto ipsec security-association lifetime seconds 120

```
  ${output} =
```

R2 end

```
  ${output} =
```

R3 configure terminal

```
  Enter configuration commands, one per line.  End with CNTL/Z.
```

R3 no crypto ipsec security-association lifetime seconds 120

```
  ${output} =
```

R3 end

```
  ${output} =
```

Teardown (this test) Restore Default SA Lifetime

08 Management Api

8.01 PASS RESTCONF Reports The Same Hostname As The CLI (0.4s, 3 commands)

Setup (suite) Open All Routers, Create RESTCONF Sessions

Acceptance criteria

- output contains **${name}** (checked 3×)

How it was checked

```
R1 show running-config | include ^hostname
  hostname c8000v-r1
R2 show running-config | include ^hostname
  hostname c8000v-r2
R3 show running-config | include ^hostname
  hostname c8000v-r3
```

Teardown (suite) Run Keyword And Ignore Error, Delete RESTCONF Loopback, Close All Routers

8.02 PASS RESTCONF Interface State Agrees With The CLI (0.1s, 1 commands)

Setup (suite) Open All Routers, Create RESTCONF Sessions

Acceptance criteria

- **${status}** equals **up**
- output matches **GigabitEthernet2\s+${R1_WAN_IP}\s+YES\s+\S+\s+up\s+up**

How it was checked

```
R1 show ip interface brief GigabitEthernet2
  Interface              IP-Address      OK? Method Status                Protocol
  GigabitEthernet2       100.64.0.1      YES NVRAM  up                    up
```

Teardown (suite) Run Keyword And Ignore Error, Delete RESTCONF Loopback, Close All Routers

8.03 PASS Config Pushed Over RESTCONF Appears In The CLI (0.3s, 1 commands)

Write path plus cross-check: RESTCONF and the CLI must be two views of one datastore, not two datastores.

Setup (suite) Open All Routers, Create RESTCONF Sessions

Acceptance criteria

- **${resp.status_code} in (201, 204)** holds
- output contains **${RC_LOOPBACK_IP}**

How it was checked

```
R1 show running-config interface Loopback200
  Building configuration...

  Current configuration : 68 bytes
  !
  interface Loopback200
   ip address 198.51.100.1 255.255.255.0
  end
```

Teardown (suite) Run Keyword And Ignore Error, Delete RESTCONF Loopback, Close All Routers

8.04 PASS Config Deleted Over RESTCONF Disappears From The CLI (0.8s, 1 commands)

Runs after the PUT test, which creates the interface; a 404 here would mean the create silently failed.

Setup (suite) Open All Routers, Create RESTCONF Sessions

Acceptance criteria

- **${resp.status_code}** equals **204**
  RESTCONF DELETE returned ${resp.status_code}; expected 204 for the loopback created by the previous test
- output does not contain **${RC_LOOPBACK_IP}**

How it was checked

```
R1 show ip interface brief Loopback200
  ^
  % Invalid input detected at '^' marker.
```

Teardown (suite) Run Keyword And Ignore Error, Delete RESTCONF Loopback, Close All Routers

8.05 PASS NETCONF Reports The Same Hostname As The CLI (0.8s, 1 commands)

Setup (suite) Open All Routers, Create RESTCONF Sessions

Acceptance criteria

- **${name}** equals **${R1_NAME}**
- output contains **${name}**

How it was checked

```
R1 show running-config | include ^hostname
  hostname c8000v-r1
```

Teardown (suite) Run Keyword And Ignore Error, Delete RESTCONF Loopback, Close All Routers

8.06 PASS NETCONF Advertises The Expected Base Capabilities (0.5s, 0 commands)

IOS-XE does not list per-module URIs such as Cisco-IOS-XE-native in the hello; modules are exposed via yang-library. That the native model is usable is proven by the hostname test above, which queries the native namespace directly.

Setup (suite) Open All Routers, Create RESTCONF Sessions

Acceptance criteria

- **${hit}** is not empty (checked 3×)

How it was checked no device commands of its own; the assertion read output captured earlier in the suite

Teardown (suite) Run Keyword And Ignore Error, Delete RESTCONF Loopback, Close All Routers

09 Bgp

9.01 PASS BGP Sessions Are Established With The Expected Peer AS (0.1s, 4 commands)

The hub holds one session per spoke; each spoke holds exactly one.

Setup (suite) Establish BGP

Acceptance criteria

- output matches **${peer}\s+4\s+${peer_asn}(\s+\d+){5}\s+\S+\s+\d+** (checked 4×)

How it was checked

R1 show bgp ipv4 unicast summary

```
BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 58, main routing table version 58
9 network entries using 2232 bytes of memory
9 path entries using 1224 bytes of memory
7/7 BGP path/bestpath attribute entries using 2072 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
6 BGP community entries using 144 bytes of memory
6 BGP route-map cache entries using 432 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 6152 total bytes of memory
BGP activity 315/306 prefixes, 336/327 paths, scan interval 60 secs
22 networks peaked at 02:27:59 Sep 8 2026 UTC (22:29:32.702 ago)

Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
... 2 more lines
```

R1 show bgp ipv4 unicast summary

```
BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 58, main routing table version 58
9 network entries using 2232 bytes of memory
9 path entries using 1224 bytes of memory
7/7 BGP path/bestpath attribute entries using 2072 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
6 BGP community entries using 144 bytes of memory
6 BGP route-map cache entries using 432 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 6152 total bytes of memory
BGP activity 315/306 prefixes, 336/327 paths, scan interval 60 secs
22 networks peaked at 02:27:59 Sep 8 2026 UTC (22:29:32.735 ago)

Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
... 2 more lines
```

R2 show bgp ipv4 unicast summary

```
BGP router identifier 172.16.2.1, local AS number 65002
BGP table version is 70, main routing table version 70
9 network entries using 2232 bytes of memory
9 path entries using 1224 bytes of memory
7/7 BGP path/bestpath attribute entries using 2072 bytes of memory
2 BGP AS-PATH entries using 64 bytes of memory
6 BGP community entries using 144 bytes of memory
6 BGP route-map cache entries using 432 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 6168 total bytes of memory
BGP activity 345/336 prefixes, 360/351 paths, scan interval 60 secs
31 networks peaked at 02:28:40 Sep 8 2026 UTC (22:28:51.764 ago)

Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
... 1 more lines
```

R3 show bgp ipv4 unicast summary

```
BGP router identifier 172.16.3.1, local AS number 65003
BGP table version is 508, main routing table version 508
9 network entries using 2232 bytes of memory
9 path entries using 1224 bytes of memory
7/7 BGP path/bestpath attribute entries using 2072 bytes of memory
2 BGP AS-PATH entries using 64 bytes of memory
6 BGP community entries using 144 bytes of memory
6 BGP route-map cache entries using 432 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 6168 total bytes of memory
BGP activity 249/240 prefixes, 258/249 paths, scan interval 60 secs
9 networks peaked at 01:35:13 Sep 8 2026 UTC (23:22:18.801 ago)

Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
... 1 more lines
```

Teardown (suite) Restore BGP Baseline, Close All Routers

9.02 PASS Each Spoke Learns The Other Spoke Prefix Through The Hub (0.1s, 2 commands)

The defining property of this topology: spokes never peer with each other, so R3's prefix can only reach R2 via the hub, and its AS path must show both hops.

Setup (suite) Establish BGP

Acceptance criteria

- output contains **${via}** (checked 2×)
- output matches **${R1_ASN}\s+${far_asn}** (checked 2×)

How it was checked

R2 show bgp ipv4 unicast 10.20.3.1

```
BGP routing table entry for 10.20.3.1/32, version 68
Paths: (1 available, best #1, table default)
  Not advertised to any peer
  Refresh Epoch 1
  65001 65003
    172.20.0.3 from 172.20.0.1 (172.16.1.1)
      Origin IGP, localpref 100, valid, external, best
      Community: 65003:1
      rx pathid: 0, tx pathid: 0x0
      Updated on Sep 9 2026 00:56:26 UTC
```

R3 show bgp ipv4 unicast 10.20.2.1

```
BGP routing table entry for 10.20.2.1/32, version 506
Paths: (1 available, best #1, table default)
  Not advertised to any peer
  Refresh Epoch 1
  65001 65002
    172.20.0.2 from 172.20.0.1 (172.16.1.1)
      Origin IGP, localpref 100, valid, external, best
      Community: 65002:1
      rx pathid: 0, tx pathid: 0x0
      Updated on Sep 9 2026 00:55:59 UTC
```

Teardown (suite) Restore BGP Baseline, Close All Routers

9.03 PASS Every Peer Is Configured With MD5 Authentication (0.4s, 3 commands)

The negative test below proves a wrong password is rejected; this proves no peer was left without one in the first place.

Setup (suite) Establish BGP

Acceptance criteria

- **${remote}** is not empty (checked 3×)
- **${remote}** matches **${secured}** exactly (checked 3×)
  msg=${r}: neighbours ${remote} but only ${secured} carry a password

How it was checked

R1 show running-config | include ^ neighbor .* remote-as|^ neighbor .* password

```
neighbor 172.20.0.2 remote-as 65002
 neighbor 172.20.0.2 password LabBgpSecret
 neighbor 172.20.0.3 remote-as 65003
 neighbor 172.20.0.3 password LabBgpSecret
```

R2 show running-config | include ^ neighbor .* remote-as|^ neighbor .* password

```
neighbor 172.20.0.1 remote-as 65001
 neighbor 172.20.0.1 password LabBgpSecret
```

R3 show running-config | include ^ neighbor .* remote-as|^ neighbor .* password

```
neighbor 172.20.0.1 remote-as 65001
 neighbor 172.20.0.1 password LabBgpSecret
```

Teardown (suite) Restore BGP Baseline, Close All Routers

9.04 PASS A BFD Session Is Up For Every BGP Peer (0.1s, 3 commands)

One BFD session per peer, on the tunnel the peer address lives on -- the hub therefore holds two.

Setup (suite) Establish BGP

Acceptance criteria

- output contains **${R2_TUNNEL_LOCAL}**
- output contains **${R3_TUNNEL_LOCAL}**
- output does not contain **Down** (checked 3×)
- output contains **${${r}_TUNNEL_HUB}** (checked 2×)

How it was checked

R1 `show bfd neighbors`

```
IPv4 Sessions
NeighAddr                              LD/RD         RH/RS     State     Int
172.20.0.2                           4098/4110       Up        Up        Tu0
172.20.0.3                           4097/4097       Up        Up        Tu0
```

R2 `show bfd neighbors`

```
IPv4 Sessions
NeighAddr                              LD/RD         RH/RS     State     Int
172.20.0.1                           4110/4098       Up        Up        Tu0
```

R3 `show bfd neighbors`

```
IPv4 Sessions
NeighAddr                              LD/RD         RH/RS     State     Int
172.20.0.1                           4097/4097       Up        Up        Tu0
```

Teardown (suite) Restore BGP Baseline, Close All Routers

9.05 PASS BGP Uses BFD For Fall-over On Every Peer (0.4s, 6 commands)

A BFD session that no protocol consumes would detect failures and change nothing, so this checks BGP is the registered client rather than just that BFD is running.

Setup (suite) Establish BGP

Acceptance criteria

- **${remote}** matches **${bfd}** exactly (checked 3×)

  msg=${r}: neighbours ${remote} but only ${bfd} use BFD fall-over
- output does not contain **Down** (checked 3×)

How it was checked

R1 `show running-config | include ^ neighbor .* remote-as|fall-over bfd`

```
neighbor 172.20.0.2 remote-as 65002
 neighbor 172.20.0.2 fall-over bfd
 neighbor 172.20.0.3 remote-as 65003
 neighbor 172.20.0.3 fall-over bfd
```

R1 `show bfd neighbors client bgp`

```
IPv4 Sessions
NeighAddr                              LD/RD         RH/RS     State     Int
172.20.0.2                           4098/4110       Up        Up        Tu0
172.20.0.3                           4097/4097       Up        Up        Tu0
```

R2 `show running-config | include ^ neighbor .* remote-as|fall-over bfd`

```
neighbor 172.20.0.1 remote-as 65001
 neighbor 172.20.0.1 fall-over bfd
```

R2 `show bfd neighbors client bgp`

```
IPv4 Sessions
NeighAddr                              LD/RD         RH/RS     State     Int
172.20.0.1                           4110/4098       Up        Up        Tu0
```

R3 `show running-config | include ^ neighbor .* remote-as|fall-over bfd`

```
neighbor 172.20.0.1 remote-as 65001
 neighbor 172.20.0.1 fall-over bfd
```

R3 `show bfd neighbors client bgp`

```
IPv4 Sessions
NeighAddr                              LD/RD         RH/RS     State     Int
172.20.0.1                           4097/4097       Up        Up        Tu0
```

Teardown (suite) Restore BGP Baseline, Close All Routers

9.06 PASS BFD Tears The Session Down Long Before The Hold Time (42.8s, 32 commands)

The test that makes BFD more than configuration. An ACL on the hub's tunnel drops only BFD control packets (UDP 3784 and 3785) and leaves BGP's TCP session untouched, so ordinary BGP liveness is unaffected. Without fall-over the session would survive until the 30s hold time expires; with it, BFD declares the peer down in about a second and BGP follows.

Setup (suite) Establish BGP

Acceptance criteria

- output matches **${peer}\s+4\s+${peer_asn}(\s+\d+){5}\s+\S+\s+\d+** (checked 4×)
- output does not match **^\d+$** (checked 3×)
- **${elapsed} < ${HOLD_TIME}** holds
- output contains **Known via "bgp** (checked 8×)

How it was checked

```
R1 show bgp ipv4 unicast summary
  BGP router identifier 172.16.1.1, local AS number 65001
  BGP table version is 58, main routing table version 58
  9 network entries using 2232 bytes of memory
  9 path entries using 1224 bytes of memory
  7/7 BGP path/bestpath attribute entries using 2072 bytes of memory
  2 BGP AS-PATH entries using 48 bytes of memory
  6 BGP community entries using 144 bytes of memory
  6 BGP route-map cache entries using 432 bytes of memory
  0 BGP filter-list cache entries using 0 bytes of memory
  BGP using 6152 total bytes of memory
  BGP activity 315/306 prefixes, 336/327 paths, scan interval 60 secs
  22 networks peaked at 02:27:59 Sep 8 2026 UTC (22:29:33.748 ago)

  Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
  ... 2 more lines
R1 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R1 ip access-list extended BLOCK-BFD
  ${output} =
R1 deny udp any any eq 3784
  ${output} =
R1 deny udp any any eq 3785
  ${output} =
R1 permit ip any any
  ${output} =
R1 end
  ${output} =
R1 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R1 interface Tunnel0
  ${output} =
R1 ip access-group BLOCK-BFD in
  ${output} =
R1 end
  ${output} =
R1 show bgp ipv4 unicast summary | include 172.20.0.2
  172.20.0.2      4        65002      17      20       58    0    0 00:01:37        3
R1 show bgp ipv4 unicast summary | include 172.20.0.2
  172.20.0.2      4        65002      17      20       58    0    0 00:01:38        3
R1 show bgp ipv4 unicast summary | include 172.20.0.2
  172.20.0.2      4        65002       0       0        1    0    0 00:00:00 Idle
R1 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R1 interface Tunnel0
  ${output} =
R1 no ip access-group BLOCK-BFD in
  ${output} =
R1 end
  ${output} =
R1 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R1 no ip access-list extended BLOCK-BFD
```

```
${output} =
```

R1 end

```
${output} =
```

R1 show bgp ipv4 unicast summary

```
BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 64, main routing table version 64
3 network entries using 744 bytes of memory
3 path entries using 408 bytes of memory
1/1 BGP path/bestpath attribute entries using 296 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 1448 total bytes of memory
BGP activity 315/312 prefixes, 336/333 paths, scan interval 60 secs
22 networks peaked at 02:27:59 Sep 8 2026 UTC (22:29:36.214 ago)

Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
172.20.0.2      4        65002       0       0        1    0    0 00:00:00 Idle
172.20.0.3      4        65003       0       0        1    0    0 00:00:01 Idle
```

R1 show bgp ipv4 unicast summary

```
BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 64, main routing table version 64
3 network entries using 744 bytes of memory
3 path entries using 408 bytes of memory
1/1 BGP path/bestpath attribute entries using 296 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 1448 total bytes of memory
BGP activity 315/312 prefixes, 336/333 paths, scan interval 60 secs
22 networks peaked at 02:27:59 Sep 8 2026 UTC (22:29:41.247 ago)

Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
172.20.0.2      4        65002       0       0        1    0    0 00:00:05 Idle
172.20.0.3      4        65003       0       0        1    0    0 00:00:06 Idle
```

R1 show bgp ipv4 unicast summary

```
BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 70, main routing table version 70
9 network entries using 2232 bytes of memory
9 path entries using 1224 bytes of memory
7/7 BGP path/bestpath attribute entries using 2072 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
6 BGP community entries using 144 bytes of memory
6 BGP route-map cache entries using 432 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 6152 total bytes of memory
BGP activity 321/312 prefixes, 342/333 paths, scan interval 60 secs
22 networks peaked at 02:27:59 Sep 8 2026 UTC (22:29:46.319 ago)

Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
... 2 more lines
```

R1 show ip route 10.20.2.1

```
Routing entry for 10.20.2.1/32
  Known via "bgp 65001", distance 20, metric 0
  Tag 65002, type external
  Last update from 172.20.0.2 00:00:03 ago
  Routing Descriptor Blocks:
  * 172.20.0.2, from 172.20.0.2, 00:00:03 ago
      opaque_ptr 0x71FE1BFD9038
      Route metric is 0, traffic share count is 1
      AS Hops 1
      Route tag 65002
      MPLS label: none
```

R3 show ip route 10.20.2.1

```
% Subnet not in table
```

R3 show ip route 10.20.2.1

```
% Subnet not in table
```

R3 show ip route 10.20.2.1

```
% Subnet not in table
```

R3 show ip route 10.20.2.1

```
% Subnet not in table
```

R3 show ip route 10.20.2.1

```
% Subnet not in table
```

R3 show ip route 10.20.2.1

```
% Subnet not in table
```

R3 show ip route 10.20.2.1

```
Routing entry for 10.20.2.1/32
```

```
Known via "bgp 65003", distance 20, metric 0
Tag 65001, type external
Last update from 172.20.0.2 00:00:03 ago
Routing Descriptor Blocks:
* 172.20.0.2, from 172.20.0.1, 00:00:03 ago
    opaque_ptr 0x7AED8E74CA10
    Route metric is 0, traffic share count is 1
    AS Hops 2
    Route tag 65001
    MPLS label: none
```

Teardown (suite) Restore BGP Baseline, Close All Routers

9.07 PASS Every Peer Has An Outbound Route-map Over Per-class Prefix-lists (0.7s, 13 commands)

One prefix-list per prefix class, and a route-map clause per class so each can carry its own community. The clauses are the whitelist: the route-map's implicit deny is what stops anything unnamed being advertised.

Setup (suite) Establish BGP

Acceptance criteria

- output contains **RM-OUT-${peer}** (checked 4×)
- output contains **PL-OWN-${cls}** (checked 9×)

How it was checked

R2 show running-config | include route-map RM-OUT-R2 out
```
neighbor 172.20.0.2 route-map RM-OUT-R2 out
```
R3 show running-config | include route-map RM-OUT-R3 out
```
neighbor 172.20.0.3 route-map RM-OUT-R3 out
```
R1 show ip prefix-list PL-OWN-LO
```
ip prefix-list PL-OWN-LO: 1 entries
   seq 5 permit 10.20.1.1/32
```
R1 show ip prefix-list PL-OWN-NAT
```
ip prefix-list PL-OWN-NAT: 1 entries
   seq 5 permit 10.30.1.0/24
```
R1 show ip prefix-list PL-OWN-LAN
```
ip prefix-list PL-OWN-LAN: 1 entries
   seq 5 permit 192.168.10.0/24
```
R1 show running-config | include route-map RM-OUT-R1 out
```
neighbor 172.20.0.1 route-map RM-OUT-R1 out
```
R2 show ip prefix-list PL-OWN-LO
```
ip prefix-list PL-OWN-LO: 1 entries
   seq 5 permit 10.20.2.1/32
```
R2 show ip prefix-list PL-OWN-NAT
```
ip prefix-list PL-OWN-NAT: 1 entries
   seq 5 permit 10.30.2.0/24
```
R2 show ip prefix-list PL-OWN-LAN
```
ip prefix-list PL-OWN-LAN: 1 entries
   seq 5 permit 192.168.20.0/24
```
R1 show running-config | include route-map RM-OUT-R1 out
```
neighbor 172.20.0.1 route-map RM-OUT-R1 out
```
R3 show ip prefix-list PL-OWN-LO
```
ip prefix-list PL-OWN-LO: 1 entries
   seq 5 permit 10.20.3.1/32
```
R3 show ip prefix-list PL-OWN-NAT
```
ip prefix-list PL-OWN-NAT: 1 entries
   seq 5 permit 10.30.3.0/24
```
R3 show ip prefix-list PL-OWN-LAN
```
ip prefix-list PL-OWN-LAN: 1 entries
   seq 5 permit 192.168.30.0/24
```

Teardown (suite) Restore BGP Baseline, Close All Routers

9.08 **PASS** Every Peer Has An Inbound Policy Matching Community And AS Path (0.6s, 11 commands)

*Both matches sit in one permit clause, so they are ANDed and the clause's implicit deny is the drop: a prefix with an unknown community *or* an unexpected AS path falls through.*

Setup (suite) Establish BGP

Acceptance criteria

- output contains **RM-IN-${peer}** (checked 4×)
- output contains **community (community-list filter): CL-VALID** (checked 4×)
- output matches **as-path \(as-path filter\): \d+** (checked 4×)
- **${entries}** has exactly **9** entries (checked 3×)

How it was checked

R2 `show running-config | include route-map RM-IN-R2 in`
```
neighbor 172.20.0.2 route-map RM-IN-R2 in
```

R2 `show route-map RM-IN-R2`
```
route-map RM-IN-R2, permit, sequence 10
  Match clauses:
    as-path (as-path filter): 10
    community (community-list filter): CL-VALID
  Set clauses:
  Policy routing matches: 0 packets, 0 bytes
```

R3 `show running-config | include route-map RM-IN-R3 in`
```
neighbor 172.20.0.3 route-map RM-IN-R3 in
```

R3 `show route-map RM-IN-R3`
```
route-map RM-IN-R3, permit, sequence 10
  Match clauses:
    as-path (as-path filter): 20
    community (community-list filter): CL-VALID
  Set clauses:
  Policy routing matches: 0 packets, 0 bytes
```

R1 `show ip community-list CL-VALID`
```
Named Community standard list CL-VALID
    permit 65001:1
    permit 65001:2
    permit 65001:3
    permit 65002:1
    permit 65002:2
    permit 65002:3
    permit 65003:1
    permit 65003:2
    permit 65003:3
```

R1 `show running-config | include route-map RM-IN-R1 in`
```
neighbor 172.20.0.1 route-map RM-IN-R1 in
```

R1 `show route-map RM-IN-R1`
```
route-map RM-IN-R1, permit, sequence 10
  Match clauses:
    as-path (as-path filter): 10
    community (community-list filter): CL-VALID
  Set clauses:
  Policy routing matches: 0 packets, 0 bytes
```

R2 `show ip community-list CL-VALID`
```
Named Community standard list CL-VALID
    permit 65001:1
    permit 65001:2
    permit 65001:3
    permit 65002:1
    permit 65002:2
    permit 65002:3
    permit 65003:1
    permit 65003:2
    permit 65003:3
```

R1 `show running-config | include route-map RM-IN-R1 in`
```
neighbor 172.20.0.1 route-map RM-IN-R1 in
```

R1 `show route-map RM-IN-R1`
```
route-map RM-IN-R1, permit, sequence 10
  Match clauses:
    as-path (as-path filter): 10
    community (community-list filter): CL-VALID
  Set clauses:
```

```
    Policy routing matches: 0 packets, 0 bytes
```

R3 show ip community-list CL-VALID

```
Named Community standard list CL-VALID
    permit 65001:1
    permit 65001:2
    permit 65001:3
    permit 65002:1
    permit 65002:2
    permit 65002:3
    permit 65003:1
    permit 65003:2
    permit 65003:3
```

Teardown (suite) Restore BGP Baseline, Close All Routers

9.09 PASS Every Received Prefix Carries The Community Of Its Origin (0.6s, 18 commands)

A prefix is tagged by the router that owns it, so the community identifies both the origin AS and the kind of prefix. Checked on the receiver, which is the only place it proves the tag survived the wire.

Setup (suite) Establish BGP

Acceptance criteria

- output contains **${expected}** (checked 18×)

How it was checked

R1 show bgp ipv4 unicast 10.20.2.1/32

```
BGP routing table entry for 10.20.2.1/32, version 70
Paths: (1 available, best #1, table default)
  Advertised to update-groups:
     87
  Refresh Epoch 1
  65002
    172.20.0.2 from 172.20.0.2 (172.16.2.1)
      Origin IGP, metric 0, localpref 100, valid, external, best
      Community: 65002:1
      rx pathid: 0, tx pathid: 0x0
      Updated on Sep 9 2026 00:57:42 UTC
```

R1 show bgp ipv4 unicast 10.20.3.1/32

```
BGP routing table entry for 10.20.3.1/32, version 67
Paths: (1 available, best #1, table default)
  Advertised to update-groups:
     88
  Refresh Epoch 1
  65003
    172.20.0.3 from 172.20.0.3 (172.16.3.1)
      Origin IGP, metric 0, localpref 100, valid, external, best
      Community: 65003:1
      rx pathid: 0, tx pathid: 0x0
      Updated on Sep 9 2026 00:57:42 UTC
```

R1 show bgp ipv4 unicast 10.30.2.0/24

```
BGP routing table entry for 10.30.2.0/24, version 69
Paths: (1 available, best #1, table default)
  Advertised to update-groups:
     87
  Refresh Epoch 1
  65002
    172.20.0.2 from 172.20.0.2 (172.16.2.1)
      Origin IGP, metric 0, localpref 100, valid, external, best
      Community: 65002:2
      rx pathid: 0, tx pathid: 0x0
      Updated on Sep 9 2026 00:57:42 UTC
```

R1 show bgp ipv4 unicast 10.30.3.0/24

```
BGP routing table entry for 10.30.3.0/24, version 66
Paths: (1 available, best #1, table default)
  Advertised to update-groups:
     88
  Refresh Epoch 1
  65003
    172.20.0.3 from 172.20.0.3 (172.16.3.1)
      Origin IGP, metric 0, localpref 100, valid, external, best
      Community: 65003:2
      rx pathid: 0, tx pathid: 0x0
      Updated on Sep 9 2026 00:57:42 UTC
```

R1 show bgp ipv4 unicast 192.168.20.0/24

```
BGP routing table entry for 192.168.20.0/24, version 68
Paths: (1 available, best #1, table default)
  Advertised to update-groups:
     87
  Refresh Epoch 1
  65002
    172.20.0.2 from 172.20.0.2 (172.16.2.1)
      Origin IGP, metric 0, localpref 100, valid, external, best
      Community: 65002:3
      rx pathid: 0, tx pathid: 0x0
      Updated on Sep 9 2026 00:57:42 UTC
```

R1 show bgp ipv4 unicast 192.168.30.0/24

```
BGP routing table entry for 192.168.30.0/24, version 65
Paths: (1 available, best #1, table default)
  Advertised to update-groups:
     88
  Refresh Epoch 1
  65003
    172.20.0.3 from 172.20.0.3 (172.16.3.1)
      Origin IGP, metric 0, localpref 100, valid, external, best
      Community: 65003:3
      rx pathid: 0, tx pathid: 0x0
      Updated on Sep 9 2026 00:57:42 UTC
```

R2 show bgp ipv4 unicast 10.20.1.1/32

```
BGP routing table entry for 10.20.1.1/32, version 79
Paths: (1 available, best #1, table default)
  Not advertised to any peer
  Refresh Epoch 1
  65001
    172.20.0.1 from 172.20.0.1 (172.16.1.1)
      Origin IGP, metric 0, localpref 100, valid, external, best
      Community: 65001:1
      rx pathid: 0, tx pathid: 0x0
      Updated on Sep 9 2026 00:57:42 UTC
```

R2 show bgp ipv4 unicast 10.20.3.1/32

```
BGP routing table entry for 10.20.3.1/32, version 80
Paths: (1 available, best #1, table default)
  Not advertised to any peer
  Refresh Epoch 1
  65001 65003
    172.20.0.3 from 172.20.0.1 (172.16.1.1)
      Origin IGP, localpref 100, valid, external, best
      Community: 65003:1
      rx pathid: 0, tx pathid: 0x0
      Updated on Sep 9 2026 00:57:42 UTC
```

R2 show bgp ipv4 unicast 10.30.1.0/24

```
BGP routing table entry for 10.30.1.0/24, version 78
Paths: (1 available, best #1, table default)
  Not advertised to any peer
  Refresh Epoch 1
  65001
    172.20.0.1 from 172.20.0.1 (172.16.1.1)
      Origin IGP, metric 0, localpref 100, valid, external, best
      Community: 65001:2
      rx pathid: 0, tx pathid: 0x0
      Updated on Sep 9 2026 00:57:42 UTC
```

R2 show bgp ipv4 unicast 10.30.3.0/24

```
BGP routing table entry for 10.30.3.0/24, version 81
Paths: (1 available, best #1, table default)
  Not advertised to any peer
  Refresh Epoch 1
  65001 65003
    172.20.0.3 from 172.20.0.1 (172.16.1.1)
      Origin IGP, localpref 100, valid, external, best
      Community: 65003:2
      rx pathid: 0, tx pathid: 0x0
      Updated on Sep 9 2026 00:57:42 UTC
```

R2 show bgp ipv4 unicast 192.168.10.0/24

```
BGP routing table entry for 192.168.10.0/24, version 77
Paths: (1 available, best #1, table default)
  Not advertised to any peer
  Refresh Epoch 1
  65001
    172.20.0.1 from 172.20.0.1 (172.16.1.1)
      Origin IGP, metric 0, localpref 100, valid, external, best
      Community: 65001:3
      rx pathid: 0, tx pathid: 0x0
      Updated on Sep 9 2026 00:57:42 UTC
```

R2 show bgp ipv4 unicast 192.168.30.0/24

```
BGP routing table entry for 192.168.30.0/24, version 82
Paths: (1 available, best #1, table default)
  Not advertised to any peer
  Refresh Epoch 1
  65001 65003
    172.20.0.3 from 172.20.0.1 (172.16.1.1)
      Origin IGP, localpref 100, valid, external, best
      Community: 65003:3
      rx pathid: 0, tx pathid: 0x0
      Updated on Sep 9 2026 00:57:42 UTC
```

R3 show bgp ipv4 unicast 10.20.1.1/32

```
BGP routing table entry for 10.20.1.1/32, version 517
Paths: (1 available, best #1, table default)
  Not advertised to any peer
  Refresh Epoch 1
  65001
    172.20.0.1 from 172.20.0.1 (172.16.1.1)
      Origin IGP, metric 0, localpref 100, valid, external, best
      Community: 65001:1
      rx pathid: 0, tx pathid: 0x0
      Updated on Sep 9 2026 00:57:42 UTC
```

R3 show bgp ipv4 unicast 10.20.2.1/32

```
BGP routing table entry for 10.20.2.1/32, version 518
Paths: (1 available, best #1, table default)
  Flag: 0x8100
  Not advertised to any peer
  Refresh Epoch 1
  65001 65002
    172.20.0.2 from 172.20.0.1 (172.16.1.1)
      Origin IGP, localpref 100, valid, external, best
      Community: 65002:1
      rx pathid: 0, tx pathid: 0x0
      Updated on Sep 9 2026 00:58:12 UTC
```

R3 show bgp ipv4 unicast 10.30.1.0/24

```
BGP routing table entry for 10.30.1.0/24, version 516
Paths: (1 available, best #1, table default)
  Not advertised to any peer
  Refresh Epoch 1
  65001
    172.20.0.1 from 172.20.0.1 (172.16.1.1)
      Origin IGP, metric 0, localpref 100, valid, external, best
      Community: 65001:2
      rx pathid: 0, tx pathid: 0x0
      Updated on Sep 9 2026 00:57:42 UTC
```

R3 show bgp ipv4 unicast 10.30.2.0/24

```
BGP routing table entry for 10.30.2.0/24, version 519
Paths: (1 available, best #1, table default)
  Flag: 0x8100
  Not advertised to any peer
  Refresh Epoch 1
  65001 65002
    172.20.0.2 from 172.20.0.1 (172.16.1.1)
      Origin IGP, localpref 100, valid, external, best
      Community: 65002:2
      rx pathid: 0, tx pathid: 0x0
      Updated on Sep 9 2026 00:58:12 UTC
```

R3 show bgp ipv4 unicast 192.168.10.0/24

```
BGP routing table entry for 192.168.10.0/24, version 515
Paths: (1 available, best #1, table default)
  Not advertised to any peer
  Refresh Epoch 1
  65001
    172.20.0.1 from 172.20.0.1 (172.16.1.1)
      Origin IGP, metric 0, localpref 100, valid, external, best
      Community: 65001:3
      rx pathid: 0, tx pathid: 0x0
      Updated on Sep 9 2026 00:57:42 UTC
```

R3 show bgp ipv4 unicast 192.168.20.0/24

```
BGP routing table entry for 192.168.20.0/24, version 520
Paths: (1 available, best #1, table default)
  Flag: 0x8100
  Not advertised to any peer
  Refresh Epoch 1
  65001 65002
    172.20.0.2 from 172.20.0.1 (172.16.1.1)
      Origin IGP, localpref 100, valid, external, best
      Community: 65002:3
```

```
    rx pathid: 0, tx pathid: 0x0
    Updated on Sep 9 2026 00:58:12 UTC
```

Teardown (suite) Restore BGP Baseline, Close All Routers

9.10 PASS Re-advertised Prefixes Keep Their Origin Community Through The Hub (0.1s, 4 commands)

A spoke-to-spoke prefix crosses the hub, which re-advertises it. If the hub re-tagged it the community would say 65001 and the origin would be lost, so this asserts the far spoke's own community survives the hop.

Setup (suite) Establish BGP

Acceptance criteria

- output contains **${expected}** (checked 4×)

How it was checked

R2 show bgp ipv4 unicast 10.20.3.1/32

```
BGP routing table entry for 10.20.3.1/32, version 80
Paths: (1 available, best #1, table default)
  Not advertised to any peer
  Refresh Epoch 1
  65001 65003
    172.20.0.3 from 172.20.0.1 (172.16.1.1)
      Origin IGP, localpref 100, valid, external, best
      Community: 65003:1
      rx pathid: 0, tx pathid: 0x0
      Updated on Sep 9 2026 00:57:42 UTC
```

R2 show bgp ipv4 unicast 192.168.30.0/24

```
BGP routing table entry for 192.168.30.0/24, version 82
Paths: (1 available, best #1, table default)
  Not advertised to any peer
  Refresh Epoch 1
  65001 65003
    172.20.0.3 from 172.20.0.1 (172.16.1.1)
      Origin IGP, localpref 100, valid, external, best
      Community: 65003:3
      rx pathid: 0, tx pathid: 0x0
      Updated on Sep 9 2026 00:57:42 UTC
```

R3 show bgp ipv4 unicast 10.20.2.1/32

```
BGP routing table entry for 10.20.2.1/32, version 518
Paths: (1 available, best #1, table default)
  Flag: 0x8100
  Not advertised to any peer
  Refresh Epoch 1
  65001 65002
    172.20.0.2 from 172.20.0.1 (172.16.1.1)
      Origin IGP, localpref 100, valid, external, best
      Community: 65002:1
      rx pathid: 0, tx pathid: 0x0
      Updated on Sep 9 2026 00:58:12 UTC
```

R3 show bgp ipv4 unicast 192.168.20.0/24

```
BGP routing table entry for 192.168.20.0/24, version 520
Paths: (1 available, best #1, table default)
  Flag: 0x8100
  Not advertised to any peer
  Refresh Epoch 1
  65001 65002
    172.20.0.2 from 172.20.0.1 (172.16.1.1)
      Origin IGP, localpref 100, valid, external, best
      Community: 65002:3
      rx pathid: 0, tx pathid: 0x0
      Updated on Sep 9 2026 00:58:12 UTC
```

Teardown (suite) Restore BGP Baseline, Close All Routers

9.11 PASS A Prefix With An Unknown Community Is Rejected Inbound (31.1s, 41 commands)

*The prefix is explicitly permitted *outbound* by a temporary clause, so the outbound whitelist cannot be what stops it -- only the inbound community match can. The test asserts R2 really did advertise it before asserting R1 refused it, otherwise it would pass whenever the advertisement failed.*

Setup (suite) Establish BGP

Acceptance criteria

- output contains **${BAD_COMM_ADDR}** (checked 7×)

  R2 is not yet advertising the test prefix
- output contains **${BAD_COMM_ADDR}**

  R2 never advertised the prefix, so this proves nothing about the inbound policy
- output contains **Network not in table**

  the hub accepted a prefix carrying community ${UNKNOWN_COMM}

How it was checked

R2 configure terminal
```
Enter configuration commands, one per line.  End with CNTL/Z.
```
R2 interface Loopback98
```
${output} =
```
R2 ip address 10.88.88.88 255.255.255.255
```
${output} =
```
R2 end
```
${output} =
```
R2 configure terminal
```
Enter configuration commands, one per line.  End with CNTL/Z.
```
R2 ip prefix-list PL-TEST seq 5 permit 10.88.88.88/32
```
${output} =
```
R2 end
```
${output} =
```
R2 configure terminal
```
Enter configuration commands, one per line.  End with CNTL/Z.
```
R2 route-map RM-OUT-R1 permit 5
```
${output} =
```
R2 match ip address prefix-list PL-TEST
```
${output} =
```
R2 set community 65500:9
```
${output} =
```
R2 end
```
${output} =
```
R2 configure terminal
```
Enter configuration commands, one per line.  End with CNTL/Z.
```
R2 router bgp 65002
```
${output} =
```
R2 address-family ipv4 unicast
```
${output} =
```
R2 network 10.88.88.88 mask 255.255.255.255
```
${output} =
```
R2 end
```
${output} =
```
R2 clear ip bgp * soft out
```
${output} =
```
R2 show bgp ipv4 unicast neighbors 172.20.0.1 advertised-routes
```
BGP table version is 82, local router ID is 172.16.2.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
              t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *>   10.20.2.1/32     0.0.0.0                  0         32768 i
 *>   10.30.2.0/24     0.0.0.0                  0         32768 i
 *>   192.168.20.0     0.0.0.0                  0         32768 i

Total number of prefixes 3
```
R2 show bgp ipv4 unicast neighbors 172.20.0.1 advertised-routes
```
BGP table version is 83, local router ID is 172.16.2.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
              t secondary path, L long-lived-stale,
```

```
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *>   10.20.2.1/32     0.0.0.0                  0         32768 i
 *>   10.30.2.0/24     0.0.0.0                  0         32768 i
 *>   192.168.20.0     0.0.0.0                  0         32768 i

Total number of prefixes 3
```

R2 show bgp ipv4 unicast neighbors 172.20.0.1 advertised-routes

```
BGP table version is 83, local router ID is 172.16.2.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
              t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *>   10.20.2.1/32     0.0.0.0                  0         32768 i
 *>   10.30.2.0/24     0.0.0.0                  0         32768 i
 *>   192.168.20.0     0.0.0.0                  0         32768 i

Total number of prefixes 3
```

R2 show bgp ipv4 unicast neighbors 172.20.0.1 advertised-routes

```
BGP table version is 83, local router ID is 172.16.2.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
              t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *>   10.20.2.1/32     0.0.0.0                  0         32768 i
 *>   10.30.2.0/24     0.0.0.0                  0         32768 i
 *>   192.168.20.0     0.0.0.0                  0         32768 i

Total number of prefixes 3
```

R2 show bgp ipv4 unicast neighbors 172.20.0.1 advertised-routes

```
BGP table version is 83, local router ID is 172.16.2.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
              t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *>   10.20.2.1/32     0.0.0.0                  0         32768 i
 *>   10.30.2.0/24     0.0.0.0                  0         32768 i
 *>   192.168.20.0     0.0.0.0                  0         32768 i

Total number of prefixes 3
```

R2 show bgp ipv4 unicast neighbors 172.20.0.1 advertised-routes

```
BGP table version is 83, local router ID is 172.16.2.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
              t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *>   10.20.2.1/32     0.0.0.0                  0         32768 i
 *>   10.30.2.0/24     0.0.0.0                  0         32768 i
 *>   192.168.20.0     0.0.0.0                  0         32768 i

Total number of prefixes 3
```

R2 show bgp ipv4 unicast neighbors 172.20.0.1 advertised-routes

```
BGP table version is 83, local router ID is 172.16.2.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
              t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
```

```
 *>   10.20.2.1/32     0.0.0.0                  0         32768 i
 *>   10.30.2.0/24     0.0.0.0                  0         32768 i
 *>   10.88.88.88/32   0.0.0.0                  0         32768 i
 *>   192.168.20.0     0.0.0.0                  0         32768 i

... 1 more lines
```

R2 **`show bgp ipv4 unicast neighbors 172.20.0.1 advertised-routes`**

```
BGP table version is 83, local router ID is 172.16.2.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
              t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *>   10.20.2.1/32     0.0.0.0                  0         32768 i
 *>   10.30.2.0/24     0.0.0.0                  0         32768 i
 *>   10.88.88.88/32   0.0.0.0                  0         32768 i
 *>   192.168.20.0     0.0.0.0                  0         32768 i

... 1 more lines
```

R1 **`show bgp ipv4 unicast 10.88.88.88`**

```
% Network not in table
```

R2 **`configure terminal`**

```
Enter configuration commands, one per line.  End with CNTL/Z.
```

R2 **`router bgp 65002`**

```
${output} =
```

R2 **`address-family ipv4 unicast`**

```
${output} =
```

R2 **`no network 10.88.88.88 mask 255.255.255.255`**

```
${output} =
```

R2 **`end`**

```
${output} =
```

R2 **`configure terminal`**

```
Enter configuration commands, one per line.  End with CNTL/Z.
```

R2 **`no route-map RM-OUT-R1 permit 5`**

```
${output} =
```

R2 **`end`**

```
${output} =
```

R2 **`configure terminal`**

```
Enter configuration commands, one per line.  End with CNTL/Z.
```

R2 **`no ip prefix-list PL-TEST`**

```
${output} =
```

R2 **`end`**

```
${output} =
```

R2 **`configure terminal`**

```
Enter configuration commands, one per line.  End with CNTL/Z.
```

R2 **`no interface Loopback98`**

```
${output} =
```

R2 **`end`**

```
${output} =
```

Teardown (this test) Remove Test Prefix From R2

9.12 PASS Every Peer Has A Maximum-prefix Limit With Headroom (0.5s, 6 commands)

Filtering controls what a peer may send; this controls how much. The limit must sit above normal load or it would trip in steady state -- a hub holds 3 prefixes per spoke, a spoke 6 -- and far below anything that could exhaust memory.

Setup (suite) Establish BGP

Acceptance criteria

- **${remote}** matches **${limited}** exactly (checked 3×)
  msg=${r}: neighbours ${remote} but only ${limited} have a maximum-prefix
- **${n} < ${MAX_PREFIX}** holds (checked 3×)

How it was checked

R1 **`show running-config | include ^ neighbor .* remote-as|maximum-prefix`**

```
neighbor 172.20.0.2 remote-as 65002
 neighbor 172.20.0.3 remote-as 65003
  neighbor 172.20.0.2 maximum-prefix 15 80
  neighbor 172.20.0.3 maximum-prefix 15 80
```

R1 show bgp ipv4 unicast summary | begin Neighbor

```
Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
172.20.0.2      4        65002      18      16       70    0    0 00:01:06        3
172.20.0.3      4        65003      13      16       70    0    0 00:01:06        3
```

R2 show running-config | include ^ neighbor .* remote-as|maximum-prefix

```
neighbor 172.20.0.1 remote-as 65001
  neighbor 172.20.0.1 maximum-prefix 15 80
```

R2 show bgp ipv4 unicast summary | begin Neighbor

```
Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
172.20.0.1      4        65001      16      18       83    0    0 00:01:06        6
```

R3 show running-config | include ^ neighbor .* remote-as|maximum-prefix

```
neighbor 172.20.0.1 remote-as 65001
  neighbor 172.20.0.1 maximum-prefix 15 80
```

R3 show bgp ipv4 unicast summary | begin Neighbor

```
Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
172.20.0.1      4        65001      16      13      520    0    0 00:01:06        6
```

Teardown (suite) Restore BGP Baseline, Close All Routers

9.13 PASS A Peer That Floods Prefixes Is Torn Down Rather Than Absorbed (32.0s, 343 commands)

R2 originates 20 extra prefixes, tagged with a valid community and carrying a valid AS path, and explicitly permitted outbound -- so neither the whitelist, the community match nor the AS-path filter can be what stops them. Only the prefix count is left. The hub must drop the session instead of accepting an unbounded table.

Setup (suite) Establish BGP

Acceptance criteria

- output does not match **^\d+$** (checked 6×)
- output contains **PfxCt**

  the session went down, but not because of the prefix count: ${state}
- output contains **${R2_TUNNEL_LOCAL}**

  the hub logged no maximum-prefix violation for ${R2_TUNNEL_LOCAL}

How it was checked

R2 configure terminal

```
Enter configuration commands, one per line.  End with CNTL/Z.
```

R2 ip prefix-list PL-FLOOD seq 5 permit 10.99.0.0/16 le 32

```
${output} =
```

R2 end

```
${output} =
```

R2 configure terminal

```
Enter configuration commands, one per line.  End with CNTL/Z.
```

R2 route-map RM-OUT-R1 permit 7

```
${output} =
```

R2 match ip address prefix-list PL-FLOOD

```
${output} =
```

R2 set community 65002:1

```
${output} =
```

R2 end

```
${output} =
```

R2 configure terminal

```
Enter configuration commands, one per line.  End with CNTL/Z.
```

R2 ip route 10.99.0.0 255.255.255.0 Null0

```
${output} =
```

R2 end

```
${output} =
```

R2 configure terminal

```
Enter configuration commands, one per line.  End with CNTL/Z.
```

R2 ip route 10.99.1.0 255.255.255.0 Null0

```
${output} =
```

R2 end
${output} =

R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R2 ip route 10.99.2.0 255.255.255.0 Null0
${output} =

R2 end
${output} =

R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R2 ip route 10.99.3.0 255.255.255.0 Null0
${output} =

R2 end
${output} =

R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R2 ip route 10.99.4.0 255.255.255.0 Null0
${output} =

R2 end
${output} =

R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R2 ip route 10.99.5.0 255.255.255.0 Null0
${output} =

R2 end
${output} =

R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R2 ip route 10.99.6.0 255.255.255.0 Null0
${output} =

R2 end
${output} =

R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R2 ip route 10.99.7.0 255.255.255.0 Null0
${output} =

R2 end
${output} =

R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R2 ip route 10.99.8.0 255.255.255.0 Null0
${output} =

R2 end
${output} =

R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R2 ip route 10.99.9.0 255.255.255.0 Null0
${output} =

R2 end
${output} =

R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R2 ip route 10.99.10.0 255.255.255.0 Null0
${output} =

R2 end
${output} =

R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R2 ip route 10.99.11.0 255.255.255.0 Null0
${output} =

R2 end
${output} =

R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R2 ip route 10.99.12.0 255.255.255.0 Null0

```
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 ip route 10.99.13.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 ip route 10.99.14.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 ip route 10.99.15.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 ip route 10.99.16.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 ip route 10.99.17.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 ip route 10.99.18.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 ip route 10.99.19.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 network 10.99.0.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 network 10.99.1.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
```

R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R2 router bgp 65002
${output} =

R2 address-family ipv4 unicast
${output} =

R2 network 10.99.2.0 mask 255.255.255.0
${output} =

R2 end
${output} =

R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R2 router bgp 65002
${output} =

R2 address-family ipv4 unicast
${output} =

R2 network 10.99.3.0 mask 255.255.255.0
${output} =

R2 end
${output} =

R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R2 router bgp 65002
${output} =

R2 address-family ipv4 unicast
${output} =

R2 network 10.99.4.0 mask 255.255.255.0
${output} =

R2 end
${output} =

R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R2 router bgp 65002
${output} =

R2 address-family ipv4 unicast
${output} =

R2 network 10.99.5.0 mask 255.255.255.0
${output} =

R2 end
${output} =

R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R2 router bgp 65002
${output} =

R2 address-family ipv4 unicast
${output} =

R2 network 10.99.6.0 mask 255.255.255.0
${output} =

R2 end
${output} =

R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R2 router bgp 65002
${output} =

R2 address-family ipv4 unicast
${output} =

R2 network 10.99.7.0 mask 255.255.255.0
${output} =

R2 end
${output} =

R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R2 router bgp 65002
${output} =

R2 address-family ipv4 unicast

${output} =

R2 network 10.99.8.0 mask 255.255.255.0
${output} =

R2 end
${output} =

R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R2 router bgp 65002
${output} =

R2 address-family ipv4 unicast
${output} =

R2 network 10.99.9.0 mask 255.255.255.0
${output} =

R2 end
${output} =

R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R2 router bgp 65002
${output} =

R2 address-family ipv4 unicast
${output} =

R2 network 10.99.10.0 mask 255.255.255.0
${output} =

R2 end
${output} =

R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R2 router bgp 65002
${output} =

R2 address-family ipv4 unicast
${output} =

R2 network 10.99.11.0 mask 255.255.255.0
${output} =

R2 end
${output} =

R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R2 router bgp 65002
${output} =

R2 address-family ipv4 unicast
${output} =

R2 network 10.99.12.0 mask 255.255.255.0
${output} =

R2 end
${output} =

R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R2 router bgp 65002
${output} =

R2 address-family ipv4 unicast
${output} =

R2 network 10.99.13.0 mask 255.255.255.0
${output} =

R2 end
${output} =

R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R2 router bgp 65002
${output} =

R2 address-family ipv4 unicast
${output} =

R2 network 10.99.14.0 mask 255.255.255.0
${output} =

R2 end
${output} =

```
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 network 10.99.15.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 network 10.99.16.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 network 10.99.17.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 network 10.99.18.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 network 10.99.19.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R1 show bgp ipv4 unicast summary | include 172.20.0.2
  172.20.0.2      4        65002      19      17       70    0    0 00:01:10        3
R1 show bgp ipv4 unicast summary | include 172.20.0.2
  172.20.0.2      4        65002      19      17       70    0    0 00:01:15        3
R1 show bgp ipv4 unicast summary | include 172.20.0.2
  172.20.0.2      4        65002      20      18       70    0    0 00:01:20        3
R1 show bgp ipv4 unicast summary | include 172.20.0.2
  172.20.0.2      4        65002      20      18       70    0    0 00:01:25        3
R1 show bgp ipv4 unicast summary | include 172.20.0.2
  172.20.0.2      4        65002      21      19       70    0    0 00:01:30        3
R1 show bgp ipv4 unicast summary | include 172.20.0.2
  172.20.0.2      4        65002       0       0        1    0    0 00:00:03 Idle (PfxCt)
R1 show bgp ipv4 unicast summary | include 172.20.0.2
  172.20.0.2      4        65002       0       0        1    0    0 00:00:03 Idle (PfxCt)
R1 show logging | include MAXPFXEXCEED
```

```
002111: Sep  8 15:33:04.012 UTC: %BGP-3-MAXPFXEXCEED: Number of prefixes received from 172.20.0.2 1/1
(afi/safi): 16 exceeds limit 15
002638: Sep  9 00:59:14.499 UTC: %BGP-3-MAXPFXEXCEED: Number of prefixes received from 172.20.0.2 1/1
(afi/safi): 16 exceeds limit 15
```

R2 configure terminal

```
Enter configuration commands, one per line.  End with CNTL/Z.
```

R2 router bgp 65002

```
${output} =
```

R2 address-family ipv4 unicast

```
${output} =
```

R2 no network 10.99.0.0 mask 255.255.255.0

```
${output} =
```

R2 end

```
${output} =
```

R2 configure terminal

```
Enter configuration commands, one per line.  End with CNTL/Z.
```

R2 no ip route 10.99.0.0 255.255.255.0 Null0

```
${output} =
```

R2 end

```
${output} =
```

R2 configure terminal

```
Enter configuration commands, one per line.  End with CNTL/Z.
```

R2 router bgp 65002

```
${output} =
```

R2 address-family ipv4 unicast

```
${output} =
```

R2 no network 10.99.1.0 mask 255.255.255.0

```
${output} =
```

R2 end

```
${output} =
```

R2 configure terminal

```
Enter configuration commands, one per line.  End with CNTL/Z.
```

R2 no ip route 10.99.1.0 255.255.255.0 Null0

```
${output} =
```

R2 end

```
${output} =
```

R2 configure terminal

```
Enter configuration commands, one per line.  End with CNTL/Z.
```

R2 router bgp 65002

```
${output} =
```

R2 address-family ipv4 unicast

```
${output} =
```

R2 no network 10.99.2.0 mask 255.255.255.0

```
${output} =
```

R2 end

```
${output} =
```

R2 configure terminal

```
Enter configuration commands, one per line.  End with CNTL/Z.
```

R2 no ip route 10.99.2.0 255.255.255.0 Null0

```
${output} =
```

R2 end

```
${output} =
```

R2 configure terminal

```
Enter configuration commands, one per line.  End with CNTL/Z.
```

R2 router bgp 65002

```
${output} =
```

R2 address-family ipv4 unicast

```
${output} =
```

R2 no network 10.99.3.0 mask 255.255.255.0

```
${output} =
```

R2 end

```
${output} =
```

R2 configure terminal

```
Enter configuration commands, one per line.  End with CNTL/Z.
```

R2 no ip route 10.99.3.0 255.255.255.0 Null0

```
${output} =
```

R2 end
${output} =

R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R2 router bgp 65002
${output} =

R2 address-family ipv4 unicast
${output} =

R2 no network 10.99.4.0 mask 255.255.255.0
${output} =

R2 end
${output} =

R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R2 no ip route 10.99.4.0 255.255.255.0 Null0
${output} =

R2 end
${output} =

R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R2 router bgp 65002
${output} =

R2 address-family ipv4 unicast
${output} =

R2 no network 10.99.5.0 mask 255.255.255.0
${output} =

R2 end
${output} =

R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R2 no ip route 10.99.5.0 255.255.255.0 Null0
${output} =

R2 end
${output} =

R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R2 router bgp 65002
${output} =

R2 address-family ipv4 unicast
${output} =

R2 no network 10.99.6.0 mask 255.255.255.0
${output} =

R2 end
${output} =

R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R2 no ip route 10.99.6.0 255.255.255.0 Null0
${output} =

R2 end
${output} =

R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R2 router bgp 65002
${output} =

R2 address-family ipv4 unicast
${output} =

R2 no network 10.99.7.0 mask 255.255.255.0
${output} =

R2 end
${output} =

R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R2 no ip route 10.99.7.0 255.255.255.0 Null0
${output} =

R2 end

```
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 no network 10.99.8.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 no ip route 10.99.8.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 no network 10.99.9.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 no ip route 10.99.9.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 no network 10.99.10.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 no ip route 10.99.10.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 no network 10.99.11.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 no ip route 10.99.11.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
```

R2 configure terminal
```
Enter configuration commands, one per line.  End with CNTL/Z.
```
R2 router bgp 65002
```
${output} =
```
R2 address-family ipv4 unicast
```
${output} =
```
R2 no network 10.99.12.0 mask 255.255.255.0
```
${output} =
```
R2 end
```
${output} =
```
R2 configure terminal
```
Enter configuration commands, one per line.  End with CNTL/Z.
```
R2 no ip route 10.99.12.0 255.255.255.0 Null0
```
${output} =
```
R2 end
```
${output} =
```
R2 configure terminal
```
Enter configuration commands, one per line.  End with CNTL/Z.
```
R2 router bgp 65002
```
${output} =
```
R2 address-family ipv4 unicast
```
${output} =
```
R2 no network 10.99.13.0 mask 255.255.255.0
```
${output} =
```
R2 end
```
${output} =
```
R2 configure terminal
```
Enter configuration commands, one per line.  End with CNTL/Z.
```
R2 no ip route 10.99.13.0 255.255.255.0 Null0
```
${output} =
```
R2 end
```
${output} =
```
R2 configure terminal
```
Enter configuration commands, one per line.  End with CNTL/Z.
```
R2 router bgp 65002
```
${output} =
```
R2 address-family ipv4 unicast
```
${output} =
```
R2 no network 10.99.14.0 mask 255.255.255.0
```
${output} =
```
R2 end
```
${output} =
```
R2 configure terminal
```
Enter configuration commands, one per line.  End with CNTL/Z.
```
R2 no ip route 10.99.14.0 255.255.255.0 Null0
```
${output} =
```
R2 end
```
${output} =
```
R2 configure terminal
```
Enter configuration commands, one per line.  End with CNTL/Z.
```
R2 router bgp 65002
```
${output} =
```
R2 address-family ipv4 unicast
```
${output} =
```
R2 no network 10.99.15.0 mask 255.255.255.0
```
${output} =
```
R2 end
```
${output} =
```
R2 configure terminal
```
Enter configuration commands, one per line.  End with CNTL/Z.
```
R2 no ip route 10.99.15.0 255.255.255.0 Null0
```
${output} =
```
R2 end
```
${output} =
```
R2 configure terminal

```
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 no network 10.99.16.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 no ip route 10.99.16.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 no network 10.99.17.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 no ip route 10.99.17.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 no network 10.99.18.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 no ip route 10.99.18.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 router bgp 65002
  ${output} =
R2 address-family ipv4 unicast
  ${output} =
R2 no network 10.99.19.0 mask 255.255.255.0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 no ip route 10.99.19.0 255.255.255.0 Null0
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
```

R2 **no route-map RM-OUT-R1 permit 7**
```
${output} =
```
R2 **end**
```
${output} =
```
R2 **configure terminal**
```
Enter configuration commands, one per line.  End with CNTL/Z.
```
R2 **no ip prefix-list PL-FLOOD**
```
${output} =
```
R2 **end**
```
${output} =
```
R1 **clear ip bgp 172.20.0.2**
```
${output} =
```
Teardown (this test) Stop Flooding From R2

9.14 PASS The Session Recovers Once The Flood Stops (25.2s, 8 commands)

A maximum-prefix teardown is deliberately sticky -- no "restart" is configured -- so recovery takes an operator clearing it. This asserts the lab returns to a converged state rather than leaving the next suite a broken topology.

Setup (suite) Establish BGP

Acceptance criteria

- output matches **${peer}\s+4\s+${peer_asn}(\s+\d+){5}\s+\S+\s+\d+** (checked 2×)
- output contains **Known via "bgp** (checked 6×)

How it was checked

R1 **show bgp ipv4 unicast summary**
```
BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 97, main routing table version 97
6 network entries using 1488 bytes of memory
6 path entries using 816 bytes of memory
8/4 BGP path/bestpath attribute entries using 2368 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
7 BGP community entries using 168 bytes of memory
7 BGP route-map cache entries using 504 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 5392 total bytes of memory
BGP activity 334/328 prefixes, 355/349 paths, scan interval 60 secs
22 networks peaked at 02:27:59 Sep 8 2026 UTC (22:31:22.072 ago)

Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
... 2 more lines
```
R1 **show bgp ipv4 unicast summary**
```
BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 100, main routing table version 100
9 network entries using 2232 bytes of memory
9 path entries using 1224 bytes of memory
8/7 BGP path/bestpath attribute entries using 2368 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
7 BGP community entries using 168 bytes of memory
7 BGP route-map cache entries using 504 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 6544 total bytes of memory
BGP activity 337/328 prefixes, 358/349 paths, scan interval 60 secs
22 networks peaked at 02:27:59 Sep 8 2026 UTC (22:31:27.108 ago)

Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
... 2 more lines
```
R1 **show ip route 10.20.2.1**
```
Routing entry for 10.20.2.1/32
  Known via "bgp 65001", distance 20, metric 0
  Tag 65002, type external
  Last update from 172.20.0.2 00:00:00 ago
  Routing Descriptor Blocks:
  * 172.20.0.2, from 172.20.0.2, 00:00:00 ago
      opaque_ptr 0x71FE1BFD9038
      Route metric is 0, traffic share count is 1
      AS Hops 1
      Route tag 65002
      MPLS label: none
```
R3 **show ip route 10.20.2.1**
```
% Subnet not in table
```

R3 **show ip route 10.20.2.1**
```
% Subnet not in table
```
R3 **show ip route 10.20.2.1**
```
% Subnet not in table
```
R3 **show ip route 10.20.2.1**
```
% Subnet not in table
```
R3 **show ip route 10.20.2.1**
```
Routing entry for 10.20.2.1/32
  Known via "bgp 65003", distance 20, metric 0
  Tag 65001, type external
  Last update from 172.20.0.2 00:00:01 ago
  Routing Descriptor Blocks:
  * 172.20.0.2, from 172.20.0.1, 00:00:01 ago
      opaque_ptr 0x7AED8E74CA10
      Route metric is 0, traffic share count is 1
      AS Hops 2
      Route tag 65001
      MPLS label: none
```
Teardown (suite) Restore BGP Baseline, Close All Routers

9.15 PASS A Prefix With An Invalid AS Path Is Rejected Inbound (10.9s, 38 commands)

Same prefix, this time with a valid community but an AS path this topology cannot produce. Only the AS-path match differs, so a rejection here isolates that filter.

Setup (suite) Establish BGP

Acceptance criteria

- output contains **${BAD_COMM_ADDR}** (checked 3×)
  R2 is not yet advertising the test prefix
- output contains **${BAD_COMM_ADDR}**
  R2 never advertised the prefix, so this proves nothing about the inbound policy
- output contains **Network not in table**
  the hub accepted a prefix whose AS path contained ${BOGUS_AS}

How it was checked

R2 **configure terminal**
```
Enter configuration commands, one per line.  End with CNTL/Z.
```
R2 **interface Loopback98**
```
${output} =
```
R2 **ip address 10.88.88.88 255.255.255.255**
```
${output} =
```
R2 **end**
```
${output} =
```
R2 **configure terminal**
```
Enter configuration commands, one per line.  End with CNTL/Z.
```
R2 **ip prefix-list PL-TEST seq 5 permit 10.88.88.88/32**
```
${output} =
```
R2 **end**
```
${output} =
```
R2 **configure terminal**
```
Enter configuration commands, one per line.  End with CNTL/Z.
```
R2 **route-map RM-OUT-R1 permit 5**
```
${output} =
```
R2 **match ip address prefix-list PL-TEST**
```
${output} =
```
R2 **set community 65002:1**
```
${output} =
```
R2 **set as-path prepend 65099**
```
${output} =
```
R2 **end**
```
${output} =
```
R2 **configure terminal**
```
Enter configuration commands, one per line.  End with CNTL/Z.
```
R2 **router bgp 65002**
```
${output} =
```

R2 address-family ipv4 unicast

```
${output} =
```

R2 network 10.88.88.88 mask 255.255.255.255

```
${output} =
```

R2 end

```
${output} =
```

R2 clear ip bgp * soft out

```
${output} =
```

R2 show bgp ipv4 unicast neighbors 172.20.0.1 advertised-routes

```
BGP table version is 136, local router ID is 172.16.2.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
              t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *>   10.20.2.1/32     0.0.0.0                  0         32768 i
 *>   10.30.2.0/24     0.0.0.0                  0         32768 i
 *>   192.168.20.0     0.0.0.0                  0         32768 i

Total number of prefixes 3
```

R2 show bgp ipv4 unicast neighbors 172.20.0.1 advertised-routes

```
BGP table version is 137, local router ID is 172.16.2.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
              t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *>   10.20.2.1/32     0.0.0.0                  0         32768 i
 *>   10.30.2.0/24     0.0.0.0                  0         32768 i
 *>   192.168.20.0     0.0.0.0                  0         32768 i

Total number of prefixes 3
```

R2 show bgp ipv4 unicast neighbors 172.20.0.1 advertised-routes

```
BGP table version is 137, local router ID is 172.16.2.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
              t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *>   10.20.2.1/32     0.0.0.0                  0         32768 i
 *>   10.30.2.0/24     0.0.0.0                  0         32768 i
 *>   10.88.88.88/32   0.0.0.0                  0         32768 i
 *>   192.168.20.0     0.0.0.0                  0         32768 i

... 1 more lines
```

R2 show bgp ipv4 unicast neighbors 172.20.0.1 advertised-routes

```
BGP table version is 137, local router ID is 172.16.2.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
              t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *>   10.20.2.1/32     0.0.0.0                  0         32768 i
 *>   10.30.2.0/24     0.0.0.0                  0         32768 i
 *>   10.88.88.88/32   0.0.0.0                  0         32768 i
 *>   192.168.20.0     0.0.0.0                  0         32768 i

... 1 more lines
```

R1 show bgp ipv4 unicast 10.88.88.88

```
% Network not in table
```

R2 configure terminal

```
Enter configuration commands, one per line.  End with CNTL/Z.
```

R2 router bgp 65002

```
${output} =
```

```
R2 address-family ipv4 unicast
  ${output} =
R2 no network 10.88.88.88 mask 255.255.255.255
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 no route-map RM-OUT-R1 permit 5
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 no ip prefix-list PL-TEST
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 no interface Loopback98
  ${output} =
R2 end
  ${output} =
```

Teardown (this test) Remove Test Prefix From R2

9.16 PASS Each Peer Is Advertised Exactly Its Whitelisted Prefixes (0.3s, 4 commands)

Not "at least" -- exactly. A whitelist that permits more than intended still passes a containment check, so the advertised set is compared for equality against what the peer should get: a spoke sends only its own prefixes, the hub sends its own plus the far spoke's.

Setup (suite) Establish BGP

Acceptance criteria

- **${actual}** matches **${expected}** exactly (checked 4×)
  msg=${from} advertises ${actual} to ${to}, expected ${expected}

How it was checked

```
R2 show bgp ipv4 unicast neighbors 172.20.0.2 advertised-routes
  BGP table version is 100, local router ID is 172.16.1.1
  Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
                r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
                x best-external, a additional-path, c RIB-compressed,
                t secondary path, L long-lived-stale,
  Origin codes: i - IGP, e - EGP, ? - incomplete
  RPKI validation codes: V valid, I invalid, N Not found

       Network          Next Hop            Metric LocPrf Weight Path
   *>   10.20.1.1/32     0.0.0.0                  0         32768 i
   *>   10.20.3.1/32     172.20.0.3               0             0 65003 i
   *>   10.30.1.0/24     0.0.0.0                  0         32768 i
   *>   10.30.3.0/24     172.20.0.3               0             0 65003 i
   *>   192.168.10.0     0.0.0.0                  0         32768 i
  ... 3 more lines
R3 show bgp ipv4 unicast neighbors 172.20.0.3 advertised-routes
  BGP table version is 100, local router ID is 172.16.1.1
  Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
                r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
                x best-external, a additional-path, c RIB-compressed,
                t secondary path, L long-lived-stale,
  Origin codes: i - IGP, e - EGP, ? - incomplete
  RPKI validation codes: V valid, I invalid, N Not found

       Network          Next Hop            Metric LocPrf Weight Path
   *>   10.20.1.1/32     0.0.0.0                  0         32768 i
   *>   10.20.2.1/32     172.20.0.2               0             0 65002 i
   *>   10.30.1.0/24     0.0.0.0                  0         32768 i
   *>   10.30.2.0/24     172.20.0.2               0             0 65002 i
   *>   192.168.10.0     0.0.0.0                  0         32768 i
```

```
... 3 more lines
```

R1 **show bgp ipv4 unicast neighbors 172.20.0.1 advertised-routes**

```
BGP table version is 138, local router ID is 172.16.2.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
              t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *>   10.20.2.1/32     0.0.0.0                  0         32768 i
 *>   10.30.2.0/24     0.0.0.0                  0         32768 i
 *>   192.168.20.0     0.0.0.0                  0         32768 i

Total number of prefixes 3
```

R1 **show bgp ipv4 unicast neighbors 172.20.0.1 advertised-routes**

```
BGP table version is 526, local router ID is 172.16.3.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
              t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *>   10.20.3.1/32     0.0.0.0                  0         32768 i
 *>   10.30.3.0/24     0.0.0.0                  0         32768 i
 *>   192.168.30.0     0.0.0.0                  0         32768 i

Total number of prefixes 3
```

Teardown (suite) Restore BGP Baseline, Close All Routers

9.17 PASS A Prefix Outside The Whitelist Is Not Advertised (10.5s, 20 commands)

The point of a whitelist. A spoke originates a prefix nobody authorised; it appears in that spoke's own BGP table, so the origination worked, and must still never reach the hub.

Setup (suite) Establish BGP

Acceptance criteria

- output contains **${OFF_LIST_ADDR}**
  R2 never originated the prefix, so this test proves nothing
- output does not contain **${OFF_LIST_ADDR}/32**
  R2 advertised an unlisted prefix to the hub
- output contains **Network not in table**
  an unlisted prefix reached the hub

How it was checked

R2 **configure terminal**

```
Enter configuration commands, one per line.  End with CNTL/Z.
```

R2 **interface Loopback99**

```
${output} =
```

R2 **ip address 10.77.77.77 255.255.255.255**

```
${output} =
```

R2 **end**

```
${output} =
```

R2 **configure terminal**

```
Enter configuration commands, one per line.  End with CNTL/Z.
```

R2 **router bgp 65002**

```
${output} =
```

R2 **address-family ipv4 unicast**

```
${output} =
```

R2 **network 10.77.77.77 mask 255.255.255.255**

```
${output} =
```

R2 **end**

```
${output} =
```

R2 **show bgp ipv4 unicast 10.77.77.77**

```
BGP routing table entry for 10.77.77.77/32, version 139
```

```
  Paths: (1 available, best #1, table default)
    Flag: 0x8100
    Not advertised to any peer
    Refresh Epoch 1
    Local
      0.0.0.0 from 0.0.0.0 (172.16.2.1)
        Origin IGP, metric 0, localpref 100, weight 32768, valid, sourced, local, best
        rx pathid: 0, tx pathid: 0x0
        Updated on Sep 9 2026 00:59:59 UTC
```

R2 show bgp ipv4 unicast neighbors 172.20.0.1 advertised-routes

```
  BGP table version is 139, local router ID is 172.16.2.1
  Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
                r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
                x best-external, a additional-path, c RIB-compressed,
                t secondary path, L long-lived-stale,
  Origin codes: i - IGP, e - EGP, ? - incomplete
  RPKI validation codes: V valid, I invalid, N Not found

       Network          Next Hop            Metric LocPrf Weight Path
   *>   10.20.2.1/32     0.0.0.0                  0         32768 i
   *>   10.30.2.0/24     0.0.0.0                  0         32768 i
   *>   192.168.20.0     0.0.0.0                  0         32768 i

  Total number of prefixes 3
```

R1 show bgp ipv4 unicast 10.77.77.77

```
  % Network not in table
```

R2 configure terminal

```
  Enter configuration commands, one per line.  End with CNTL/Z.
```

R2 router bgp 65002

```
  ${output} =
```

R2 address-family ipv4 unicast

```
  ${output} =
```

R2 no network 10.77.77.77 mask 255.255.255.255

```
  ${output} =
```

R2 end

```
  ${output} =
```

R2 configure terminal

```
  Enter configuration commands, one per line.  End with CNTL/Z.
```

R2 no interface Loopback99

```
  ${output} =
```

R2 end

```
  ${output} =
```

Teardown (suite) Restore BGP Baseline, Close All Routers

9.18 PASS Each Router Receives The Peer Prefix (0.1s, 2 commands)

Setup (suite) Establish BGP

Acceptance criteria

- output matches ***>\s+${R2_BGP_PREFIX}/32\s+${R2_TUNNEL_LOCAL}**
- output contains **${R2_ASN}**
- output matches ***>\s+${R1_BGP_PREFIX}/32\s+${R2_TUNNEL_HUB}**
- output contains **${R1_ASN}**

How it was checked

R1 show bgp ipv4 unicast

```
  BGP table version is 100, local router ID is 172.16.1.1
  Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
                r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
                x best-external, a additional-path, c RIB-compressed,
                t secondary path, L long-lived-stale,
  Origin codes: i - IGP, e - EGP, ? - incomplete
  RPKI validation codes: V valid, I invalid, N Not found

       Network          Next Hop            Metric LocPrf Weight Path
   *>   10.20.1.1/32     0.0.0.0                  0         32768 i
   *>   10.20.2.1/32     172.20.0.2               0             0 65002 i
   *>   10.20.3.1/32     172.20.0.3               0             0 65003 i
   *>   10.30.1.0/24     0.0.0.0                  0         32768 i
   *>   10.30.2.0/24     172.20.0.2               0             0 65002 i
```

```
  ... 4 more lines
```

R2 show bgp ipv4 unicast

```
  BGP table version is 140, local router ID is 172.16.2.1
  Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
                r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
                x best-external, a additional-path, c RIB-compressed,
                t secondary path, L long-lived-stale,
  Origin codes: i - IGP, e - EGP, ? - incomplete
  RPKI validation codes: V valid, I invalid, N Not found

       Network          Next Hop            Metric LocPrf Weight Path
   *>   10.20.1.1/32     172.20.0.1               0             0 65001 i
   *>   10.20.2.1/32     0.0.0.0                  0         32768 i
   *>   10.20.3.1/32     172.20.0.3                             0 65001 65003 i
   *>   10.30.1.0/24     172.20.0.1               0             0 65001 i
   *>   10.30.2.0/24     0.0.0.0                  0         32768 i
  ... 4 more lines
```

Teardown (suite) Restore BGP Baseline, Close All Routers

9.19 PASS Peer Prefix Is Learned Via BGP And Not A Static Route (0.0s, 1 commands)

Distinguishes a genuinely BGP-installed route from one that merely happens to exist. eBGP is distance 20; a static is 1.

Setup (suite) Establish BGP

Acceptance criteria

- output contains **Known via "bgp ${R1_ASN}"**
- output contains **distance 20**
- output contains **type external**

How it was checked

R1 show ip route 10.20.2.1

```
  Routing entry for 10.20.2.1/32
    Known via "bgp 65001", distance 20, metric 0
    Tag 65002, type external
    Last update from 172.20.0.2 00:00:42 ago
    Routing Descriptor Blocks:
    * 172.20.0.2, from 172.20.0.2, 00:00:42 ago
        opaque_ptr 0x71FE1BFD9038
        Route metric is 0, traffic share count is 1
        AS Hops 1
        Route tag 65002
        MPLS label: none
```

Teardown (suite) Restore BGP Baseline, Close All Routers

9.20 PASS BGP Route Resolves Out The Encrypted Tunnel (0.0s, 1 commands)

The next hop is a tunnel address, so the forwarding path must recurse to Tunnel0 rather than the cleartext link.

Setup (suite) Establish BGP

Acceptance criteria

- output contains **${R2_TUNNEL_LOCAL}**
- output contains **Tunnel0**

How it was checked

R1 show ip cef 10.20.2.1

```
  10.20.2.1/32
    nexthop 172.20.0.2 Tunnel0
```

Teardown (suite) Restore BGP Baseline, Close All Routers

9.21 PASS Traffic Over The BGP-Learned Route Reaches The Peer (0.0s, 1 commands)

End to end: this only succeeds because BGP installed the route.

Setup (suite) Establish BGP

Acceptance criteria

- output contains **Success rate is 100 percent**

How it was checked

R1 ping 10.20.2.1 source Loopback1 repeat 5

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.20.2.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.1.1
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
```

Teardown (suite) Restore BGP Baseline, Close All Routers

9.22 PASS BGP-Routed Traffic Is Encrypted (5.7s, 7 commands)

Setup (suite) Establish BGP

Acceptance criteria

- output contains **Success rate is 100 percent**
- **${r1_enc} >= ${expected}** holds (checked 2×)
- **${r1_dec} >= ${expected}** holds (checked 2×)
- **${r2_enc} >= ${expected}** holds (checked 2×)
- **${r2_dec} >= ${expected}** holds (checked 2×)

How it was checked

R1 show crypto ipsec sa

```
interface: Tunnel0
    Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
   remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
   current_peer 100.64.0.3 port 500
     PERMIT, flags={origin_is_acl,}
    #pkts encaps: 971, #pkts encrypt: 971, #pkts digest: 971
    #pkts decaps: 967, #pkts decrypt: 967, #pkts verify: 967
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0
... 78 more lines
```

R2 show crypto ipsec sa

```
interface: Tunnel0
    Crypto map tag: Tunnel0-head-0, local addr 100.64.0.2

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
   remote ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
   current_peer 100.64.0.1 port 500
     PERMIT, flags={origin_is_acl,}
    #pkts encaps: 193408, #pkts encrypt: 193408, #pkts digest: 193408
    #pkts decaps: 193373, #pkts decrypt: 193373, #pkts verify: 193373
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0
... 33 more lines
```

R1 ping 10.20.2.1 source Loopback1 repeat 20

```
Type escape sequence to abort.
Sending 20, 100-byte ICMP Echos to 10.20.2.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.1.1
!!!!!!!!!!!!!!!!!!!!
Success rate is 100 percent (20/20), round-trip min/avg/max = 1/1/2 ms
```

R1 show crypto ipsec sa

```
interface: Tunnel0
    Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
   remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
   current_peer 100.64.0.3 port 500
     PERMIT, flags={origin_is_acl,}
    #pkts encaps: 971, #pkts encrypt: 971, #pkts digest: 971
    #pkts decaps: 967, #pkts decrypt: 967, #pkts verify: 967
```

```
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0
... 78 more lines
```

R2 show crypto ipsec sa

```
interface: Tunnel0
    Crypto map tag: Tunnel0-head-0, local addr 100.64.0.2

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
   remote ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
   current_peer 100.64.0.1 port 500
     PERMIT, flags={origin_is_acl,}
    #pkts encaps: 193438, #pkts encrypt: 193438, #pkts digest: 193438
    #pkts decaps: 193403, #pkts decrypt: 193403, #pkts verify: 193403
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0
... 33 more lines
```

R1 show crypto ipsec sa

```
interface: Tunnel0
    Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
   remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
   current_peer 100.64.0.3 port 500
     PERMIT, flags={origin_is_acl,}
    #pkts encaps: 1001, #pkts encrypt: 1001, #pkts digest: 1001
    #pkts decaps: 1003, #pkts decrypt: 1003, #pkts verify: 1003
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0
... 78 more lines
```

R2 show crypto ipsec sa

```
interface: Tunnel0
    Crypto map tag: Tunnel0-head-0, local addr 100.64.0.2

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
   remote ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
   current_peer 100.64.0.1 port 500
     PERMIT, flags={origin_is_acl,}
    #pkts encaps: 193466, #pkts encrypt: 193466, #pkts digest: 193466
    #pkts decaps: 193430, #pkts decrypt: 193430, #pkts verify: 193430
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0
... 33 more lines
```

Teardown (suite) Restore BGP Baseline, Close All Routers

9.23 PASS BGP Control Plane Traffic Is Encrypted On The Wire (18.8s, 8 commands)

The reason for peering over the tunnel. Captures a keepalive window and asserts the session is invisible: no TCP, no peering addresses, no advertised prefixes -- only ESP.

Setup (suite) Establish BGP

Acceptance criteria

- output contains **ESP**
- output does not contain **TCP**
- output does not contain **${R2_TUNNEL_HUB}**
- output does not contain **${R2_TUNNEL_LOCAL}**
- output does not contain **${R1_BGP_PREFIX}**
- output does not contain **${R2_BGP_PREFIX}**

How it was checked

R1 no monitor capture BGPCAP

```
Capture does not exist
```

R1 **monitor capture BGPCAP interface GigabitEthernet2 both**

```
${output} =
```

R1 **monitor capture BGPCAP match ipv4 any any**

```
${output} =
```

R1 **monitor capture BGPCAP buffer size 5**

```
${output} =
```

R1 **monitor capture BGPCAP start**

```
Started capture point : BGPCAP
```

R1 **monitor capture BGPCAP stop**

```
Stopped capture point : BGPCAP
```

R1 **show monitor capture BGPCAP buffer brief**

```
 ----------------------------------------------------------------------------
  #   size   timestamp     source             destination      dscp    protocol
 ----------------------------------------------------------------------------
    0  138    0.000000   100.64.0.3       ->  100.64.0.1       48 CS6  ESP
    1  138    0.000000   100.64.0.1       ->  100.64.0.3       48 CS6  ESP
    2  138    0.011993   100.64.0.2       ->  100.64.0.1       48 CS6  ESP
    3  138    0.011993   100.64.0.1       ->  100.64.0.2       48 CS6  ESP
    4  138    0.222040   100.64.0.1       ->  100.64.0.2       48 CS6  ESP
    5  138    0.222040   100.64.0.2       ->  100.64.0.1       48 CS6  ESP
    6  138    0.242028   100.64.0.1       ->  100.64.0.3       48 CS6  ESP
    7  138    0.256035   100.64.0.1       ->  100.64.0.3       48 CS6  ESP
    8  138    0.257027   100.64.0.3       ->  100.64.0.1       48 CS6  ESP
    9  138    0.421020   100.64.0.2       ->  100.64.0.1       48 CS6  ESP
   10  138    0.421020   100.64.0.1       ->  100.64.0.2       48 CS6  ESP
... 322 more lines
```

R1 **no monitor capture BGPCAP**

```
${output} =
```

Teardown (suite) Restore BGP Baseline, Close All Routers

9.24 PASS Withdrawing A Prefix Removes It From The Peer (60.5s, 23 commands)

Shuts the advertised interface on R2 and requires R1 to lose both the BGP path and the route, then restores it.

Setup (suite) Establish BGP

Acceptance criteria

- output contains **% Subnet not in table** (checked 7×)
- output contains **Known via "bgp** (checked 8×)

How it was checked

R2 **configure terminal**

```
Enter configuration commands, one per line.  End with CNTL/Z.
```

R2 **interface Loopback1**

```
${output} =
```

R2 **shutdown**

```
${output} =
```

R2 **end**

```
${output} =
```

R1 **show ip route 10.20.2.1**

```
Routing entry for 10.20.2.1/32
  Known via "bgp 65001", distance 20, metric 0
  Tag 65002, type external
  Last update from 172.20.0.2 00:01:07 ago
  Routing Descriptor Blocks:
  * 172.20.0.2, from 172.20.0.2, 00:01:07 ago
      opaque_ptr 0x71FE1BFD9038
      Route metric is 0, traffic share count is 1
      AS Hops 1
      Route tag 65002
      MPLS label: none
```

R1 **show ip route 10.20.2.1**

```
Routing entry for 10.20.2.1/32
  Known via "bgp 65001", distance 20, metric 0
  Tag 65002, type external
  Last update from 172.20.0.2 00:01:12 ago
  Routing Descriptor Blocks:
  * 172.20.0.2, from 172.20.0.2, 00:01:12 ago
      opaque_ptr 0x71FE1BFD9038
      Route metric is 0, traffic share count is 1
      AS Hops 1
```

```
      Route tag 65002
      MPLS label: none
```

R1 show ip route 10.20.2.1

```
Routing entry for 10.20.2.1/32
  Known via "bgp 65001", distance 20, metric 0
  Tag 65002, type external
  Last update from 172.20.0.2 00:01:17 ago
  Routing Descriptor Blocks:
  * 172.20.0.2, from 172.20.0.2, 00:01:17 ago
      opaque_ptr 0x71FE1BFD9038
      Route metric is 0, traffic share count is 1
      AS Hops 1
      Route tag 65002
      MPLS label: none
```

R1 show ip route 10.20.2.1

```
Routing entry for 10.20.2.1/32
  Known via "bgp 65001", distance 20, metric 0
  Tag 65002, type external
  Last update from 172.20.0.2 00:01:22 ago
  Routing Descriptor Blocks:
  * 172.20.0.2, from 172.20.0.2, 00:01:22 ago
      opaque_ptr 0x71FE1BFD9038
      Route metric is 0, traffic share count is 1
      AS Hops 1
      Route tag 65002
      MPLS label: none
```

R1 show ip route 10.20.2.1

```
Routing entry for 10.20.2.1/32
  Known via "bgp 65001", distance 20, metric 0
  Tag 65002, type external
  Last update from 172.20.0.2 00:01:27 ago
  Routing Descriptor Blocks:
  * 172.20.0.2, from 172.20.0.2, 00:01:27 ago
      opaque_ptr 0x71FE1BFD9038
      Route metric is 0, traffic share count is 1
      AS Hops 1
      Route tag 65002
      MPLS label: none
```

R1 show ip route 10.20.2.1

```
Routing entry for 10.20.2.1/32
  Known via "bgp 65001", distance 20, metric 0
  Tag 65002, type external
  Last update from 172.20.0.2 00:01:32 ago
  Routing Descriptor Blocks:
  * 172.20.0.2, from 172.20.0.2, 00:01:32 ago
      opaque_ptr 0x71FE1BFD9038
      Route metric is 0, traffic share count is 1
      AS Hops 1
      Route tag 65002
      MPLS label: none
```

R1 show ip route 10.20.2.1

```
% Subnet not in table
```

R2 configure terminal

```
Enter configuration commands, one per line.  End with CNTL/Z.
```

R2 interface Loopback1

```
${output} =
```

R2 no shutdown

```
${output} =
```

R2 end

```
${output} =
```

R1 show ip route 10.20.2.1

```
% Subnet not in table
```

R1 show ip route 10.20.2.1

```
% Subnet not in table
```

R1 show ip route 10.20.2.1

```
% Subnet not in table
```

R1 show ip route 10.20.2.1

```
% Subnet not in table
```

R1 show ip route 10.20.2.1

```
% Subnet not in table
```

R1 show ip route 10.20.2.1

```
% Subnet not in table
```

R1 show ip route 10.20.2.1

```
Routing entry for 10.20.2.1/32
  Known via "bgp 65001", distance 20, metric 0
  Tag 65002, type external
  Last update from 172.20.0.2 00:00:04 ago
  Routing Descriptor Blocks:
  * 172.20.0.2, from 172.20.0.2, 00:00:04 ago
      opaque_ptr 0x71FE1BFD9038
      Route metric is 0, traffic share count is 1
      AS Hops 1
      Route tag 65002
      MPLS label: none
```

R2 show ip route 10.20.1.1

```
Routing entry for 10.20.1.1/32
  Known via "bgp 65002", distance 20, metric 0
  Tag 65001, type external
  Last update from 172.20.0.1 00:02:07 ago
  Routing Descriptor Blocks:
  * 172.20.0.1, from 172.20.0.1, 00:02:07 ago
      opaque_ptr 0x798416842588
      Route metric is 0, traffic share count is 1
      AS Hops 1
      Route tag 65001
      MPLS label: none
```

Teardown (suite) Restore BGP Baseline, Close All Routers

9.25 PASS Session Recovers After A Hard Clear (10.2s, 5 commands)

Setup (suite) Establish BGP

Acceptance criteria

- output matches **${peer}\s+4\s+${peer_asn}(\s+\d+){5}\s+\S+\s+\d+** (checked 3×)
- output contains **Known via "bgp**

How it was checked

R1 clear ip bgp *

```
${output} =
```

R1 show bgp ipv4 unicast summary

```
BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 1, main routing table version 1
3 network entries using 744 bytes of memory
3 path entries using 408 bytes of memory
2/0 BGP path/bestpath attribute entries using 592 bytes of memory
1 BGP AS-PATH entries using 24 bytes of memory
1 BGP community entries using 24 bytes of memory
1 BGP route-map cache entries using 72 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 1864 total bytes of memory
BGP activity 341/338 prefixes, 362/359 paths, scan interval 60 secs
22 networks peaked at 02:27:59 Sep 8 2026 UTC (22:33:34.050 ago)

Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
... 2 more lines
```

R1 show bgp ipv4 unicast summary

```
BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 1, main routing table version 1
3 network entries using 744 bytes of memory
3 path entries using 408 bytes of memory
2/0 BGP path/bestpath attribute entries using 592 bytes of memory
1 BGP AS-PATH entries using 24 bytes of memory
1 BGP community entries using 24 bytes of memory
1 BGP route-map cache entries using 72 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 1864 total bytes of memory
BGP activity 341/338 prefixes, 362/359 paths, scan interval 60 secs
22 networks peaked at 02:27:59 Sep 8 2026 UTC (22:33:39.090 ago)

Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
... 2 more lines
```

R1 show bgp ipv4 unicast summary

```
BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 10, main routing table version 10
9 network entries using 2232 bytes of memory
9 path entries using 1224 bytes of memory
7/7 BGP path/bestpath attribute entries using 2072 bytes of memory
```

```
2 BGP AS-PATH entries using 48 bytes of memory
6 BGP community entries using 144 bytes of memory
6 BGP route-map cache entries using 432 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 6152 total bytes of memory
BGP activity 347/338 prefixes, 368/359 paths, scan interval 60 secs
22 networks peaked at 02:27:59 Sep 8 2026 UTC (22:33:44.126 ago)

Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
... 2 more lines
```

R1 show ip route 10.20.2.1

```
Routing entry for 10.20.2.1/32
  Known via "bgp 65001", distance 20, metric 0
  Tag 65002, type external
  Last update from 172.20.0.2 00:00:01 ago
  Routing Descriptor Blocks:
  * 172.20.0.2, from 172.20.0.2, 00:00:01 ago
      opaque_ptr 0x71FE1BFD9038
      Route metric is 0, traffic share count is 1
      AS Hops 1
      Route tag 65002
      MPLS label: none
```

Teardown (suite) Restore BGP Baseline, Close All Routers

9.26 PASS Mismatched MD5 Password Prevents The Session Coming Up (50.4s, 19 commands)

Negative control on BGP authentication, mirroring the IPsec wrong-PSK test: without it, a session that accepted any peer would still pass everything above.

Setup (suite) Establish BGP

Acceptance criteria

- output does not match **${peer}(\s+\d+){6}\s+\S+\s+\d+\s*$**
- output matches **${peer}\s+4\s+${peer_asn}(\s+\d+){5}\s+\S+\s+\d+** (checked 2×)

How it was checked

R2 configure terminal

```
Enter configuration commands, one per line.  End with CNTL/Z.
```

R2 router bgp 65002

```
${output} =
```

R2 neighbor 172.20.0.1 password WrongBgpSecret999

```
${output} =
```

R2 end

```
${output} =
```

R1 clear ip bgp *

```
${output} =
```

R2 clear ip bgp *

```
${output} =
```

R1 show bgp ipv4 unicast summary

```
BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 1, main routing table version 1
6 network entries using 1488 bytes of memory
6 path entries using 816 bytes of memory
7/0 BGP path/bestpath attribute entries using 2072 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
6 BGP community entries using 144 bytes of memory
6 BGP route-map cache entries using 432 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 5000 total bytes of memory
BGP activity 353/347 prefixes, 374/368 paths, scan interval 60 secs
22 networks peaked at 02:27:59 Sep 8 2026 UTC (22:34:29.295 ago)

Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
... 2 more lines
```

R2 configure terminal

```
Enter configuration commands, one per line.  End with CNTL/Z.
```

R2 router bgp 65002

```
${output} =
```

R2 neighbor 172.20.0.1 password LabBgpSecret

```
${output} =
```

R2 end

```
${output} =
```

R2 configure terminal

```
Enter configuration commands, one per line.  End with CNTL/Z.
```

R2 interface Loopback1

```
${output} =
```

R2 no shutdown

```
${output} =
```

R2 end

```
${output} =
```

R1 clear ip bgp *

```
${output} =
```

R2 clear ip bgp *

```
${output} =
```

R1 show bgp ipv4 unicast summary

```
BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 1, main routing table version 1
3 network entries using 744 bytes of memory
3 path entries using 408 bytes of memory
7/0 BGP path/bestpath attribute entries using 2072 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
6 BGP community entries using 144 bytes of memory
6 BGP route-map cache entries using 432 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 3848 total bytes of memory
BGP activity 356/353 prefixes, 377/374 paths, scan interval 60 secs
22 networks peaked at 02:27:59 Sep 8 2026 UTC (22:34:29.496 ago)

Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
... 2 more lines
```

R1 show bgp ipv4 unicast summary

```
BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 1, main routing table version 1
3 network entries using 744 bytes of memory
3 path entries using 408 bytes of memory
7/0 BGP path/bestpath attribute entries using 2072 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
6 BGP community entries using 144 bytes of memory
6 BGP route-map cache entries using 432 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 3848 total bytes of memory
BGP activity 356/353 prefixes, 377/374 paths, scan interval 60 secs
22 networks peaked at 02:27:59 Sep 8 2026 UTC (22:34:34.531 ago)

Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
... 2 more lines
```

Teardown (this test) Restore BGP Baseline

10 Host To Host

10.01 PASS Every Host Holds Its LAN Address (0.1s, 3 commands)

Setup (suite) Open All Routers, Open All Hosts, Wait For LAN Routes, Warm Up Host Paths

Acceptance criteria

- output contains **${HOST_IP}[${h}]** (checked 3×)

How it was checked

H1 ip addr show eth1

```
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 52:54:00:aa:01:02 brd ff:ff:ff:ff:ff:ff
    inet 192.168.10.10/24 scope global eth1
       valid_lft forever preferred_lft forever
    inet 169.254.229.55/16 brd 169.254.255.255 scope global noprefixroute eth1
       valid_lft forever preferred_lft forever
    inet 192.168.10.11/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.10.12/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.10.13/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.10.14/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
... 70 more lines
```

H2 ip addr show eth1

```
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 52:54:00:aa:02:02 brd ff:ff:ff:ff:ff:ff
    inet 192.168.20.10/24 scope global eth1
       valid_lft forever preferred_lft forever
    inet 169.254.199.41/16 brd 169.254.255.255 scope global noprefixroute eth1
       valid_lft forever preferred_lft forever
    inet 192.168.20.11/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.20.12/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.20.13/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.20.14/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
... 70 more lines
```

H3 ip addr show eth1

```
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 52:54:00:aa:03:02 brd ff:ff:ff:ff:ff:ff
    inet 192.168.30.10/24 scope global eth1
       valid_lft forever preferred_lft forever
    inet 169.254.207.242/16 brd 169.254.255.255 scope global noprefixroute eth1
       valid_lft forever preferred_lft forever
    inet 192.168.30.11/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.30.12/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.30.13/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.30.14/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
... 70 more lines
```

Teardown (suite) Run Keyword And Ignore Error, Delete Capture, R1, HOSTCAP, Close All Connections

10.02 PASS Every Host Routes The Other Two LANs Via Its Own Router (0.0s, 3 commands)

Setup (suite) Open All Routers, Open All Hosts, Wait For LAN Routes, Warm Up Host Paths

Acceptance criteria

- output matches **${net}/24 via ${HOST_GW}[${h}]** (checked 9×)

How it was checked

H1 ip route

```
default via 10.0.2.2 dev eth0  src 10.0.2.15  metric 1002
10.0.2.0/24 dev eth0 scope link  src 10.0.2.15  metric 1002
10.30.0.0/16 via 192.168.10.1 dev eth1
```

```
169.254.0.0/16 dev eth1 scope link  src 169.254.229.55  metric 1003
192.168.10.0/24 dev eth1 scope link  src 192.168.10.10
192.168.20.0/24 via 192.168.10.1 dev eth1
192.168.30.0/24 via 192.168.10.1 dev eth1
```

H2 ip route

```
default via 10.0.2.2 dev eth0  src 10.0.2.15  metric 1002
10.0.2.0/24 dev eth0 scope link  src 10.0.2.15  metric 1002
10.30.0.0/16 via 192.168.20.1 dev eth1
192.168.30.10 via 192.168.20.1 dev eth1
169.254.0.0/16 dev eth1 scope link  src 169.254.199.41  metric 1003
192.168.10.0/24 via 192.168.20.1 dev eth1
192.168.30.10 via 192.168.20.1 dev eth1
192.168.20.0/24 dev eth1 scope link  src 192.168.20.10
192.168.30.0/24 via 192.168.20.1 dev eth1
192.168.30.10 via 192.168.20.1 dev eth1
```

H3 ip route

```
default via 10.0.2.2 dev eth0  src 10.0.2.15  metric 1002
10.0.2.0/24 dev eth0 scope link  src 10.0.2.15  metric 1002
10.30.0.0/16 via 192.168.30.1 dev eth1
169.254.0.0/16 dev eth1 scope link  src 169.254.207.242  metric 1003
192.168.10.0/24 via 192.168.30.1 dev eth1
192.168.20.0/24 via 192.168.30.1 dev eth1
192.168.30.0/24 dev eth1 scope link  src 192.168.30.10
```

Teardown (suite) Run Keyword And Ignore Error, Delete Capture, R1, HOSTCAP, Close All Connections

10.03 PASS Every Host Can Reach Its Default Gateway (12.3s, 3 commands)

Isolates the host-to-router leg so a failure further along is not misread as a broken LAN.

Setup (suite) Open All Routers, Open All Hosts, Wait For LAN Routes, Warm Up Host Paths

Acceptance criteria

- output contains **, 0% packet loss** (checked 3×)

  ping from ${alias} to ${target} lost packets: ${out}

How it was checked

H1 ping -c 5 192.168.10.1

```
PING 192.168.10.1 (192.168.10.1) 56(84) bytes of data.
64 bytes from 192.168.10.1: icmp_seq=1 ttl=255 time=0.579 ms
64 bytes from 192.168.10.1: icmp_seq=2 ttl=255 time=0.455 ms
64 bytes from 192.168.10.1: icmp_seq=3 ttl=255 time=0.460 ms
64 bytes from 192.168.10.1: icmp_seq=4 ttl=255 time=0.459 ms
64 bytes from 192.168.10.1: icmp_seq=5 ttl=255 time=0.403 ms

--- 192.168.10.1 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4073ms
rtt min/avg/max/mdev = 0.403/0.471/0.579/0.057 ms
```

H2 ping -c 5 192.168.20.1

```
PING 192.168.20.1 (192.168.20.1) 56(84) bytes of data.
64 bytes from 192.168.20.1: icmp_seq=1 ttl=255 time=0.455 ms
64 bytes from 192.168.20.1: icmp_seq=2 ttl=255 time=0.424 ms
64 bytes from 192.168.20.1: icmp_seq=3 ttl=255 time=0.496 ms
64 bytes from 192.168.20.1: icmp_seq=4 ttl=255 time=0.437 ms
64 bytes from 192.168.20.1: icmp_seq=5 ttl=255 time=0.418 ms

--- 192.168.20.1 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4079ms
rtt min/avg/max/mdev = 0.418/0.446/0.496/0.028 ms
```

H3 ping -c 5 192.168.30.1

```
PING 192.168.30.1 (192.168.30.1) 56(84) bytes of data.
64 bytes from 192.168.30.1: icmp_seq=1 ttl=255 time=0.430 ms
64 bytes from 192.168.30.1: icmp_seq=2 ttl=255 time=0.406 ms
64 bytes from 192.168.30.1: icmp_seq=3 ttl=255 time=0.447 ms
64 bytes from 192.168.30.1: icmp_seq=4 ttl=255 time=0.357 ms
64 bytes from 192.168.30.1: icmp_seq=5 ttl=255 time=0.392 ms

--- 192.168.30.1 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4082ms
rtt min/avg/max/mdev = 0.357/0.406/0.447/0.031 ms
```

Teardown (suite) Run Keyword And Ignore Error, Delete Capture, R1, HOSTCAP, Close All Connections

10.04 PASS Every Host Can Reach Every Other Host (24.1s, 6 commands)

The full mesh, including h2 <-> h3 which never touches a direct link and can only work if the hub re-advertises between spokes.

Setup (suite) Open All Routers, Open All Hosts, Wait For LAN Routes, Warm Up Host Paths

Acceptance criteria

- output contains **, 0% packet loss** (checked 6×)
  ping from ${alias} to ${target} lost packets: ${out}

How it was checked

H2 ping -c 5 192.168.20.10

```
PING 192.168.20.10 (192.168.20.10) 56(84) bytes of data.
64 bytes from 192.168.20.10: icmp_seq=1 ttl=62 time=1.14 ms
64 bytes from 192.168.20.10: icmp_seq=2 ttl=62 time=1.19 ms
64 bytes from 192.168.20.10: icmp_seq=3 ttl=62 time=1.51 ms
64 bytes from 192.168.20.10: icmp_seq=4 ttl=62 time=1.23 ms
64 bytes from 192.168.20.10: icmp_seq=5 ttl=62 time=1.26 ms

--- 192.168.20.10 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4006ms
rtt min/avg/max/mdev = 1.143/1.266/1.511/0.128 ms
```

H3 ping -c 5 192.168.30.10

```
PING 192.168.30.10 (192.168.30.10) 56(84) bytes of data.
64 bytes from 192.168.30.10: icmp_seq=1 ttl=62 time=1.16 ms
64 bytes from 192.168.30.10: icmp_seq=2 ttl=62 time=1.15 ms
64 bytes from 192.168.30.10: icmp_seq=3 ttl=62 time=1.20 ms
64 bytes from 192.168.30.10: icmp_seq=4 ttl=62 time=1.22 ms
64 bytes from 192.168.30.10: icmp_seq=5 ttl=62 time=1.65 ms

--- 192.168.30.10 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4006ms
rtt min/avg/max/mdev = 1.146/1.276/1.651/0.189 ms
```

H1 ping -c 5 192.168.10.10

```
PING 192.168.10.10 (192.168.10.10) 56(84) bytes of data.
64 bytes from 192.168.10.10: icmp_seq=1 ttl=62 time=1.13 ms
64 bytes from 192.168.10.10: icmp_seq=2 ttl=62 time=1.17 ms
64 bytes from 192.168.10.10: icmp_seq=3 ttl=62 time=1.27 ms
64 bytes from 192.168.10.10: icmp_seq=4 ttl=62 time=1.22 ms
64 bytes from 192.168.10.10: icmp_seq=5 ttl=62 time=1.26 ms

--- 192.168.10.10 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4006ms
rtt min/avg/max/mdev = 1.132/1.211/1.273/0.054 ms
```

H3 ping -c 5 192.168.30.10

```
PING 192.168.30.10 (192.168.30.10) 56(84) bytes of data.
64 bytes from 192.168.30.10: icmp_seq=1 ttl=62 time=1.33 ms
64 bytes from 192.168.30.10: icmp_seq=2 ttl=62 time=1.17 ms
64 bytes from 192.168.30.10: icmp_seq=3 ttl=62 time=1.48 ms
64 bytes from 192.168.30.10: icmp_seq=4 ttl=62 time=1.19 ms
64 bytes from 192.168.30.10: icmp_seq=5 ttl=62 time=1.18 ms

--- 192.168.30.10 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4006ms
rtt min/avg/max/mdev = 1.172/1.270/1.481/0.120 ms
```

H1 ping -c 5 192.168.10.10

```
PING 192.168.10.10 (192.168.10.10) 56(84) bytes of data.
64 bytes from 192.168.10.10: icmp_seq=1 ttl=62 time=1.30 ms
64 bytes from 192.168.10.10: icmp_seq=2 ttl=62 time=1.18 ms
64 bytes from 192.168.10.10: icmp_seq=3 ttl=62 time=1.27 ms
64 bytes from 192.168.10.10: icmp_seq=4 ttl=62 time=1.13 ms
64 bytes from 192.168.10.10: icmp_seq=5 ttl=62 time=1.74 ms

--- 192.168.10.10 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4006ms
rtt min/avg/max/mdev = 1.133/1.323/1.743/0.217 ms
```

H2 ping -c 5 192.168.20.10

```
PING 192.168.20.10 (192.168.20.10) 56(84) bytes of data.
64 bytes from 192.168.20.10: icmp_seq=1 ttl=62 time=1.16 ms
64 bytes from 192.168.20.10: icmp_seq=2 ttl=62 time=1.21 ms
64 bytes from 192.168.20.10: icmp_seq=3 ttl=62 time=1.28 ms
64 bytes from 192.168.20.10: icmp_seq=4 ttl=62 time=1.27 ms
64 bytes from 192.168.20.10: icmp_seq=5 ttl=62 time=1.22 ms

--- 192.168.20.10 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4006ms
rtt min/avg/max/mdev = 1.162/1.227/1.282/0.042 ms
```

Teardown (suite) Run Keyword And Ignore Error, Delete Capture, R1, HOSTCAP, Close All Connections

10.05 PASS Spoke To Spoke Traffic Is Routed Through The Hub (0.1s, 3 commands)

R2 must reach R3's LAN via the hub's tunnel address, not by any direct path -- there is no spoke-to-spoke link.

Setup (suite) Open All Routers, Open All Hosts, Wait For LAN Routes, Warm Up Host Paths

Acceptance criteria

- output contains **Known via "bgp ${R2_ASN}"**
- output contains **${R2_TUNNEL_HUB}**
- output contains **Tunnel0** (checked 2×)

How it was checked

```
R2 show ip route 192.168.30.0
  Routing entry for 192.168.30.0/24
    Known via "bgp 65002", distance 20, metric 0
    Tag 65001, type external
    Last update from 172.20.0.3 00:01:26 ago
    Routing Descriptor Blocks:
    * 172.20.0.3, from 172.20.0.1, 00:01:26 ago
        opaque_ptr 0x798416842588
        Route metric is 0, traffic share count is 1
        AS Hops 2
        Route tag 65001
        MPLS label: none
R2 show ip cef 192.168.30.10
  192.168.30.0/24
    nexthop 172.20.0.3 Tunnel0
R3 show ip cef 192.168.20.10
  192.168.20.0/24
    nexthop 172.20.0.2 Tunnel0
```

Teardown (suite) Run Keyword And Ignore Error, Delete Capture, R1, HOSTCAP, Close All Connections

10.06 PASS Hub Reaches Both Spoke LANs On Separate Tunnels (0.0s, 2 commands)

Setup (suite) Open All Routers, Open All Hosts, Wait For LAN Routes, Warm Up Host Paths

Acceptance criteria

- output contains **${DMVPN_TUNNEL}** (checked 2×)

How it was checked

```
R1 show ip cef 192.168.20.10
  192.168.20.0/24
    nexthop 172.20.0.2 Tunnel0
R1 show ip cef 192.168.30.10
  192.168.30.0/24
    nexthop 172.20.0.3 Tunnel0
```

Teardown (suite) Run Keyword And Ignore Error, Delete Capture, R1, HOSTCAP, Close All Connections

10.07 PASS Host Traffic Is Encrypted In Transit (19.6s, 5 commands)

No host runs crypto: the routers encrypt for them, so the ESP counters must move by the traffic the hosts generated.

Setup (suite) Open All Routers, Open All Hosts, Wait For LAN Routes, Warm Up Host Paths

Acceptance criteria

- output contains **, 0% packet loss**
  ping from ${alias} to ${target} lost packets: ${out}
- **${r1_enc} >= ${expected}** holds
- **${r1_dec} >= ${expected}** holds
- **${r2_enc} >= ${expected}** holds
- **${r2_dec} >= ${expected}** holds

How it was checked

R1 show crypto ipsec sa

```
interface: Tunnel0
    Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
   remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
   current_peer 100.64.0.3 port 500
     PERMIT, flags={origin_is_acl,}
    #pkts encaps: 2523, #pkts encrypt: 2523, #pkts digest: 2523
    #pkts decaps: 2521, #pkts decrypt: 2521, #pkts verify: 2521
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0
... 78 more lines
```

R2 show crypto ipsec sa

```
interface: Tunnel0
    Crypto map tag: Tunnel0-head-0, local addr 100.64.0.2

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
   remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
   current_peer 100.64.0.3 port 500
     PERMIT, flags={origin_is_acl,}
    #pkts encaps: 255, #pkts encrypt: 255, #pkts digest: 255
    #pkts decaps: 258, #pkts decrypt: 258, #pkts verify: 258
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0
... 78 more lines
```

H1 ping -c 20 192.168.20.10

```
PING 192.168.20.10 (192.168.20.10) 56(84) bytes of data.
64 bytes from 192.168.20.10: icmp_seq=1 ttl=62 time=1.27 ms
64 bytes from 192.168.20.10: icmp_seq=2 ttl=62 time=1.27 ms
64 bytes from 192.168.20.10: icmp_seq=3 ttl=62 time=1.21 ms
64 bytes from 192.168.20.10: icmp_seq=4 ttl=62 time=1.23 ms
64 bytes from 192.168.20.10: icmp_seq=5 ttl=62 time=1.35 ms
64 bytes from 192.168.20.10: icmp_seq=6 ttl=62 time=1.17 ms
64 bytes from 192.168.20.10: icmp_seq=7 ttl=62 time=1.18 ms
64 bytes from 192.168.20.10: icmp_seq=8 ttl=62 time=1.23 ms
64 bytes from 192.168.20.10: icmp_seq=9 ttl=62 time=1.26 ms
64 bytes from 192.168.20.10: icmp_seq=10 ttl=62 time=1.48 ms
64 bytes from 192.168.20.10: icmp_seq=11 ttl=62 time=1.38 ms
64 bytes from 192.168.20.10: icmp_seq=12 ttl=62 time=1.35 ms
64 bytes from 192.168.20.10: icmp_seq=13 ttl=62 time=1.54 ms
... 11 more lines
```

R1 show crypto ipsec sa

```
interface: Tunnel0
    Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
   remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
   current_peer 100.64.0.3 port 500
     PERMIT, flags={origin_is_acl,}
    #pkts encaps: 2635, #pkts encrypt: 2635, #pkts digest: 2635
    #pkts decaps: 2639, #pkts decrypt: 2639, #pkts verify: 2639
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0
... 78 more lines
```

R2 show crypto ipsec sa

```
interface: Tunnel0
    Crypto map tag: Tunnel0-head-0, local addr 100.64.0.2

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
   remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
   current_peer 100.64.0.3 port 500
     PERMIT, flags={origin_is_acl,}
    #pkts encaps: 365, #pkts encrypt: 365, #pkts digest: 365
    #pkts decaps: 372, #pkts decrypt: 372, #pkts verify: 372
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0
```

```
... 78 more lines
```

Teardown (suite) Run Keyword And Ignore Error, Delete Capture, R1, HOSTCAP, Close All Connections

10.08 PASS Host Addresses Never Appear In Cleartext On The Spoke Link (11.6s, 9 commands)

Capture the R1<->R2 link while the hosts talk: only ESP between the link addresses should be visible.

Setup (suite) Open All Routers, Open All Hosts, Wait For LAN Routes, Warm Up Host Paths

Acceptance criteria

- output contains **, 0% packet loss**
  ping from ${alias} to ${target} lost packets: ${out}
- output contains **ESP**
- output does not contain **${H1_IP}**
- output does not contain **${H2_IP}**
- output does not contain **ICMP**

How it was checked

R1 no monitor capture HOSTCAP

```
Capture does not exist
```

R1 monitor capture HOSTCAP interface GigabitEthernet2 both

```
${output} =
```

R1 monitor capture HOSTCAP match ipv4 any any

```
${output} =
```

R1 monitor capture HOSTCAP buffer size 5

```
${output} =
```

R1 monitor capture HOSTCAP start

```
Started capture point : HOSTCAP
```

H1 ping -c 10 192.168.20.10

```
PING 192.168.20.10 (192.168.20.10) 56(84) bytes of data.
64 bytes from 192.168.20.10: icmp_seq=1 ttl=62 time=1.34 ms
64 bytes from 192.168.20.10: icmp_seq=2 ttl=62 time=1.12 ms
64 bytes from 192.168.20.10: icmp_seq=3 ttl=62 time=1.20 ms
64 bytes from 192.168.20.10: icmp_seq=4 ttl=62 time=1.20 ms
64 bytes from 192.168.20.10: icmp_seq=5 ttl=62 time=1.27 ms
64 bytes from 192.168.20.10: icmp_seq=6 ttl=62 time=1.13 ms
64 bytes from 192.168.20.10: icmp_seq=7 ttl=62 time=1.15 ms
64 bytes from 192.168.20.10: icmp_seq=8 ttl=62 time=1.32 ms
64 bytes from 192.168.20.10: icmp_seq=9 ttl=62 time=1.22 ms
64 bytes from 192.168.20.10: icmp_seq=10 ttl=62 time=1.31 ms

--- 192.168.20.10 ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9013ms
... 1 more lines
```

R1 monitor capture HOSTCAP stop

```
Stopped capture point : HOSTCAP
```

R1 show monitor capture HOSTCAP buffer brief

```
----------------------------------------------------------------------------
 #   size   timestamp     source             destination      dscp    protocol
----------------------------------------------------------------------------
   0  170    0.000000   100.64.0.1       ->  100.64.0.2       0  BE   ESP
   1  170    0.001007   100.64.0.2       ->  100.64.0.1       0  BE   ESP
   2  138    0.014998   100.64.0.3       ->  100.64.0.1       48 CS6  ESP
   3  138    0.014998   100.64.0.1       ->  100.64.0.3       48 CS6  ESP
   4  138    0.021010   100.64.0.1       ->  100.64.0.3       48 CS6  ESP
   5  138    0.094996   100.64.0.1       ->  100.64.0.2       48 CS6  ESP
   6  138    0.094996   100.64.0.2       ->  100.64.0.1       48 CS6  ESP
   7  138    0.150993   100.64.0.1       ->  100.64.0.3       48 CS6  ESP
   8  138    0.150993   100.64.0.3       ->  100.64.0.1       48 CS6  ESP
   9  138    0.276993   100.64.0.2       ->  100.64.0.1       48 CS6  ESP
  10  138    0.276993   100.64.0.1       ->  100.64.0.2       48 CS6  ESP
... 223 more lines
```

R1 no monitor capture HOSTCAP

```
${output} =
```

Teardown (suite) Run Keyword And Ignore Error, Delete Capture, R1, HOSTCAP, Close All Connections

10.09 PASS Spoke To Spoke Traffic Is Encrypted On The Direct Path (11.6s, 11 commands)

The VTI lab this mirrors captured the hub's second link, because h2 to h3 crossed two tunnels and was decrypted and re-encrypted in between. Under DMVPN there is no second leg to capture: once the shortcut forms the hub is not on the path at all, so the capture moves to the spoke's own WAN interface -- the only place that traffic now appears. It should be ESP there, with neither host address visible.

Setup (suite) Open All Routers, Open All Hosts, Wait For LAN Routes, Warm Up Host Paths

Acceptance criteria

- output contains **${R3_WAN_IP}**
  R2 has not resolved R3 directly, so no shortcut is in place
- output contains **, 0% packet loss**
  ping from ${alias} to ${target} lost packets: ${out}
- output contains **ESP**
- output does not contain **${H2_IP}**
- output does not contain **${H3_IP}**
- output does not contain **ICMP**

How it was checked

R2 ping 10.20.3.1 source Loopback1 repeat 5
```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.2.1
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
```

R2 show ip nhrp 172.20.0.3
```
172.20.0.3/32 via 172.20.0.3
   Tunnel0 created 00:01:14, expire 00:03:45
   Type: dynamic, Flags: router nhop bfd
   NBMA address: 100.64.0.3
```

R2 no monitor capture HOSTCAP
```
Capture does not exist
```

R2 monitor capture HOSTCAP interface GigabitEthernet2 both
```
${output} =
```

R2 monitor capture HOSTCAP match ipv4 any any
```
${output} =
```

R2 monitor capture HOSTCAP buffer size 5
```
${output} =
```

R2 monitor capture HOSTCAP start
```
Started capture point : HOSTCAP
```

H2 ping -c 10 192.168.30.10
```
PING 192.168.30.10 (192.168.30.10) 56(84) bytes of data.
64 bytes from 192.168.30.10: icmp_seq=1 ttl=62 time=1.43 ms
64 bytes from 192.168.30.10: icmp_seq=2 ttl=62 time=1.32 ms
64 bytes from 192.168.30.10: icmp_seq=3 ttl=62 time=1.32 ms
64 bytes from 192.168.30.10: icmp_seq=4 ttl=62 time=1.24 ms
64 bytes from 192.168.30.10: icmp_seq=5 ttl=62 time=1.21 ms
64 bytes from 192.168.30.10: icmp_seq=6 ttl=62 time=1.19 ms
64 bytes from 192.168.30.10: icmp_seq=7 ttl=62 time=1.30 ms
64 bytes from 192.168.30.10: icmp_seq=8 ttl=62 time=1.08 ms
64 bytes from 192.168.30.10: icmp_seq=9 ttl=62 time=1.23 ms
64 bytes from 192.168.30.10: icmp_seq=10 ttl=62 time=1.11 ms

--- 192.168.30.10 ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9013ms
... 1 more lines
```

R2 monitor capture HOSTCAP stop
```
Stopped capture point : HOSTCAP
```

R2 show monitor capture HOSTCAP buffer brief
```
----------------------------------------------------------------------------
 #   size   timestamp     source             destination      dscp    protocol
 ----------------------------------------------------------------------------
   0  170    0.000000   100.64.0.2       ->  100.64.0.3       0  BE   ESP
   1  170    0.001007   100.64.0.3       ->  100.64.0.2       0  BE   ESP
   2  138    0.006012   100.64.0.2       ->  100.64.0.1       48 CS6  ESP
   3  138    0.007004   100.64.0.1       ->  100.64.0.2       48 CS6  ESP
   4  138    0.103999   100.64.0.2       ->  100.64.0.3       48 CS6  ESP
   5  138    0.115000   100.64.0.3       ->  100.64.0.2       48 CS6  ESP
   6  138    0.115000   100.64.0.2       ->  100.64.0.3       48 CS6  ESP
   7  138    0.255999   100.64.0.2       ->  100.64.0.3       48 CS6  ESP
```

```
   8  138    0.255999   100.64.0.3       ->  100.64.0.2       48 CS6  ESP
   9  138    0.322982   100.64.0.1       ->  100.64.0.2       48 CS6  ESP
  10  138    0.322982   100.64.0.2       ->  100.64.0.1       48 CS6  ESP
... 219 more lines
```

R2 no monitor capture HOSTCAP

```
${output} =
```

Teardown (suite) Run Keyword And Ignore Error, Delete Capture, R1, HOSTCAP, Close All Connections

10.10 PASS The Hub Is Not On The Spoke To Spoke Path (0.0s, 2 commands)

The routing consequence of the shortcut, from the hosts' point of view rather than the routers'.

Setup (suite) Open All Routers, Open All Hosts, Wait For LAN Routes, Warm Up Host Paths

Acceptance criteria

- output contains **${R3_WAN_IP}**

  R2 has not resolved R3 directly, so no shortcut is in place
- output does not contain **${R1_DMVPN_IP}**

  host traffic between the spokes still transits the hub:\n${trace}

How it was checked

R2 show ip nhrp 172.20.0.3

```
172.20.0.3/32 via 172.20.0.3
   Tunnel0 created 00:01:26, expire 00:03:33
   Type: dynamic, Flags: router nhop bfd
   NBMA address: 100.64.0.3
```

H2 traceroute -n -q 1 -w 1 192.168.30.10

```
traceroute to 192.168.30.10 (192.168.30.10), 30 hops max, 60 byte packets
 1  192.168.20.1  0.880 ms
 2  172.20.0.3  2.003 ms
 3  192.168.30.10  2.284 ms
```

Teardown (suite) Run Keyword And Ignore Error, Delete Capture, R1, HOSTCAP, Close All Connections

11 Throughput

11.01 PASS Hub To Spoke Hosts Sustain The Target Throughput (32.0s, 2 commands)

h1 -> h2 crosses one tunnel.

Setup (suite) Open All Routers, Open All Hosts, Wait For Fabric

Acceptance criteria

- **${m}** is not empty
- **${mbps} >= ${TP_TARGET}** holds

How it was checked

```
H1 nc -l -p 5001 > /dev/null 2>&1 &
  ${output} =
H1 S=$(date +%s); dd if=/dev/zero bs=64k count=1024 2>/dev/null | nc -w 1 192.168.20.10 5001; echo SECS=$(( $(date
+%s) - S ))
  SECS=31
```

Teardown (suite) Close All Connections

11.02 PASS Spoke To Spoke Hosts Sustain The Target Throughput (32.1s, 2 commands)

h2 -> h3 crosses two tunnels, being decrypted and re-encrypted at the hub, so it is the more demanding path.

Setup (suite) Open All Routers, Open All Hosts, Wait For Fabric

Acceptance criteria

- **${m}** is not empty
- **${mbps} >= ${TP_TARGET}** holds

How it was checked

```
H2 nc -l -p 5001 > /dev/null 2>&1 &
  ${output} =
H2 S=$(date +%s); dd if=/dev/zero bs=64k count=1024 2>/dev/null | nc -w 1 192.168.30.10 5001; echo SECS=$(( $(date
+%s) - S ))
  SECS=31
```

Teardown (suite) Close All Connections

11.03 PASS Bulk Throughput Traffic Is Encrypted (17.3s, 6 commands)

Rate alone says nothing about protection. A transfer of this size must move thousands of packets through ESP, so assert the counters rose in proportion to the bytes actually sent.

Setup (suite) Open All Routers, Open All Hosts, Wait For Fabric

Acceptance criteria

- **${m}** is not empty
- **${data_out} >= ${expected}** holds
- **${data_in} >= ${expected}** holds
- **${ack_in} > 0** holds
- **${ack_out} > 0** holds

How it was checked

```
R1 show crypto ipsec sa
  interface: Tunnel0
      Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

     protected vrf: (none)
     local  ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
     remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
     current_peer 100.64.0.3 port 500
       PERMIT, flags={origin_is_acl,}
      #pkts encaps: 3155, #pkts encrypt: 3155, #pkts digest: 3155
      #pkts decaps: 3161, #pkts decrypt: 3161, #pkts verify: 3161
      #pkts compressed: 0, #pkts decompressed: 0
```

```
     #pkts not compressed: 0, #pkts compr. failed: 0
     #pkts not decompressed: 0, #pkts decompress failed: 0
     #send errors 0, #recv errors 0
 ... 78 more lines
```

R2 show crypto ipsec sa

```
 interface: Tunnel0
     Crypto map tag: Tunnel0-head-0, local addr 100.64.0.2

    protected vrf: (none)
    local  ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
    remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
    current_peer 100.64.0.3 port 500
      PERMIT, flags={origin_is_acl,}
     #pkts encaps: 50734, #pkts encrypt: 50734, #pkts digest: 50734
     #pkts decaps: 26344, #pkts decrypt: 26344, #pkts verify: 26344
     #pkts compressed: 0, #pkts decompressed: 0
     #pkts not compressed: 0, #pkts compr. failed: 0
     #pkts not decompressed: 0, #pkts decompress failed: 0
     #send errors 0, #recv errors 0
 ... 78 more lines
```

H1 nc -l -p 5001 > /dev/null 2>&1 &

```
 [1]+  Stopped (tty input)        nc -l -p 5001 1>/dev/null 2>&1
```

H1 S=$(date +%s); dd if=/dev/zero bs=64k count=512 2>/dev/null | nc -w 1 192.168.20.10 5001; echo SECS=$(($(date +%s) - S))

```
 SECS=16
```

R1 show crypto ipsec sa

```
 interface: Tunnel0
     Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

    protected vrf: (none)
    local  ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
    remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
    current_peer 100.64.0.3 port 500
      PERMIT, flags={origin_is_acl,}
     #pkts encaps: 3254, #pkts encrypt: 3254, #pkts digest: 3254
     #pkts decaps: 3261, #pkts decrypt: 3261, #pkts verify: 3261
     #pkts compressed: 0, #pkts decompressed: 0
     #pkts not compressed: 0, #pkts compr. failed: 0
     #pkts not decompressed: 0, #pkts decompress failed: 0
     #send errors 0, #recv errors 0
 ... 78 more lines
```

R1 show crypto ipsec sa

```
 interface: Tunnel0
     Crypto map tag: Tunnel0-head-0, local addr 100.64.0.2

    protected vrf: (none)
    local  ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
    remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
    current_peer 100.64.0.3 port 500
      PERMIT, flags={origin_is_acl,}
     #pkts encaps: 50831, #pkts encrypt: 50831, #pkts digest: 50831
     #pkts decaps: 26441, #pkts decrypt: 26441, #pkts verify: 26441
     #pkts compressed: 0, #pkts decompressed: 0
     #pkts not compressed: 0, #pkts compr. failed: 0
     #pkts not decompressed: 0, #pkts decompress failed: 0
     #send errors 0, #recv errors 0
 ... 78 more lines
```

Teardown (suite) Close All Connections

12 Nat

12.01 PASS Every Router Has A Static Source NAT Mapping For Its Host (0.3s, 3 commands)

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Warm Up NAT Paths

Acceptance criteria

- output contains **${HOST_IP}[${h}]** (checked 3×)
- output contains **${HOST_NAT}[${h}]** (checked 3×)
- output contains **route-map NAT-TO-DOMAIN** (checked 3×)

How it was checked

H1 show running-config | include ip nat inside source

```
ip nat inside source static 192.168.10.10 10.30.1.10 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.11 10.30.1.11 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.12 10.30.1.12 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.13 10.30.1.13 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.14 10.30.1.14 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.15 10.30.1.15 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.16 10.30.1.16 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.17 10.30.1.17 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.18 10.30.1.18 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.19 10.30.1.19 route-map NAT-TO-DOMAIN
ip nat inside source list NAT-POOL-SRC pool SITE-POOL
```

H2 show running-config | include ip nat inside source

```
ip nat inside source static 192.168.20.10 10.30.2.10 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.11 10.30.2.11 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.12 10.30.2.12 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.13 10.30.2.13 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.14 10.30.2.14 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.15 10.30.2.15 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.16 10.30.2.16 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.17 10.30.2.17 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.18 10.30.2.18 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.19 10.30.2.19 route-map NAT-TO-DOMAIN
ip nat inside source list NAT-POOL-SRC pool SITE-POOL
```

H3 show running-config | include ip nat inside source

```
ip nat inside source static 192.168.30.10 10.30.3.10 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.11 10.30.3.11 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.12 10.30.3.12 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.13 10.30.3.13 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.14 10.30.3.14 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.15 10.30.3.15 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.16 10.30.3.16 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.17 10.30.3.17 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.18 10.30.3.18 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.19 10.30.3.19 route-map NAT-TO-DOMAIN
ip nat inside source list NAT-POOL-SRC pool SITE-POOL
```

Teardown (suite) Close All Connections

12.02 PASS NAT Inside And Outside Interfaces Are Marked Correctly (0.3s, 7 commands)

The LAN faces inside, the tunnels face outside. Getting this backwards silently disables translation rather than erroring.

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Warm Up NAT Paths

Acceptance criteria

- output contains **ip nat inside** (checked 3×)
- output contains **ip nat outside** (checked 4×)

How it was checked

R1 show running-config interface GigabitEthernet3

```
Building configuration...

Current configuration : 145 bytes
!
interface GigabitEthernet3
 description LAN toward the local host
 ip address 192.168.10.1 255.255.255.0
 ip nat inside
 negotiation auto
```

```
end
```

R1 show running-config interface Tunnel0

```
Building configuration...

Current configuration : 399 bytes
!
interface Tunnel0
 description DMVPN hub
 ip address 172.20.0.1 255.255.255.0
 no ip redirects
 ip nat outside
 ip nhrp authentication LabNhrp
 ip nhrp network-id 100
 ip nhrp holdtime 300
 ip nhrp redirect
 ip tcp adjust-mss 1360
... 6 more lines
```

R2 show running-config interface GigabitEthernet3

```
Building configuration...

Current configuration : 145 bytes
!
interface GigabitEthernet3
 description LAN toward the local host
 ip address 192.168.20.1 255.255.255.0
 ip nat inside
 negotiation auto
end
```

R2 show running-config interface Tunnel0

```
Building configuration...

Current configuration : 433 bytes
!
interface Tunnel0
 description DMVPN spoke
 ip address 172.20.0.2 255.255.255.0
 no ip redirects
 ip nat outside
 ip nhrp authentication LabNhrp
 ip nhrp network-id 100
 ip nhrp holdtime 300
 ip nhrp nhs 172.20.0.1 nbma 100.64.0.1 multicast
 ip tcp adjust-mss 1360
... 6 more lines
```

R3 show running-config interface GigabitEthernet3

```
Building configuration...

Current configuration : 145 bytes
!
interface GigabitEthernet3
 description LAN toward the local host
 ip address 192.168.30.1 255.255.255.0
 ip nat inside
 negotiation auto
end
```

R3 show running-config interface Tunnel0

```
Building configuration...

Current configuration : 433 bytes
!
interface Tunnel0
 description DMVPN spoke
 ip address 172.20.0.3 255.255.255.0
 no ip redirects
 ip nat outside
 ip nhrp authentication LabNhrp
 ip nhrp network-id 100
 ip nhrp holdtime 300
 ip nhrp nhs 172.20.0.1 nbma 100.64.0.1 multicast
 ip tcp adjust-mss 1360
... 6 more lines
```

R1 show running-config interface Tunnel0

```
Building configuration...

Current configuration : 399 bytes
!
interface Tunnel0
 description DMVPN hub
```

```
  ip address 172.20.0.1 255.255.255.0
  no ip redirects
  ip nat outside
  ip nhrp authentication LabNhrp
  ip nhrp network-id 100
  ip nhrp holdtime 300
  ip nhrp redirect
  ip tcp adjust-mss 1360
... 6 more lines
```

Teardown (suite) Close All Connections

12.03 PASS Mapped Prefixes Are Advertised Across The Fabric (0.1s, 3 commands)

Return traffic can only find its way home if each site's NAT range is reachable, so the mapped prefixes ride BGP like the real ones.

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Warm Up NAT Paths

Acceptance criteria

- output contains **Known via "bgp ${R2_ASN}"** (checked 2×)
- output contains **Known via "bgp ${R3_ASN}"**

How it was checked

R2 show ip route 10.30.1.0

```
Routing entry for 10.30.1.0/24
  Known via "bgp 65002", distance 20, metric 0
  Tag 65001, type external
  Last update from 172.20.0.1 00:03:44 ago
  Routing Descriptor Blocks:
  * 172.20.0.1, from 172.20.0.1, 00:03:44 ago
      opaque_ptr 0x798416842948
      Route metric is 0, traffic share count is 1
      AS Hops 1
      Route tag 65001
      MPLS label: none
```

R2 show ip route 10.30.3.0

```
Routing entry for 10.30.3.0/24
  Known via "bgp 65002", distance 20, metric 0
  Tag 65001, type external
  Last update from 172.20.0.3 00:03:44 ago
  Routing Descriptor Blocks:
  * 172.20.0.3, from 172.20.0.1, 00:03:44 ago
      opaque_ptr 0x7984168426C8
      Route metric is 0, traffic share count is 1
      AS Hops 2
      Route tag 65001
      MPLS label: none
```

R3 show ip route 10.30.1.0

```
Routing entry for 10.30.1.0/24
  Known via "bgp 65003", distance 20, metric 0
  Tag 65001, type external
  Last update from 172.20.0.1 00:03:44 ago
  Routing Descriptor Blocks:
  * 172.20.0.1, from 172.20.0.1, 00:03:44 ago
      opaque_ptr 0x7AED8E74C510
      Route metric is 0, traffic share count is 1
      AS Hops 1
      Route tag 65001
      MPLS label: none
```

Teardown (suite) Close All Connections

12.04 PASS Every Host Reaches Every Other Host At Its Mapped Address (24.1s, 6 commands)

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Warm Up NAT Paths

Acceptance criteria

- output contains **, 0% packet loss** (checked 6×)
  ping from ${alias} to ${target} lost packets: ${out}

How it was checked

H2 ping -c 5 10.30.2.10

```
PING 10.30.2.10 (10.30.2.10) 56(84) bytes of data.
```

```
64 bytes from 10.30.2.10: icmp_seq=1 ttl=62 time=1.71 ms
64 bytes from 10.30.2.10: icmp_seq=2 ttl=62 time=1.31 ms
64 bytes from 10.30.2.10: icmp_seq=3 ttl=62 time=1.16 ms
64 bytes from 10.30.2.10: icmp_seq=4 ttl=62 time=1.39 ms
64 bytes from 10.30.2.10: icmp_seq=5 ttl=62 time=1.27 ms

--- 10.30.2.10 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4006ms
rtt min/avg/max/mdev = 1.162/1.367/1.710/0.186 ms
```

H3 ping -c 5 10.30.3.10

```
PING 10.30.3.10 (10.30.3.10) 56(84) bytes of data.
64 bytes from 10.30.3.10: icmp_seq=1 ttl=62 time=1.34 ms
64 bytes from 10.30.3.10: icmp_seq=2 ttl=62 time=1.26 ms
64 bytes from 10.30.3.10: icmp_seq=3 ttl=62 time=1.45 ms
64 bytes from 10.30.3.10: icmp_seq=4 ttl=62 time=1.23 ms
64 bytes from 10.30.3.10: icmp_seq=5 ttl=62 time=1.08 ms

--- 10.30.3.10 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4006ms
rtt min/avg/max/mdev = 1.080/1.270/1.447/0.122 ms
```

H1 ping -c 5 10.30.1.10

```
PING 10.30.1.10 (10.30.1.10) 56(84) bytes of data.
64 bytes from 10.30.1.10: icmp_seq=1 ttl=62 time=1.20 ms
64 bytes from 10.30.1.10: icmp_seq=2 ttl=62 time=1.31 ms
64 bytes from 10.30.1.10: icmp_seq=3 ttl=62 time=1.27 ms
64 bytes from 10.30.1.10: icmp_seq=4 ttl=62 time=1.30 ms
64 bytes from 10.30.1.10: icmp_seq=5 ttl=62 time=1.30 ms

--- 10.30.1.10 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4006ms
rtt min/avg/max/mdev = 1.204/1.277/1.310/0.038 ms
```

H3 ping -c 5 10.30.3.10

```
PING 10.30.3.10 (10.30.3.10) 56(84) bytes of data.
64 bytes from 10.30.3.10: icmp_seq=1 ttl=62 time=1.30 ms
64 bytes from 10.30.3.10: icmp_seq=2 ttl=62 time=1.24 ms
64 bytes from 10.30.3.10: icmp_seq=3 ttl=62 time=1.10 ms
64 bytes from 10.30.3.10: icmp_seq=4 ttl=62 time=1.17 ms
64 bytes from 10.30.3.10: icmp_seq=5 ttl=62 time=1.18 ms

--- 10.30.3.10 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4006ms
rtt min/avg/max/mdev = 1.103/1.199/1.296/0.065 ms
```

H1 ping -c 5 10.30.1.10

```
PING 10.30.1.10 (10.30.1.10) 56(84) bytes of data.
64 bytes from 10.30.1.10: icmp_seq=1 ttl=62 time=1.58 ms
64 bytes from 10.30.1.10: icmp_seq=2 ttl=62 time=1.22 ms
64 bytes from 10.30.1.10: icmp_seq=3 ttl=62 time=1.17 ms
64 bytes from 10.30.1.10: icmp_seq=4 ttl=62 time=1.18 ms
64 bytes from 10.30.1.10: icmp_seq=5 ttl=62 time=1.15 ms

--- 10.30.1.10 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4006ms
rtt min/avg/max/mdev = 1.151/1.257/1.578/0.161 ms
```

H2 ping -c 5 10.30.2.10

```
PING 10.30.2.10 (10.30.2.10) 56(84) bytes of data.
64 bytes from 10.30.2.10: icmp_seq=1 ttl=62 time=1.31 ms
64 bytes from 10.30.2.10: icmp_seq=2 ttl=62 time=1.15 ms
64 bytes from 10.30.2.10: icmp_seq=3 ttl=62 time=1.25 ms
64 bytes from 10.30.2.10: icmp_seq=4 ttl=62 time=1.13 ms
64 bytes from 10.30.2.10: icmp_seq=5 ttl=62 time=1.19 ms

--- 10.30.2.10 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4006ms
rtt min/avg/max/mdev = 1.128/1.203/1.314/0.068 ms
```

Teardown (suite) Close All Connections

12.05 PASS Translations Show The Host Rewritten To Its Mapped Address (4.2s, 2 commands)

The inside-local / inside-global pair is the direct evidence that the source address was rewritten.

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Warm Up NAT Paths

Acceptance criteria

- output contains **, 0% packet loss**

  ping from ${alias} to ${target} lost packets: ${out}

- output matches **icmp\s+${H1_NAT_IP}:\d+\s+${H1_IP}:\d+\s+${H2_NAT_IP}**

How it was checked

H1 ping -c 5 10.30.2.10

```
PING 10.30.2.10 (10.30.2.10) 56(84) bytes of data.
64 bytes from 10.30.2.10: icmp_seq=1 ttl=62 time=1.47 ms
64 bytes from 10.30.2.10: icmp_seq=2 ttl=62 time=1.10 ms
64 bytes from 10.30.2.10: icmp_seq=3 ttl=62 time=1.25 ms
64 bytes from 10.30.2.10: icmp_seq=4 ttl=62 time=1.35 ms
64 bytes from 10.30.2.10: icmp_seq=5 ttl=62 time=1.04 ms

--- 10.30.2.10 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4006ms
rtt min/avg/max/mdev = 1.039/1.241/1.470/0.159 ms
```

R1 show ip nat translations

```
Pro  Inside global         Inside local          Outside local         Outside global
---  10.30.1.128           192.168.10.88         ---                   ---
---  10.30.1.129           192.168.10.67         ---                   ---
---  10.30.1.113           192.168.10.72         ---                   ---
---  10.30.1.114           192.168.10.80         ---                   ---
---  10.30.1.116           192.168.10.87         ---                   ---
---  10.30.1.119           192.168.10.85         ---                   ---
---  10.30.1.123           192.168.10.71         ---                   ---
---  10.30.1.120           192.168.10.93         ---                   ---
---  10.30.1.126           192.168.10.75         ---                   ---
---  10.30.1.125           192.168.10.84         ---                   ---
---  10.30.1.101           192.168.10.91         ---                   ---
---  10.30.1.102           192.168.10.78         ---                   ---
---  10.30.1.111           192.168.10.70         ---                   ---
... 35 more lines
```

Teardown (suite) Close All Connections

12.06 PASS The Far Host Sees The Mapped Source Not The Real One (4.2s, 2 commands)

On the receiving router the peer appears as the sender's mapped address, which is the whole point of source NAT.

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Warm Up NAT Paths

Acceptance criteria

- output contains **, 0% packet loss**

  ping from ${alias} to ${target} lost packets: ${out}
- output contains **${H1_NAT_IP}**
- output does not contain **${H1_IP}**

How it was checked

H1 ping -c 5 10.30.2.10

```
PING 10.30.2.10 (10.30.2.10) 56(84) bytes of data.
64 bytes from 10.30.2.10: icmp_seq=1 ttl=62 time=1.75 ms
64 bytes from 10.30.2.10: icmp_seq=2 ttl=62 time=1.14 ms
64 bytes from 10.30.2.10: icmp_seq=3 ttl=62 time=1.25 ms
64 bytes from 10.30.2.10: icmp_seq=4 ttl=62 time=1.34 ms
64 bytes from 10.30.2.10: icmp_seq=5 ttl=62 time=1.24 ms

--- 10.30.2.10 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4007ms
rtt min/avg/max/mdev = 1.141/1.344/1.754/0.213 ms
```

R2 show ip nat translations

```
Pro  Inside global         Inside local          Outside local         Outside global
---  10.30.2.128           192.168.20.80         ---                   ---
---  10.30.2.129           192.168.20.75         ---                   ---
---  10.30.2.106           192.168.20.81         ---                   ---
---  10.30.2.105           192.168.20.74         ---                   ---
---  10.30.2.111           192.168.20.89         ---                   ---
---  10.30.2.108           192.168.20.82         ---                   ---
---  10.30.2.101           192.168.20.78         ---                   ---
---  10.30.2.102           192.168.20.90         ---                   ---
---  10.30.2.126           192.168.20.77         ---                   ---
---  10.30.2.125           192.168.20.79         ---                   ---
---  10.30.2.123           192.168.20.87         ---                   ---
---  10.30.2.120           192.168.20.68         ---                   ---
---  10.30.2.116           192.168.20.93         ---                   ---
... 37 more lines
```

Teardown (suite) Close All Connections

12.07 PASS Spoke To Spoke NAT Works Through The Hub (4.4s, 3 commands)

h2 -> h3 by mapped address crosses two tunnels and two translations, with no direct spoke-to-spoke path.

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Warm Up NAT Paths

Acceptance criteria

- output contains **, 0% packet loss**

  ping from ${alias} to ${target} lost packets: ${out}
- output contains **${H3_NAT_IP}**
- output contains **${H2_NAT_IP}**

How it was checked

H2 ping -c 5 10.30.3.10

```
PING 10.30.3.10 (10.30.3.10) 56(84) bytes of data.
64 bytes from 10.30.3.10: icmp_seq=1 ttl=62 time=1.40 ms
64 bytes from 10.30.3.10: icmp_seq=2 ttl=62 time=1.24 ms
64 bytes from 10.30.3.10: icmp_seq=3 ttl=62 time=1.28 ms
64 bytes from 10.30.3.10: icmp_seq=4 ttl=62 time=1.28 ms
64 bytes from 10.30.3.10: icmp_seq=5 ttl=62 time=1.30 ms

--- 10.30.3.10 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4006ms
rtt min/avg/max/mdev = 1.242/1.300/1.402/0.054 ms
```

R2 show ip nat translations

```
Pro  Inside global         Inside local          Outside local         Outside global
---  10.30.2.128           192.168.20.80         ---                   ---
---  10.30.2.129           192.168.20.75         ---                   ---
---  10.30.2.106           192.168.20.81         ---                   ---
---  10.30.2.105           192.168.20.74         ---                   ---
---  10.30.2.111           192.168.20.89         ---                   ---
---  10.30.2.108           192.168.20.82         ---                   ---
---  10.30.2.101           192.168.20.78         ---                   ---
---  10.30.2.102           192.168.20.90         ---                   ---
---  10.30.2.126           192.168.20.77         ---                   ---
---  10.30.2.125           192.168.20.79         ---                   ---
---  10.30.2.123           192.168.20.87         ---                   ---
---  10.30.2.120           192.168.20.68         ---                   ---
---  10.30.2.116           192.168.20.93         ---                   ---
... 38 more lines
```

R3 show ip nat translations

```
Pro  Inside global         Inside local          Outside local         Outside global
---  10.30.3.104           192.168.30.93         ---                   ---
---  10.30.3.107           192.168.30.89         ---                   ---
---  10.30.3.109           192.168.30.92         ---                   ---
---  10.30.3.110           192.168.30.91         ---                   ---
---  10.30.3.103           192.168.30.86         ---                   ---
---  10.30.3.100           192.168.30.90         ---                   ---
---  10.30.3.124           192.168.30.84         ---                   ---
---  10.30.3.127           192.168.30.82         ---                   ---
---  10.30.3.121           192.168.30.85         ---                   ---
---  10.30.3.122           192.168.30.78         ---                   ---
---  10.30.3.118           192.168.30.76         ---                   ---
---  10.30.3.117           192.168.30.80         ---                   ---
---  10.30.3.115           192.168.30.73         ---                   ---
... 37 more lines
```

Teardown (suite) Close All Connections

12.08 PASS NATted Traffic Is Still Encrypted (19.6s, 5 commands)

Translation happens before encryption, so NATted traffic must still raise the ESP counters.

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Warm Up NAT Paths

Acceptance criteria

- output contains **, 0% packet loss**

  ping from ${alias} to ${target} lost packets: ${out}
- **${r1_enc} >= ${expected}** holds
- **${r1_dec} >= ${expected}** holds

- **${r2_enc} >= ${expected}** holds
- **${r2_dec} >= ${expected}** holds

How it was checked

R1 show crypto ipsec sa

```
interface: Tunnel0
    Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
   remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
   current_peer 100.64.0.3 port 500
     PERMIT, flags={origin_is_acl,}
    #pkts encaps: 3570, #pkts encrypt: 3570, #pkts digest: 3570
    #pkts decaps: 3571, #pkts decrypt: 3571, #pkts verify: 3571
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0
... 78 more lines
```

R2 show crypto ipsec sa

```
interface: Tunnel0
    Crypto map tag: Tunnel0-head-0, local addr 100.64.0.2

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
   remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
   current_peer 100.64.0.3 port 500
     PERMIT, flags={origin_is_acl,}
    #pkts encaps: 51138, #pkts encrypt: 51138, #pkts digest: 51138
    #pkts decaps: 26750, #pkts decrypt: 26750, #pkts verify: 26750
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0
... 78 more lines
```

H1 ping -c 20 10.30.2.10

```
PING 10.30.2.10 (10.30.2.10) 56(84) bytes of data.
64 bytes from 10.30.2.10: icmp_seq=1 ttl=62 time=1.43 ms
64 bytes from 10.30.2.10: icmp_seq=2 ttl=62 time=1.19 ms
64 bytes from 10.30.2.10: icmp_seq=3 ttl=62 time=1.07 ms
64 bytes from 10.30.2.10: icmp_seq=4 ttl=62 time=1.24 ms
64 bytes from 10.30.2.10: icmp_seq=5 ttl=62 time=1.35 ms
64 bytes from 10.30.2.10: icmp_seq=6 ttl=62 time=1.10 ms
64 bytes from 10.30.2.10: icmp_seq=7 ttl=62 time=1.10 ms
64 bytes from 10.30.2.10: icmp_seq=8 ttl=62 time=1.09 ms
64 bytes from 10.30.2.10: icmp_seq=9 ttl=62 time=1.20 ms
64 bytes from 10.30.2.10: icmp_seq=10 ttl=62 time=1.17 ms
64 bytes from 10.30.2.10: icmp_seq=11 ttl=62 time=1.25 ms
64 bytes from 10.30.2.10: icmp_seq=12 ttl=62 time=1.19 ms
64 bytes from 10.30.2.10: icmp_seq=13 ttl=62 time=1.14 ms
... 11 more lines
```

R1 show crypto ipsec sa

```
interface: Tunnel0
    Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
   remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
   current_peer 100.64.0.3 port 500
     PERMIT, flags={origin_is_acl,}
    #pkts encaps: 3683, #pkts encrypt: 3683, #pkts digest: 3683
    #pkts decaps: 3684, #pkts decrypt: 3684, #pkts verify: 3684
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0
... 78 more lines
```

R2 show crypto ipsec sa

```
interface: Tunnel0
    Crypto map tag: Tunnel0-head-0, local addr 100.64.0.2

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
   remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
   current_peer 100.64.0.3 port 500
     PERMIT, flags={origin_is_acl,}
    #pkts encaps: 51248, #pkts encrypt: 51248, #pkts digest: 51248
```

```
    #pkts decaps: 26858, #pkts decrypt: 26858, #pkts verify: 26858
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0
... 78 more lines
```

Teardown (suite) Close All Connections

12.09 PASS Mapped Addresses Never Appear In Cleartext On The WAN (13.9s, 18 commands)

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Warm Up NAT Paths

Acceptance criteria

- output contains **, 0% packet loss**

  ping from ${alias} to ${target} lost packets: ${out}
- output contains **ESP**
- output does not contain **${H1_NAT_IP}**
- output does not contain **${H2_NAT_IP}**
- output does not contain **${H1_IP}**

How it was checked

R1 **`no monitor capture NATCAP`**
```
Capture does not exist
```
R1 **`monitor capture NATCAP interface GigabitEthernet2 both`**
```
${output} =
```
R1 **`monitor capture NATCAP match ipv4 any any`**
```
${output} =
```
R1 **`monitor capture NATCAP buffer size 5`**
```
${output} =
```
R1 **`monitor capture NATCAP start`**
```
Started capture point : NATCAP
```
R1 **`ping 100.64.0.2 repeat 2`**
```
Type escape sequence to abort.
Sending 2, 100-byte ICMP Echos to 100.64.0.2, timeout is 2 seconds:
!!
Success rate is 100 percent (2/2), round-trip min/avg/max = 1/1/1 ms
```
R1 **`monitor capture NATCAP stop`**
```
Stopped capture point : NATCAP
```
R1 **`show monitor capture NATCAP buffer brief`**
```
----------------------------------------------------------------------------
 #   size   timestamp     source             destination      dscp    protocol
 ----------------------------------------------------------------------------
   0  114    0.000000   100.64.0.1       ->  100.64.0.2       0  BE   ICMP
   1  114    0.000000   100.64.0.2       ->  100.64.0.1       0  BE   ICMP
   2  114    0.000992   100.64.0.1       ->  100.64.0.2       0  BE   ICMP
   3  114    0.000992   100.64.0.2       ->  100.64.0.1       0  BE   ICMP
   4  138    0.022002   100.64.0.2       ->  100.64.0.1       48 CS6  ESP
   5  138    0.022002   100.64.0.1       ->  100.64.0.2       48 CS6  ESP
   6  138    0.046995   100.64.0.1       ->  100.64.0.2       48 CS6  ESP
   7  138    0.048002   100.64.0.2       ->  100.64.0.1       48 CS6  ESP
   8  138    0.072003   100.64.0.2       ->  100.64.0.1       48 CS6  ESP
   9  138    0.162986   100.64.0.3       ->  100.64.0.1       48 CS6  ESP
  10  138    0.260988   100.64.0.3       ->  100.64.0.1       48 CS6  ESP
... 42 more lines
```
R1 **`no monitor capture NATCAP`**
```
${output} =
```
R1 **`no monitor capture NATCAP`**
```
Capture does not exist
```
R1 **`monitor capture NATCAP interface GigabitEthernet2 both`**
```
${output} =
```
R1 **`monitor capture NATCAP match ipv4 any any`**
```
${output} =
```
R1 **`monitor capture NATCAP buffer size 5`**
```
${output} =
```
R1 **`monitor capture NATCAP start`**
```
Started capture point : NATCAP
```
H1 **`ping -c 10 10.30.2.10`**
```
PING 10.30.2.10 (10.30.2.10) 56(84) bytes of data.
```

```
64 bytes from 10.30.2.10: icmp_seq=1 ttl=62 time=1.37 ms
64 bytes from 10.30.2.10: icmp_seq=2 ttl=62 time=1.25 ms
64 bytes from 10.30.2.10: icmp_seq=3 ttl=62 time=1.20 ms
64 bytes from 10.30.2.10: icmp_seq=4 ttl=62 time=1.37 ms
64 bytes from 10.30.2.10: icmp_seq=5 ttl=62 time=1.16 ms
64 bytes from 10.30.2.10: icmp_seq=6 ttl=62 time=1.30 ms
64 bytes from 10.30.2.10: icmp_seq=7 ttl=62 time=1.28 ms
64 bytes from 10.30.2.10: icmp_seq=8 ttl=62 time=1.21 ms
64 bytes from 10.30.2.10: icmp_seq=9 ttl=62 time=1.11 ms
64 bytes from 10.30.2.10: icmp_seq=10 ttl=62 time=1.24 ms

--- 10.30.2.10 ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9013ms
... 1 more lines
```

R1 monitor capture NATCAP stop

```
Stopped capture point : NATCAP
```

R1 show monitor capture NATCAP buffer brief

```
----------------------------------------------------------------------------
 #   size   timestamp     source             destination      dscp    protocol
 ----------------------------------------------------------------------------
   0  170    0.000000   100.64.0.1       ->  100.64.0.2       0  BE   ESP
   1  170    0.000992   100.64.0.2       ->  100.64.0.1       0  BE   ESP
   2  154    0.076992   100.64.0.2       ->  100.64.0.1       48 CS6  ESP
   3  138    0.190984   100.64.0.2       ->  100.64.0.1       48 CS6  ESP
   4  138    0.212986   100.64.0.1       ->  100.64.0.3       48 CS6  ESP
   5  138    0.212986   100.64.0.3       ->  100.64.0.1       48 CS6  ESP
   6  138    0.260988   100.64.0.1       ->  100.64.0.2       48 CS6  ESP
   7  138    0.260988   100.64.0.2       ->  100.64.0.1       48 CS6  ESP
   8  138    0.308974   100.64.0.2       ->  100.64.0.1       48 CS6  ESP
   9  138    0.308974   100.64.0.1       ->  100.64.0.2       48 CS6  ESP
  10  170    0.343976   100.64.0.2       ->  100.64.0.1       48 CS6  ESP
... 222 more lines
```

R1 no monitor capture NATCAP

```
${output} =
```

Teardown (suite) Close All Connections

12.10 PASS The Untranslated LAN Path Still Works (8.0s, 2 commands)

Guards the route-map scoping. If translation ever widened to all traffic, replies on this path would arrive from a mapped address the sender never contacted, and this would fail.

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Warm Up NAT Paths

Acceptance criteria

- output contains **, 0% packet loss** (checked 2×)

  ping from ${alias} to ${target} lost packets: ${out}

How it was checked

H1 ping -c 5 192.168.20.10

```
PING 192.168.20.10 (192.168.20.10) 56(84) bytes of data.
64 bytes from 192.168.20.10: icmp_seq=1 ttl=62 time=1.35 ms
64 bytes from 192.168.20.10: icmp_seq=2 ttl=62 time=1.28 ms
64 bytes from 192.168.20.10: icmp_seq=3 ttl=62 time=1.35 ms
64 bytes from 192.168.20.10: icmp_seq=4 ttl=62 time=1.60 ms
64 bytes from 192.168.20.10: icmp_seq=5 ttl=62 time=1.12 ms

--- 192.168.20.10 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4006ms
rtt min/avg/max/mdev = 1.120/1.338/1.599/0.154 ms
```

H2 ping -c 5 192.168.30.10

```
PING 192.168.30.10 (192.168.30.10) 56(84) bytes of data.
64 bytes from 192.168.30.10: icmp_seq=1 ttl=62 time=1.36 ms
64 bytes from 192.168.30.10: icmp_seq=2 ttl=62 time=1.19 ms
64 bytes from 192.168.30.10: icmp_seq=3 ttl=62 time=1.13 ms
64 bytes from 192.168.30.10: icmp_seq=4 ttl=62 time=1.15 ms
64 bytes from 192.168.30.10: icmp_seq=5 ttl=62 time=1.13 ms

--- 192.168.30.10 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4006ms
rtt min/avg/max/mdev = 1.129/1.190/1.355/0.084 ms
```

Teardown (suite) Close All Connections

13 Nat Pool

13.01 PASS Every Host Carries Its Full Block Of Addresses (0.1s, 6 commands)

10 statically mapped plus 30 pool-eligible, on one interface.

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Reset Translations

Acceptance criteria

- output contains **${lan}.${o}/24** (checked 12×)
- **${n}** equals **40** (checked 3×)

How it was checked

H1 ip addr show eth1

```
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 52:54:00:aa:01:02 brd ff:ff:ff:ff:ff:ff
    inet 192.168.10.10/24 scope global eth1
       valid_lft forever preferred_lft forever
    inet 169.254.229.55/16 brd 169.254.255.255 scope global noprefixroute eth1
       valid_lft forever preferred_lft forever
    inet 192.168.10.11/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.10.12/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.10.13/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.10.14/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
... 70 more lines
```

H1 ip addr show eth1 | grep -c 'inet 192.168.10.'

```
40
```

H2 ip addr show eth1

```
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 52:54:00:aa:02:02 brd ff:ff:ff:ff:ff:ff
    inet 192.168.20.10/24 scope global eth1
       valid_lft forever preferred_lft forever
    inet 169.254.199.41/16 brd 169.254.255.255 scope global noprefixroute eth1
       valid_lft forever preferred_lft forever
    inet 192.168.20.11/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.20.12/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.20.13/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.20.14/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
... 70 more lines
```

H2 ip addr show eth1 | grep -c 'inet 192.168.20.'

```
40
```

H3 ip addr show eth1

```
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 52:54:00:aa:03:02 brd ff:ff:ff:ff:ff:ff
    inet 192.168.30.10/24 scope global eth1
       valid_lft forever preferred_lft forever
    inet 169.254.207.242/16 brd 169.254.255.255 scope global noprefixroute eth1
       valid_lft forever preferred_lft forever
    inet 192.168.30.11/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.30.12/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.30.13/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.30.14/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
... 70 more lines
```

H3 ip addr show eth1 | grep -c 'inet 192.168.30.'

```
40
```

Teardown (suite) Close All Connections

13.02 PASS Every Router Configures Ten Static Mappings And A Thirty Address Pool (0.4s, 6 commands)

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Reset Translations

Acceptance criteria

- output contains **= 10** (checked 3×)
- output contains **total addresses 30** (checked 3×)

How it was checked

H1 show running-config | count ^ip nat inside source static
```
Number of lines which match regexp = 10
```

H1 show ip nat statistics
```
Total active translations: 10 (10 static, 0 dynamic; 0 extended)
Outside interfaces:
  Tunnel0
Inside interfaces:
  GigabitEthernet3
Hits: 0  Misses: 0
 Reserved port setting disabled provisioned no
 Dynamic overload mapping configured: 0
Expired translations: 0
Dynamic mappings:
-- Inside Source
[Id: 2] access-list NAT-POOL-SRC pool SITE-POOL refcount 0
 pool SITE-POOL: id 2, netmask 255.255.255.0
        start 10.30.1.100 end 10.30.1.129
... 9 more lines
```

H2 show running-config | count ^ip nat inside source static
```
Number of lines which match regexp = 10
```

H2 show ip nat statistics
```
Total active translations: 10 (10 static, 0 dynamic; 0 extended)
Outside interfaces:
  Tunnel0
Inside interfaces:
  GigabitEthernet3
Hits: 0  Misses: 0
 Reserved port setting disabled provisioned no
 Dynamic overload mapping configured: 0
Expired translations: 0
Dynamic mappings:
-- Inside Source
[Id: 2] access-list NAT-POOL-SRC pool SITE-POOL refcount 0
 pool SITE-POOL: id 2, netmask 255.255.255.0
        start 10.30.2.100 end 10.30.2.129
... 9 more lines
```

H3 show running-config | count ^ip nat inside source static
```
Number of lines which match regexp = 10
```

H3 show ip nat statistics
```
Total active translations: 10 (10 static, 0 dynamic; 0 extended)
Outside interfaces:
  Tunnel0
Inside interfaces:
  GigabitEthernet3
Hits: 0  Misses: 0
 Reserved port setting disabled provisioned no
 Dynamic overload mapping configured: 0
Expired translations: 0
Dynamic mappings:
-- Inside Source
[Id: 2] access-list NAT-POOL-SRC pool SITE-POOL refcount 0
 pool SITE-POOL: id 2, netmask 255.255.255.0
        start 10.30.3.100 end 10.30.3.129
... 9 more lines
```

Teardown (suite) Close All Connections

13.03 PASS The Pool Access List Is One Range And Excludes The Static Block (0.1s, 3 commands)

A single /27 range rather than thirty host entries, and it must not overlap the statically mapped addresses -- otherwise which rule wins depends on evaluation order.

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Reset Translations

Acceptance criteria

- **${permits}** has exactly **1** entries (checked 3×)

  NAT-POOL-SRC should hold one range entry, found ${permits}
- output contains **${lan}.${DYN_OCTET_FIRST} 0.0.0.31** (checked 3×)
- output does not contain **host ${lan}.${o}** (checked 6×)

How it was checked

H1 show ip access-lists NAT-POOL-SRC
```
Extended IP access list NAT-POOL-SRC
    10 permit ip 192.168.10.64 0.0.0.31 10.30.0.0 0.0.255.255
```
H2 show ip access-lists NAT-POOL-SRC
```
Extended IP access list NAT-POOL-SRC
    10 permit ip 192.168.20.64 0.0.0.31 10.30.0.0 0.0.255.255
```
H3 show ip access-lists NAT-POOL-SRC
```
Extended IP access list NAT-POOL-SRC
    10 permit ip 192.168.30.64 0.0.0.31 10.30.0.0 0.0.255.255
```
Teardown (suite) Close All Connections

13.04 PASS All Ten Static Addresses Translate To Their Own Fixed Mapping (9.5s, 2 commands)

Static NAT is one-to-one and deterministic: inside .10-.19 must appear outside as .10-.19, each to its own.

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Reset Translations

Acceptance criteria

- **${static}** has exactly **10** entries

  only ${static} of the 10 static addresses were translated
- **${d}** equals **10**

  the 10 static addresses shared outside addresses; they must be one-to-one
- **${aligned}** holds

How it was checked

H1 for i in $(seq 10 19) $(seq 64 93); do ping -c 2 -W 2 -I 192.168.10.$i 10.30.2.10 >/dev/null 2>&1 & done; wait; echo SENT
```
SENT
[40]+  Done                    ping -c 2 -W 2 -I 192.168.10.${i} 10.30.2.10 1>/dev/null 2>&1
[39]+  Done                    ping -c 2 -W 2 -I 192.168.10.${i} 10.30.2.10 1>/dev/null 2>&1
[38]+  Done                    ping -c 2 -W 2 -I 192.168.10.${i} 10.30.2.10 1>/dev/null 2>&1
[37]+  Done                    ping -c 2 -W 2 -I 192.168.10.${i} 10.30.2.10 1>/dev/null 2>&1
[36]+  Done                    ping -c 2 -W 2 -I 192.168.10.${i} 10.30.2.10 1>/dev/null 2>&1
[35]+  Done                    ping -c 2 -W 2 -I 192.168.10.${i} 10.30.2.10 1>/dev/null 2>&1
[34]+  Done                    ping -c 2 -W 2 -I 192.168.10.${i} 10.30.2.10 1>/dev/null 2>&1
[33]+  Done                    ping -c 2 -W 2 -I 192.168.10.${i} 10.30.2.10 1>/dev/null 2>&1
[32]+  Done                    ping -c 2 -W 2 -I 192.168.10.${i} 10.30.2.10 1>/dev/null 2>&1
[31]+  Done                    ping -c 2 -W 2 -I 192.168.10.${i} 10.30.2.10 1>/dev/null 2>&1
[30]+  Done                    ping -c 2 -W 2 -I 192.168.10.${i} 10.30.2.10 1>/dev/null 2>&1
[29]+  Done                    ping -c 2 -W 2 -I 192.168.10.${i} 10.30.2.10 1>/dev/null 2>&1
[28]+  Done                    ping -c 2 -W 2 -I 192.168.10.${i} 10.30.2.10 1>/dev/null 2>&1
... 27 more lines
```
R1 show ip nat translations
```
Pro  Inside global      Inside local       Outside local      Outside global
---  10.30.1.128        192.168.10.88      ---                ---
---  10.30.1.129        192.168.10.66      ---                ---
---  10.30.1.113        192.168.10.92      ---                ---
---  10.30.1.114        192.168.10.81      ---                ---
---  10.30.1.116        192.168.10.74      ---                ---
---  10.30.1.119        192.168.10.83      ---                ---
---  10.30.1.123        192.168.10.85      ---                ---
---  10.30.1.120        192.168.10.77      ---                ---
---  10.30.1.126        192.168.10.93      ---                ---
---  10.30.1.125        192.168.10.73      ---                ---
---  10.30.1.101        192.168.10.79      ---                ---
---  10.30.1.102        192.168.10.91      ---                ---
---  10.30.1.111        192.168.10.68      ---                ---
... 68 more lines
```
Teardown (suite) Close All Connections

13.05 PASS All Thirty Pool Addresses Translate Into The Pool And Are Distinct (0.5s, 1 commands)

Without overload each client takes its own pool address, so 30 clients must yield 30 different outside addresses.

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Reset Translations

Acceptance criteria

- **${dyn}** has exactly **30** entries
  only ${dyn} of the 30 pool clients were translated
- **${d}** equals **30**
  pool clients shared outside addresses; that is overload, not a 1:1 pool
- **${within}** holds

How it was checked

R1 show ip nat translations

```
Pro  Inside global         Inside local          Outside local         Outside global
---  10.30.1.128           192.168.10.88         ---                   ---
---  10.30.1.129           192.168.10.66         ---                   ---
---  10.30.1.113           192.168.10.92         ---                   ---
---  10.30.1.114           192.168.10.81         ---                   ---
---  10.30.1.116           192.168.10.74         ---                   ---
---  10.30.1.119           192.168.10.83         ---                   ---
---  10.30.1.123           192.168.10.85         ---                   ---
---  10.30.1.120           192.168.10.77         ---                   ---
---  10.30.1.126           192.168.10.93         ---                   ---
---  10.30.1.125           192.168.10.73         ---                   ---
---  10.30.1.101           192.168.10.79         ---                   ---
---  10.30.1.102           192.168.10.91         ---                   ---
---  10.30.1.111           192.168.10.68         ---                   ---
... 68 more lines
```

Teardown (suite) Close All Connections

13.06 PASS The Pool Is Exactly Consumed With No Misses (0.0s, 1 commands)

30 clients against a 30 address pool: full allocation and no misses is the evidence that none of them shared or failed.

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Reset Translations

Acceptance criteria

- output contains **allocated 30 (100%)**
- output matches **misses 0**

How it was checked

R1 show ip nat statistics

```
Total active translations: 80 (10 static, 70 dynamic; 40 extended)
Outside interfaces:
  Tunnel0
Inside interfaces:
  GigabitEthernet3
Hits: 120  Misses: 40
 Reserved port setting disabled provisioned no
 Dynamic overload mapping configured: 0
Expired translations: 0
Dynamic mappings:
-- Inside Source
[Id: 2] access-list NAT-POOL-SRC pool SITE-POOL refcount 60
 pool SITE-POOL: id 2, netmask 255.255.255.0
        start 10.30.1.100 end 10.30.1.129
... 9 more lines
```

Teardown (suite) Close All Connections

13.07 PASS A Thirty-first Client Gets No Translation Once The Pool Is Exhausted (9.3s, 8 commands)

The boundary the pool sizing never otherwise reaches. The pool holds exactly as many addresses as there are clients, which means the exhaustion path is never exercised by a normal run -- and an off-by-one in pool bounds would look identical to a correct configuration. So bring up one more client inside the pool ACL and require the router to refuse it: no translation, no borrowing of another client's address, and the miss counted. Without "overload" there is nothing to share, so silently reusing an address would be a correctness bug, not graceful degradation. Run against R1/H1, because it is R1's pool that the tests above fill: the suite drives traffic from H1 first and only reaches the other routers later. Asserting exhaustion on a router whose pool is still empty would test nothing.

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Reset Translations

Acceptance criteria

- output contains **allocated 30 (100%)**
- **${n}** is not empty (checked 3×)
- output contains **100% packet loss**

  the thirty-first client was translated even though the pool was exhausted
- **${xlate}** is empty
- **${after} > ${before}** holds (checked 2×)

How it was checked

R1 show ip nat statistics | include allocated
```
type generic, total addresses 30, allocated 30 (100%), misses 0
```

R1 show ip nat statistics
```
Total active translations: 80 (10 static, 70 dynamic; 40 extended)
Outside interfaces:
  Tunnel0
Inside interfaces:
  GigabitEthernet3
Hits: 120  Misses: 40
 Reserved port setting disabled provisioned no
 Dynamic overload mapping configured: 0
Expired translations: 0
Dynamic mappings:
-- Inside Source
[Id: 2] access-list NAT-POOL-SRC pool SITE-POOL refcount 60
 pool SITE-POOL: id 2, netmask 255.255.255.0
        start 10.30.1.100 end 10.30.1.129
... 9 more lines
```

H1 sudo ip addr add 192.168.10.94/24 dev eth1
```
${output} =
```

H1 ping -c 3 -W 2 -I 192.168.10.94 10.30.2.10
```
PING 10.30.2.10 (10.30.2.10) from 192.168.10.94 : 56(84) bytes of data.
From 192.168.10.1 icmp_seq=2 Destination Host Unreachable
From 192.168.10.1 icmp_seq=3 Destination Host Unreachable

--- 10.30.2.10 ping statistics ---
3 packets transmitted, 0 received, +2 errors, 100% packet loss, time 2036ms
```

R1 show ip nat translations | include 192.168.10.94
```
${output} =
```

R1 show ip nat statistics
```
Total active translations: 80 (10 static, 70 dynamic; 40 extended)
Outside interfaces:
  Tunnel0
Inside interfaces:
  GigabitEthernet3
Hits: 120  Misses: 40
 Reserved port setting disabled provisioned no
 Dynamic overload mapping configured: 0
Expired translations: 0
Dynamic mappings:
-- Inside Source
[Id: 2] access-list NAT-POOL-SRC pool SITE-POOL refcount 60
 pool SITE-POOL: id 2, netmask 255.255.255.0
        start 10.30.1.100 end 10.30.1.129
... 9 more lines
```

R1 show ip nat statistics
```
Total active translations: 80 (10 static, 70 dynamic; 40 extended)
Outside interfaces:
  Tunnel0
```

```
Inside interfaces:
  GigabitEthernet3
Hits: 120  Misses: 40
 Reserved port setting disabled provisioned no
 Dynamic overload mapping configured: 0
Expired translations: 0
Dynamic mappings:
-- Inside Source
[Id: 2] access-list NAT-POOL-SRC pool SITE-POOL refcount 60
 pool SITE-POOL: id 2, netmask 255.255.255.0
        start 10.30.1.100 end 10.30.1.129
... 9 more lines
```

H1 sudo ip addr del 192.168.10.94/24 dev eth1

```
${output} =
```

Teardown (this test) Remove The Spare Client From H1

13.08 PASS The Pool Is Still Intact After The Exhaustion Attempt (0.5s, 2 commands)

The refusal must cost nothing: the thirty legitimate clients keep the addresses they held, and none was evicted to make room for the one that was turned away.

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Reset Translations

Acceptance criteria

- output contains **allocated 30 (100%)**
- **${dyn}** has exactly **30** entries

  only ${dyn} of the original thirty pool clients still hold a translation
- **${d}** equals **30**

  the refused client caused two survivors to share an address

How it was checked

R1 show ip nat statistics | include allocated

```
type generic, total addresses 30, allocated 30 (100%), misses 3
```

R1 show ip nat translations

```
Pro  Inside global         Inside local          Outside local         Outside global
---  10.30.1.128           192.168.10.88         ---                   ---
---  10.30.1.129           192.168.10.66         ---                   ---
---  10.30.1.113           192.168.10.92         ---                   ---
---  10.30.1.114           192.168.10.81         ---                   ---
---  10.30.1.116           192.168.10.74         ---                   ---
---  10.30.1.119           192.168.10.83         ---                   ---
---  10.30.1.123           192.168.10.85         ---                   ---
---  10.30.1.120           192.168.10.77         ---                   ---
---  10.30.1.126           192.168.10.93         ---                   ---
---  10.30.1.125           192.168.10.73         ---                   ---
---  10.30.1.101           192.168.10.79         ---                   ---
---  10.30.1.102           192.168.10.91         ---                   ---
---  10.30.1.111           192.168.10.68         ---                   ---
... 68 more lines
```

Teardown (suite) Close All Connections

13.09 PASS Static And Pool Translations Coexist Across The Whole Block (0.5s, 1 commands)

All 40 of a host's addresses in flight, translated by two different mechanisms, with no outside address reused.

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Reset Translations

Acceptance criteria

- **${all}** has exactly **40** entries

  expected 40 translated addresses, found ${all}
- **${d}** equals **40**

  the 40 inside addresses did not receive 40 distinct outside addresses

How it was checked

R1 show ip nat translations

```
Pro  Inside global         Inside local          Outside local         Outside global
---  10.30.1.128           192.168.10.88         ---                   ---
---  10.30.1.129           192.168.10.66         ---                   ---
---  10.30.1.113           192.168.10.92         ---                   ---
```

```
---  10.30.1.114          192.168.10.81         ---                   ---
---  10.30.1.116          192.168.10.74         ---                   ---
---  10.30.1.119          192.168.10.83         ---                   ---
---  10.30.1.123          192.168.10.85         ---                   ---
---  10.30.1.120          192.168.10.77         ---                   ---
---  10.30.1.126          192.168.10.93         ---                   ---
---  10.30.1.125          192.168.10.73         ---                   ---
---  10.30.1.101          192.168.10.79         ---                   ---
---  10.30.1.102          192.168.10.91         ---                   ---
---  10.30.1.111          192.168.10.68         ---                   ---
... 68 more lines
```

Teardown (suite) Close All Connections

13.10 PASS Both Mechanisms Work On Every Router (19.5s, 4 commands)

The spokes carry the same scheme, not just the hub.

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Reset Translations

Acceptance criteria

- **${static}** has exactly **10** entries (checked 2×)
- **${dyn}** has exactly **30** entries (checked 2×)
- **${aligned}** holds (checked 2×)

How it was checked

H2 for i in $(seq 10 19) $(seq 64 93); do ping -c 2 -W 2 -I 192.168.20.$i 10.30.1.10 >/dev/null 2>&1 & done; wait; echo SENT

```
SENT
[40]+  Done                    ping -c 2 -W 2 -I 192.168.20.${i} 10.30.1.10 1>/dev/null 2>&1
[39]+  Done                    ping -c 2 -W 2 -I 192.168.20.${i} 10.30.1.10 1>/dev/null 2>&1
[38]+  Done                    ping -c 2 -W 2 -I 192.168.20.${i} 10.30.1.10 1>/dev/null 2>&1
[37]+  Done                    ping -c 2 -W 2 -I 192.168.20.${i} 10.30.1.10 1>/dev/null 2>&1
[36]+  Done                    ping -c 2 -W 2 -I 192.168.20.${i} 10.30.1.10 1>/dev/null 2>&1
[35]+  Done                    ping -c 2 -W 2 -I 192.168.20.${i} 10.30.1.10 1>/dev/null 2>&1
[34]+  Done                    ping -c 2 -W 2 -I 192.168.20.${i} 10.30.1.10 1>/dev/null 2>&1
[33]+  Done                    ping -c 2 -W 2 -I 192.168.20.${i} 10.30.1.10 1>/dev/null 2>&1
[32]+  Done                    ping -c 2 -W 2 -I 192.168.20.${i} 10.30.1.10 1>/dev/null 2>&1
[31]+  Done                    ping -c 2 -W 2 -I 192.168.20.${i} 10.30.1.10 1>/dev/null 2>&1
[30]+  Done                    ping -c 2 -W 2 -I 192.168.20.${i} 10.30.1.10 1>/dev/null 2>&1
[29]+  Done                    ping -c 2 -W 2 -I 192.168.20.${i} 10.30.1.10 1>/dev/null 2>&1
[28]+  Done                    ping -c 2 -W 2 -I 192.168.20.${i} 10.30.1.10 1>/dev/null 2>&1
... 27 more lines
```

H2 show ip nat translations

```
Pro  Inside global         Inside local          Outside local         Outside global
---  10.30.2.128          192.168.20.87         ---                   ---
---  10.30.2.129          192.168.20.75         ---                   ---
---  10.30.2.106          192.168.20.81         ---                   ---
---  10.30.2.105          192.168.20.79         ---                   ---
---  10.30.2.111          192.168.20.78         ---                   ---
---  10.30.2.108          192.168.20.80         ---                   ---
---  10.30.2.101          192.168.20.70         ---                   ---
---  10.30.2.102          192.168.20.77         ---                   ---
---  10.30.2.126          192.168.20.71         ---                   ---
---  10.30.2.125          192.168.20.91         ---                   ---
---  10.30.2.123          192.168.20.82         ---                   ---
---  10.30.2.120          192.168.20.66         ---                   ---
---  10.30.2.116          192.168.20.86         ---                   ---
... 108 more lines
```

H3 for i in $(seq 10 19) $(seq 64 93); do ping -c 2 -W 2 -I 192.168.30.$i 10.30.1.10 >/dev/null 2>&1 & done; wait; echo SENT

```
SENT
[40]+  Done                    ping -c 2 -W 2 -I 192.168.30.${i} 10.30.1.10 1>/dev/null 2>&1
[39]+  Done                    ping -c 2 -W 2 -I 192.168.30.${i} 10.30.1.10 1>/dev/null 2>&1
[38]+  Done                    ping -c 2 -W 2 -I 192.168.30.${i} 10.30.1.10 1>/dev/null 2>&1
[37]+  Done                    ping -c 2 -W 2 -I 192.168.30.${i} 10.30.1.10 1>/dev/null 2>&1
[36]+  Done                    ping -c 2 -W 2 -I 192.168.30.${i} 10.30.1.10 1>/dev/null 2>&1
[35]+  Done                    ping -c 2 -W 2 -I 192.168.30.${i} 10.30.1.10 1>/dev/null 2>&1
[34]+  Done                    ping -c 2 -W 2 -I 192.168.30.${i} 10.30.1.10 1>/dev/null 2>&1
[33]+  Done                    ping -c 2 -W 2 -I 192.168.30.${i} 10.30.1.10 1>/dev/null 2>&1
[32]+  Done                    ping -c 2 -W 2 -I 192.168.30.${i} 10.30.1.10 1>/dev/null 2>&1
[31]+  Done                    ping -c 2 -W 2 -I 192.168.30.${i} 10.30.1.10 1>/dev/null 2>&1
[30]+  Done                    ping -c 2 -W 2 -I 192.168.30.${i} 10.30.1.10 1>/dev/null 2>&1
[29]+  Done                    ping -c 2 -W 2 -I 192.168.30.${i} 10.30.1.10 1>/dev/null 2>&1
[28]+  Done                    ping -c 2 -W 2 -I 192.168.30.${i} 10.30.1.10 1>/dev/null 2>&1
... 27 more lines
```

```
H3 show ip nat translations
  Pro  Inside global         Inside local          Outside local         Outside global
  ---  10.30.3.104           192.168.30.93         ---                   ---
  ---  10.30.3.107           192.168.30.91         ---                   ---
  ---  10.30.3.109           192.168.30.92         ---                   ---
  ---  10.30.3.110           192.168.30.89         ---                   ---
  ---  10.30.3.103           192.168.30.88         ---                   ---
  ---  10.30.3.100           192.168.30.90         ---                   ---
  ---  10.30.3.124           192.168.30.84         ---                   ---
  ---  10.30.3.127           192.168.30.81         ---                   ---
  ---  10.30.3.121           192.168.30.87         ---                   ---
  ---  10.30.3.122           192.168.30.76         ---                   ---
  ---  10.30.3.118           192.168.30.78         ---                   ---
  ---  10.30.3.117           192.168.30.86         ---                   ---
  ---  10.30.3.115           192.168.30.74         ---                   ---
  ... 68 more lines
```

Teardown (suite) Close All Connections

13.11 PASS Only The Static Mappings Live In The Configuration (0.1s, 1 commands)

Static mappings are configuration and survive a reload; pool allocations are runtime state and do not. Confusing the two is how NAT surprises people after a reboot.

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Reset Translations

Acceptance criteria

- output contains **static ${H1_IP} ${H1_NAT_IP}**
- output does not contain **${NAT24}[R1].${POOL_OCTET_FIRST}**
- output contains **ip nat inside source list NAT-POOL-SRC pool SITE-POOL**

How it was checked

```
R1 show running-config | include ip nat inside source
  ip nat inside source static 192.168.10.10 10.30.1.10 route-map NAT-TO-DOMAIN
  ip nat inside source static 192.168.10.11 10.30.1.11 route-map NAT-TO-DOMAIN
  ip nat inside source static 192.168.10.12 10.30.1.12 route-map NAT-TO-DOMAIN
  ip nat inside source static 192.168.10.13 10.30.1.13 route-map NAT-TO-DOMAIN
  ip nat inside source static 192.168.10.14 10.30.1.14 route-map NAT-TO-DOMAIN
  ip nat inside source static 192.168.10.15 10.30.1.15 route-map NAT-TO-DOMAIN
  ip nat inside source static 192.168.10.16 10.30.1.16 route-map NAT-TO-DOMAIN
  ip nat inside source static 192.168.10.17 10.30.1.17 route-map NAT-TO-DOMAIN
  ip nat inside source static 192.168.10.18 10.30.1.18 route-map NAT-TO-DOMAIN
  ip nat inside source static 192.168.10.19 10.30.1.19 route-map NAT-TO-DOMAIN
  ip nat inside source list NAT-POOL-SRC pool SITE-POOL
```

Teardown (suite) Close All Connections

13.12 PASS Traffic From Both Mechanisms Is Encrypted (38.6s, 6 commands)

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Reset Translations

Acceptance criteria

- output contains **, 0% packet loss** (checked 2×)
  ping from ${source} on ${alias} to ${target} lost packets
- **${r1_enc} >= ${expected}** holds
- **${r1_dec} >= ${expected}** holds
- **${r2_enc} >= ${expected}** holds
- **${r2_dec} >= ${expected}** holds

How it was checked

```
R1 show crypto ipsec sa
  interface: Tunnel0
      Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

     protected vrf: (none)
     local  ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
     remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
     current_peer 100.64.0.3 port 500
       PERMIT, flags={origin_is_acl,}
      #pkts encaps: 4138, #pkts encrypt: 4138, #pkts digest: 4138
```

```
     #pkts decaps: 4141, #pkts decrypt: 4141, #pkts verify: 4141
     #pkts compressed: 0, #pkts decompressed: 0
     #pkts not compressed: 0, #pkts compr. failed: 0
     #pkts not decompressed: 0, #pkts decompress failed: 0
     #send errors 0, #recv errors 0
 ... 78 more lines
```

R2 show crypto ipsec sa

```
interface: Tunnel0
    Crypto map tag: Tunnel0-head-0, local addr 100.64.0.2

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
   remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
   current_peer 100.64.0.3 port 500
     PERMIT, flags={origin_is_acl,}
    #pkts encaps: 51619, #pkts encrypt: 51619, #pkts digest: 51619
    #pkts decaps: 27229, #pkts decrypt: 27229, #pkts verify: 27229
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0
... 78 more lines
```

H1 ping -c 20 -I 192.168.10.10 10.30.2.10

```
PING 10.30.2.10 (10.30.2.10) from 192.168.10.10 : 56(84) bytes of data.
64 bytes from 10.30.2.10: icmp_seq=1 ttl=62 time=1.35 ms
64 bytes from 10.30.2.10: icmp_seq=2 ttl=62 time=1.20 ms
64 bytes from 10.30.2.10: icmp_seq=3 ttl=62 time=1.17 ms
64 bytes from 10.30.2.10: icmp_seq=4 ttl=62 time=1.48 ms
64 bytes from 10.30.2.10: icmp_seq=5 ttl=62 time=1.24 ms
64 bytes from 10.30.2.10: icmp_seq=6 ttl=62 time=1.17 ms
64 bytes from 10.30.2.10: icmp_seq=7 ttl=62 time=1.13 ms
64 bytes from 10.30.2.10: icmp_seq=8 ttl=62 time=1.11 ms
64 bytes from 10.30.2.10: icmp_seq=9 ttl=62 time=1.18 ms
64 bytes from 10.30.2.10: icmp_seq=10 ttl=62 time=1.14 ms
64 bytes from 10.30.2.10: icmp_seq=11 ttl=62 time=1.27 ms
64 bytes from 10.30.2.10: icmp_seq=12 ttl=62 time=1.20 ms
64 bytes from 10.30.2.10: icmp_seq=13 ttl=62 time=1.11 ms
... 11 more lines
```

H1 ping -c 20 -I 192.168.10.64 10.30.2.10

```
PING 10.30.2.10 (10.30.2.10) from 192.168.10.64 : 56(84) bytes of data.
64 bytes from 10.30.2.10: icmp_seq=1 ttl=62 time=1.09 ms
64 bytes from 10.30.2.10: icmp_seq=2 ttl=62 time=1.56 ms
64 bytes from 10.30.2.10: icmp_seq=3 ttl=62 time=1.43 ms
64 bytes from 10.30.2.10: icmp_seq=4 ttl=62 time=1.20 ms
64 bytes from 10.30.2.10: icmp_seq=5 ttl=62 time=1.22 ms
64 bytes from 10.30.2.10: icmp_seq=6 ttl=62 time=1.23 ms
64 bytes from 10.30.2.10: icmp_seq=7 ttl=62 time=1.10 ms
64 bytes from 10.30.2.10: icmp_seq=8 ttl=62 time=1.16 ms
64 bytes from 10.30.2.10: icmp_seq=9 ttl=62 time=1.24 ms
64 bytes from 10.30.2.10: icmp_seq=10 ttl=62 time=1.13 ms
64 bytes from 10.30.2.10: icmp_seq=11 ttl=62 time=1.14 ms
64 bytes from 10.30.2.10: icmp_seq=12 ttl=62 time=1.15 ms
64 bytes from 10.30.2.10: icmp_seq=13 ttl=62 time=1.24 ms
... 11 more lines
```

R1 show crypto ipsec sa

```
interface: Tunnel0
    Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
   remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
   current_peer 100.64.0.3 port 500
     PERMIT, flags={origin_is_acl,}
    #pkts encaps: 4368, #pkts encrypt: 4368, #pkts digest: 4368
    #pkts decaps: 4366, #pkts decrypt: 4366, #pkts verify: 4366
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0
... 78 more lines
```

R2 show crypto ipsec sa

```
interface: Tunnel0
    Crypto map tag: Tunnel0-head-0, local addr 100.64.0.2

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
   remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
   current_peer 100.64.0.3 port 500
     PERMIT, flags={origin_is_acl,}
```

```
      #pkts encaps: 51836, #pkts encrypt: 51836, #pkts digest: 51836
      #pkts decaps: 27445, #pkts decrypt: 27445, #pkts verify: 27445
      #pkts compressed: 0, #pkts decompressed: 0
      #pkts not compressed: 0, #pkts compr. failed: 0
      #pkts not decompressed: 0, #pkts decompress failed: 0
      #send errors 0, #recv errors 0
  ... 78 more lines
```

Teardown (suite) Close All Connections

13.13 PASS Neither Real Nor Translated Addresses Appear In Cleartext (11.5s, 9 commands)

Setup (suite) Open All Routers, Open All Hosts, Wait For NAT Routes, Reset Translations

Acceptance criteria

- output contains **, 0% packet loss**

  ping from ${source} on ${alias} to ${target} lost packets
- output contains **ESP**
- output does not contain **${LAN24}[H1].${DYN_OCTET_FIRST}**
- output does not contain **${NAT24}[R1].${POOL_OCTET_FIRST}**

How it was checked

R1 no monitor capture POOLCAP
```
Capture does not exist
```

R1 monitor capture POOLCAP interface GigabitEthernet2 both
```
${output} =
```

R1 monitor capture POOLCAP match ipv4 any any
```
${output} =
```

R1 monitor capture POOLCAP buffer size 5
```
${output} =
```

R1 monitor capture POOLCAP start
```
Started capture point : POOLCAP
```

H1 ping -c 10 -I 192.168.10.64 10.30.2.10
```
PING 10.30.2.10 (10.30.2.10) from 192.168.10.64 : 56(84) bytes of data.
64 bytes from 10.30.2.10: icmp_seq=1 ttl=62 time=1.44 ms
64 bytes from 10.30.2.10: icmp_seq=2 ttl=62 time=1.26 ms
64 bytes from 10.30.2.10: icmp_seq=3 ttl=62 time=1.32 ms
64 bytes from 10.30.2.10: icmp_seq=4 ttl=62 time=1.35 ms
64 bytes from 10.30.2.10: icmp_seq=5 ttl=62 time=1.18 ms
64 bytes from 10.30.2.10: icmp_seq=6 ttl=62 time=1.71 ms
64 bytes from 10.30.2.10: icmp_seq=7 ttl=62 time=1.27 ms
64 bytes from 10.30.2.10: icmp_seq=8 ttl=62 time=1.25 ms
64 bytes from 10.30.2.10: icmp_seq=9 ttl=62 time=1.30 ms
64 bytes from 10.30.2.10: icmp_seq=10 ttl=62 time=1.91 ms

--- 10.30.2.10 ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9014ms
... 1 more lines
```

R1 monitor capture POOLCAP stop
```
Stopped capture point : POOLCAP
```

R1 show monitor capture POOLCAP buffer brief
```
----------------------------------------------------------------------------
 #   size   timestamp     source             destination      dscp    protocol
 ----------------------------------------------------------------------------
   0  170    0.000000   100.64.0.1       ->  100.64.0.2       0  BE   ESP
   1  170    0.000992   100.64.0.2       ->  100.64.0.1       0  BE   ESP
   2  138    0.098994   100.64.0.2       ->  100.64.0.1       48 CS6  ESP
   3  138    0.098994   100.64.0.1       ->  100.64.0.2       48 CS6  ESP
   4  138    0.267976   100.64.0.3       ->  100.64.0.1       48 CS6  ESP
   5  138    0.267976   100.64.0.1       ->  100.64.0.3       48 CS6  ESP
   6  138    0.303985   100.64.0.2       ->  100.64.0.1       48 CS6  ESP
   7  138    0.335981   100.64.0.1       ->  100.64.0.2       48 CS6  ESP
   8  138    0.335981   100.64.0.2       ->  100.64.0.1       48 CS6  ESP
   9  138    0.397975   100.64.0.1       ->  100.64.0.3       48 CS6  ESP
  10  138    0.398966   100.64.0.3       ->  100.64.0.1       48 CS6  ESP
... 221 more lines
```

R1 no monitor capture POOLCAP
```
${output} =
```

Teardown (suite) Close All Connections

14 Ntp

14.01 PASS Every Router Is An NTP Client Of The NMS Over The Management VRF (0.3s, 3 commands)

Setup (suite) Open All Routers

Acceptance criteria

- output contains **ntp source ${OOB_INTF}[${r}]** (checked 3×)
- output contains **ntp server vrf ${MGMT_VRF} ${NMS_OOB_IP}** (checked 3×)
- output does not contain **pool.ntp.org** (checked 3×)

How it was checked

```
R1 show running-config | include ^ntp
  ntp source GigabitEthernet4
  ntp server vrf MGMT 192.168.99.10
R2 show running-config | include ^ntp
  ntp source GigabitEthernet4
  ntp server vrf MGMT 192.168.99.10
R3 show running-config | include ^ntp
  ntp source GigabitEthernet4
  ntp server vrf MGMT 192.168.99.10
```

Teardown (suite) Close Nms Session, Close All Connections

14.02 PASS The NMS Itself Synchronises To The Public Pool (2.4s, 2 commands)

The precondition the rest of this suite rests on. chrony is configured "local stratum 10", so it serves time whether or not it has an upstream -- meaning the routers can agree perfectly on time that came from nowhere. This requires a real upstream: a reference that is not the local clock, and a stratum that proves it came from somewhere else.

Setup (suite) Open All Routers

Acceptance criteria

- output does not contain **7F7F0101**

  the NMS is serving its own local clock, so lab time is fabricated
- **${m}** is not empty
- **0 < ${m}[0] < ${MAX_STRATUM}** holds
- output matches **\^*\s+\d+\.\d+\.\d+\.\d+**

  the NMS has selected no upstream source: ${sources}

How it was checked

```
NMS chronyc tracking
  Reference ID    : C0CF37FE (192.207.55.254)
  Stratum         : 2
  Ref time (UTC)  : Wed Sep 09 00:54:46 2026
  System time     : 0.000339688 seconds fast of NTP time
  Last offset     : +0.000057765 seconds
  RMS offset      : 0.000284445 seconds
  Frequency       : 8.617 ppm fast
  Residual freq   : +0.001 ppm
  Skew            : 0.087 ppm
  Root delay      : 0.052005194 seconds
  Root dispersion : 0.001374164 seconds
  Update interval : 1033.9 seconds
  Leap status     : Normal
NMS chronyc -n sources
  MS Name/IP address         Stratum Poll Reach LastRx Last sample
  ===============================================================================
  ^+ 96.19.94.82                   1  10   377   804  -1190us[-1190us] +/-   34ms
  ^- 172.233.157.223               2  10   377   815  -5912us[-5912us] +/-  118ms
  ^- 66.118.229.14                 2  10   377   445  -2165us[-2165us] +/-  145ms
  ^+ 38.45.64.130                  2  10   377   845   +945us[ +945us] +/-   27ms
  ^* 192.207.55.254                1  10   377   889   +853us[ +911us] +/-   26ms
  ^- 69.10.208.170                 2  10   377   265  +4064us[+4064us] +/-   79ms
```

Teardown (suite) Close Nms Session, Close All Connections

14.03 PASS Every Router Appears As A Client Of The NMS (0.1s, 1 commands)

Seen from the server, which is where a missing client shows up as an absence rather than as a router quietly using something else.

Setup (suite) Open All Routers

Acceptance criteria

- output contains **${${r}_OOB_IP}** (checked 3×)

How it was checked

NMS sudo chronyc clients

```
Hostname                      NTP   Drop Int IntL Last     Cmd   Drop Int  Last
===============================================================================
192.168.99.2                 1203      0   6   -   854       0      0   -     -
192.168.99.3                 1019      0   6   -     0       0      0   -     -
192.168.99.1                  901      0   6   -   467       0      0   -     -
localhost                       0      0   -   -     -      32      0  10     0
```

Teardown (suite) Close Nms Session, Close All Connections

14.04 PASS Every Router Forms Associations With Pool Servers (0.1s, 3 commands)

Setup (suite) Open All Routers

Acceptance criteria

- **${rows}** is not empty (checked 3×)

How it was checked

R1 show ntp associations

```
address         ref clock       st   when   poll reach  delay  offset   disp
*~192.168.99.10   192.207.55.254   2    467   1024   377  0.913 -12.265  1.058
 * sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

R2 show ntp associations

```
address         ref clock       st   when   poll reach  delay  offset   disp
*~192.168.99.10   192.207.55.254   2    854   1024   377  0.899  -7.120  1.063
 * sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

R3 show ntp associations

```
address         ref clock       st   when   poll reach  delay  offset   disp
*~192.168.99.10   192.207.55.254   2      0    256   377  0.909 -10.326  7.131
 * sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

Teardown (suite) Close Nms Session, Close All Connections

14.05 PASS Associations Become Reachable (0.1s, 3 commands)

A non-zero reach register means packets are actually coming back, which distinguishes a working client from one that has merely been configured.

Setup (suite) Open All Routers

Acceptance criteria

- **${up}** is not empty (checked 3×)

How it was checked

R1 show ntp associations

```
address         ref clock       st   when   poll reach  delay  offset   disp
*~192.168.99.10   192.207.55.254   2    467   1024   377  0.913 -12.265  1.058
 * sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

R2 show ntp associations

```
address         ref clock       st   when   poll reach  delay  offset   disp
*~192.168.99.10   192.207.55.254   2    854   1024   377  0.899  -7.120  1.063
 * sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

R3 show ntp associations

```
address         ref clock       st   when   poll reach  delay  offset   disp
*~192.168.99.10   192.207.55.254   2      0    256   377  0.909 -10.326  7.131
 * sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

Teardown (suite) Close Nms Session, Close All Connections

14.06 PASS Every Router Selects A System Peer (0.1s, 3 commands)

Setup (suite) Open All Routers

Acceptance criteria

- **${peer}** differs from **${None}** (checked 3×)

How it was checked

R1 show ntp associations
```
address         ref clock       st   when   poll reach  delay  offset   disp
*~192.168.99.10   192.207.55.254   2    467   1024   377  0.913 -12.265  1.058
 * sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```
R2 show ntp associations
```
address         ref clock       st   when   poll reach  delay  offset   disp
*~192.168.99.10   192.207.55.254   2    854   1024   377  0.899  -7.120  1.063
 * sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```
R3 show ntp associations
```
address         ref clock       st   when   poll reach  delay  offset   disp
*~192.168.99.10   192.207.55.254   2      0    256   377  0.909 -10.326  7.131
 * sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

Teardown (suite) Close Nms Session, Close All Connections

14.07 PASS Every Router Clock Is Synchronised (0.1s, 3 commands)

The end state: IOS declares the clock synchronised. This lags real accuracy by a long way. IOS holds the flag back until its loopfilter leaves the 'FREQ' drift-measurement phase, measured on this platform at about 17 minutes from a cold NTP start -- while the offset is already within milliseconds well before that. The wait below is bounded to cover a freshly provisioned lab; on a lab that has been up it returns on the first poll and costs nothing.

Setup (suite) Open All Routers

Acceptance criteria

- output contains **Clock is synchronized** (checked 3×)

How it was checked

R1 show ntp status
```
Clock is synchronized, stratum 3, reference is 192.168.99.10
nominal freq is 250.0000 Hz, actual freq is 250.0020 Hz, precision is 2**10
ntp uptime is 8489600 (1/100 of seconds), resolution is 4000
reference time is EE4B201F.E45A1F20 (00:00:31.892 UTC Wed Sep 9 2026)
clock offset is -12.2655 msec, root delay is 52.69 msec
root dispersion is 75.60 msec, peer dispersion is 1.05 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is -0.000008126 s/s
system poll interval is 1024, last update was 4145 sec ago.
```
R2 show ntp status
```
Clock is synchronized, stratum 3, reference is 192.168.99.10
nominal freq is 250.0000 Hz, actual freq is 250.0022 Hz, precision is 2**10
ntp uptime is 8489600 (1/100 of seconds), resolution is 4000
reference time is EE4B0A01.03D70A48 (22:26:09.015 UTC Tue Sep 8 2026)
clock offset is -7.1208 msec, root delay is 51.81 msec
root dispersion is 152.29 msec, peer dispersion is 1.06 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is -0.000008817 s/s
system poll interval is 1024, last update was 9807 sec ago.
```
R3 show ntp status
```
Clock is synchronized, stratum 3, reference is 192.168.99.10
nominal freq is 250.0000 Hz, actual freq is 250.0020 Hz, precision is 2**10
ntp uptime is 8489600 (1/100 of seconds), resolution is 4000
reference time is EE4B21E7.4A3D7170 (00:08:07.290 UTC Wed Sep 9 2026)
clock offset is -10.3269 msec, root delay is 52.69 msec
root dispersion is 74.91 msec, peer dispersion is 7.13 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is -0.000008070 s/s
system poll interval is 256, last update was 3689 sec ago.
```

Teardown (suite) Close Nms Session, Close All Connections

14.08 PASS Stratum Is Sane On Every Router (0.1s, 3 commands)

Stratum 16 means unsynchronised. A client of a stratum 1-2 pool server should land well below the limit.

Setup (suite) Open All Routers

Acceptance criteria

- **${m}** is not empty (checked 3×)
- **${m}[0] < ${MAX_STRATUM}** holds (checked 3×)

How it was checked

R1 show ntp status

```
Clock is synchronized, stratum 3, reference is 192.168.99.10
nominal freq is 250.0000 Hz, actual freq is 250.0020 Hz, precision is 2**10
ntp uptime is 8489600 (1/100 of seconds), resolution is 4000
reference time is EE4B201F.E45A1F20 (00:00:31.892 UTC Wed Sep 9 2026)
clock offset is -12.2655 msec, root delay is 52.69 msec
root dispersion is 75.60 msec, peer dispersion is 1.05 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is -0.000008126 s/s
system poll interval is 1024, last update was 4145 sec ago.
```

R2 show ntp status

```
Clock is synchronized, stratum 3, reference is 192.168.99.10
nominal freq is 250.0000 Hz, actual freq is 250.0022 Hz, precision is 2**10
ntp uptime is 8489600 (1/100 of seconds), resolution is 4000
reference time is EE4B0A01.03D70A48 (22:26:09.015 UTC Tue Sep 8 2026)
clock offset is -7.1208 msec, root delay is 51.81 msec
root dispersion is 152.29 msec, peer dispersion is 1.06 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is -0.000008817 s/s
system poll interval is 1024, last update was 9807 sec ago.
```

R3 show ntp status

```
Clock is synchronized, stratum 3, reference is 192.168.99.10
nominal freq is 250.0000 Hz, actual freq is 250.0020 Hz, precision is 2**10
ntp uptime is 8489600 (1/100 of seconds), resolution is 4000
reference time is EE4B21E7.4A3D7170 (00:08:07.290 UTC Wed Sep 9 2026)
clock offset is -10.3269 msec, root delay is 52.69 msec
root dispersion is 74.93 msec, peer dispersion is 7.13 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is -0.000008070 s/s
system poll interval is 256, last update was 3689 sec ago.
```

Teardown (suite) Close Nms Session, Close All Connections

14.09 PASS The Reference Is One Of The Peers The Router Actually Polls (0.1s, 6 commands)

Guards against a router calling itself synchronised to something that is not in its association list.

Setup (suite) Open All Routers

Acceptance criteria

- **${ref}** is not empty (checked 3×)
- output contains **${ref}[0]** (checked 3×)

How it was checked

R1 show ntp status

```
Clock is synchronized, stratum 3, reference is 192.168.99.10
nominal freq is 250.0000 Hz, actual freq is 250.0020 Hz, precision is 2**10
ntp uptime is 8489600 (1/100 of seconds), resolution is 4000
reference time is EE4B201F.E45A1F20 (00:00:31.892 UTC Wed Sep 9 2026)
clock offset is -12.2655 msec, root delay is 52.69 msec
root dispersion is 75.60 msec, peer dispersion is 1.05 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is -0.000008126 s/s
system poll interval is 1024, last update was 4145 sec ago.
```

R1 show ntp associations

```
address         ref clock       st   when   poll reach  delay  offset   disp
*~192.168.99.10   192.207.55.254   2    467   1024   377  0.913 -12.265  1.058
 * sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

R2 show ntp status

```
Clock is synchronized, stratum 3, reference is 192.168.99.10
nominal freq is 250.0000 Hz, actual freq is 250.0022 Hz, precision is 2**10
ntp uptime is 8489600 (1/100 of seconds), resolution is 4000
reference time is EE4B0A01.03D70A48 (22:26:09.015 UTC Tue Sep 8 2026)
clock offset is -7.1208 msec, root delay is 51.81 msec
root dispersion is 152.29 msec, peer dispersion is 1.06 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is -0.000008817 s/s
system poll interval is 1024, last update was 9807 sec ago.
```

R2 show ntp associations

```
address         ref clock       st   when   poll reach  delay  offset   disp
*~192.168.99.10   192.207.55.254   2    854   1024   377  0.899  -7.120  1.063
 * sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

R3 show ntp status

```
Clock is synchronized, stratum 3, reference is 192.168.99.10
nominal freq is 250.0000 Hz, actual freq is 250.0020 Hz, precision is 2**10
ntp uptime is 8489600 (1/100 of seconds), resolution is 4000
reference time is EE4B21E7.4A3D7170 (00:08:07.290 UTC Wed Sep 9 2026)
clock offset is -10.3269 msec, root delay is 52.69 msec
root dispersion is 74.93 msec, peer dispersion is 7.13 msec
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is -0.000008070 s/s
system poll interval is 256, last update was 3689 sec ago.
```

R3 show ntp associations

```
address         ref clock       st   when   poll reach  delay  offset   disp
*~192.168.99.10   192.207.55.254   2      1    256    377  0.909 -10.326  7.131
 * sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

Teardown (suite) Close Nms Session, Close All Connections

14.10 PASS Clock Offset Is Small On Every Router (0.1s, 3 commands)

Setup (suite) Open All Routers

Acceptance criteria

- **${peer}** differs from **${None}** (checked 3×)
- **${offset} < ${MAX_OFFSET_MS}** holds (checked 3×)

How it was checked

R1 show ntp associations

```
address         ref clock       st   when   poll reach  delay  offset   disp
*~192.168.99.10   192.207.55.254   2    467   1024   377  0.913 -12.265  1.058
 * sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

R2 show ntp associations

```
address         ref clock       st   when   poll reach  delay  offset   disp
*~192.168.99.10   192.207.55.254   2    854   1024   377  0.899  -7.120  1.063
 * sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

R3 show ntp associations

```
address         ref clock       st   when   poll reach  delay  offset   disp
*~192.168.99.10   192.207.55.254   2      1    256    377  0.909 -10.326  7.131
 * sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

Teardown (suite) Close Nms Session, Close All Connections

14.11 PASS All Routers Agree On The Time (0.0s, 3 commands)

The practical payoff: three independently synchronised clocks should read the same, which is what makes logs across the fabric correlatable.

Setup (suite) Open All Routers

Acceptance criteria

- **${s12} < ${MAX_SKEW_SECONDS}** holds
- **${s13} < ${MAX_SKEW_SECONDS}** holds

How it was checked

R1 show clock

```
01:09:36.542 UTC Wed Sep 9 2026
```

R2 show clock

```
01:09:36.548 UTC Wed Sep 9 2026
```

R3 show clock

```
01:09:36.562 UTC Wed Sep 9 2026
```

Teardown (suite) Close Nms Session, Close All Connections

14.12 PASS The Clock Is No Longer At The Boot Epoch (0.1s, 6 commands)

An unsynchronised IOS device reports a 1993 or 1900 clock; a synchronised one reports now. This is the crude check that catches a clock that never moved at all.

Setup (suite) Open All Routers

Acceptance criteria

- **${year} >= 2026** holds (checked 3×)
- output does not contain ***** (checked 3×)

How it was checked

R1 **show clock**

```
01:09:36.579 UTC Wed Sep 9 2026
```

R1 **show clock**

```
01:09:36.593 UTC Wed Sep 9 2026
```

R2 **show clock**

```
01:09:36.600 UTC Wed Sep 9 2026
```

R2 **show clock**

```
01:09:36.612 UTC Wed Sep 9 2026
```

R3 **show clock**

```
01:09:36.627 UTC Wed Sep 9 2026
```

R3 **show clock**

```
01:09:36.640 UTC Wed Sep 9 2026
```

Teardown (suite) Close Nms Session, Close All Connections

15 Syslog

15.01 PASS The Collector Is A Service On The NMS Serving Every Router On One Port (0.1s, 2 commands)

What the move bought: one rsyslog listener for all three routers instead of one busybox listener per router, and a system service rather than something a suite starts -- so no run can leave a stale listener swallowing another's messages.

Setup (suite) Start Collector And Routers

Acceptance criteria

- **${state}** equals **active**
- output contains **:${SYSLOG_PORT}**
- output contains **rsyslogd**

How it was checked

```
NMS systemctl is-active rsyslog
  active
NMS sudo ss -lunp | grep ':514 ' || echo NONE
  UNCONN 0      0            0.0.0.0:514       0.0.0.0:*    users:(("rsyslogd",pid=1126,fd=6))
  UNCONN 0      0               [::]:514          [::]:*    users:(("rsyslogd",pid=1126,fd=7))
```

Teardown (suite) Close Collector, Close All Connections

15.02 PASS Every Router Is Configured To Log To The Collector (0.4s, 3 commands)

Setup (suite) Start Collector And Routers

Acceptance criteria

- output contains **logging host ${SYSLOG_HOST}** (checked 3×)
- output contains **logging source-interface ${OOB_INTF}[${r}] vrf ${MGMT_VRF}** (checked 3×)
- output contains **service sequence-numbers** (checked 3×)

How it was checked

```
R1 show running-config | include ^logging|^service (timestamps|sequence)
  service timestamps debug datetime msec
  service timestamps log datetime msec show-timezone
  service sequence-numbers
  logging source-interface GigabitEthernet4 vrf MGMT
  logging host 192.168.99.10 vrf MGMT
R2 show running-config | include ^logging|^service (timestamps|sequence)
  service timestamps debug datetime msec
  service timestamps log datetime msec show-timezone
  service sequence-numbers
  logging source-interface GigabitEthernet4 vrf MGMT
  logging host 192.168.99.10 vrf MGMT
R3 show running-config | include ^logging|^service (timestamps|sequence)
  service timestamps debug datetime msec
  service timestamps log datetime msec show-timezone
  service sequence-numbers
  logging source-interface GigabitEthernet4 vrf MGMT
  logging host 192.168.99.10 vrf MGMT
```

Teardown (suite) Close Collector, Close All Connections

15.03 PASS Every Router Reports The Collector As An Active Logging Host (0.2s, 3 commands)

The router's own view: it should name the collector and show a non-zero count of messages sent to it.

Setup (suite) Start Collector And Routers

Acceptance criteria

- output contains **${SYSLOG_HOST}** (checked 3×)
- output contains **level ${TRAP_LEVEL}** (checked 3×)

How it was checked

```
R1 show logging | include Trap logging|Logging to 192.168.99.10
  Trap logging: level informational, 2726 message lines logged
          Logging to 192.168.99.10  (MGMT) (udp port 514, audit disabled,
```

```
R2 show logging | include Trap logging|Logging to 192.168.99.10
  Trap logging: level informational, 3397 message lines logged
          Logging to 192.168.99.10  (MGMT) (udp port 514, audit disabled,
```

```
R3 show logging | include Trap logging|Logging to 192.168.99.10
  Trap logging: level informational, 2372 message lines logged
          Logging to 192.168.99.10  (MGMT) (udp port 514, audit disabled,
```

Teardown (suite) Close Collector, Close All Connections

15.04 PASS The Hub Message Reaches The Collector (4.0s, 1 commands)

End to end over the LAN: emit a uniquely marked message and find it at the collector.

Setup (suite) Start Collector And Routers

Acceptance criteria

- **${n} > 0** holds

How it was checked

```
NMS sudo grep -c -- 'LABSYSLOG-HUB-210937' /var/log/lab/192.168.99.1.log 2>/dev/null || echo 0
  1
```

Teardown (suite) Close Collector, Close All Connections

15.05 PASS Each Spoke Message Reaches The Collector Across The Tunnel (8.1s, 2 commands)

The spokes have no direct path to h1's LAN; their messages are carried by the tunnels and the hub's routing.

Setup (suite) Start Collector And Routers

Acceptance criteria

- **${n} > 0** holds (checked 2×)

How it was checked

```
NMS sudo grep -c -- 'LABSYSLOG-R2-210937' /var/log/lab/192.168.99.2.log 2>/dev/null || echo 0
  1
```

```
NMS sudo grep -c -- 'LABSYSLOG-R3-210937' /var/log/lab/192.168.99.3.log 2>/dev/null || echo 0
  1
```

Teardown (suite) Close Collector, Close All Connections

15.06 PASS Received Messages Carry Facility Severity And Mnemonic (4.1s, 2 commands)

Setup (suite) Start Collector And Routers

Acceptance criteria

- **${n} > 0** holds
- **${m}[facility]** equals **FACCHECK**
- **${m}[mnemonic]** equals **MNEMCHECK**
- **${m}[sev]** equals **5**

How it was checked

```
NMS sudo grep -c -- 'LABSYSLOG-FIELDS-210937' /var/log/lab/192.168.99.1.log 2>/dev/null || echo 0
  1
```

```
NMS sudo cat /var/log/lab/192.168.99.1.log 2>/dev/null || true
  <189>207: 000203: *Sep  7 22:17:21.168 UTC: %SSH-5-SSH2_SESSION: SSH2 Session request from 10.0.2.2 (tty =
  0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
  <189>208: 000204: *Sep  7 22:17:21.214 UTC: %SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: lab] [Source:
  10.0.2.2] [localport: 22] at 22:17:21 UTC Mon Sep 7 2026
  <189>209: 000205: *Sep  7 22:17:21.214 UTC: %SSH-5-SSH2_USERAUTH: User 'lab' authentication for SSH2 Session
  from 10.0.2.2 (tty = 0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
  <190>210: 000206: *Sep  7 22:17:56.590 UTC: %SYS-6-LOGOUT: User lab has exited tty session 434(10.0.2.2)
  <189>211: 000207: *Sep  7 22:17:56.591 UTC: %SSH-5-SSH2_CLOSE: SSH2 Session from 10.0.2.2 (tty = 0) for user
  '' using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' closed
  <189>212: 000208: *Sep  7 22:18:27.361 UTC: %SSH-5-SSH2_SESSION: SSH2 Session request from 10.0.2.2 (tty =
```

```
0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<189>213: 000209: *Sep  7 22:18:27.410 UTC: %SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: lab] [Source:
10.0.2.2] [localport: 22] at 22:18:27 UTC Mon Sep 7 2026
<189>214: 000210: *Sep  7 22:18:27.410 UTC: %SSH-5-SSH2_USERAUTH: User 'lab' authentication for SSH2 Session
from 10.0.2.2 (tty = 0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<189>215: 000211: *Sep  7 22:18:27.605 UTC: %SYS-5-CONFIG_I: Configured from console by lab on vty0
(10.0.2.2)
<190>216: 000212: *Sep  7 22:18:27.747 UTC: %SYS-6-LOGOUT: User lab has exited tty session 434(10.0.2.2)
<189>217: 000213: *Sep  7 22:18:27.747 UTC: %SSH-5-SSH2_CLOSE: SSH2 Session from 10.0.2.2 (tty = 0) for user
'' using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' closed
<187>218: 000214: *Sep  7 22:19:26.134 UTC: %DMVPN-3-DMVPN_NHRP_ERROR:  Tunnel0: Recieved wrong
authentication string for  Registration Request , Reason:  authentication failure (11) on (Tunnel:
172.20.0.3 NBMA: 100.64.0.1)
<187>219: 000215: *Sep  7 22:19:26.227 UTC: %DMVPN-3-DMVPN_NHRP_ERROR:  Tunnel0: Recieved wrong
authentication string for  Registration Request , Reason:  authentication failure (11) on (Tunnel:
172.20.0.2 NBMA: 100.64.0.1)
<187>220: 000216: *Sep  7 22:19:26.906 UTC: %DMVPN-3-DMVPN_NHRP_ERROR:  Tunnel0: Recieved wrong
authentication string for  Registration Request , Reason:  authentication failure (11) on (Tunnel:
172.20.0.3 NBMA: 100.64.0.1)
... 3631 more lines
```

Teardown (suite) Close Collector, Close All Connections

15.07 PASS Received Messages Carry Sequence Numbers (0.1s, 1 commands)

'service sequence-numbers' is configured, so every message should be numbered -- which is how a collector detects loss.

Setup (suite) Start Collector And Routers

Acceptance criteria

- **${msgs}** is not empty
- **${uniq} > 1** holds
- **min($seqs) >= 0** holds

How it was checked

NMS **sudo cat /var/log/lab/192.168.99.1.log 2>/dev/null || true**

```
<189>207: 000203: *Sep  7 22:17:21.168 UTC: %SSH-5-SSH2_SESSION: SSH2 Session request from 10.0.2.2 (tty =
0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<189>208: 000204: *Sep  7 22:17:21.214 UTC: %SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: lab] [Source:
10.0.2.2] [localport: 22] at 22:17:21 UTC Mon Sep 7 2026
<189>209: 000205: *Sep  7 22:17:21.214 UTC: %SSH-5-SSH2_USERAUTH: User 'lab' authentication for SSH2 Session
from 10.0.2.2 (tty = 0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<190>210: 000206: *Sep  7 22:17:56.590 UTC: %SYS-6-LOGOUT: User lab has exited tty session 434(10.0.2.2)
<189>211: 000207: *Sep  7 22:17:56.591 UTC: %SSH-5-SSH2_CLOSE: SSH2 Session from 10.0.2.2 (tty = 0) for user
'' using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' closed
<189>212: 000208: *Sep  7 22:18:27.361 UTC: %SSH-5-SSH2_SESSION: SSH2 Session request from 10.0.2.2 (tty =
0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<189>213: 000209: *Sep  7 22:18:27.410 UTC: %SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: lab] [Source:
10.0.2.2] [localport: 22] at 22:18:27 UTC Mon Sep 7 2026
<189>214: 000210: *Sep  7 22:18:27.410 UTC: %SSH-5-SSH2_USERAUTH: User 'lab' authentication for SSH2 Session
from 10.0.2.2 (tty = 0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<189>215: 000211: *Sep  7 22:18:27.605 UTC: %SYS-5-CONFIG_I: Configured from console by lab on vty0
(10.0.2.2)
<190>216: 000212: *Sep  7 22:18:27.747 UTC: %SYS-6-LOGOUT: User lab has exited tty session 434(10.0.2.2)
<189>217: 000213: *Sep  7 22:18:27.747 UTC: %SSH-5-SSH2_CLOSE: SSH2 Session from 10.0.2.2 (tty = 0) for user
'' using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' closed
<187>218: 000214: *Sep  7 22:19:26.134 UTC: %DMVPN-3-DMVPN_NHRP_ERROR:  Tunnel0: Recieved wrong
authentication string for  Registration Request , Reason:  authentication failure (11) on (Tunnel:
172.20.0.3 NBMA: 100.64.0.1)
<187>219: 000215: *Sep  7 22:19:26.227 UTC: %DMVPN-3-DMVPN_NHRP_ERROR:  Tunnel0: Recieved wrong
authentication string for  Registration Request , Reason:  authentication failure (11) on (Tunnel:
172.20.0.2 NBMA: 100.64.0.1)
<187>220: 000216: *Sep  7 22:19:26.906 UTC: %DMVPN-3-DMVPN_NHRP_ERROR:  Tunnel0: Recieved wrong
authentication string for  Registration Request , Reason:  authentication failure (11) on (Tunnel:
172.20.0.3 NBMA: 100.64.0.1)
... 3631 more lines
```

Teardown (suite) Close Collector, Close All Connections

15.08 PASS Message Timestamps Agree With The Router Clock (4.1s, 3 commands)

Ties syslog to NTP: a message's timestamp should match the sending router's clock. Skewed timestamps make a collector's correlation across devices worthless.

Setup (suite) Start Collector And Routers

Acceptance criteria

- **${n} > 0** holds
- **${skew} < ${MAX_TS_SKEW}** holds

How it was checked

NMS `sudo grep -c -- 'LABSYSLOG-TIME-210937' /var/log/lab/192.168.99.1.log 2>/dev/null || echo 0`

```
1
```

R1 `show clock`

```
01:10:00.094 UTC Wed Sep 9 2026
```

NMS `sudo cat /var/log/lab/192.168.99.1.log 2>/dev/null || true`

```
<189>207: 000203: *Sep  7 22:17:21.168 UTC: %SSH-5-SSH2_SESSION: SSH2 Session request from 10.0.2.2 (tty =
0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<189>208: 000204: *Sep  7 22:17:21.214 UTC: %SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: lab] [Source:
10.0.2.2] [localport: 22] at 22:17:21 UTC Mon Sep 7 2026
<189>209: 000205: *Sep  7 22:17:21.214 UTC: %SSH-5-SSH2_USERAUTH: User 'lab' authentication for SSH2 Session
from 10.0.2.2 (tty = 0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<190>210: 000206: *Sep  7 22:17:56.590 UTC: %SYS-6-LOGOUT: User lab has exited tty session 434(10.0.2.2)
<189>211: 000207: *Sep  7 22:17:56.591 UTC: %SSH-5-SSH2_CLOSE: SSH2 Session from 10.0.2.2 (tty = 0) for user
'' using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' closed
<189>212: 000208: *Sep  7 22:18:27.361 UTC: %SSH-5-SSH2_SESSION: SSH2 Session request from 10.0.2.2 (tty =
0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<189>213: 000209: *Sep  7 22:18:27.410 UTC: %SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: lab] [Source:
10.0.2.2] [localport: 22] at 22:18:27 UTC Mon Sep 7 2026
<189>214: 000210: *Sep  7 22:18:27.410 UTC: %SSH-5-SSH2_USERAUTH: User 'lab' authentication for SSH2 Session
from 10.0.2.2 (tty = 0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<189>215: 000211: *Sep  7 22:18:27.605 UTC: %SYS-5-CONFIG_I: Configured from console by lab on vty0
(10.0.2.2)
<190>216: 000212: *Sep  7 22:18:27.747 UTC: %SYS-6-LOGOUT: User lab has exited tty session 434(10.0.2.2)
<189>217: 000213: *Sep  7 22:18:27.747 UTC: %SSH-5-SSH2_CLOSE: SSH2 Session from 10.0.2.2 (tty = 0) for user
'' using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' closed
<187>218: 000214: *Sep  7 22:19:26.134 UTC: %DMVPN-3-DMVPN_NHRP_ERROR:  Tunnel0: Recieved wrong
authentication string for  Registration Request , Reason:  authentication failure (11) on (Tunnel:
172.20.0.3 NBMA: 100.64.0.1)
<187>219: 000215: *Sep  7 22:19:26.227 UTC: %DMVPN-3-DMVPN_NHRP_ERROR:  Tunnel0: Recieved wrong
authentication string for  Registration Request , Reason:  authentication failure (11) on (Tunnel:
172.20.0.2 NBMA: 100.64.0.1)
<187>220: 000216: *Sep  7 22:19:26.906 UTC: %DMVPN-3-DMVPN_NHRP_ERROR:  Tunnel0: Recieved wrong
authentication string for  Registration Request , Reason:  authentication failure (11) on (Tunnel:
172.20.0.3 NBMA: 100.64.0.1)
... 3632 more lines
```

Teardown (suite) Close Collector, Close All Connections

15.09 PASS Messages Below The Trap Level Are Not Forwarded (14.0s, 1 commands)

Negative control. Trap level is informational (6), so a debug-severity (7) message must not reach the collector. If this ever passes traffic, the level filter is not working and the positive tests above prove much less than they appear to.

Setup (suite) Start Collector And Routers

Acceptance criteria

- **${n}** equals **0**

  a debug-severity message reached the collector despite trap level ${TRAP_LEVEL}

How it was checked

NMS `sudo grep -c -- 'LABSYSLOG-DEBUG-210937' /var/log/lab/192.168.99.1.log 2>/dev/null || echo 0`

```
0
0
```

Teardown (suite) Close Collector, Close All Connections

15.10 PASS All Three Routers Are Represented At The Collector (12.1s, 3 commands)

Messages from every router, distinguishable at the collector, which is the point of a central log host.

Setup (suite) Start Collector And Routers

Acceptance criteria

- **${n} > 0** holds (checked 3×)

How it was checked

```
NMS sudo grep -c -- 'LABSYSLOG-ALL-R1-210937' /var/log/lab/192.168.99.1.log 2>/dev/null || echo 0
  1
NMS sudo grep -c -- 'LABSYSLOG-ALL-R2-210937' /var/log/lab/192.168.99.2.log 2>/dev/null || echo 0
  1
NMS sudo grep -c -- 'LABSYSLOG-ALL-R3-210937' /var/log/lab/192.168.99.3.log 2>/dev/null || echo 0
  1
```

Teardown (suite) Close Collector, Close All Connections

16 Snmpv3

16.01 PASS Both Trap Receivers Are Services On The NMS (0.1s, 2 commands)

snmptrapd on the standard port and the raw capture on a second one, both system services rather than listeners a suite starts -- so no run can leave one behind, which is how a negative test once passed while nothing was arriving.

Setup (suite) Start Snmp Collectors

Acceptance criteria

- **${state}** equals **active active**
- output contains **:${TRAP_PORT}**
- output contains **:${TRAP_RAW_PORT}**

How it was checked

```
NMS systemctl is-active snmptrapd lab-trapcap | tr '\n' ' '
  active active
```

```
NMS sudo ss -lunp | grep -E ':(162|1162) ' || echo NONE
  UNCONN 0      0             0.0.0.0:1162      0.0.0.0:*    users:(("python3",pid=1288,fd=3))
  UNCONN 0      0             0.0.0.0:162       0.0.0.0:*
  users:(("snmptrapd",pid=1247,fd=3),("systemd",pid=1,fd=101))
  UNCONN 0      0                [::]:162          [::]:*
  users:(("snmptrapd",pid=1247,fd=4),("systemd",pid=1,fd=102))
```

Teardown (suite) Close Trap Collector, Close All Connections

16.02 PASS Every Router Has An SNMPv3 User With Authentication And Privacy (0.1s, 3 commands)

v3 without auth and priv is barely better than v2c, so the protocols in use are asserted, not just the user's existence.

Setup (suite) Start Snmp Collectors

Acceptance criteria

- output contains **User name: ${SNMP_USER}** (checked 3×)
- output contains **Authentication Protocol: SHA** (checked 3×)
- output contains **Privacy Protocol: AES128** (checked 3×)
- output contains **Group-name: ${SNMP_GROUP}** (checked 3×)

How it was checked

```
R1 show snmp user
  User name: labmon
  Engine ID: 800000090300525400C80101
  storage-type: nonvolatile        active
  Authentication Protocol: SHA
  Privacy Protocol: AES128
  Group-name: LABGRP
```

```
R2 show snmp user
  User name: labmon
  Engine ID: 800000090300525400C80201
  storage-type: nonvolatile        active
  Authentication Protocol: SHA
  Privacy Protocol: AES128
  Group-name: LABGRP
```

```
R3 show snmp user
  User name: labmon
  Engine ID: 800000090300525400C80301
  storage-type: nonvolatile        active
  Authentication Protocol: SHA
  Privacy Protocol: AES128
  Group-name: LABGRP
```

Teardown (suite) Close Trap Collector, Close All Connections

16.03 PASS Every Router Has A v3 Group Requiring Privacy (0.1s, 3 commands)

Setup (suite) Start Snmp Collectors

Acceptance criteria

- output contains **groupname: ${SNMP_GROUP}** (checked 3×)
- output contains **security model:v3 priv** (checked 3×)

How it was checked

R1 **show snmp group**

```
groupname: ILMI                            security model:v1
contextname: <no context specified>        storage-type: permanent
readview : *ilmi                           writeview: *ilmi
notifyview: <no notifyview specified>
row status: active

groupname: ILMI                            security model:v2c
contextname: <no context specified>        storage-type: permanent
readview : *ilmi                           writeview: *ilmi
notifyview: <no notifyview specified>
row status: active

groupname: LABGRP                          security model:v3 priv
contextname: <no context specified>        storage-type: nonvolatile
... 3 more lines
```

R2 **show snmp group**

```
groupname: ILMI                            security model:v1
contextname: <no context specified>        storage-type: permanent
readview : *ilmi                           writeview: *ilmi
notifyview: <no notifyview specified>
row status: active

groupname: ILMI                            security model:v2c
contextname: <no context specified>        storage-type: permanent
readview : *ilmi                           writeview: *ilmi
notifyview: <no notifyview specified>
row status: active

groupname: LABGRP                          security model:v3 priv
contextname: <no context specified>        storage-type: nonvolatile
... 3 more lines
```

R3 **show snmp group**

```
groupname: ILMI                            security model:v1
contextname: <no context specified>        storage-type: permanent
readview : *ilmi                           writeview: *ilmi
notifyview: <no notifyview specified>
row status: active

groupname: ILMI                            security model:v2c
contextname: <no context specified>        storage-type: permanent
readview : *ilmi                           writeview: *ilmi
notifyview: <no notifyview specified>
row status: active

groupname: LABGRP                          security model:v3 priv
contextname: <no context specified>        storage-type: nonvolatile
... 3 more lines
```

Teardown (suite) Close Trap Collector, Close All Connections

16.04 PASS No Router Accepts SNMPv1 Or v2c Communities (0.4s, 3 commands)

Security posture: configuring v3 is pointless if a community string still gets you in.

Setup (suite) Start Snmp Collectors

Acceptance criteria

- **${out}** is empty (checked 3×)

How it was checked

R1 **show running-config | include ^snmp-server community**

```
${output} =
```

R2 **show running-config | include ^snmp-server community**

```
${output} =
```

R3 **show running-config | include ^snmp-server community**

```
${output} =
```

Teardown (suite) Close Trap Collector, Close All Connections

16.05 PASS Every Router Targets The Collector With v3 Privacy (0.1s, 3 commands)

Setup (suite) Start Snmp Collectors

Acceptance criteria

- output contains **${COLLECTOR}** (checked 3×)
- output contains **security model: v3 priv** (checked 3×)
- output contains **user: ${SNMP_USER}** (checked 3×)

How it was checked

R1 show snmp host
```
Notification host: 192.168.99.10        udp-port: 1162  VRFName: MGMT   type: trap
user: labmon    security model: v3 priv

Notification host: 192.168.99.10        udp-port: 162   VRFName: MGMT   type: trap
user: labmon    security model: v3 priv
```
R2 show snmp host
```
Notification host: 192.168.99.10        udp-port: 1162  VRFName: MGMT   type: trap
user: labmon    security model: v3 priv

Notification host: 192.168.99.10        udp-port: 162   VRFName: MGMT   type: trap
user: labmon    security model: v3 priv
```
R3 show snmp host
```
Notification host: 192.168.99.10        udp-port: 1162  VRFName: MGMT   type: trap
user: labmon    security model: v3 priv

Notification host: 192.168.99.10        udp-port: 162   VRFName: MGMT   type: trap
user: labmon    security model: v3 priv
```
Teardown (suite) Close Trap Collector, Close All Connections

16.06 PASS Global Trap Sending Is Enabled (0.1s, 3 commands)

Enabling individual trap types leaves the global switch off: the router builds trap PDUs and never sends them, which looks exactly like a broken collector.

Setup (suite) Start Snmp Collectors

Acceptance criteria

- output contains **SNMP global trap: enabled** (checked 3×)

How it was checked

R1 show snmp | include global trap
```
SNMP global trap: enabled
```
R2 show snmp | include global trap
```
SNMP global trap: enabled
```
R3 show snmp | include global trap
```
SNMP global trap: enabled
```
Teardown (suite) Close Trap Collector, Close All Connections

16.07 PASS Traps From Every Router Reach The Collector (13.4s, 51 commands)

Setup (suite) Start Snmp Collectors

Acceptance criteria

- **${traps}** is not empty (checked 3×)

How it was checked

R1 configure terminal
```
Enter configuration commands, one per line.  End with CNTL/Z.
```
R1 interface Loopback270
```
${output} =
```
R1 ip address 10.99.97.1 255.255.255.255
```
${output} =
```
R1 snmp trap link-status

${output} =

R1 end
${output} =

R1 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R1 interface Loopback270
${output} =

R1 shutdown
${output} =

R1 end
${output} =

R1 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R1 interface Loopback270
${output} =

R1 no shutdown
${output} =

R1 end
${output} =

R1 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R1 no interface Loopback270
${output} =

R1 end
${output} =

R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R2 interface Loopback270
${output} =

R2 ip address 10.99.97.1 255.255.255.255
${output} =

R2 snmp trap link-status
${output} =

R2 end
${output} =

R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R2 interface Loopback270
${output} =

R2 shutdown
${output} =

R2 end
${output} =

R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R2 interface Loopback270
${output} =

R2 no shutdown
${output} =

R2 end
${output} =

R2 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R2 no interface Loopback270
${output} =

R2 end
${output} =

R3 configure terminal
Enter configuration commands, one per line. End with CNTL/Z.

R3 interface Loopback270
${output} =

R3 ip address 10.99.97.1 255.255.255.255
${output} =

R3 snmp trap link-status
${output} =

R3 **end**

```
${output} =
```

R3 **configure terminal**

```
Enter configuration commands, one per line.  End with CNTL/Z.
```

R3 **interface Loopback270**

```
${output} =
```

R3 **shutdown**

```
${output} =
```

R3 **end**

```
${output} =
```

R3 **configure terminal**

```
Enter configuration commands, one per line.  End with CNTL/Z.
```

R3 **interface Loopback270**

```
${output} =
```

R3 **no shutdown**

```
${output} =
```

R3 **end**

```
${output} =
```

R3 **configure terminal**

```
Enter configuration commands, one per line.  End with CNTL/Z.
```

R3 **no interface Loopback270**

```
${output} =
```

R3 **end**

```
${output} =
```

NMS **sudo cat /var/log/lab/traps-raw.log 2>/dev/null || true**


```
1788819441.732 192.168.99.1 3081f1020103300d020136020205dc04010302010304373035040c800000090300525400c8010102
0102020200a204066c61626d6f6e040c0330a96d73b02577bd14002704089b184efb1b39c2180481a31e0cb257972ac1650a9c016737
d7c3450c8b69b1810963c1c351af94e662bb62948440836307965026c2d246d66d8c3bf10781cd8fcab6521f98448783a6d9bf665ade
b66e773f6cb948e6c84fbc1c22baffabc9a7c820adbbccc0607f9fe235937749231d4e94bb8fdcc059dcffbba830d801e015c85863d7
158e72f510620b1c9e484b7417b4d3a185d046ea549f3642f878698207c3f818a4b10c5927a8e880fcba
1788819441.768 192.168.99.2 3081f1020103300d02013e020205dc04010302010304373035040c800000090300525400c8020102
0102020200a104066c61626d6f6e040c280bc8173afb140f39665d400408bdf4dc46c97074930481a3485df5849e5eeeb771b44c093a
ef6ddaf4b7ba8b009f4cb725b19f5e751850a70e9cd62e9471b527848f3350df4234902e59c6ba9cd696e552e207ef62af971e8908ef
ae317dcaed8d6702a20fa8a0f5d3535538a2851e16b3bc0d2979168bf60354d2878e5cd7ac03c1daa98fd24eb8a451e70e624bdffc96
8ee7b7c9e197db97da0102a62640bc8d2abf526e9a3bd5db09b0713ae471d14774a711e1c49770a339c6
1788819441.806 192.168.99.3 3081f1020103300d02013a020205dc04010302010304373035040c800000090300525400c8030102
0102020200a104066c61626d6f6e040c1a06b8620cf2bdfae082242d0408a7c7be9c6efe40570481a3f13563f7a9d2b23f9d6aedd6db
b3ee8cb9bfc314d080afa0b0fb48addca5f604a787508df6374629f81bf9c5e8ed322c4fbc9fd5daf2e665dcbc312cc01027c101b15d
95d74b6c51ef9faf06c0670481620af82d566719918aab054f908d5415c32688a26757db8041390a1fd440c938835bf8a171ee3321ef
59f0d1a550a2e55083266bba89add3ddb3176843bbfc92d3de40fb901ba83f7d8c7124f27c7fe6db1dce
1788819441.982 192.168.99.1 3081f1020103300d020138020205dc04010302010304373035040c800000090300525400c8010102
0102020200a204066c61626d6f6e040c54e0b089dfb6d3286639954d04089b184efb1b39c21a0481a38399f66019ba392443b719d26a
fd64cb17b4f7e47e6efc18540d7a161bb4df34a5d637d8baf88d7959b126d7f786e1521239899f10d3d27949f04880309d9a8e456a78
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2b44a96e3e05c10c5df1deb3e7c8f3b1022054494304e4456fad68c0f554b6016aa90535c790e85d921b
1788819442.019 192.168.99.2 3081f1020103300d020140020205dc04010302010304373035040c800000090300525400c8020102
0102020200a204066c61626d6f6e040cc5e940312135bbb74315f07f0408bdf4dc46c97074950481a3d122cd69208ea63a33c15acfa8
7a45102a3d13ff4011a6fe0077d1190d97e78309f972e8bd391feaf8f439f142fa831bc51566d9eb8017d4cb97e7436f0ff3c8ef79ef
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dcbea7abdb77818b788f02536a47b9893019f437a968d65cf16414aaec7588c7acf31617d6f2d26c240f
1788819442.056 192.168.99.3 3081f1020103300d02013c020205dc04010302010304373035040c800000090300525400c8030102
0102020200a204066c61626d6f6e040c97f59d812586c8cbc82aeec80408a7c7be9c6efe40590481a3944f05c35a695e8ab698a77fc0
679d4817a25d2867b284b3553e5b9560d1abeb0194ef14399db882e9fa2c2b7a2e5a3a8776478141012d370690d28f11c26bfc8aa1fd
53215e87e900d9dda1de7d41080b83bf95c9c0743949bd5c4af9ea11abc6a118939fad85eb0735991d6b9d8e891ccf0bd9cb99abf172
805a8c9b5e5d778a279e1a0384713bb102e8ba0b0be1d56a9527722ef00de7e6baabfe3aae5f497a44f4
1788819442.233 192.168.99.1 3081f1020103300d02013a020205dc04010302010304373035040c800000090300525400c8010102
0102020200a204066c61626d6f6e040ce97d7d93c11dac8ea0caaafe04089b184efb1b39c21c0481a3ee52a2e7caf58003f3aeaef04a
dcaabcf43bd1c13a51d4087e88c6a9a9cfb270aec98cf2cbdfc6fd803608b8bc6591cde59384e91da399d6f1080102f98fe25152158b
139ebf2f59507f8016a328a4c7a9d1b80a4219982fe2f74978cda2b2ef1bf7d7faf9d835a5cf3f9042d5c74a4e898d39833cc1dbe052
02b505f5d02dbbe48944635bbb890fd08e79baaef5523fd4bd8e7aa0f86d71427b43736a5d9d7bef4403
1788819442.267 192.168.99.2 3081f1020103300d020142020205dc04010302010304373035040c800000090300525400c8020102
0102020200a204066c61626d6f6e040cd6f8a0f7f93b069d069759170408bdf4dc46c97074970481a33116ac4348d5c39ada28f96d78
e74d155c5cc9bffa42fe9fdadacbc9f298c97a79d1335ef72ad9631dc12834db86976c08bc0949678ce7926afec969d51ff59beb6fd1
7b912bab7415c3068dc730665587dd829e838b378fbcd81aeaa7be721b39cc8ec406c7f91f5bf3e02ef0f84d884b62698b64590b35f1
8921e3a8316c7d66a27f4300e3b40e92dcf8449782b483b7ebb1c0a3da45fbe4c24622a7d600f36d7023
1788819442.306 192.168.99.3 3081f1020103300d02013e020205dc04010302010304373035040c800000090300525400c8030102
0102020200a204066c61626d6f6e040c3bb1f3582527a19034ec5d610408a7c7be9c6efe405b0481a3fa3381e2249898d509d85e1d9b
3389b3128c7d34aaa0145cd77854357e6d69cd388db0d86c165a4fdf8a3d87f9bb961854ba187a1984c45dbbf58adca5541a5485c344
583a562913d9b44704b325f28a9b0c3c74d18b4c3bd08fc9130c877165dc4c6d22a6e4226cd8546bac4ed78bd313ac2d76bfad8165bf
d95c8ff1fe7981815276c5c81bab3f027acbecb251449958991c430e2e4f57b98530410546f515e98124
1788819442.482 192.168.99.1 3081f1020103300d02013c020205dc04010302010304373035040c800000090300525400c8010102
0102020200a204066c61626d6f6e040c73a76bc49e552aab0d08f64d04089b184efb1b39c21e0481a3b1a6978d7875aefa7484d41426
fcac7bd01a45e729fb041e4e0053bcf6e1e7b4453849a0950cb51754d5aa8bb1e683b91e18dedd70c8eeb400fd5392669c3139475a2e
f9cea2e5f556ab286affc24b299f7257a7f9aaa3362d185225a4037aefe861fb964fcd2e3f759405bf4911bb9c8e340b9c3c32a39f9b
caf86df01c6728b43aa549379361193e7dccae3f3f0abd1c6f25b53985eb169cb05adfaa725d29e505e5
```

```
1788819442.518 192.168.99.2 3081f1020103300d020144020205dc04010302010304373035040c800000090300525400c8020102
0102020200a204066c61626d6f6e040ce22c3e1cccc3a0816b2de3090408bdf4dc46c97074990481a362ccc6cfc3fcb1f5dfe6a36ba9
061eb6d5f8e4cea2b22f8283e04ff27fe21962da8a88d91b1c4fbdc6a5d339e9dd070feed2193e628e3bdc3aeaea01f49bc99930a11e
69a68b57bc34e6500c2a923e4766fa1263512636869017b82cec5982625feded62ba54a93598fd3604d189683a16346138220d45f3a5
2b61fdeb71554fcc5d3fa447dc8681234e12ab5d5157c513e3981ae8abd5e8e564fe10aecfc7bd4560ef
1788819442.556 192.168.99.3 308201db020103300d020140020205dc04010302010304373035040c800000090300525400c80301
020102020200a204066c61626d6f6e040cd0e347d7a518a7d9e2f37c1b0408a7c7be9c6efe405d0482018cd8efa54000ad401bbc326d
2bbcde825ab5152edf654545b7ab79bde2ffadf60703003bf05fd372b55c6598a7edececfd536ad432095fb9260748bd46f1cc973fce
7d0c5d461a74567d653f8915b5eb8f3625a9acd114f2efc9c842fde25afe53f39a555ecb3dcf7d27a08432dc9340b766d913d297b4bb
07970a2510babd3db4f4ce0ebb07f01f9ffd45e6fb29ef8feac58621e676b92935f9ce739373e633a53e74e506575e6f46825bb060ff
e5b0c8526d79831062a4e95bdec7dd7039dd603bf20ecded96990e1d2ab506b0cb8e915f9e5b37d553f38e99197198e6c070324af943
30b1625bd961d67078f27e4290be8c1b30f9fdd4f9da38629eaf2dd334b0e0a26532c0549a3fc8782afe71d694abf3ac5084fb3ae4b9
34af1b612e329beb9c7e1163aa36dc290f9784b5cc2f25a844f00d69d85b8798ee860408950a0e0a5c9d4a79f2784d07c6aca87f8a39
ea45e822985914ea2cb93b21d583ae4f38bbbb056238ff969149d9ebbe2171e6f569e3e2f1e75f5a94e133e9fca802228704d2f319ba
aef92a4fd087aa
1788819442.733 192.168.99.1 3081f1020103300d02013e020205dc04010302010304373035040c800000090300525400c8010102
0102020200a204066c61626d6f6e040cf1375d27c9b13e3bc630dda704089b184efb1b39c2200481a30198f4b559036be6e04256f622
03359df9f4b249e253bfc47820d97178378d92b51f24de8d8455b5342f2d76797f484d9471253faea500058a89ac76482836c4d83314
b637ad7fc13ded93b704de011795c6d883a9036f15a8f96224a3162a8d83a10fe88c1e9da55743953b410a155c88ebec8dc5ed1f85e8
d6719b4b8a6db27cd5500b799d76f31678bbff135288e19a3de00a02a2b05ec9f77a51e2e82d417f4fd9
1788819442.769 192.168.99.2 3081bb020103300d020146020205dc04010302010304373035040c800000090300525400c8020102
0102020200a204066c61626d6f6e040c3df77a9a6185cfb1cfecf88a0408bdf4dc46c970749b046ef296014ded7c0ce032945da3386e
1ebd3a6f6a183b3c2e4ebf6a7517d3b4fbe0d2965d0cedde29f5e6919fa5f14ad44ce1b2ffc828265b33e24d5f8c4f415511af6b9294
f9d44b5f12d56f7e84a1eea2197ad6b4e73721c87bfd23d6a00a98a878bc9f8f206e82d8a3606be17431
... 11907 more lines
```

NMS sudo cat /var/log/lab/traps-raw.log 2>/dev/null || true

```
1788819441.732 192.168.99.1 3081f1020103300d020136020205dc04010302010304373035040c800000090300525400c8010102
0102020200a204066c61626d6f6e040c0330a96d73b02577bd14002704089b184efb1b39c2180481a31e0cb257972ac1650a9c016737
d7c3450c8b69b1810963c1c351af94e662bb629484408363079650 26c2d246d66d8c3bf10781cd8fcab6521f98448783a6d9bf665ade
b66e773f6cb948e6c84fbc1c22baffabc9a7c820adbbccc0607f9fe235937749231d4e94bb8fdcc059dcffbba830d801e015c85863d7
158e72f510620b1c9e484b7417b4d3a185d046ea549f3642f878698207c3f818a4b10c5927a8e880fcba
1788819441.768 192.168.99.2 3081f1020103300d02013e020205dc04010302010304373035040c800000090300525400c8020102
0102020200a104066c61626d6f6e040c280bc8173afb140f39665d400408bdf4dc46c97074930481a3485df5849e5eeeb771b44c093a
ef6ddaf4b7ba8b009f4cb725b19f5e751850a70e9cd62e9471b527848f3350df4234902e59c6ba9cd696e552e207ef62af971e8908ef
ae317dcaed8d6702a20fa8a0f5d3535538a2851e16b3bc0d2979168bf60354d2878e5cd7ac03c1daa98fd24eb8a451e70e624bdffc96
8ee7b7c9e197db97da0102a62640bc8d2abf526e9a3bd5db09b0713ae471d14774a711e1c49770a339c6
1788819441.806 192.168.99.3 3081f1020103300d02013a020205dc04010302010304373035040c800000090300525400c8030102
0102020200a104066c61626d6f6e040c1a06b8620cf2bdfae082242d0408a7c7be9c6efe40570481a3f13563f7a9d2b23f9d6aedd6db
b3ee8cb9bfc314d080afa0b0fb48addca5f604a787508df6374629f81bf9c5e8ed322c4fbc9fd5daf2e665dcbc312cc01027c101b15d
95d74b6c51ef9faf06c0670481620af82d566719918aab054f908d5415c32688a26757db8041390a1fd440c938835bf8a171ee3321ef
59f0d1a550a2e55083266bba89add3ddb3176843bbfc92d3de40fb901ba83f7d8c7124f27c7fe6db1dce
1788819441.982 192.168.99.1 3081f1020103300d020138020205dc04010302010304373035040c800000090300525400c8010102
0102020200a204066c61626d6f6e040c54e0b089dfb6d3286639954d04089b184efb1b39c21a0481a38399f66019ba392443b719d26a
fd64cb17b4f7e47e6efc18540d7a161bb4df34a5d637d8baf88d7959b126d7f786e1521239899f10d3d27949f04880309d9a8e456a78
4b512b0558a7215ca4ba9844e8fac890bae9434f944d06cc273d3d6bb9fd842b28c44aa4b6f7723acfc5a8d378b7f6d6019f1c750dbd
2b44a96e3e05c10c5df1deb3e7c8f3b1022054494304e4456fad68c0f554b6016aa90535c790e85d921b
1788819442.019 192.168.99.2 3081f1020103300d020140020205dc04010302010304373035040c800000090300525400c8020102
0102020200a204066c61626d6f6e040cc5e940312135bbb74315f07f0408bdf4dc46c97074950481a3d122cd69208ea63a33c15acfa8
7a45102a3d13ff4011a6fe0077d1190d97e78309f972e8bd391feaf8f439f142fa831bc51566d9eb8017d4cb97e7436f0ff3c8ef79ef
ea5d0c4c1aab31cc9b7f8cff78355da4f4fcfa92bddd9281133bec7fae933da59e755762b226e899cb3c6998c3fb887f3397554d5152
dcbea7abdb77818b788f02536a47b9893019f437a968d65cf16414aaec7588c7acf31617d6f2d26c240f
1788819442.056 192.168.99.3 3081f1020103300d02013c020205dc04010302010304373035040c800000090300525400c8030102
0102020200a204066c61626d6f6e040c97f59d812586c8cbc82aeec80408a7c7be9c6efe40590481a3944f05c35a695e8ab698a77fc0
679d4817a25d2867b284b3553e5b9560d1abeb0194ef14399db882e9fa2c2b7a2e5a3a8776478141012d370690d28f11c26bfc8aa1fd
53215e87e900d9dda1de7d41080b83bf95c9c0743949bd5c4af9ea11abc6a118939fad85eb0735991d6b9d8e891ccf0bd9cb99abf172
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1788819442.233 192.168.99.1 3081f1020103300d02013a020205dc04010302010304373035040c800000090300525400c8010102
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```

```
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... 11908 more lines
```

NMS **sudo cat /var/log/lab/traps-raw.log 2>/dev/null || true**

```
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1788819442.306 192.168.99.3 3081f1020103300d02013e020205dc04010302010304373035040c800000090300525400c8030102
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1788819442.482 192.168.99.1 3081f1020103300d02013c020205dc04010302010304373035040c800000090300525400c8010102
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1788819442.556 192.168.99.3 308201db020103300d020140020205dc04010302010304373035040c800000090300525400c80301
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aef92a4fd087aa
1788819442.733 192.168.99.1 3081f1020103300d02013e020205dc040103020103043730350400c800000090300525400c8010102
0102020200a204066c61626d6f6e040cf1375d27c9b13e3bc630dda704089b184efb1b39c2200481a30198f4b559036be6e04256f622
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1788819442.769 192.168.99.2 3081bb020103300d020146020205dc040103020103043730350400c800000090300525400c8020102
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f9d44b5f12d56f7e84a1eea2197ad6b4e73721c87bfd23d6a00a98a878bc9f8f206e82d8a3606be17431
... 11909 more lines
```


Teardown (suite) Close Trap Collector, Close All Connections

16.08 PASS Received Traps Are SNMP Version 3 (0.6s, 3 commands)

Reads the version field out of the datagram itself rather than trusting the configuration.

Setup (suite) Start Snmp Collectors

Acceptance criteria

- **${traps}** is not empty (checked 3×)
- **${versions}** equals **${[3]}** (checked 3×)

How it was checked

NMS **`sudo cat /var/log/lab/traps-raw.log 2>/dev/null || true`**


```
1788819441.732 192.168.99.1 3081f1020103300d020136020205dc040103020103043730350400c800000090300525400c8010102
0102020200a204066c61626d6f6e040c0330a96d73b02577bd14002704089b184efb1b39c2180481a31e0cb257972ac1650a9c016737
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158e72f510620b1c9e484b7417b4d3a185d046ea549f3642f878698207c3f818a4b10c5927a8e880fcba
1788819441.768 192.168.99.2 3081f1020103300d02013e020205dc040103020103043730350400c800000090300525400c8020102
0102020200a104066c61626d6f6e040c280bc8173afb140f39665d400408bdf4dc46c97074930481a3485df5849e5eeeb771b44c093a
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8ee7b7c9e197db97da0102a62640bc8d2abf526e9a3bd5db09b0713ae471d14774a711e1c49770a339c6
1788819441.806 192.168.99.3 3081f1020103300d02013a020205dc040103020103043730350400c800000090300525400c8030102
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... 11910 more lines
```

NMS sudo cat /var/log/lab/traps-raw.log 2>/dev/null || true


```
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... 11911 more lines
```


Teardown (suite) Close Trap Collector, Close All Connections

16.09 PASS Received Traps Carry The Configured User And Privacy Flags (0.6s, 3 commands)

Setup (suite) Start Snmp Collectors

Acceptance criteria

- **${users}** equals **${{['${SNMP_USER}']}}** (checked 3×)
- **${all_priv}** holds (checked 3×)

How it was checked

NMS **sudo cat /var/log/lab/traps-raw.log 2>/dev/null || true**


```
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NMS sudo cat /var/log/lab/traps-raw.log 2>/dev/null || true

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NMS sudo cat /var/log/lab/traps-raw.log 2>/dev/null || true

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2b61fdeb71554fcc5d3fa447dc8681234e12ab5d5157c513e3981ae8abd5e8e564fe10aecfc7bd4560ef
1788819442.556 192.168.99.3 308201db020103300d020140020205dc04010302010304373035040c800000090300525400c80301
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```

```
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1788819442.733 192.168.99.1 3081f1020103300d02013e020205dc040103020103043730350 40c800000090300525400c8010102
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1788819442.769 192.168.99.2 3081bb020103300d020146020205dc040103020103043730350 40c800000090300525400c8020102
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... 11911 more lines
```


Teardown (suite) Close Trap Collector, Close All Connections

16.10 PASS Trap Payloads Are Encrypted (0.5s, 3 commands)

With privacy in force the scoped PDU arrives as an opaque OCTET STRING. If it were a readable SEQUENCE the varbinds would be on the wire in clear.

Setup (suite) Start Snmp Collectors

Acceptance criteria

- **${encrypted}** holds (checked 3×)

How it was checked

NMS `sudo cat /var/log/lab/traps-raw.log 2>/dev/null || true`


```
1788819441.732 192.168.99.1 3081f1020103300d020136020205dc0401030201030437303504 0c800000090300525400c8010102
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1788819441.768 192.168.99.2 3081f1020103300d02013e020205dc0401030201030437303504 0c800000090300525400c8020102
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1788819442.306 192.168.99.3 3081f1020103300d02013e020205dc0401030201030437303504 0c800000090300525400c8030102
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```

```
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1788819442.556 192.168.99.3 308201db020103300d020140020205dc04010302010304373035040c800000090300525400c80301
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1788819442.769 192.168.99.2 3081bb020103300d020146020205dc04010302010304373035040c800000090300525400c8020102
0102020200a204066c61626d6f6e040c3df77a9a6185cfb1cfecf88a0408bdf4dc46c970749b046ef296014ded7c0ce032945da3386e
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... 11911 more lines
```

NMS sudo cat /var/log/lab/traps-raw.log 2>/dev/null || true

```
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1788819441.768 192.168.99.2 3081f1020103300d02013e020205dc04010302010304373035040c800000090300525400c8020102
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1788819441.806 192.168.99.3 3081f1020103300d02013a020205dc04010302010304373035040c800000090300525400c8030102
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1788819441.982 192.168.99.1 3081f1020103300d020138020205dc04010302010304373035040c800000090300525400c8010102
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... 11911 more lines
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iso.3.6.1.4.1.9.9.171.1.3.3.1.5.2.1 "64 40 00 03 "      iso.3.6.1.4.1.9.9.171.1.3.3.1.9.2.1 1
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2026-09-08 01:45:06 <UNKNOWN> [UDP: [192.168.99.2]:51414->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:10:34.66        iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.171.2.0.7
iso.3.6.1.4.1.9.9.171.1.3.2.1.9.2 3600  iso.3.6.1.4.1.9.9.171.1.3.2.1.8.2 4608000
iso.3.6.1.4.1.9.9.171.1.3.3.1.3.2.1 1   iso.3.6.1.4.1.9.9.171.1.3.3.1.4.2.1 "64 40 00 02 "
iso.3.6.1.4.1.9.9.171.1.3.3.1.5.2.1 "64 40 00 02 "      iso.3.6.1.4.1.9.9.171.1.3.3.1.9.2.1 1
iso.3.6.1.4.1.9.9.171.1.3.3.1.10.2.1 "64 40 00 03 "     iso.3.6.1.4.1.9.9.171.1.3.3.1.11.2.1 "64 40 00 03 "
iso.3.6.1.4.1.9.9.171.1.3.3.1.6.2.1 47  iso.3.6.1.4.1.9.9.171.1.3.3.1.7.2.1 0
2026-09-08 01:45:06 <UNKNOWN> [UDP: [192.168.99.3]:50504->[192.168.99.10]:162]:
... 17047 more lines
```

NMS **sudo cat /var/log/lab/snmptrapd.log 2>/dev/null || true**

```
NET-SNMP version 5.9.4.pre2
2026-09-08 01:45:03 <UNKNOWN> [UDP: [192.168.99.3]:50504->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:10:32.39        iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.10.62.2.0.3
iso.3.6.1.4.1.9.10.62.1.2.3.3.1.2.14.84.117.110.110.101.108.48.45.104.101.97.100.45.48.65538 2
iso.3.6.1.4.1.9.10.62.1.2.3.1.1.2.14.84.117.110.110.101.108.48.45.104.101.97.100.45.48 3
2026-09-08 01:45:05 <UNKNOWN> [UDP: [192.168.99.2]:51414->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:10:34.65        iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.10.62.2.0.3
iso.3.6.1.4.1.9.10.62.1.2.3.3.1.2.14.84.117.110.110.101.108.48.45.104.101.97.100.45.48.65538 2
iso.3.6.1.4.1.9.10.62.1.2.3.1.1.2.14.84.117.110.110.101.108.48.45.104.101.97.100.45.48 3
2026-09-08 01:45:05 <UNKNOWN> [UDP: [192.168.99.3]:50504->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:10:34.29        iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.171.2.0.1
iso.3.6.1.4.1.9.9.171.1.2.2.1.6.1.4.100.64.0.3.1.4.100.64.0.2.2 "64 40 00 03 "
iso.3.6.1.4.1.9.9.171.1.2.2.1.7.1.4.100.64.0.3.1.4.100.64.0.2.2 "64 40 00 02 "
iso.3.6.1.4.1.9.9.171.1.2.3.1.15.2 86400
2026-09-08 01:45:05 <UNKNOWN> [UDP: [192.168.99.2]:51414->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:10:34.65        iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.171.2.0.1
iso.3.6.1.4.1.9.9.171.1.2.2.1.6.1.4.100.64.0.2.1.4.100.64.0.3.2 "64 40 00 02 "
iso.3.6.1.4.1.9.9.171.1.2.2.1.7.1.4.100.64.0.2.1.4.100.64.0.3.2 "64 40 00 03 "
iso.3.6.1.4.1.9.9.171.1.2.3.1.15.2 86400
2026-09-08 01:45:05 <UNKNOWN> [UDP: [192.168.99.3]:50504->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:10:34.30        iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.171.2.0.7
iso.3.6.1.4.1.9.9.171.1.3.2.1.9.2 3600  iso.3.6.1.4.1.9.9.171.1.3.2.1.8.2 4608000
iso.3.6.1.4.1.9.9.171.1.3.3.1.3.2.1 1   iso.3.6.1.4.1.9.9.171.1.3.3.1.4.2.1 "64 40 00 03 "
iso.3.6.1.4.1.9.9.171.1.3.3.1.5.2.1 "64 40 00 03 "      iso.3.6.1.4.1.9.9.171.1.3.3.1.9.2.1 1
iso.3.6.1.4.1.9.9.171.1.3.3.1.10.2.1 "64 40 00 02 "     iso.3.6.1.4.1.9.9.171.1.3.3.1.11.2.1 "64 40 00 02 "
iso.3.6.1.4.1.9.9.171.1.3.3.1.6.2.1 47  iso.3.6.1.4.1.9.9.171.1.3.3.1.7.2.1 0
2026-09-08 01:45:06 <UNKNOWN> [UDP: [192.168.99.2]:51414->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:0:10:34.66        iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.171.2.0.7
iso.3.6.1.4.1.9.9.171.1.3.2.1.9.2 3600  iso.3.6.1.4.1.9.9.171.1.3.2.1.8.2 4608000
iso.3.6.1.4.1.9.9.171.1.3.3.1.3.2.1 1   iso.3.6.1.4.1.9.9.171.1.3.3.1.4.2.1 "64 40 00 02 "
iso.3.6.1.4.1.9.9.171.1.3.3.1.5.2.1 "64 40 00 02 "      iso.3.6.1.4.1.9.9.171.1.3.3.1.9.2.1 1
iso.3.6.1.4.1.9.9.171.1.3.3.1.10.2.1 "64 40 00 03 "     iso.3.6.1.4.1.9.9.171.1.3.3.1.11.2.1 "64 40 00 03 "
iso.3.6.1.4.1.9.9.171.1.3.3.1.6.2.1 47  iso.3.6.1.4.1.9.9.171.1.3.3.1.7.2.1 0
2026-09-08 01:45:06 <UNKNOWN> [UDP: [192.168.99.3]:50504->[192.168.99.10]:162]:
... 17047 more lines
```

Teardown (suite) Close Trap Collector, Close All Connections

16.12 PASS One Raw Listener Receives Every Router (0.1s, 1 commands)

The property the per-router-port workaround existed to dodge: a single socket now serves all three senders.

Setup (suite) Start Snmp Collectors

Acceptance criteria

- output contains **${R1_OOB_IP}**
- output contains **${R2_OOB_IP}**
- output contains **${R3_OOB_IP}**

How it was checked

NMS sudo cat /var/log/lab/traps-raw.log 2>/dev/null || true


```
1788819441.732 192.168.99.1 3081f1020103300d020136020205dc040103020103043730350 40c800000090300525400c8010102
0102020200a204066c61626d6f6e040c0330a96d73b02577bd14002704089b184efb1b39c2180481a31e0cb257972ac1650a9c016737
d7c3450c8b69b1810963c1c351af94e662bb6294844083630796502 6c2d246d66d8c3bf10781cd8fcab6521f98448783a6d9bf665ade
b66e773f6cb948e6c84fbc1c22baffabc9a7c820adbbccc0607f9fe235937749231d4e94bb8fdcc059dcffbba830d801e015c85863d7
158e72f510620b1c9e484b7417b4d3a185d046ea549f3642f878698207c3f818a4b10c5927a8e880fcba
1788819441.768 192.168.99.2 3081f1020103300d02013e020205dc040103020103043730350 40c800000090300525400c8020102
0102020200a104066c61626d6f6e040c280bc8173afb140f39665d400408bdf4dc46c97074930481a3485df5849e5eeeb771b44c093a
ef6ddaf4b7ba8b009f4cb725b19f5e751850a70e9cd62e9471b527848f3350df4234902e59c6ba9cd696e552e207ef62af971e8908ef
ae317dcaed8d6702a20fa8a0f5d3535538a2851e16b3bc0d2979168bf60354d2878e5cd7ac03c1daa98fd24eb8a451e70e624bdffc96
8ee7b7c9e197db97da0102a62640bc8d2abf526e9a3bd5db09b0713ae471d14774a711e1c49770a339c6
1788819441.806 192.168.99.3 3081f1020103300d02013a020205dc040103020103043730350 40c800000090300525400c8030102
0102020200a104066c61626d6f6e040c1a06b8620cf2bdfae082242d0408a7c7be9c6efe40570481a3f13563f7a9d2b23f9d6aedd6db
b3ee8cb9bfc314d080afa0b0fb48addca5f604a787508df6374629f81bf9c5e8ed322c4fbc9fd5daf2e665dcbc312cc01027c101b15d
95d74b6c51ef9faf06c0670481620af82d566719918aab054f908d5415c32688a26757db8041390a1fd440c938835bf8a171ee3321ef
59f0d1a550a2e55083266bba89add3ddb3176843bbfc92d3de40fb901ba83f7d8c7124f27c7fe6db1dce
1788819441.982 192.168.99.1 3081f1020103300d020138020205dc040103020103043730350 40c800000090300525400c8010102
0102020200a204066c61626d6f6e040c54e0b089dfb6d3286639954d04089b184efb1b39c21a0481a38399f66019ba392443b719d26a
fd64cb17b4f7e47e6efc18540d7a161bb4df34a5d637d8baf88d7959b126d7f786e1521239899f10d3d27949f04880309d9a8e456a78
4b512b0558a7215ca4ba9844e8fac890bae9434f944d06cc273d3d6bb9fd842b28c44aa4b6f7723acfc5a8d378b7f6d6019f1c750dbd
2b44a96e3e05c10c5df1deb3e7c8f3b1022054494304e4456fad68c0f554b6016aa90535c790e85d921b
1788819442.019 192.168.99.2 3081f1020103300d020140020205dc040103020103043730350 40c800000090300525400c8020102
0102020200a204066c61626d6f6e040cc5e940312135bbb74315f07f0408bdf4dc46c97074950481a3d122cd69208ea63a33c15acfa8
7a45102a3d13ff4011a6fe0077d1190d97e78309f972e8bd391feaf8f439f142fa831bc51566d9eb8017d4cb97e7436f0ff3c8ef79ef
ea5d0c4c1aab31cc9b7f8cff78355da4f4fcfa92bddd9281133bec7fae933da59e755762b226e899cb3c6998c3fb887f3397554d5152
dcbea7abdb77818b788f02536a47b9893019f437a968d65cf16414aaec7588c7acf31617d6f2d26c240f
1788819442.056 192.168.99.3 3081f1020103300d02013c020205dc040103020103043730350 40c800000090300525400c8030102
0102020200a204066c61626d6f6e040c97f59d812586c8cbc82aeec80408a7c7be9c6efe40590481a3944f05c35a695e8ab698a77fc0
679d4817a25d2867b284b3553e5b9560d1abeb0194ef14399db882e9fa2c2b7a2e5a3a8776478141012d370690d28f11c26bfc8aa1fd
53215e87e900d9dda1de7d41080b83bf95c9c0743949bd5c4af9ea11abc6a118939fad85eb0735991d6b9d8e891ccf0bd9cb99abf172
805a8c9b5e5d778a279e1a0384713bb102e8ba0b0be1d56a9527722ef00de7e6baabfe3aae5f497a44f4
1788819442.233 192.168.99.1 3081f1020103300d02013a020205dc040103020103043730350 40c800000090300525400c8010102
0102020200a204066c61626d6f6e040ce97d7d93c11dac8ea0caaafe04089b184efb1b39c21c0481a3ee52a2e7caf58003f3aeaef04a
dcaabcf43bd1c13a51d4087e88c6a9a9cfb270aec98cf2cbdfc6fd803608b8bc6591cde59384e91da399d6f1080102f98fe25152158b
139ebf2f59507f8016a328a4c7a9d1b80a4219982fe2f74978cda2b2ef1bf7d7faf9d835a5cf3f9042d5c74a4e898d39833cc1dbe052
02b505f5d02dbbe48944635bbb890fd08e79baaef5523fd4bd8e7aa0f86d71427b43736a5d9d7bef4403
1788819442.267 192.168.99.2 3081f1020103300d020142020205dc040103020103043730350 40c800000090300525400c8020102
0102020200a204066c61626d6f6e040cd6f8a0f7f93b069d069759170408bdf4dc46c97074970481a33116ac4348d5c39ada28f96d78
e74d155c5cc9bffa42fe9fdadacbc9f298c97a79d1335ef72ad9631dc12834db86976c08bc0949678ce7926afec969d51ff59beb6fd1
7b912bab7415c3068dc730665587dd829e838b378fbcd81aeaa7be721b39cc8ec406c7f91f5bf3e02ef0f84d884b62698b64590b35f1
8921e3a8316c7d66a27f4300e3b40e92dcf8449782b483b7ebb1c0a3da45fbe4c24622a7d600f36d7023
1788819442.306 192.168.99.3 3081f1020103300d02013e020205dc040103020103043730350 40c800000090300525400c8030102
0102020200a204066c61626d6f6e040c3bb1f3582527a19034ec5d610408a7c7be9c6efe405b0481a3fa3381e2249898d509d85e1d9b
3389b3128c7d34aaa0145cd77854357e6d69cd388db0d86c165a4fdf8a3d87f9bb961854ba187a1984c45dbbf58adca5541a5485c344
583a562913d9b44704b325f28a9b0c3c74d18b4c3bd08fc9130c877165dc4c6d22a6e4226cd8546bac4ed78bd313ac2d76bfad8165bf
d95c8ff1fe7981815276c5c81bab3f027acbecb251449958991c430e2e4f57b98530410546f515e98124
1788819442.482 192.168.99.1 3081f1020103300d02013c020205dc040103020103043730350 40c800000090300525400c8010102
0102020200a204066c61626d6f6e040c73a76bc49e552aab0d08f64d04089b184efb1b39c21e0481a3b1a6978d7875aefa7484d41426
fcac7bd01a45e729fb041e4e0053bcf6e1e7b4453849a0950cb51754d5aa8bb1e683b91e18dedd70c8eeb400fd5392669c3139475a2e
f9cea2e5f556ab286affc24b299f7257a7f9aaa3362d185225a4037aefe861fb964fcd2e3f759405bf4911bb9c8e340b9c3c32a39f9b
caf86df01c6728b43aa549379361193e7dccae3f3f0abd1c6f25b53985eb169cb05adfaa725d29e505e5
1788819442.518 192.168.99.2 3081f1020103300d020144020205dc040103020103043730350 40c800000090300525400c8020102
0102020200a204066c61626d6f6e040ce22c3e1cccc3a0816b2de3090408bdf4dc46c97074990481a362ccc6cfc3fcb1f5dfe6a36ba9
061eb6d5f8e4cea2b22f8283e04ff27fe21962da8a88d91b1c4fbdc6a5d339e9dd070feed2193e628e3bdc3aeaea01f49bc99930a11e
69a68b57bc34e6500c2a923e4766fa1263512636869017b82cec5982625feded62ba54a93598fd3604d189683a16346138220d45f3a5
2b61fdeb71554fcc5d3fa447dc8681234e12ab5d5157c513e3981ae8abd5e8e564fe10aecfc7bd4560ef
1788819442.556 192.168.99.3 308201db020103300d020140020205dc040103020103043730350 40c800000090300525400c80301
020102020200a204066c61626d6f6e040cd0e347d7a518a7d9e2f37c1b0408a7c7be9c6efe405d0482018cd8efa54000ad401bbc326d
2bbcde825ab5152edf654545b7ab79bde2ffadf60703003bf05fd372b55c6598a7edececfd536ad432095fb9260748bd46f1cc973fce
7d0c5d461a74567d653f8915b5eb8f3625a9acd114f2efc9c842fde25afe53f39a555ecb3dcf7d27a08432dc9340b766d913d297b4bb
07970a2510babd3db4f4ce0ebb07f01f9ffd45e6fb29ef8feac58621e676b92935f9ce739373e633a53e74e506575e6f46825bb060ff
e5b0c8526d79831062a4e95bdec7dd7039dd603bf20ecded96990e1d2ab506b0cb8e915f9e5b37d553f38e99197198e6c070324af943
30b1625bd961d67078f27e4290be8c1b30f9fdd4f9da38629eaf2dd334b0e0a26532c0549a3fc8782afe71d694abf3ac5084fb3ae4b9
34af1b612e329beb9c7e1163aa36dc290f9784b5cc2f25a844f00d69d85b8798ee860408950a0e0a5c9d4a79f2784d07c6aca87f8a39
ea45e822985914ea2cb93b21d583ae4f38bbbb056238ff969149d9ebbe2171e6f569e3e2f1e75f5a94e133e9fca802228704d2f319ba
aef92a4fd087aa
1788819442.733 192.168.99.1 3081f1020103300d02013e020205dc040103020103043730350 40c800000090300525400c8010102
0102020200a204066c61626d6f6e040cf1375d27c9b13e3bc630dda704089b184efb1b39c2200481a30198f4b559036be6e04256f622
03359df9f4b249e253bfc47820d97178378d92b51f24de8d8455b5342f2d76797f484d9471253faea500058a89ac76482836c4d83314
b637ad7fc13ded93b704de011795c6d883a9036f15a8f96224a3162a8d83a10fe88c1e9da55743953b410a155c88ebec8dc5ed1f85e8
d6719b4b8a6db27cd5500b799d76f31678bbff135288e19a3de00a02a2b05ec9f77a51e2e82d417f4fd9
1788819442.769 192.168.99.2 3081bb020103300d020146020205dc040103020103043730350 40c800000090300525400c8020102
0102020200a204066c61626d6f6e040c3df77a9a6185cfb1cfecf88a0408bdf4dc46c970749b046ef296014ded7c0ce032945da3386e
1ebd3a6f6a183b3c2e4ebf6a7517d3b4fbe0d2965d0cedde29f5e6919fa5f14ad44ce1b2ffc828265b33e24d5f8c4f415511af6b9294
```

```
  f9d44b5f12d56f7e84a1eea2197ad6b4e73721c87bfd23d6a00a98a878bc9f8f206e82d8a3606be17431
  ... 11911 more lines
```

Teardown (suite) Close Trap Collector, Close All Connections

16.13 PASS Each Router Identifies Itself By Its Own Engine ID (0.6s, 6 commands)

The engine ID in the received trap must match the one the router reports, and must differ between routers -- that is what lets a collector attribute a trap to a device.

Setup (suite) Start Snmp Collectors

Acceptance criteria

- **${local}** is not empty (checked 3×)
- **${ids}** has exactly **1** entries (checked 3×)
- **${ids}[0]** equals **${expected}** (checked 3×)
- **${distinct}** equals **3**

  the three routers did not present three distinct engine IDs: ${seen}

How it was checked

R1 show snmp engineID | include Local

```
  Local SNMP engineID: 800000090300525400C80101
```

NMS sudo cat /var/log/lab/traps-raw.log 2>/dev/null || true


```
  1788819441.732 192.168.99.1 3081f1020103300d020136020205dc04010302010304373035040c800000090300525400c8010102
  0102020200a204066c61626d6f6e040c0330a96d73b02577bd14002704089b184efb1b39c2180481a31e0cb257972ac1650a9c016737
  d7c3450c8b69b1810963c1c351af94e662bb62948440836307965026c2d246d66d8c3bf10781cd8fcab6521f98448783a6d9bf665ade
  b66e773f6cb948e6c84fbc1c22baffabc9a7c820adbbccc0607f9fe235937749231d4e94bb8fdcc059dcffbba830d801e015c85863d7
  158e72f510620b1c9e484b7417b4d3a185d046ea549f3642f878698207c3f818a4b10c5927a8e880fcba
  1788819441.768 192.168.99.2 3081f1020103300d02013e020205dc04010302010304373035040c800000090300525400c8020102
  0102020200a104066c61626d6f6e040c280bc8173afb140f39665d400408bdf4dc46c97074930481a3485df5849e5eeeb771b44c093a
  ef6ddaf4b7ba8b009f4cb725b19f5e751850a70e9cd62e9471b527848f3350df4234902e59c6ba9cd696e552e207ef62af971e8908ef
  ae317dcaed8d6702a20fa8a0f5d3535538a2851e16b3bc0d2979168bf60354d2878e5cd7ac03c1daa98fd24eb8a451e70e624bdffc96
  8ee7b7c9e197db97da0102a62640bc8d2abf526e9a3bd5db09b0713ae471d14774a711e1c49770a339c6
  1788819441.806 192.168.99.3 3081f1020103300d02013a020205dc04010302010304373035040c800000090300525400c8030102
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... 11911 more lines
```


R2 show snmp engineID | include Local

```
Local SNMP engineID: 800000090300525400C80201
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NMS sudo cat /var/log/lab/traps-raw.log 2>/dev/null || true


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d6719b4b8a6db27cd5500b799d76f31678bbff135288e19a3de00a02a2b05ec9f77a51e2e82d417f4fd9
1788819442.769 192.168.99.2 3081bb020103300d020146020205dc04010302010304373035040c800000090300525400c8020102
0102020200a204066c61626d6f6e040c3df77a9a6185cfb1cfecf88a0408bdf4dc46c970749b046ef296014ded7c0ce032945da3386e
1ebd3a6f6a183b3c2e4ebf6a7517d3b4fbe0d2965d0cedde29f5e6919fa5f14ad44ce1b2ffc828265b33e24d5f8c4f415511af6b9294
f9d44b5f12d56f7e84a1eea2197ad6b4e73721c87bfd23d6a00a98a878bc9f8f206e82d8a3606be17431
... 11911 more lines
```


R3 **show snmp engineID | include Local**

```
Local SNMP engineID: 800000090300525400C80301
```

NMS **sudo cat /var/log/lab/traps-raw.log 2>/dev/null || true**


```
1788819441.732 192.168.99.1 3081f1020103300d020136020205dc04010302010304373035040c800000090300525400c8010102
0102020200a204066c61626d6f6e040c0330a96d73b02577bd14002704089b184efb1b39c2180481a31e0cb257972ac1650a9c016737
d7c3450c8b69b1810963c1c351af94e662bb6294844083630796502 6c2d246d66d8c3bf10781cd8fcab6521f98448783a6d9bf665ade
b66e773f6cb948e6c84fbc1c22baffabc9a7c820adbbccc0607f9fe235937749231d4e94bb8fdcc059dcffbba830d801e015c85863d7
158e72f510620b1c9e484b7417b4d3a185d046ea549f3642f878698207c3f818a4b10c5927a8e880fcba
1788819441.768 192.168.99.2 3081f1020103300d02013e020205dc04010302010304373035040c800000090300525400c8020102
0102020200a104066c61626d6f6e040c280bc8173afb140f39665d400408bdf4dc46c97074930481a3485df5849e5eeeb771b44c093a
ef6ddaf4b7ba8b009f4cb725b19f5e751850a70e9cd62e9471b527848f3350df4234902e59c6ba9cd696e552e207ef62af971e8908ef
ae317dcaed8d6702a20fa8a0f5d3535538a2851e16b3bc0d2979168bf60354d2878e5cd7ac03c1daa98fd24eb8a451e70e624bdffc96
8ee7b7c9e197db97da0102a62640bc8d2abf526e9a3bd5db09b0713ae471d14774a711e1c49770a339c6
1788819441.806 192.168.99.3 3081f1020103300d02013a020205dc04010302010304373035040c800000090300525400c8030102
0102020200a104066c61626d6f6e040c1a06b8620cf2bdfae082242d0408a7c7be9c6efe40570481a3f13563f7a9d2b23f9d6aedd6db
b3ee8cb9bfc314d080afa0b0fb48addca5f604a787508df6374629f81bf9c5e8ed322c4fbc9fd5daf2e665dcbc312cc01027c101b15d
95d74b6c51ef9faf06c0670481620af82d566719918aab054f908d5415c32688a26757db8041390a1fd440c938835bf8a171ee3321ef
59f0d1a550a2e55083266bba89add3ddb3176843bbfc92d3de40fb901ba83f7d8c7124f27c7fe6db1dce
1788819441.982 192.168.99.1 3081f1020103300d020138020205dc04010302010304373035040c800000090300525400c8010102
0102020200a204066c61626d6f6e040c54e0b089dfb6d3286639954d04089b184efb1b39c21a0481a38399f66019ba392443b719d26a
fd64cb17b4f7e47e6efc18540d7a161bb4df34a5d637d8baf88d7959b126d7f786e1521239899f10d3d27949f04880309d9a8e456a78
4b512b0558a7215ca4ba9844e8fac890bae9434f944d06cc273d3d6bb9fd842b28c44aa4b6f7723acfc5a8d378b7f6d6019f1c750dbd
2b44a96e3e05c10c5df1deb3e7c8f3b1022054494304e4456fad68c0f554b6016aa90535c790e85d921b
1788819442.019 192.168.99.2 3081f1020103300d020140020205dc04010302010304373035040c800000090300525400c8020102
0102020200a204066c61626d6f6e040cc5e940312135bbb74315f07f0408bdf4dc46c97074950481a3d122cd69208ea63a33c15acfa8
7a45102a3d13ff4011a6fe0077d1190d97e78309f972e8bd391feaf8f439f142fa831bc51566d9eb8017d4cb97e7436f0ff3c8ef79ef
ea5d0c4c1aab31cc9b7f8cff78355da4f4fcfa92bddd9281133bec7fae933da59e755762b226e899cb3c6998c3fb887f3397554d5152
dcbea7abdb77818b788f02536a47b9893019f437a968d65cf16414aaec7588c7acf31617d6f2d26c240f
1788819442.056 192.168.99.3 3081f1020103300d02013c020205dc04010302010304373035040c800000090300525400c8030102
0102020200a204066c61626d6f6e040c97f59d812586c8cbc82aeec80408a7c7be9c6efe40590481a3944f05c35a695e8ab698a77fc0
679d4817a25d2867b284b3553e5b9560d1abeb0194ef14399db882e9fa2c2b7a2e5a3a8776478141012d370690d28f11c26bfc8aa1fd
53215e87e900d9dda1de7d41080b83bf95c9c0743949bd5c4af9ea11abc6a118939fad85eb0735991d6b9d8e891ccf0bd9cb99abf172
805a8c9b5e5d778a279e1a0384713bb102e8ba0b0be1d56a9527722ef00de7e6baabfe3aae5f497a44f4
1788819442.233 192.168.99.1 3081f1020103300d02013a020205dc04010302010304373035040c800000090300525400c8010102
0102020200a204066c61626d6f6e040ce97d7d93c11dac8ea0caaafe04089b184efb1b39c21c0481a3ee52a2e7caf58003f3aeaef04a
dcaabcf43bd1c13a51d4087e88c6a9a9cfb270aec98cf2cbdfc6fd803608b8bc6591cde59384e91da399d6f1080102f98fe25152158b
139ebf2f59507f8016a328a4c7a9d1b80a4219982fe2f74978cda2b2ef1bf7d7faf9d835a5cf3f9042d5c74a4e898d39833cc1dbe052
02b505f5d02dbbe48944635bbb890fd08e79baaef5523fd4bd8e7aa0f86d71427b43736a5d9d7bef4403
1788819442.267 192.168.99.2 3081f1020103300d020142020205dc04010302010304373035040c800000090300525400c8020102
0102020200a204066c61626d6f6e040cd6f8a0f7f93b069d069759170408bdf4dc46c97074970481a33116ac4348d5c39ada28f96d78
e74d155c5cc9bffa42fe9fdadacbc9f298c97a79d1335ef72ad9631dc12834db86976c08bc0949678ce7926afec969d51ff59beb6fd1
7b912bab7415c3068dc730665587dd829e838b378fbcd81aeaa7be721b39cc8ec406c7f91f5bf3e02ef0f84d884b62698b64590b35f1
8921e3a8316c7d66a27f4300e3b40e92dcf8449782b483b7ebb1c0a3da45fbe4c24622a7d600f36d7023
1788819442.306 192.168.99.3 3081f1020103300d02013e020205dc04010302010304373035040c800000090300525400c8030102
0102020200a204066c61626d6f6e040c3bb1f3582527a19034ec5d610408a7c7be9c6efe405b0481a3fa3381e2249898d509d85e1d9b
3389b3128c7d34aaa0145cd77854357e6d69cd388db0d86c165a4fdf8a3d87f9bb961854ba187a1984c45dbbf58adca5541a5485c344
583a562913d9b44704b325f28a9b0c3c74d18b4c3bd08fc9130c877165dc4c6d22a6e4226cd8546bac4ed78bd313ac2d76bfad8165bf
d95c8ff1fe7981815276c5c81bab3f027acbecb251449958991c430e2e4f57b98530410546f515e98124
1788819442.482 192.168.99.1 3081f1020103300d02013c020205dc04010302010304373035040c800000090300525400c8010102
0102020200a204066c61626d6f6e040c73a76bc49e552aab0d08f64d04089b184efb1b39c21e0481a3b1a6978d7875aefa7484d41426
fcac7bd01a45e729fb041e4e0053bcf6e1e7b4453849a0950cb51754d5aa8bb1e683b91e18dedd70c8eeb400fd5392669c3139475a2e
f9cea2e5f556ab286affc24b299f7257a7f9aaa3362d185225a4037aefe861fb964fcd2e3f759405bf4911bb9c8e340b9c3c32a39f9b
caf86df01c6728b43aa549379361193e7dccae3f3f0abd1c6f25b53985eb169cb05adfaa725d29e505e5
1788819442.518 192.168.99.2 3081f1020103300d020144020205dc04010302010304373035040c800000090300525400c8020102
0102020200a204066c61626d6f6e040ce22c3e1cccc3a0816b2de3090408bdf4dc46c97074990481a362ccc6cfc3fcb1f5dfe6a36ba9
061eb6d5f8e4cea2b22f8283e04ff27fe21962da8a88d91b1c4fbdc6a5d339e9dd070feed2193e628e3bdc3aeaea01f49bc99930a11e
69a68b57bc34e6500c2a923e4766fa1263512636869017b82cec5982625feded62ba54a93598fd3604d189683a16346138220d45f3a5
2b61fdeb71554fcc5d3fa447dc8681234e12ab5d5157c513e3981ae8abd5e8e564fe10aecfc7bd4560ef
1788819442.556 192.168.99.3 308201db020103300d020140020205dc04010302010304373035040c800000090300525400c80301
```

```
020102020200a204066c61626d6f6e040cd0e347d7a518a7d9e2f37c1b0408a7c7be9c6efe405d0482018cd8efa54000ad401bbc326d
2bbcde825ab5152edf654545b7ab79bde2ffadf60703003bf05fd372b55c6598a7edececfd536ad432095fb9260748bd46f1cc973fce
7d0c5d461a74567d653f8915b5eb8f3625a9acd114f2efc9c842fde25afe53f39a555ecb3dcf7d27a08432dc9340b766d913d297b4bb
07970a2510babd3db4f4ce0ebb07f01f9ffd45e6fb29ef8feac58621e676b92935f9ce739373e633a53e74e506575e6f46825bb060ff
e5b0c8526d79831062a4e95bdec7dd7039dd603bf20ecded96990e1d2ab506b0cb8e915f9e5b37d553f38e99197198e6c070324af943
30b1625bd961d67078f27e4290be8c1b30f9fdd4f9da38629eaf2dd334b0e0a26532c0549a3fc8782afe71d694abf3ac5084fb3ae4b9
34af1b612e329beb9c7e1163aa36dc290f9784b5cc2f25a844f00d69d85b8798ee860408950a0e0a5c9d4a79f2784d07c6aca87f8a39
ea45e822985914ea2cb93b21d583ae4f38bbbb056238ff969149d9ebbe2171e6f569e3e2f1e75f5a94e133e9fca802228704d2f319ba
aef92a4fd087aa
1788819442.733 192.168.99.1 3081f1020103300d02013e020205dc040103020103043730350400c800000090300525400c8010102
0102020200a204066c61626d6f6e040cf1375d27c9b13e3bc630dda704089b184efb1b39c2200481a30198f4b559036be6e04256f622
03359df9f4b249e253bfc47820d97178378d92b51f24de8d8455b5342f2d76797f484d9471253faea500058a89ac76482836c4d83314
b637ad7fc13ded93b704de011795c6d883a9036f15a8f96224a3162a8d83a10fe88c1e9da55743953b410a155c88ebec8dc5ed1f85e8
d6719b4b8a6db27cd5500b799d76f31678bbff135288e19a3de00a02a2b05ec9f77a51e2e82d417f4fd9
1788819442.769 192.168.99.2 3081bb020103300d020146020205dc040103020103043730350400c800000090300525400c8020102
0102020200a204066c61626d6f6e040c3df77a9a6185cfb1cfecf88a0408bdf4dc46c970749b046ef296014ded7c0ce032945da3386e
1ebd3a6f6a183b3c2e4ebf6a7517d3b4fbe0d2965d0cedde29f5e6919fa5f14ad44ce1b2ffc828265b33e24d5f8c4f415511af6b9294
f9d44b5f12d56f7e84a1eea2197ad6b4e73721c87bfd23d6a00a98a878bc9f8f206e82d8a3606be17431
... 11911 more lines
```


Teardown (suite) Close Trap Collector, Close All Connections

17 Snmp Polling

17.01 PASS The NMS Runs Real Net-SNMP Tooling (0.1s, 2 commands)

The point of this VM: unlike the cirros hosts it has a genuine SNMP client, so the polling below is the real thing rather than something hand rolled for the test.

Setup (suite) Open All Routers, Open Nms

Acceptance criteria

- output contains **NET-SNMP version**
- output contains **snmpget**
- output contains **snmpwalk**

How it was checked

```
NMS snmpget --version 2>&1 | head -2; snmpwalk --version 2>&1 | head -1
  NET-SNMP version: 5.9.4.pre2
  NET-SNMP version: 5.9.4.pre2
NMS command -v snmpget snmpwalk snmptrapd
  /usr/bin/snmpget
  /usr/bin/snmpwalk
  /usr/sbin/snmptrapd
```

Teardown (suite) Close Nms, Close All Connections

17.02 PASS Every Router Answers An SNMPv3 authPriv Poll (0.2s, 3 commands)

The agent must be reachable and must accept the credentials the provisioner configured.

Setup (suite) Open All Routers, Open Nms

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked

```
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.1
1.3.6.1.2.1.1.5.0
  .1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r1.lab.local"
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.1.5.0
  .1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3
1.3.6.1.2.1.1.5.0
  .1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r3.lab.local"
```

Teardown (suite) Close Nms, Close All Connections

17.03 PASS Polled sysName Matches The Router's Own Hostname (0.4s, 6 commands)

Cross-check: the value SNMP returns is compared against what the device says on its CLI, so a plausible-looking answer from the wrong box cannot pass.

Setup (suite) Open All Routers, Open Nms

Acceptance criteria

- **${polled}** starts with **${expected}[0]** (checked 3×)
  msg=SNMP sysName '${polled}' does not match CLI hostname '${expected}[0]'

How it was checked

```
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.1
1.3.6.1.2.1.1.5.0
  .1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r1.lab.local"
R1 show running-config | include ^hostname
  hostname c8000v-r1
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.1.5.0
  .1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
```

R2 **show running-config | include ^hostname**
```
hostname c8000v-r2
```
NMS **snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3 1.3.6.1.2.1.1.5.0**
```
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r3.lab.local"
```
R3 **show running-config | include ^hostname**
```
hostname c8000v-r3
```
Teardown (suite) Close Nms, Close All Connections

17.04 PASS Polled sysDescr Identifies The Platform (0.2s, 6 commands)

Setup (suite) Open All Routers, Open Nms

Acceptance criteria

- output matches **Cisco IOS[-]XE Software|Cisco IOS Software \[IOSXE\]** (checked 3×)

How it was checked

NMS **snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.1 1.3.6.1.2.1.1.1.0**
```
.1.3.6.1.2.1.1.1.0 = STRING: "Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-
UNIVERSALK9-M), Version 17.15.6, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2026 by Cisco Systems, Inc.
Compiled Wed 22-Jul-26 03:39 b"
```
R1 **show version | include Version**
```
Cisco IOS XE Software, Version 17.15.06
Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-UNIVERSALK9-M), Version 17.15.6, RELEASE
SOFTWARE (fc2)
licensed under the GNU General Public License ("GPL") Version 2.0.  The
software code licensed under GPL Version 2.0 is free software that comes
GPL code under the terms of GPL Version 2.0.  For more details, see the
```
NMS **snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2 1.3.6.1.2.1.1.1.0**
```
.1.3.6.1.2.1.1.1.0 = STRING: "Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-
UNIVERSALK9-M), Version 17.15.6, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2026 by Cisco Systems, Inc.
Compiled Wed 22-Jul-26 03:39 b"
```
R2 **show version | include Version**
```
Cisco IOS XE Software, Version 17.15.06
Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-UNIVERSALK9-M), Version 17.15.6, RELEASE
SOFTWARE (fc2)
licensed under the GNU General Public License ("GPL") Version 2.0.  The
software code licensed under GPL Version 2.0 is free software that comes
GPL code under the terms of GPL Version 2.0.  For more details, see the
```
NMS **snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3 1.3.6.1.2.1.1.1.0**
```
.1.3.6.1.2.1.1.1.0 = STRING: "Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-
UNIVERSALK9-M), Version 17.15.6, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2026 by Cisco Systems, Inc.
Compiled Wed 22-Jul-26 03:39 b"
```
R3 **show version | include Version**
```
Cisco IOS XE Software, Version 17.15.06
Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-UNIVERSALK9-M), Version 17.15.6, RELEASE
SOFTWARE (fc2)
licensed under the GNU General Public License ("GPL") Version 2.0.  The
software code licensed under GPL Version 2.0 is free software that comes
GPL code under the terms of GPL Version 2.0.  For more details, see the
```
Teardown (suite) Close Nms, Close All Connections

17.05 PASS sysUpTime Advances Between Two Polls (5.0s, 2 commands)

Proves the answers come from a live agent rather than a cached or replayed response.

Setup (suite) Open All Routers, Open Nms

Acceptance criteria

- **${second} > ${first}** holds

How it was checked

```
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.1
1.3.6.1.2.1.1.3.0
  .1.3.6.1.2.1.1.3.0 = Timeticks: (8499540) 23:36:35.40
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.1
1.3.6.1.2.1.1.3.0
  .1.3.6.1.2.1.1.3.0 = Timeticks: (8500042) 23:36:40.42
```

Teardown (suite) Close Nms, Close All Connections

17.06 PASS Walking The Interface Table Returns Every Interface The CLI Shows (0.2s, 6 commands)

A walk, not a get: the agent must return the whole subtree. Every physical interface named by the CLI must appear in it.

Setup (suite) Open All Routers, Open Nms

Acceptance criteria

- **${walked}** contains **${intf}** (checked 12×)
  msg=${r}: ifDescr walk is missing ${intf}

How it was checked

```
NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.1
1.3.6.1.2.1.2.2.1.2
  .1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
  .1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
  .1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
  .1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
  .1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"
  .1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"
  .1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"
  .1.3.6.1.2.1.2.2.1.2.8 = STRING: "Loopback1"
  .1.3.6.1.2.1.2.2.1.2.9 = STRING: "Tunnel0"
R1 show ip interface brief
  Interface              IP-Address      OK? Method Status                Protocol
  GigabitEthernet1       10.0.2.15       YES DHCP   up                    up
  GigabitEthernet2       100.64.0.1      YES NVRAM  up                    up
  GigabitEthernet3       192.168.10.1    YES NVRAM  up                    up
  GigabitEthernet4       192.168.99.1    YES NVRAM  up                    up
  Loopback0              172.16.1.1      YES NVRAM  up                    up
  Loopback1              10.20.1.1       YES NVRAM  up                    up
  Tunnel0                172.20.0.1      YES NVRAM  up                    up
NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.2.2.1.2
  .1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
  .1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
  .1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
  .1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
  .1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"
  .1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"
  .1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"
  .1.3.6.1.2.1.2.2.1.2.8 = STRING: "Loopback1"
  .1.3.6.1.2.1.2.2.1.2.9 = STRING: "Tunnel0"
R2 show ip interface brief
  Interface              IP-Address      OK? Method Status                Protocol
  GigabitEthernet1       10.0.2.15       YES DHCP   up                    up
  GigabitEthernet2       100.64.0.2      YES NVRAM  up                    up
  GigabitEthernet3       192.168.20.1    YES NVRAM  up                    up
  GigabitEthernet4       192.168.99.2    YES NVRAM  up                    up
  Loopback0              172.16.2.1      YES NVRAM  up                    up
  Loopback1              10.20.2.1       YES NVRAM  up                    up
  Tunnel0                172.20.0.2      YES NVRAM  up                    up
NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3
1.3.6.1.2.1.2.2.1.2
  .1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
  .1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
  .1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
  .1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
  .1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"
  .1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"
  .1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"
  .1.3.6.1.2.1.2.2.1.2.8 = STRING: "Loopback1"
  .1.3.6.1.2.1.2.2.1.2.9 = STRING: "Tunnel0"
```

```
R3 show ip interface brief
  Interface              IP-Address      OK? Method Status                Protocol
  GigabitEthernet1       10.0.2.15       YES DHCP   up                    up
  GigabitEthernet2       100.64.0.3      YES NVRAM  up                    up
  GigabitEthernet3       192.168.30.1    YES NVRAM  up                    up
  GigabitEthernet4       192.168.99.3    YES NVRAM  up                    up
  Loopback0              172.16.3.1      YES NVRAM  up                    up
  Loopback1              10.20.3.1       YES NVRAM  up                    up
  Tunnel0                172.20.0.3      YES NVRAM  up                    up
```

Teardown (suite) Close Nms, Close All Connections

17.07 PASS Polled Interface Count Agrees With The Interface Table (0.3s, 6 commands)

ifNumber is a scalar the agent maintains separately from the table, so the two disagreeing would mean a broken agent.

Setup (suite) Open All Routers, Open Nms

Acceptance criteria

- **${number}** equals **${rows}** (checked 3×)

  msg=${r}: ifNumber says ${number} but the table has ${rows} rows

How it was checked

```
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.1
1.3.6.1.2.1.2.1.0
  .1.3.6.1.2.1.2.1.0 = INTEGER: 9
NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.1
1.3.6.1.2.1.2.2.1.2
  .1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
  .1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
  .1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
  .1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
  .1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"
  .1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"
  .1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"
  .1.3.6.1.2.1.2.2.1.2.8 = STRING: "Loopback1"
  .1.3.6.1.2.1.2.2.1.2.9 = STRING: "Tunnel0"
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.2.1.0
  .1.3.6.1.2.1.2.1.0 = INTEGER: 9
NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.2.2.1.2
  .1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
  .1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
  .1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
  .1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
  .1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"
  .1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"
  .1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"
  .1.3.6.1.2.1.2.2.1.2.8 = STRING: "Loopback1"
  .1.3.6.1.2.1.2.2.1.2.9 = STRING: "Tunnel0"
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3
1.3.6.1.2.1.2.1.0
  .1.3.6.1.2.1.2.1.0 = INTEGER: 9
NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3
1.3.6.1.2.1.2.2.1.2
  .1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
  .1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
  .1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
  .1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
  .1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"
  .1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"
  .1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"
  .1.3.6.1.2.1.2.2.1.2.8 = STRING: "Loopback1"
  .1.3.6.1.2.1.2.2.1.2.9 = STRING: "Tunnel0"
```

Teardown (suite) Close Nms, Close All Connections

17.08 PASS The LAN Interface Is Reported Up By Both SNMP And The CLI (0.4s, 9 commands)

ifOperStatus 1 is up(1). Checked against the CLI line protocol state for the same interface.

Setup (suite) Open All Routers, Open Nms

Acceptance criteria

- **${index}** differs from **-1** (checked 3×)
  msg=${r}: no Gi3 in the walk
- output contains **1** (checked 3×)
  msg=${r}: SNMP says Gi3 is not up
- output contains **line protocol is up** (checked 3×)

How it was checked

NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.1 1.3.6.1.2.1.2.2.1.2

```
.1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
.1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
.1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
.1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
.1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"
.1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"
.1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"
.1.3.6.1.2.1.2.2.1.2.8 = STRING: "Loopback1"
.1.3.6.1.2.1.2.2.1.2.9 = STRING: "Tunnel0"
```

NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.1 1.3.6.1.2.1.2.2.1.8

```
.1.3.6.1.2.1.2.2.1.8.1 = INTEGER: 1
.1.3.6.1.2.1.2.2.1.8.2 = INTEGER: 1
.1.3.6.1.2.1.2.2.1.8.3 = INTEGER: 1
.1.3.6.1.2.1.2.2.1.8.4 = INTEGER: 1
.1.3.6.1.2.1.2.2.1.8.5 = INTEGER: 1
.1.3.6.1.2.1.2.2.1.8.6 = INTEGER: 1
.1.3.6.1.2.1.2.2.1.8.7 = INTEGER: 1
.1.3.6.1.2.1.2.2.1.8.8 = INTEGER: 1
.1.3.6.1.2.1.2.2.1.8.9 = INTEGER: 1
```

R1 show interfaces GigabitEthernet3 | include line protocol

```
GigabitEthernet3 is up, line protocol is up
```

NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2 1.3.6.1.2.1.2.2.1.2

```
.1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
.1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
.1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
.1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
.1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"
.1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"
.1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"
.1.3.6.1.2.1.2.2.1.2.8 = STRING: "Loopback1"
.1.3.6.1.2.1.2.2.1.2.9 = STRING: "Tunnel0"
```

NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2 1.3.6.1.2.1.2.2.1.8

```
.1.3.6.1.2.1.2.2.1.8.1 = INTEGER: 1
.1.3.6.1.2.1.2.2.1.8.2 = INTEGER: 1
.1.3.6.1.2.1.2.2.1.8.3 = INTEGER: 1
.1.3.6.1.2.1.2.2.1.8.4 = INTEGER: 1
.1.3.6.1.2.1.2.2.1.8.5 = INTEGER: 1
.1.3.6.1.2.1.2.2.1.8.6 = INTEGER: 1
.1.3.6.1.2.1.2.2.1.8.7 = INTEGER: 1
.1.3.6.1.2.1.2.2.1.8.8 = INTEGER: 1
.1.3.6.1.2.1.2.2.1.8.9 = INTEGER: 1
```

R2 show interfaces GigabitEthernet3 | include line protocol

```
GigabitEthernet3 is up, line protocol is up
```

NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3 1.3.6.1.2.1.2.2.1.2

```
.1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
.1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
.1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
.1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
.1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"
.1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"
.1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"
.1.3.6.1.2.1.2.2.1.2.8 = STRING: "Loopback1"
.1.3.6.1.2.1.2.2.1.2.9 = STRING: "Tunnel0"
```

NMS snmpwalk -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3 1.3.6.1.2.1.2.2.1.8

```
.1.3.6.1.2.1.2.2.1.8.1 = INTEGER: 1
.1.3.6.1.2.1.2.2.1.8.2 = INTEGER: 1
.1.3.6.1.2.1.2.2.1.8.3 = INTEGER: 1
```

```
.1.3.6.1.2.1.2.2.1.8.4 = INTEGER: 1
.1.3.6.1.2.1.2.2.1.8.5 = INTEGER: 1
.1.3.6.1.2.1.2.2.1.8.6 = INTEGER: 1
.1.3.6.1.2.1.2.2.1.8.7 = INTEGER: 1
.1.3.6.1.2.1.2.2.1.8.8 = INTEGER: 1
.1.3.6.1.2.1.2.2.1.8.9 = INTEGER: 1
```

R3 show interfaces GigabitEthernet3 | include line protocol

```
GigabitEthernet3 is up, line protocol is up
```

Teardown (suite) Close Nms, Close All Connections

17.09 PASS Spokes Are Polled Across The Encrypted Tunnel (6.5s, 18 commands)

The NMS shares a LAN with the hub only. Traffic to R2 and R3 crosses the IPsec tunnels, so a burst of polling must show up on the ESP counters at both ends.

Setup (suite) Open All Routers, Open Nms

Acceptance criteria

- **${a_enc} >= ${expected}** holds (checked 2×)
- **${a_dec} >= ${expected}** holds (checked 2×)
- **${b_enc} >= ${expected}** holds (checked 2×)
- **${b_dec} >= ${expected}** holds (checked 2×)

How it was checked

R1 show crypto ipsec sa

```
interface: Tunnel0
    Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
   remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
   current_peer 100.64.0.3 port 500
     PERMIT, flags={origin_is_acl,}
    #pkts encaps: 5030, #pkts encrypt: 5030, #pkts digest: 5030
    #pkts decaps: 5031, #pkts decrypt: 5031, #pkts verify: 5031
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0
... 78 more lines
```

R2 show crypto ipsec sa

```
interface: Tunnel0
    Crypto map tag: Tunnel0-head-0, local addr 100.64.0.2

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
   remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
   current_peer 100.64.0.3 port 500
     PERMIT, flags={origin_is_acl,}
    #pkts encaps: 52471, #pkts encrypt: 52471, #pkts digest: 52471
    #pkts decaps: 28081, #pkts decrypt: 28081, #pkts verify: 28081
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0
... 78 more lines
```

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2 1.3.6.1.2.1.1.5.0

```
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
```

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2 1.3.6.1.2.1.1.5.0

```
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
```

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2 1.3.6.1.2.1.1.5.0

```
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
```

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2 1.3.6.1.2.1.1.5.0

```
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
```

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2 1.3.6.1.2.1.1.5.0

```
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
```

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2 1.3.6.1.2.1.1.5.0

```
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
```

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2 1.3.6.1.2.1.1.5.0

```
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
```

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2 1.3.6.1.2.1.1.5.0

```
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
```

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2 1.3.6.1.2.1.1.5.0

```
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
```

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2 1.3.6.1.2.1.1.5.0

```
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
```

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2 1.3.6.1.2.1.1.5.0

```
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
```

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2 1.3.6.1.2.1.1.5.0

```
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
```

R1 show crypto ipsec sa

```
interface: Tunnel0
    Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
   remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
   current_peer 100.64.0.3 port 500
     PERMIT, flags={origin_is_acl,}
    #pkts encaps: 5036, #pkts encrypt: 5036, #pkts digest: 5036
    #pkts decaps: 5036, #pkts decrypt: 5036, #pkts verify: 5036
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0
... 78 more lines
```

R1 show crypto ipsec sa

```
interface: Tunnel0
    Crypto map tag: Tunnel0-head-0, local addr 100.64.0.2

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
   remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
   current_peer 100.64.0.3 port 500
     PERMIT, flags={origin_is_acl,}
    #pkts encaps: 52477, #pkts encrypt: 52477, #pkts digest: 52477
    #pkts decaps: 28086, #pkts decrypt: 28086, #pkts verify: 28086
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0
... 78 more lines
```

R1 show crypto ipsec sa

```
interface: Tunnel0
    Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
   remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
   current_peer 100.64.0.3 port 500
     PERMIT, flags={origin_is_acl,}
    #pkts encaps: 5066, #pkts encrypt: 5066, #pkts digest: 5066
    #pkts decaps: 5066, #pkts decrypt: 5066, #pkts verify: 5066
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0
... 78 more lines
```

R1 show crypto ipsec sa

```
interface: Tunnel0
    Crypto map tag: Tunnel0-head-0, local addr 100.64.0.2

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (100.64.0.2/255.255.255.255/47/0)
```

```
    remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
    current_peer 100.64.0.3 port 500
      PERMIT, flags={origin_is_acl,}
     #pkts encaps: 52506, #pkts encrypt: 52506, #pkts digest: 52506
     #pkts decaps: 28121, #pkts decrypt: 28121, #pkts verify: 28121
     #pkts compressed: 0, #pkts decompressed: 0
     #pkts not compressed: 0, #pkts compr. failed: 0
     #pkts not decompressed: 0, #pkts decompress failed: 0
     #send errors 0, #recv errors 0
... 78 more lines
```

Teardown (suite) Close Nms, Close All Connections

17.10 PASS A Wrong Authentication Password Is Refused (0.1s, 2 commands)

Negative control. The agent answers the correct credentials first, so a refusal here cannot be a dead router.

Setup (suite) Open All Routers, Open Nms

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.1 1.3.6.1.2.1.1.5.0

```
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r1.lab.local"
```

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A NotTheRightPassword -x AES -X LabPrivPass123 -t 2 -r 0 192.168.99.1 1.3.6.1.2.1.1.5.0

```
snmpget: Authentication failure (incorrect password, community or key)
```

Teardown (suite) Close Nms, Close All Connections

17.11 PASS A Wrong Privacy Password Is Refused (2.2s, 3 commands)

Authentication succeeds but the payload will not decrypt, so the router drops the request silently and the client times out. Bracketed by a good poll to prove it is still alive.

Setup (suite) Open All Routers, Open Nms

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.1 1.3.6.1.2.1.1.5.0

```
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r1.lab.local"
```

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X NotTheRightPassword -t 2 -r 0 192.168.99.1 1.3.6.1.2.1.1.5.0

```
Timeout: No Response from 192.168.99.1.
```

NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.1 1.3.6.1.2.1.1.5.0

```
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r1.lab.local"
```

Teardown (suite) Close Nms, Close All Connections

17.12 PASS Polling Without Privacy Is Refused By Every Router (0.3s, 6 commands)

The group is configured "v3 priv". Correct credentials at a lower security level must still be denied, otherwise the privacy requirement is decorative.

Setup (suite) Open All Routers, Open Nms

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked

NMS snmpget -On -v3 -l noAuthNoPriv -u labmon -t 2 -r 0 192.168.99.1 1.3.6.1.2.1.1.5.0

```
Error in packet
Reason: authorizationError (access denied to that object)
```

NMS snmpget -On -v3 -l authNoPriv -u labmon -a SHA -A LabAuthPass123 -t 2 -r 0 192.168.99.1 1.3.6.1.2.1.1.5.0

```
Error in packet
Reason: authorizationError (access denied to that object)
```

NMS **snmpget -On -v3 -l noAuthNoPriv -u labmon -t 2 -r 0 192.168.99.2 1.3.6.1.2.1.1.5.0**

```
Error in packet
Reason: authorizationError (access denied to that object)
```

NMS **snmpget -On -v3 -l authNoPriv -u labmon -a SHA -A LabAuthPass123 -t 2 -r 0 192.168.99.2 1.3.6.1.2.1.1.5.0**

```
Error in packet
Reason: authorizationError (access denied to that object)
```

NMS **snmpget -On -v3 -l noAuthNoPriv -u labmon -t 2 -r 0 192.168.99.3 1.3.6.1.2.1.1.5.0**

```
Error in packet
Reason: authorizationError (access denied to that object)
```

NMS **snmpget -On -v3 -l authNoPriv -u labmon -a SHA -A LabAuthPass123 -t 2 -r 0 192.168.99.3 1.3.6.1.2.1.1.5.0**

```
Error in packet
Reason: authorizationError (access denied to that object)
```

Teardown (suite) Close Nms, Close All Connections

17.13 PASS An Unknown User Is Refused (0.1s, 1 commands)

Only the provisioned user exists in the USM table.

Setup (suite) Open All Routers, Open Nms

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked

NMS **snmpget -On -v3 -l authPriv -u intruder -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 2 -r 0 192.168.99.1 1.3.6.1.2.1.1.5.0**

```
snmpget: Unknown user name
```

Teardown (suite) Close Nms, Close All Connections

17.14 PASS SNMPv2c Gets No Answer At The Wire (6.3s, 6 commands)

Suite 16 asserts no community is configured. This asserts the consequence on the network: a v2c poll gets nothing back, while a v3 poll to the same address succeeds.

Setup (suite) Open All Routers, Open Nms

Acceptance criteria

- output contains **Timeout** (checked 3×)

How it was checked

NMS **snmpget -On -v2c -c public -t 2 -r 0 192.168.99.1 1.3.6.1.2.1.1.5.0**

```
Timeout: No Response from 192.168.99.1.
```

NMS **snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.1 1.3.6.1.2.1.1.5.0**

```
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r1.lab.local"
```

NMS **snmpget -On -v2c -c public -t 2 -r 0 192.168.99.2 1.3.6.1.2.1.1.5.0**

```
Timeout: No Response from 192.168.99.2.
```

NMS **snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2 1.3.6.1.2.1.1.5.0**

```
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
```

NMS **snmpget -On -v2c -c public -t 2 -r 0 192.168.99.3 1.3.6.1.2.1.1.5.0**

```
Timeout: No Response from 192.168.99.3.
```

NMS **snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3 1.3.6.1.2.1.1.5.0**

```
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r3.lab.local"
```

Teardown (suite) Close Nms, Close All Connections

18 Mtu

18.01 PASS Every Tunnel Reports The Expected Transport MTU (0.3s, 9 commands)

The tunnel MTU must be below the physical one by the ESP overhead. Equal would mean encryption costs nothing, which would mean it is not happening.

Setup (suite) Open All Routers, Open All Hosts

Acceptance criteria

- **${tunnels}** is not empty (checked 3×)
- output contains **${TUNNEL_MTU} bytes** (checked 3×)
- output contains **MTU is ${TUNNEL_MTU} bytes** (checked 3×)
- **${TUNNEL_MTU} < ${PHYSICAL_MTU}** holds

How it was checked

```
R1 show ip interface brief | include ^Tunnel
  Tunnel0                172.20.0.1      YES NVRAM  up                    up
R1 show interfaces Tunnel0 | include Tunnel transport MTU
  Tunnel transport MTU 1472 bytes
R1 show ip interface Tunnel0 | include MTU
  MTU is 1472 bytes
R2 show ip interface brief | include ^Tunnel
  Tunnel0                172.20.0.2      YES NVRAM  up                    up
R2 show interfaces Tunnel0 | include Tunnel transport MTU
  Tunnel transport MTU 1472 bytes
R2 show ip interface Tunnel0 | include MTU
  MTU is 1472 bytes
R3 show ip interface brief | include ^Tunnel
  Tunnel0                172.20.0.3      YES NVRAM  up                    up
R3 show interfaces Tunnel0 | include Tunnel transport MTU
  Tunnel transport MTU 1472 bytes
R3 show ip interface Tunnel0 | include MTU
  MTU is 1472 bytes
```

Teardown (suite) Close All Connections

18.02 PASS The Overhead Is What GRE With A Tunnel Key Costs (0.0s, 0 commands)

Records the number rather than assuming it: 20 bytes of outer IP, 4 of GRE and 4 for the tunnel key. If the encapsulation changes -- dropping the key, or moving IPsec from transport to tunnel mode -- this is the test that should fail first.

Setup (suite) Open All Routers, Open All Hosts

Acceptance criteria

- **${overhead}** equals **28**

How it was checked no device commands of its own; the assertion read output captured earlier in the suite

Teardown (suite) Close All Connections

18.03 PASS A Packet Exactly At The Tunnel MTU Crosses With DF Set (0.1s, 2 commands)

The permissive half of the boundary. Sourced from Loopback1 so the traffic is inside the protected selector. Warmed up first: the first packet between two spokes is consumed establishing the shortcut and its security association, so a cold measurement here reports 80 percent and says nothing about the MTU. That first-packet loss is DMVPN working, not failing.

Setup (suite) Open All Routers, Open All Hosts

Acceptance criteria

- output contains **Success rate is 100 percent**
  a packet of exactly the tunnel MTU (${TUNNEL_MTU}) did not cross

How it was checked

```
R2 ping 10.20.3.1 source Loopback1 repeat 5
```

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.2.1
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
```

R2 ping 10.20.3.1 source Loopback1 size 1472 df-bit repeat 5

```
Type escape sequence to abort.
Sending 5, 1472-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.2.1
Packet sent with the DF bit set
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
```

Teardown (suite) Close All Connections

18.04 PASS One Byte Over The Tunnel MTU Does Not Cross With DF Set (10.0s, 1 commands)

The restrictive half. Without this the test above would pass just as well against a tunnel with no MTU limit at all.

Setup (suite) Open All Routers, Open All Hosts

Acceptance criteria

- output contains **Success rate is 0 percent**

  a packet one byte over the tunnel MTU crossed with DF set

How it was checked

R2 ping 10.20.3.1 source Loopback1 size 1473 df-bit repeat 5

```
Type escape sequence to abort.
Sending 5, 1473-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.2.1
Packet sent with the DF bit set
.....
Success rate is 0 percent (0/5)
```

Teardown (suite) Close All Connections

18.05 PASS Oversize Traffic Still Crosses When Fragmentation Is Allowed (0.1s, 2 commands)

Distinguishes an MTU limit from a blackhole: the same sizes that fail with DF set must succeed without it.

Setup (suite) Open All Routers, Open All Hosts

Acceptance criteria

- output contains **Success rate is 100 percent** (checked 2×)

  a ${size}-byte packet failed even with fragmentation allowed

How it was checked

R2 ping 10.20.3.1 source Loopback1 size 1500 repeat 5

```
Type escape sequence to abort.
Sending 5, 1500-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.2.1
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/3 ms
```

R2 ping 10.20.3.1 source Loopback1 size 2000 repeat 5

```
Type escape sequence to abort.
Sending 5, 2000-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.2.1
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
```

Teardown (suite) Close All Connections

18.06 PASS Hosts Pass Sub-MTU Traffic With DF Set (4.0s, 1 commands)

End to end across two encrypt/decrypt hops, from machines that know nothing about the tunnel.

Setup (suite) Open All Routers, Open All Hosts

Acceptance criteria

• output contains **, 0% packet loss**

a host could not pass ${HOST_PAYLOAD} bytes with DF set

How it was checked

```
H2 ping -c 5 -M do -s 1444 192.168.30.10
  PING 192.168.30.10 (192.168.30.10) 1444(1472) bytes of data.
  1452 bytes from 192.168.30.10: icmp_seq=1 ttl=62 time=1.52 ms
  1452 bytes from 192.168.30.10: icmp_seq=2 ttl=62 time=1.38 ms
  1452 bytes from 192.168.30.10: icmp_seq=3 ttl=62 time=1.33 ms
  1452 bytes from 192.168.30.10: icmp_seq=4 ttl=62 time=1.39 ms
  1452 bytes from 192.168.30.10: icmp_seq=5 ttl=62 time=1.37 ms

  --- 192.168.30.10 ping statistics ---
  5 packets transmitted, 5 received, 0% packet loss, time 4006ms
  rtt min/avg/max/mdev = 1.333/1.396/1.516/0.062 ms
```

Teardown (suite) Close All Connections

18.07 PASS An End Host Learns The Tunnel Path MTU (2.1s, 1 commands)

The test that matters most in practice. The host is two hops and an encryption boundary away from the constraint, yet it must end up knowing the real limit -- otherwise a client retransmits into a black hole instead of backing off. The host reporting the exact tunnel MTU is proof that path MTU discovery survived the tunnel.

Setup (suite) Open All Routers, Open All Hosts

Acceptance criteria

• output contains **mtu=${TUNNEL_MTU}**

the host did not learn the ${TUNNEL_MTU}-byte path MTU: ${out}

How it was checked

```
H2 ping -c 2 -M do -s 1472 192.168.30.10
  PING 192.168.30.10 (192.168.30.10) 1472(1500) bytes of data.
  From 192.168.20.1 icmp_seq=1 Frag needed and DF set (mtu = 1472)
  ping: local error: message too long, mtu=1472

  --- 192.168.30.10 ping statistics ---
  2 packets transmitted, 0 received, +2 errors, 100% packet loss, time 1029ms
```

Teardown (suite) Close All Connections

18.08 PASS Hosts Pass Oversize Traffic Without DF Set (4.0s, 1 commands)

The counterpart: the same payload that was refused with DF set gets through when it may be fragmented.

Setup (suite) Open All Routers, Open All Hosts

Acceptance criteria

• output contains **, 0% packet loss**

oversize host traffic failed even with fragmentation allowed

How it was checked

```
H2 ping -c 5 -s 1472 192.168.30.10
  PING 192.168.30.10 (192.168.30.10) 1472(1500) bytes of data.
  1480 bytes from 192.168.30.10: icmp_seq=1 ttl=62 time=1.49 ms
  1480 bytes from 192.168.30.10: icmp_seq=2 ttl=62 time=1.70 ms
  1480 bytes from 192.168.30.10: icmp_seq=3 ttl=62 time=1.53 ms
  1480 bytes from 192.168.30.10: icmp_seq=4 ttl=62 time=1.91 ms
  1480 bytes from 192.168.30.10: icmp_seq=5 ttl=62 time=1.97 ms

  --- 192.168.30.10 ping statistics ---
  5 packets transmitted, 5 received, 0% packet loss, time 4006ms
  rtt min/avg/max/mdev = 1.487/1.720/1.974/0.195 ms
```

Teardown (suite) Close All Connections

19 Oob

19.01 PASS Every Router Has A Management VRF (0.0s, 3 commands)

Setup (suite) Open All Routers, Open Nms, Open Collector

Acceptance criteria

- output contains **${MGMT_VRF}** (checked 3×)
- output contains **ipv4** (checked 3×)

How it was checked

R1 show vrf
```
Name                              Default RD            Protocols   Interfaces
  MGMT                              <not set>             ipv4        Gi4
```
R2 show vrf
```
Name                              Default RD            Protocols   Interfaces
  MGMT                              <not set>             ipv4        Gi4
```
R3 show vrf
```
Name                              Default RD            Protocols   Interfaces
  MGMT                              <not set>             ipv4        Gi4
```
Teardown (suite) Restore Tunnels, Close Nms, Close Collector, Close All Connections

19.02 PASS Only The OOB Interface Is In The Management VRF (0.1s, 3 commands)

A VRF containing a data interface would defeat the point, so this asserts membership is exactly one interface.

Setup (suite) Open All Routers, Open Nms, Open Collector

Acceptance criteria

- output contains **${short}** (checked 3×)
- output does not contain **${data}** (checked 9×)

How it was checked

R1 show vrf MGMT
```
Name                              Default RD            Protocols   Interfaces
  MGMT                              <not set>             ipv4        Gi4
```
R2 show vrf MGMT
```
Name                              Default RD            Protocols   Interfaces
  MGMT                              <not set>             ipv4        Gi4
```
R3 show vrf MGMT
```
Name                              Default RD            Protocols   Interfaces
  MGMT                              <not set>             ipv4        Gi4
```
Teardown (suite) Restore Tunnels, Close Nms, Close Collector, Close All Connections

19.03 PASS Every Router Holds Its OOB Address On The Flat Network (0.1s, 3 commands)

Setup (suite) Open All Routers, Open Nms, Open Collector

Acceptance criteria

- output contains **${OOB_IP}[${r}]** (checked 3×)
- output contains **up** (checked 3×)

How it was checked

R1 show ip interface brief | include GigabitEthernet4
```
GigabitEthernet4       192.168.99.1    YES NVRAM  up                    up
```
R2 show ip interface brief | include GigabitEthernet4
```
GigabitEthernet4       192.168.99.2    YES NVRAM  up                    up
```
R3 show ip interface brief | include GigabitEthernet4
```
GigabitEthernet4       192.168.99.3    YES NVRAM  up                    up
```
Teardown (suite) Restore Tunnels, Close Nms, Close Collector, Close All Connections

19.04 PASS The Management VRF Routing Table Holds Only The Flat Network (0.1s, 3 commands)

The isolation, stated as routing: the VRF must know how to reach the management network and nothing else. A default route or a data prefix here would be a leak.

Setup (suite) Open All Routers, Open Nms, Open Collector

Acceptance criteria

- output contains **${OOB_NET}** (checked 3×)
- output does not contain **${data}** (checked 12×)

How it was checked

R1 show ip route vrf MGMT

```
Routing Table: MGMT
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
       n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       H - NHRP, G - NHRP registered, g - NHRP registration summary
       o - ODR, P - periodic downloaded static route, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR
       & - replicated local route overrides by connected

... 5 more lines
```

R2 show ip route vrf MGMT

```
Routing Table: MGMT
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
       n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       H - NHRP, G - NHRP registered, g - NHRP registration summary
       o - ODR, P - periodic downloaded static route, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR
       & - replicated local route overrides by connected

... 5 more lines
```

R3 show ip route vrf MGMT

```
Routing Table: MGMT
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
       n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       H - NHRP, G - NHRP registered, g - NHRP registration summary
       o - ODR, P - periodic downloaded static route, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR
       & - replicated local route overrides by connected

... 5 more lines
```

Teardown (suite) Restore Tunnels, Close Nms, Close Collector, Close All Connections

19.05 PASS The Global Table Cannot Reach The Management Network (0.1s, 3 commands)

The other direction. Traffic in the global table must have no path to the OOB network, otherwise the VRF is decorative.

Setup (suite) Open All Routers, Open Nms, Open Collector

Acceptance criteria

- output contains **not in table** (checked 3×)

How it was checked

```
R1 show ip route 192.168.99.0
  % Network not in table
R2 show ip route 192.168.99.0
  % Network not in table
R3 show ip route 192.168.99.0
  % Network not in table
```

Teardown (suite) Restore Tunnels, Close Nms, Close Collector, Close All Connections

19.06 PASS Every Router Answers SNMP On Its OOB Address (0.4s, 6 commands)

Setup (suite) Open All Routers, Open Nms, Open Collector

Acceptance criteria

- **${name}** starts with **${expected}[0]** (checked 3×)
  polling ${OOB_IP}[${r}] answered as '${name}', not ${r}

How it was checked

```
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.1
1.3.6.1.2.1.1.5.0
  .1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r1.lab.local"
R1 show running-config | include ^hostname
  hostname c8000v-r1
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.1.5.0
  .1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
R2 show running-config | include ^hostname
  hostname c8000v-r2
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3
1.3.6.1.2.1.1.5.0
  .1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r3.lab.local"
R3 show running-config | include ^hostname
  hostname c8000v-r3
```

Teardown (suite) Restore Tunnels, Close Nms, Close Collector, Close All Connections

19.07 PASS Management Traffic Never Touches The Data Path (3.3s, 28 commands)

The negative property that makes this out-of-band. Polling a spoke used to cross an IPsec tunnel and was asserted to advance the ESP counters; now it must do the opposite. The comparison is against an idle window of the same length, because BGP and BFD tick those counters continuously regardless -- so an absolute delta proves nothing.

Setup (suite) Open All Routers, Open Nms, Open Collector

Acceptance criteria

- **${attributable} < ${POLL_BURST}** holds

How it was checked

```
R1 show crypto ipsec sa
  interface: Tunnel0
      Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

     protected vrf: (none)
     local  ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
     remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
     current_peer 100.64.0.3 port 500
       PERMIT, flags={origin_is_acl,}
      #pkts encaps: 5281, #pkts encrypt: 5281, #pkts digest: 5281
      #pkts decaps: 5280, #pkts decrypt: 5280, #pkts verify: 5280
      #pkts compressed: 0, #pkts decompressed: 0
      #pkts not compressed: 0, #pkts compr. failed: 0
      #pkts not decompressed: 0, #pkts decompress failed: 0
      #send errors 0, #recv errors 0
  ... 78 more lines
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.1.5.0
  .1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
```

```
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3
1.3.6.1.2.1.1.5.0
  .1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r3.lab.local"
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.1.5.0
  .1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3
1.3.6.1.2.1.1.5.0
  .1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r3.lab.local"
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.1.5.0
  .1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3
1.3.6.1.2.1.1.5.0
  .1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r3.lab.local"
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.1.5.0
  .1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3
1.3.6.1.2.1.1.5.0
  .1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r3.lab.local"
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.1.5.0
  .1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3
1.3.6.1.2.1.1.5.0
  .1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r3.lab.local"
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.1.5.0
  .1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3
1.3.6.1.2.1.1.5.0
  .1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r3.lab.local"
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.1.5.0
  .1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3
1.3.6.1.2.1.1.5.0
  .1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r3.lab.local"
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.1.5.0
  .1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3
1.3.6.1.2.1.1.5.0
  .1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r3.lab.local"
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.1.5.0
  .1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3
1.3.6.1.2.1.1.5.0
  .1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r3.lab.local"
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.1.5.0
  .1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3
1.3.6.1.2.1.1.5.0
  .1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r3.lab.local"
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.1.5.0
  .1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3
1.3.6.1.2.1.1.5.0
  .1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r3.lab.local"
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2
1.3.6.1.2.1.1.5.0
  .1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
```

```
NMS snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.3
1.3.6.1.2.1.1.5.0
  .1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r3.lab.local"
```

```
R1 show crypto ipsec sa
  interface: Tunnel0
      Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

     protected vrf: (none)
     local  ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
     remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
     current_peer 100.64.0.3 port 500
       PERMIT, flags={origin_is_acl,}
      #pkts encaps: 5286, #pkts encrypt: 5286, #pkts digest: 5286
      #pkts decaps: 5290, #pkts decrypt: 5290, #pkts verify: 5290
      #pkts compressed: 0, #pkts decompressed: 0
      #pkts not compressed: 0, #pkts compr. failed: 0
      #pkts not decompressed: 0, #pkts decompress failed: 0
      #send errors 0, #recv errors 0
  ... 78 more lines
```

```
R1 show crypto ipsec sa
  interface: Tunnel0
      Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

     protected vrf: (none)
     local  ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
     remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
     current_peer 100.64.0.3 port 500
       PERMIT, flags={origin_is_acl,}
      #pkts encaps: 5286, #pkts encrypt: 5286, #pkts digest: 5286
      #pkts decaps: 5290, #pkts decrypt: 5290, #pkts verify: 5290
      #pkts compressed: 0, #pkts decompressed: 0
      #pkts not compressed: 0, #pkts compr. failed: 0
      #pkts not decompressed: 0, #pkts decompress failed: 0
      #send errors 0, #recv errors 0
  ... 78 more lines
```

```
R1 show crypto ipsec sa
  interface: Tunnel0
      Crypto map tag: Tunnel0-head-0, local addr 100.64.0.1

     protected vrf: (none)
     local  ident (addr/mask/prot/port): (100.64.0.1/255.255.255.255/47/0)
     remote ident (addr/mask/prot/port): (100.64.0.3/255.255.255.255/47/0)
     current_peer 100.64.0.3 port 500
       PERMIT, flags={origin_is_acl,}
      #pkts encaps: 5298, #pkts encrypt: 5298, #pkts digest: 5298
      #pkts decaps: 5297, #pkts decrypt: 5297, #pkts verify: 5297
      #pkts compressed: 0, #pkts decompressed: 0
      #pkts not compressed: 0, #pkts compr. failed: 0
      #pkts not decompressed: 0, #pkts decompress failed: 0
      #send errors 0, #recv errors 0
  ... 78 more lines
```

Teardown (suite) Restore Tunnels, Close Nms, Close Collector, Close All Connections

19.08 PASS Syslog Reaches The Collector From The OOB Address (12.2s, 4 commands)

rsyslog files are named for the sender. A file named after a LAN address would mean the router is logging from the data path despite the configuration. Each router is made to emit a marked message rather than waiting on ambient traffic: whether a router happens to have logged recently is not what this test is about, and depending on it makes the result a matter of timing.

Setup (suite) Open All Routers, Open Nms, Open Collector

Acceptance criteria

- output contains **${OOB_IP}[${r}]** (checked 3×)
- **${n} > 0** holds (checked 3×)
- output does not contain **${${r}_LAN_IP}.log** (checked 3×)

How it was checked

```
NMS sudo grep -c -- 'LABOOB-SRC-R1-211157' /var/log/lab/192.168.99.1.log 2>/dev/null || echo 0
  1
```

```
NMS sudo grep -c -- 'LABOOB-SRC-R2-211157' /var/log/lab/192.168.99.2.log 2>/dev/null || echo 0
  1
```

```
NMS sudo grep -c -- 'LABOOB-SRC-R3-211157' /var/log/lab/192.168.99.3.log 2>/dev/null || echo 0
```

```
  1
```

NMS `sudo ls -l /var/log/lab`

```
  total 11420
  -rw-r--r-- 1 syslog syslog  564368 Sep  9 01:12 192.168.99.1.log
  -rw-r--r-- 1 syslog syslog  697825 Sep  9 01:12 192.168.99.2.log
  -rw-r--r-- 1 syslog syslog  479658 Sep  9 01:12 192.168.99.3.log
  -rw-r--r-- 1 root   root   3102008 Sep  9 01:11 snmptrapd.log
  -rw-r--r-- 1 root   root   6818873 Sep  9 01:11 traps-raw.log
```

Teardown (suite) Restore Tunnels, Close Nms, Close Collector, Close All Connections

19.09 PASS A Spoke Stays Manageable With Its Tunnel Down (19.4s, 16 commands)

The reason out-of-band management exists. Shutting R2's tunnel removes every data path between the hub and the spoke -- BGP drops, the LAN becomes unreachable -- and the spoke must still answer SNMP and keep logging, because none of that ever used the tunnel.

Setup (suite) Open All Routers, Open Nms, Open Collector

Acceptance criteria

- output contains **not in table** (checked 2×)
  the hub can still reach R2's LAN, so the data path is not down yet
- **${before}** equals **${after}**
  R2 stopped answering SNMP once its tunnel went down, so management is not out-of-band
- **${n} > 0** holds
- output matches **${peer}\s+4\s+${peer_asn}(\s+\d+){5}\s+\S+\s+\d+** (checked 3×)

How it was checked

NMS `snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2 1.3.6.1.2.1.1.5.0`

```
  .1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
```

R2 `configure terminal`

```
  Enter configuration commands, one per line.  End with CNTL/Z.
```

R2 `interface Tunnel0`

```
  ${output} =
```

R2 `shutdown`

```
  ${output} =
```

R2 `end`

```
  ${output} =
```

R1 `show ip route 192.168.20.0`

```
  Routing entry for 192.168.20.0/24
    Known via "bgp 65001", distance 20, metric 0
    Tag 65002, type external
    Last update from 172.20.0.2 00:08:37 ago
    Routing Descriptor Blocks:
    * 172.20.0.2, from 172.20.0.2, 00:08:37 ago
        opaque_ptr 0x71FE1BFD8DB8
        Route metric is 0, traffic share count is 1
        AS Hops 1
        Route tag 65002
        MPLS label: none
```

R1 `show ip route 192.168.20.0`

```
  % Network not in table
```

NMS `snmpget -On -v3 -l authPriv -u labmon -a SHA -A LabAuthPass123 -x AES -X LabPrivPass123 -t 3 -r 1 192.168.99.2 1.3.6.1.2.1.1.5.0`

```
  .1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
```

NMS `sudo grep -c -- 'LABOOB-211215' /var/log/lab/192.168.99.2.log 2>/dev/null || echo 0`

```
  1
```

R2 `configure terminal`

```
  Enter configuration commands, one per line.  End with CNTL/Z.
```

R2 `interface Tunnel0`

```
  ${output} =
```

R2 `no shutdown`

```
  ${output} =
```

R2 `end`

```
  ${output} =
```

R1 `show bgp ipv4 unicast summary`

```
  BGP router identifier 172.16.1.1, local AS number 65001
```

```
BGP table version is 13, main routing table version 13
6 network entries using 1488 bytes of memory
6 path entries using 816 bytes of memory
4/4 BGP path/bestpath attribute entries using 1184 bytes of memory
1 BGP AS-PATH entries using 24 bytes of memory
3 BGP community entries using 72 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 3584 total bytes of memory
BGP activity 365/359 prefixes, 386/380 paths, scan interval 60 secs
22 networks peaked at 02:27:59 Sep 8 2026 UTC (22:44:20.389 ago)

Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
... 2 more lines
```

R1 **show bgp ipv4 unicast summary**

```
BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 13, main routing table version 13
6 network entries using 1488 bytes of memory
6 path entries using 816 bytes of memory
4/4 BGP path/bestpath attribute entries using 1184 bytes of memory
1 BGP AS-PATH entries using 24 bytes of memory
3 BGP community entries using 72 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 3584 total bytes of memory
BGP activity 365/359 prefixes, 386/380 paths, scan interval 60 secs
22 networks peaked at 02:27:59 Sep 8 2026 UTC (22:44:25.425 ago)

Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
... 2 more lines
```

R1 **show bgp ipv4 unicast summary**

```
BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 16, main routing table version 16
9 network entries using 2232 bytes of memory
9 path entries using 1224 bytes of memory
7/7 BGP path/bestpath attribute entries using 2072 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
6 BGP community entries using 144 bytes of memory
3 BGP route-map cache entries using 216 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 5936 total bytes of memory
BGP activity 368/359 prefixes, 389/380 paths, scan interval 60 secs
22 networks peaked at 02:27:59 Sep 8 2026 UTC (22:44:30.462 ago)

Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
... 2 more lines
```

Teardown (this test) Restore Tunnels

20 Tacacs

20.01 PASS The TACACS Server Runs On The NMS (0.1s, 2 commands)

Setup (suite) Open All Routers, Open Tacacs Server

Acceptance criteria

- **${state}** equals **active**

  tac_plus is not running on the NMS
- output contains **:${TAC_PORT}**
- output contains **tac_plus**

How it was checked

```
NMS systemctl is-active tacacs-lab
  active
NMS sudo ss -ltnp | grep ':49 ' || echo NONE
  LISTEN 0      128          0.0.0.0:49        0.0.0.0:*    users:(("tac_plus",pid=1460,fd=4))
  LISTEN 0      128             [::]:49           [::]:*    users:(("tac_plus",pid=1460,fd=5))
```

Teardown (suite) Restore Oob Interfaces, Close Tacacs Server, Close All Connections

20.02 PASS The Server Holds Both Remote Users And The Automation Account (0.1s, 1 commands)

The automation account has to exist on the server as well as locally on each router. Without it the first router to get AAA becomes unmanageable by this very test suite, while the server sits there perfectly healthy.

Setup (suite) Open All Routers, Open Tacacs Server

Acceptance criteria

- output contains **${ADMIN_USER}**
- output contains **${OPS_USER}**
- output contains **${USERNAME}**

  the automation account is missing from the server, which would lock the suite out

How it was checked

```
NMS sudo grep -oE '^user = [A-Za-z0-9_-]+' /etc/tacacs+/tac_plus.conf
  user = netadmin
  user = netops
  user = lab
```

Teardown (suite) Restore Oob Interfaces, Close Tacacs Server, Close All Connections

20.03 PASS Every Router Points At The Server Through The Management VRF (0.4s, 3 commands)

Setup (suite) Open All Routers, Open Tacacs Server

Acceptance criteria

- output contains **aaa new-model** (checked 3×)
- output contains **address ipv4 ${NMS_OOB_IP}** (checked 3×)
- output contains **ip vrf forwarding ${MGMT_VRF}** (checked 3×)
- output contains **ip tacacs source-interface ${OOB_INTF}[${r}]** (checked 3×)

How it was checked

```
R1 show running-config | include ^aaa |^tacacs server|^ address ipv4|^ ip vrf forwarding|^ ip tacacs source
  aaa new-model
  aaa group server tacacs+ LABTACGRP
   ip vrf forwarding MGMT
   ip tacacs source-interface GigabitEthernet4
  aaa authentication login default group LABTACGRP local
  aaa authentication enable default group LABTACGRP enable
  aaa authorization exec default group LABTACGRP local if-authenticated
  aaa accounting exec default start-stop group LABTACGRP
```

```
aaa accounting commands 15 default start-stop group LABTACGRP
aaa session-id common
tacacs server LAB-TAC
 address ipv4 192.168.99.10
```

R2 show running-config | include ^aaa |^tacacs server|^ address ipv4|^ ip vrf forwarding|^ ip tacacs source

```
aaa new-model
aaa group server tacacs+ LABTACGRP
 ip vrf forwarding MGMT
 ip tacacs source-interface GigabitEthernet4
aaa authentication login default group LABTACGRP local
aaa authentication enable default group LABTACGRP enable
aaa authorization exec default group LABTACGRP local if-authenticated
aaa accounting exec default start-stop group LABTACGRP
aaa accounting commands 15 default start-stop group LABTACGRP
aaa session-id common
tacacs server LAB-TAC
 address ipv4 192.168.99.10
```

R3 show running-config | include ^aaa |^tacacs server|^ address ipv4|^ ip vrf forwarding|^ ip tacacs source

```
aaa new-model
aaa group server tacacs+ LABTACGRP
 ip vrf forwarding MGMT
 ip tacacs source-interface GigabitEthernet4
aaa authentication login default group LABTACGRP local
aaa authentication enable default group LABTACGRP enable
aaa authorization exec default group LABTACGRP local if-authenticated
aaa accounting exec default start-stop group LABTACGRP
aaa accounting commands 15 default start-stop group LABTACGRP
aaa session-id common
tacacs server LAB-TAC
 address ipv4 192.168.99.10
```

Teardown (suite) Restore Oob Interfaces, Close Tacacs Server, Close All Connections

20.04 PASS Every Method List Falls Back To The Local Database (0.4s, 3 commands)

A method list with no local fallback locks the lab out the moment the server is unreachable -- including from the console, which is the one way back in.

Setup (suite) Open All Routers, Open Tacacs Server

Acceptance criteria

- output contains **login default group ${TAC_GROUP} local** (checked 3×)
- output contains **exec default group ${TAC_GROUP} local** (checked 3×)

How it was checked

R1 show running-config | include ^aaa authentication login|^aaa authorization exec

```
aaa authentication login default group LABTACGRP local
aaa authorization exec default group LABTACGRP local if-authenticated
```

R2 show running-config | include ^aaa authentication login|^aaa authorization exec

```
aaa authentication login default group LABTACGRP local
aaa authorization exec default group LABTACGRP local if-authenticated
```

R3 show running-config | include ^aaa authentication login|^aaa authorization exec

```
aaa authentication login default group LABTACGRP local
aaa authorization exec default group LABTACGRP local if-authenticated
```

Teardown (suite) Restore Oob Interfaces, Close Tacacs Server, Close All Connections

20.05 PASS Every Router Reports The Server Alive (0.1s, 3 commands)

Setup (suite) Open All Routers, Open Tacacs Server

Acceptance criteria

- output contains **Server Status: Alive** (checked 3×)
- output contains **${NMS_OOB_IP}** (checked 3×)

How it was checked

R1 show tacacs

```
Tacacs+ Server -  public  :
               Server name: LAB-TAC
            Server address: 192.168.99.10
               Server port: 49
```

```
              Socket opens:       2336
             Socket closes:       2336
             Socket aborts:          0
             Socket errors:          0
           Socket Timeouts:          0
   Failed Connect Attempts:          0
        Total Packets Sent:       2226
        Total Packets Recv:       2226
             Server Status: Alive
Continous Authc fail count:          3
... 1 more lines
```

R2 **show tacacs**

```
Tacacs+ Server -  public  :
               Server name: LAB-TAC
            Server address: 192.168.99.10
               Server port: 49
              Socket opens:       3609
             Socket closes:       3609
             Socket aborts:          0
             Socket errors:          0
           Socket Timeouts:          0
   Failed Connect Attempts:          0
        Total Packets Sent:       3506
        Total Packets Recv:       3506
             Server Status: Alive
Continous Authc fail count:          3
... 1 more lines
```

R3 **show tacacs**

```
Tacacs+ Server -  public  :
               Server name: LAB-TAC
            Server address: 192.168.99.10
               Server port: 49
              Socket opens:       1594
             Socket closes:       1594
             Socket aborts:          0
             Socket errors:          0
           Socket Timeouts:          0
   Failed Connect Attempts:          0
        Total Packets Sent:       1517
        Total Packets Recv:       1517
             Server Status: Alive
Continous Authc fail count:          3
... 1 more lines
```

Teardown (suite) Restore Oob Interfaces, Close Tacacs Server, Close All Connections

20.06 PASS A Remote Administrator Authenticates And Is Given Privilege 15 (0.2s, 0 commands)

Setup (suite) Open All Routers, Open Tacacs Server

Acceptance criteria

- **${priv}** equals **${ADMIN_PRIV}** (checked 3×)

How it was checked no device commands of its own; the assertion read output captured earlier in the suite

Teardown (suite) Restore Oob Interfaces, Close Tacacs Server, Close All Connections

20.07 PASS A Remote Operator Authenticates And Is Held At Privilege 1 (0.2s, 0 commands)

The pair that proves the level comes from the server. Two users, same router, same method list, different levels -- which cannot happen if the router is applying a default.

Setup (suite) Open All Routers, Open Tacacs Server

Acceptance criteria

- **${priv}** equals **${OPS_PRIV}** (checked 3×)

How it was checked no device commands of its own; the assertion read output captured earlier in the suite

Teardown (suite) Restore Oob Interfaces, Close Tacacs Server, Close All Connections

20.08 PASS A Wrong Password Is Refused (2.0s, 0 commands)

Setup (suite) Open All Routers, Open Tacacs Server

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked no device commands of its own; the assertion read output captured earlier in the suite

Teardown (suite) Restore Oob Interfaces, Close Tacacs Server, Close All Connections

20.09 PASS An Unknown User Is Refused (2.1s, 0 commands)

The server rejects the user and the router honours that rather than quietly trying the local database.

Setup (suite) Open All Routers, Open Tacacs Server

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked no device commands of its own; the assertion read output captured earlier in the suite

Teardown (suite) Restore Oob Interfaces, Close Tacacs Server, Close All Connections

20.10 PASS The Server Records Authentication In Its Accounting Log (0.1s, 1 commands)

Accounting is what makes remote authentication auditable; without it the router knows who logged in and nobody else ever finds out.

Setup (suite) Open All Routers, Open Tacacs Server

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked

```
NMS sudo tail -200 /var/log/tac_plus.acct 2>/dev/null || true
  Sep  9 01:09:32 192.168.99.3    lab     tty434  10.0.2.2        stop    task_id=5108    timezone=UTC
  service=shell   start_time=1788916081   disc-cause=1    disc-cause-ext=9        pre-session-time=0
  elapsed_time=91 stop_time=1788916172
  Sep  9 01:09:32 192.168.99.2    lab     tty435  10.0.2.2        stop    task_id=7074    timezone=UTC
  service=shell   start_time=1788916081   disc-cause=1    disc-cause-ext=9        pre-session-time=0
  elapsed_time=91 stop_time=1788916172
  Sep  9 01:09:32 192.168.99.1    lab     tty434  10.0.2.2        stop    task_id=5815    timezone=UTC
  service=shell   start_time=1788916081   disc-cause=1    disc-cause-ext=9        pre-session-time=0
  elapsed_time=92 stop_time=1788916173
  Sep  9 01:09:33 192.168.99.1    lab     tty435  10.0.2.2        start   task_id=5836    timezone=UTC
  service=shell   start_time=1788916173
  Sep  9 01:09:33 192.168.99.2    lab     tty434  10.0.2.2        start   task_id=7082    timezone=UTC
  service=shell   start_time=1788916173
  Sep  9 01:09:33 192.168.99.3    lab     tty434  10.0.2.2        start   task_id=5114    timezone=UTC
  service=shell   start_time=1788916173
  Sep  9 01:09:33 192.168.99.1    lab     tty435  10.0.2.2        stop    task_id=5836    timezone=UTC
  service=shell   start_time=1788916173   priv-lvl=15     cmd=show running-config <cr>
  Sep  9 01:09:33 192.168.99.2    lab     tty434  10.0.2.2        stop    task_id=7082    timezone=UTC
  service=shell   start_time=1788916173   priv-lvl=15     cmd=show running-config <cr>
  Sep  9 01:09:33 192.168.99.3    lab     tty434  10.0.2.2        stop    task_id=5114    timezone=UTC
  service=shell   start_time=1788916173   priv-lvl=15     cmd=show running-config <cr>
  Sep  9 01:09:36 192.168.99.1    lab     tty435  10.0.2.2        stop    task_id=5836    timezone=UTC
  service=shell   start_time=1788916173   disc-cause=1    disc-cause-ext=9        pre-session-time=0
  elapsed_time=3  stop_time=1788916176
  Sep  9 01:09:36 192.168.99.2    lab     tty434  10.0.2.2        stop    task_id=7082    timezone=UTC
  service=shell   start_time=1788916173   disc-cause=1    disc-cause-ext=9        pre-session-time=0
  elapsed_time=3  stop_time=1788916176
  Sep  9 01:09:36 192.168.99.3    lab     tty434  10.0.2.2        stop    task_id=5114    timezone=UTC
  service=shell   start_time=1788916173   disc-cause=1    disc-cause-ext=9        pre-session-time=0
  elapsed_time=3  stop_time=1788916176
  Sep  9 01:09:36 192.168.99.1    lab     tty434  10.0.2.2        start   task_id=5838    timezone=UTC
  service=shell   start_time=1788916176
  Sep  9 01:09:36 192.168.99.2    lab     tty435  10.0.2.2        start   task_id=7084    timezone=UTC
  service=shell   start_time=1788916176
  ... 186 more lines
```

Teardown (suite) Restore Oob Interfaces, Close Tacacs Server, Close All Connections

20.11 PASS Accounting Records Carry The Privilege Level And Come From The OOB Address (0.1s, 1 commands)

Setup (suite) Open All Routers, Open Tacacs Server

Acceptance criteria

- output contains **service=shell**
- output contains **${R3_OOB_IP}**

  the record did not come from R3's out-of-band address

How it was checked

NMS **sudo tail -200 /var/log/tac_plus.acct 2>/dev/null || true**
```
Sep  9 01:09:32 192.168.99.3    lab     tty434  10.0.2.2        stop    task_id=5108    timezone=UTC
service=shell   start_time=1788916081   disc-cause=1    disc-cause-ext=9        pre-session-time=0
elapsed_time=91 stop_time=1788916172
Sep  9 01:09:32 192.168.99.2    lab     tty435  10.0.2.2        stop    task_id=7074    timezone=UTC
service=shell   start_time=1788916081   disc-cause=1    disc-cause-ext=9        pre-session-time=0
elapsed_time=91 stop_time=1788916172
Sep  9 01:09:32 192.168.99.1    lab     tty434  10.0.2.2        stop    task_id=5815    timezone=UTC
service=shell   start_time=1788916081   disc-cause=1    disc-cause-ext=9        pre-session-time=0
elapsed_time=92 stop_time=1788916173
Sep  9 01:09:33 192.168.99.1    lab     tty435  10.0.2.2        start   task_id=5836    timezone=UTC
service=shell   start_time=1788916173
Sep  9 01:09:33 192.168.99.2    lab     tty434  10.0.2.2        start   task_id=7082    timezone=UTC
service=shell   start_time=1788916173
Sep  9 01:09:33 192.168.99.3    lab     tty434  10.0.2.2        start   task_id=5114    timezone=UTC
service=shell   start_time=1788916173
Sep  9 01:09:33 192.168.99.1    lab     tty435  10.0.2.2        stop    task_id=5836    timezone=UTC
service=shell   start_time=1788916173   priv-lvl=15     cmd=show running-config <cr>
Sep  9 01:09:33 192.168.99.2    lab     tty434  10.0.2.2        stop    task_id=7082    timezone=UTC
service=shell   start_time=1788916173   priv-lvl=15     cmd=show running-config <cr>
Sep  9 01:09:33 192.168.99.3    lab     tty434  10.0.2.2        stop    task_id=5114    timezone=UTC
service=shell   start_time=1788916173   priv-lvl=15     cmd=show running-config <cr>
Sep  9 01:09:36 192.168.99.1    lab     tty435  10.0.2.2        stop    task_id=5836    timezone=UTC
service=shell   start_time=1788916173   disc-cause=1    disc-cause-ext=9        pre-session-time=0
elapsed_time=3  stop_time=1788916176
Sep  9 01:09:36 192.168.99.2    lab     tty434  10.0.2.2        stop    task_id=7082    timezone=UTC
service=shell   start_time=1788916173   disc-cause=1    disc-cause-ext=9        pre-session-time=0
elapsed_time=3  stop_time=1788916176
Sep  9 01:09:36 192.168.99.3    lab     tty434  10.0.2.2        stop    task_id=5114    timezone=UTC
service=shell   start_time=1788916173   disc-cause=1    disc-cause-ext=9        pre-session-time=0
elapsed_time=3  stop_time=1788916176
Sep  9 01:09:36 192.168.99.1    lab     tty434  10.0.2.2        start   task_id=5838    timezone=UTC
service=shell   start_time=1788916176
Sep  9 01:09:36 192.168.99.2    lab     tty435  10.0.2.2        start   task_id=7084    timezone=UTC
service=shell   start_time=1788916176
... 186 more lines
```

Teardown (suite) Restore Oob Interfaces, Close Tacacs Server, Close All Connections

20.12 PASS The Local Account Still Works When The Server Is Unreachable (21.5s, 21 commands)

The resilience property. R2's out-of-band interface is shut, which is the only path to the server, so TACACS becomes unreachable rather than merely unwilling. The automation account must still get in through the local database, and a remote-only user must not -- it exists nowhere else. The precondition is a ping inside the VRF, not "show tacacs". That field reports cached state -- it keeps saying Alive until a transaction actually fails -- so waiting on it would either hang or, worse, let this test run with the server still reachable and prove nothing.

Setup (suite) Open All Routers, Open Tacacs Server

Acceptance criteria

- output matches **Success rate is 0 percent|does not have a usable source address**
- **${priv}** equals **15**

  the local account could not get in with the server unreachable
- output contains **Success rate is 100 percent** (checked 4×)

How it was checked

R2 **configure terminal**
```
Enter configuration commands, one per line.  End with CNTL/Z.
```

R2 **interface GigabitEthernet4**
```
${output} =
```

R2 **shutdown**
```
${output} =
```

R2 **end**

```
${output} =
```

R2 **ping vrf MGMT 192.168.99.10 repeat 2**

```
% VRF MGMT does not have a usable source address
```

R1 **configure terminal**

```
Enter configuration commands, one per line.  End with CNTL/Z.
```

R1 **interface GigabitEthernet4**

```
${output} =
```

R1 **no shutdown**

```
${output} =
```

R1 **end**

```
${output} =
```

R2 **configure terminal**

```
Enter configuration commands, one per line.  End with CNTL/Z.
```

R2 **interface GigabitEthernet4**

```
${output} =
```

R2 **no shutdown**

```
${output} =
```

R2 **end**

```
${output} =
```

R3 **configure terminal**

```
Enter configuration commands, one per line.  End with CNTL/Z.
```

R3 **interface GigabitEthernet4**

```
${output} =
```

R3 **no shutdown**

```
${output} =
```

R3 **end**

```
${output} =
```

R2 **ping vrf MGMT 192.168.99.10 repeat 5**

```
% VRF MGMT does not have a usable source address
```

R2 **ping vrf MGMT 192.168.99.10 repeat 5**

```
% VRF MGMT does not have a usable source address
```

R2 **ping vrf MGMT 192.168.99.10 repeat 5**

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.99.10, timeout is 2 seconds:
..!!!
Success rate is 60 percent (3/5), round-trip min/avg/max = 1/1/1 ms
```

R2 **ping vrf MGMT 192.168.99.10 repeat 5**

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.99.10, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
```

Teardown (this test) Restore Oob Interfaces

21 Vty Acl

21.01 PASS Every Router Applies The ACL Inbound On All VTY Lines (0.3s, 3 commands)

Setup (suite) Open All Routers, Open Tacacs Server, Open Collector

Acceptance criteria

- output contains **access-class ${VTY_ACL} in vrf-also** (checked 3×)

How it was checked

R1 show running-config | section ^line vty

```
line vty 0 4
 access-class VTY-MGMT in vrf-also
 exec-timeout 0 0
 activation-character 13
 transport input ssh
line vty 5 15
 access-class VTY-MGMT in vrf-also
 exec-timeout 0 0
 activation-character 13
 transport input ssh
```

R2 show running-config | section ^line vty

```
line vty 0 4
 access-class VTY-MGMT in vrf-also
 exec-timeout 0 0
 activation-character 13
 transport input ssh
line vty 5 15
 access-class VTY-MGMT in vrf-also
 exec-timeout 0 0
 activation-character 13
 transport input ssh
```

R3 show running-config | section ^line vty

```
line vty 0 4
 access-class VTY-MGMT in vrf-also
 exec-timeout 0 0
 activation-character 13
 transport input ssh
line vty 5 15
 access-class VTY-MGMT in vrf-also
 exec-timeout 0 0
 activation-character 13
 transport input ssh
```

Teardown (suite) Close Tacacs Server, Close Collector, Close All Connections

21.02 PASS Only SSH Is Accepted On The VTY Lines (0.3s, 3 commands)

An ACL restricting who may connect is worth less if telnet is still an option for those who may.

Setup (suite) Open All Routers, Open Tacacs Server, Open Collector

Acceptance criteria

- output contains **transport input ssh** (checked 3×)
- output does not contain **transport input all** (checked 3×)
- output does not contain **transport input telnet** (checked 3×)

How it was checked

R1 show running-config | section ^line vty

```
line vty 0 4
 access-class VTY-MGMT in vrf-also
 exec-timeout 0 0
 activation-character 13
 transport input ssh
line vty 5 15
 access-class VTY-MGMT in vrf-also
 exec-timeout 0 0
 activation-character 13
 transport input ssh
```

R2 show running-config | section ^line vty

```
line vty 0 4
 access-class VTY-MGMT in vrf-also
```

```
 exec-timeout 0 0
 activation-character 13
 transport input ssh
line vty 5 15
 access-class VTY-MGMT in vrf-also
 exec-timeout 0 0
 activation-character 13
 transport input ssh
```

R3 show running-config | section ^line vty

```
line vty 0 4
 access-class VTY-MGMT in vrf-also
 exec-timeout 0 0
 activation-character 13
 transport input ssh
line vty 5 15
 access-class VTY-MGMT in vrf-also
 exec-timeout 0 0
 activation-character 13
 transport input ssh
```

Teardown (suite) Close Tacacs Server, Close Collector, Close All Connections

21.03 PASS The ACL Permits The Management Network And Denies The Rest (0.1s, 3 commands)

Setup (suite) Open All Routers, Open Tacacs Server, Open Collector

Acceptance criteria

- output contains **permit ${OOB_NET}, wildcard bits 0.0.0.255** (checked 3×)
- output contains **deny** (checked 3×)
  any log

How it was checked

R1 show ip access-lists VTY-MGMT

```
Standard IP access list VTY-MGMT
    10 permit 192.168.99.0, wildcard bits 0.0.0.255 (6 matches)
    20 permit 10.0.2.0, wildcard bits 0.0.0.255 (274 matches)
    30 deny   any log (9 matches)
```

R2 show ip access-lists VTY-MGMT

```
Standard IP access list VTY-MGMT
    10 permit 192.168.99.0, wildcard bits 0.0.0.255
    20 permit 10.0.2.0, wildcard bits 0.0.0.255 (322 matches)
    30 deny   any log (3 matches)
```

R3 show ip access-lists VTY-MGMT

```
Standard IP access list VTY-MGMT
    10 permit 192.168.99.0, wildcard bits 0.0.0.255
    20 permit 10.0.2.0, wildcard bits 0.0.0.255 (274 matches)
    30 deny   any log (3 matches)
```

Teardown (suite) Close Tacacs Server, Close Collector, Close All Connections

21.04 PASS The Documented Exception Is Present And Is Only The Hypervisor Path (0.1s, 3 commands)

Pinned deliberately. This permit is the one concession in the policy, so it should be visible in a test rather than only in a comment -- and widening it should break something.

Setup (suite) Open All Routers, Open Tacacs Server, Open Collector

Acceptance criteria

- output contains **permit ${MGMT_NAT_NET}, wildcard bits 0.0.0.255** (checked 3×)
- **${permits}** has exactly **2** entries (checked 3×)

How it was checked

R1 show ip access-lists VTY-MGMT

```
Standard IP access list VTY-MGMT
    10 permit 192.168.99.0, wildcard bits 0.0.0.255 (6 matches)
    20 permit 10.0.2.0, wildcard bits 0.0.0.255 (274 matches)
    30 deny   any log (9 matches)
```

```
R2 show ip access-lists VTY-MGMT
  Standard IP access list VTY-MGMT
      10 permit 192.168.99.0, wildcard bits 0.0.0.255
      20 permit 10.0.2.0, wildcard bits 0.0.0.255 (322 matches)
      30 deny   any log (3 matches)
R3 show ip access-lists VTY-MGMT
  Standard IP access list VTY-MGMT
      10 permit 192.168.99.0, wildcard bits 0.0.0.255
      20 permit 10.0.2.0, wildcard bits 0.0.0.255 (274 matches)
      30 deny   any log (3 matches)
```

Teardown (suite) Close Tacacs Server, Close Collector, Close All Connections

21.05 PASS SSH From The Management Network Is Allowed (0.0s, 1 commands)

Setup (suite) Open All Routers, Open Tacacs Server, Open Collector

Acceptance criteria

- **${banner}** starts with **SSH-**

How it was checked

```
NMS python3 -c "import socket,sys
  s=socket.socket()
  s.settimeout(6)
  s.bind(('192.168.99.10',0))
  try:
      s.connect(('192.168.99.1',22))
      print(s.recv(64).decode(errors='replace').strip() or 'EMPTY')
  except Exception as e:
      print(f'REFUSED {type(e).__name__}: {e}')
  "
  SSH-2.0-Cisco-1.25
```

Teardown (suite) Close Tacacs Server, Close Collector, Close All Connections

21.06 PASS SSH From Outside The Management Network Is Refused (0.1s, 1 commands)

The same machine, a different source address, and a target it is directly connected to -- so nothing but the ACL can account for the difference. Before the access-class was applied this path returned a banner just like the one above.

Setup (suite) Open All Routers, Open Tacacs Server, Open Collector

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked

```
NMS python3 -c "import socket,sys
  s=socket.socket()
  s.settimeout(6)
  s.bind(('192.168.10.20',0))
  try:
      s.connect(('192.168.10.1',22))
      print(s.recv(64).decode(errors='replace').strip() or 'EMPTY')
  except Exception as e:
      print(f'REFUSED {type(e).__name__}: {e}')
  "
  REFUSED ConnectionRefusedError: [Errno 111] Connection refused
```

Teardown (suite) Close Tacacs Server, Close Collector, Close All Connections

21.07 PASS Every Router Refuses A Login From Outside The Management Network (0.2s, 3 commands)

Setup (suite) Open All Routers, Open Tacacs Server, Open Collector

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked

```
NMS python3 -c "import socket,sys
  s=socket.socket()
  s.settimeout(6)
```

```
s.bind(('192.168.10.20',0))
try:
    s.connect(('192.168.10.1',22))
    print(s.recv(64).decode(errors='replace').strip() or 'EMPTY')
except Exception as e:
    print(f'REFUSED {type(e).__name__}: {e}')
"
REFUSED ConnectionRefusedError: [Errno 111] Connection refused
```

NMS python3 -c "import socket,sys

```
s=socket.socket()
s.settimeout(6)
s.bind(('192.168.10.20',0))
try:
    s.connect(('192.168.20.1',22))
    print(s.recv(64).decode(errors='replace').strip() or 'EMPTY')
except Exception as e:
    print(f'REFUSED {type(e).__name__}: {e}')
"
REFUSED ConnectionRefusedError: [Errno 111] Connection refused
```

NMS python3 -c "import socket,sys

```
s=socket.socket()
s.settimeout(6)
s.bind(('192.168.10.20',0))
try:
    s.connect(('192.168.30.1',22))
    print(s.recv(64).decode(errors='replace').strip() or 'EMPTY')
except Exception as e:
    print(f'REFUSED {type(e).__name__}: {e}')
"
REFUSED ConnectionRefusedError: [Errno 111] Connection refused
```

Teardown (suite) Close Tacacs Server, Close Collector, Close All Connections

21.08 PASS The Refusal Is Logged And Reaches The Collector (0.1s, 2 commands)

"deny any log" makes a rejected connection auditable rather than merely blocked, and the message travels the OOB network to the collector like every other log.

Setup (suite) Open All Routers, Open Tacacs Server, Open Collector

Acceptance criteria

- output matches **SEC-6-IPACCESSLOG\S***

  no access-list denial logged by ${router} reached the collector

How it was checked

NMS python3 -c "import socket,sys

```
s=socket.socket()
s.settimeout(6)
s.bind(('192.168.10.20',0))
try:
    s.connect(('192.168.10.1',22))
    print(s.recv(64).decode(errors='replace').strip() or 'EMPTY')
except Exception as e:
    print(f'REFUSED {type(e).__name__}: {e}')
"
REFUSED ConnectionRefusedError: [Errno 111] Connection refused
```

NMS sudo cat /var/log/lab/192.168.99.1.log 2>/dev/null || true

```
<189>207: 000203: *Sep  7 22:17:21.168 UTC: %SSH-5-SSH2_SESSION: SSH2 Session request from 10.0.2.2 (tty =
0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<189>208: 000204: *Sep  7 22:17:21.214 UTC: %SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: lab] [Source:
10.0.2.2] [localport: 22] at 22:17:21 UTC Mon Sep 7 2026
<189>209: 000205: *Sep  7 22:17:21.214 UTC: %SSH-5-SSH2_USERAUTH: User 'lab' authentication for SSH2 Session
from 10.0.2.2 (tty = 0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<190>210: 000206: *Sep  7 22:17:56.590 UTC: %SYS-6-LOGOUT: User lab has exited tty session 434(10.0.2.2)
<189>211: 000207: *Sep  7 22:17:56.591 UTC: %SSH-5-SSH2_CLOSE: SSH2 Session from 10.0.2.2 (tty = 0) for user
'' using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' closed
<189>212: 000208: *Sep  7 22:18:27.361 UTC: %SSH-5-SSH2_SESSION: SSH2 Session request from 10.0.2.2 (tty =
0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<189>213: 000209: *Sep  7 22:18:27.410 UTC: %SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: lab] [Source:
10.0.2.2] [localport: 22] at 22:18:27 UTC Mon Sep 7 2026
<189>214: 000210: *Sep  7 22:18:27.410 UTC: %SSH-5-SSH2_USERAUTH: User 'lab' authentication for SSH2 Session
from 10.0.2.2 (tty = 0) using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' Succeeded
<189>215: 000211: *Sep  7 22:18:27.605 UTC: %SYS-5-CONFIG_I: Configured from console by lab on vty0
(10.0.2.2)
<190>216: 000212: *Sep  7 22:18:27.747 UTC: %SYS-6-LOGOUT: User lab has exited tty session 434(10.0.2.2)
<189>217: 000213: *Sep  7 22:18:27.747 UTC: %SSH-5-SSH2_CLOSE: SSH2 Session from 10.0.2.2 (tty = 0) for user
```

```
'' using crypto cipher 'aes128-ctr', hmac 'hmac-sha2-256' closed
<187>218: 000214: *Sep  7 22:19:26.134 UTC: %DMVPN-3-DMVPN_NHRP_ERROR:  Tunnel0: Recieved wrong
authentication string for  Registration Request , Reason:  authentication failure (11) on (Tunnel:
172.20.0.3 NBMA: 100.64.0.1)
<187>219: 000215: *Sep  7 22:19:26.227 UTC: %DMVPN-3-DMVPN_NHRP_ERROR:  Tunnel0: Recieved wrong
authentication string for  Registration Request , Reason:  authentication failure (11) on (Tunnel:
172.20.0.2 NBMA: 100.64.0.1)
<187>220: 000216: *Sep  7 22:19:26.906 UTC: %DMVPN-3-DMVPN_NHRP_ERROR:  Tunnel0: Recieved wrong
authentication string for  Registration Request , Reason:  authentication failure (11) on (Tunnel:
172.20.0.3 NBMA: 100.64.0.1)
... 3708 more lines
```

Teardown (suite) Close Tacacs Server, Close Collector, Close All Connections

21.09 PASS The Harness Path Still Works (0.1s, 3 commands)

The reason the second permit exists. If this fails the lab is unreachable by every other suite, so it is asserted here rather than discovered in the next run. Asserted from the ACL's own hit counter rather than from "show users". That column prints a resolved name when reverse DNS answers -- inside the packaged appliance the same session shows as "_gateway" instead of 10.0.2.2 -- so matching on the rendered address made this depend on name resolution rather than on the thing being tested. A hit on that ACE means a session was admitted by exactly the permit under discussion.

Setup (suite) Open All Routers, Open Tacacs Server, Open Collector

Acceptance criteria

- **${hits}** is not empty (checked 3×)
- **${hits}[0] > 0** holds (checked 3×)

How it was checked

R1 show ip access-lists VTY-MGMT

```
Standard IP access list VTY-MGMT
    10 permit 192.168.99.0, wildcard bits 0.0.0.255 (8 matches)
    20 permit 10.0.2.0, wildcard bits 0.0.0.255 (274 matches)
    30 deny   any log (12 matches)
```

R2 show ip access-lists VTY-MGMT

```
Standard IP access list VTY-MGMT
    10 permit 192.168.99.0, wildcard bits 0.0.0.255
    20 permit 10.0.2.0, wildcard bits 0.0.0.255 (322 matches)
    30 deny   any log (4 matches)
```

R3 show ip access-lists VTY-MGMT

```
Standard IP access list VTY-MGMT
    10 permit 192.168.99.0, wildcard bits 0.0.0.255
    20 permit 10.0.2.0, wildcard bits 0.0.0.255 (274 matches)
    30 deny   any log (4 matches)
```

Teardown (suite) Close Tacacs Server, Close Collector, Close All Connections

22 Banner

22.01 PASS Every Router Has A Login Banner Configured (0.3s, 3 commands)

Setup (suite) Open All Routers

Acceptance criteria

- output contains **banner login** (checked 3×)

How it was checked

```
R1 show running-config | include ^banner
  banner login ^C
R2 show running-config | include ^banner
  banner login ^C
R3 show running-config | include ^banner
  banner login ^C
```

Teardown (suite) Close All Connections

22.02 PASS The Banner Is Presented Before Authentication (6.2s, 0 commands)

Captured from a login that is refused, so the banner demonstrably reaches an unauthenticated client.

Setup (suite) Open All Routers

Acceptance criteria

- **${banner}** is not empty (checked 3×)

How it was checked no device commands of its own; the assertion read output captured earlier in the suite

Teardown (suite) Close All Connections

22.03 PASS The Presented Banner Matches The Source File Exactly (6.3s, 0 commands)

Line for line, not a substring match: a device carrying a stale or truncated copy should fail rather than pass on one memorable phrase still being present.

Setup (suite) Open All Routers

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked no device commands of its own; the assertion read output captured earlier in the suite

Teardown (suite) Close All Connections

22.04 PASS The Banner States The Warning It Exists To Give (2.0s, 0 commands)

A banner that does not actually warn is decoration. These are the elements that make it one: restricted access, a statement of monitoring, and consent following from use.

Setup (suite) Open All Routers

Acceptance criteria

- output contains **${phrase}** (checked 4×)
  the banner does not state: ${phrase}

How it was checked no device commands of its own; the assertion read output captured earlier in the suite

Teardown (suite) Close All Connections

22.05 PASS The Banner Discloses Nothing About The Platform (6.2s, 0 commands)

An unauthenticated visitor should learn that they are not welcome, and nothing else. A banner naming the vendor, model, software or hostname helps whoever should not be there more than it helps anyone who should.

Setup (suite) Open All Routers

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked no device commands of its own; the assertion read output captured earlier in the suite

Teardown (suite) Close All Connections

22.06 PASS The Banner Survives A Configuration Save (0.1s, 3 commands)

Guards the case where a banner is applied to the running configuration and never written -- present until the next reload, and absent exactly when it would be needed.

Setup (suite) Open All Routers

Acceptance criteria

- output contains **banner login** (checked 3×)

How it was checked

```
R1 show startup-config | include ^banner
  banner login ^C
R2 show startup-config | include ^banner
  banner login ^C
R3 show startup-config | include ^banner
  banner login ^C
```

Teardown (suite) Close All Connections

23 Destructive

23.01 PASS A Shortcut Outlives The Hub, But Its Routing Does Not (67.7s, 30 commands)

The claim worth testing rather than assuming. A spoke-to- spoke tunnel is genuinely direct: the NHRP mapping and the IPsec session are between the spokes and the hub is not on the path. So the hub going away should not break it. Except that it does, one layer up. Both spokes learn each other's prefixes from BGP, and both peer only with the hub. Lose the hub and BFD tears those sessions down in about two seconds, the prefixes withdraw, and there is no route to put on the tunnel -- which is still sitting there, perfectly healthy, carrying nothing. So this asserts both halves: the tunnel address still answers, and the loopback behind it does not. The data plane is hub-independent here; the control plane is not.

Setup (suite) Open All Routers, Open All Hosts

Acceptance criteria

- output contains **${${peer}_WAN_IP}**
- output contains **not in table** (checked 2×)
- output contains **${R3_WAN_IP}**

  the direct NHRP mapping did not survive the hub, so the tunnel was never direct
- output contains **${R3_WAN_IP}**

  the spoke-to-spoke IPsec session did not survive the hub
- output contains **Success rate is 100 percent**

  the direct tunnel stopped carrying traffic when the hub went away
- output contains **Success rate is 0 percent**

  the spoke still routes to the far loopback with the hub down, so the
- output contains **${${r}_WAN_IP}** (checked 8×)
- output contains **Known via** (checked 9×)
- output contains **Success rate is 100 percent**

  the spokes cannot reach each other end to end, so the fabric is not converged

How it was checked

R2 ping 10.20.3.1 source Loopback1 repeat 10
```
Type escape sequence to abort.
Sending 10, 100-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.2.1
.!!!!!!!!!
Success rate is 90 percent (9/10), round-trip min/avg/max = 1/1/2 ms
```

R2 show ip nhrp 172.20.0.3
```
172.20.0.3/32 via 172.20.0.3
   Tunnel0 created 00:00:02, expire 00:04:58
   Type: dynamic, Flags: router nhop bfd
   NBMA address: 100.64.0.3
```

R1 configure terminal
```
Enter configuration commands, one per line.  End with CNTL/Z.
```

R1 interface Tunnel0
```
${output} =
```

R1 shutdown
```
${output} =
```

R1 end
```
${output} =
```

R2 show ip route 10.20.3.1
```
Routing entry for 10.20.3.1/32
  Known via "bgp 65002", distance 20, metric 0
  Tag 65001, type external
  Last update from 172.20.0.3 00:01:05 ago
  Routing Descriptor Blocks:
  * 172.20.0.3, from 172.20.0.1, 00:01:05 ago
      opaque_ptr 0x798416842808
      Route metric is 0, traffic share count is 1
      AS Hops 2
      Route tag 65001
      MPLS label: none
```

R2 show ip route 10.20.3.1
```
% Subnet not in table
```

R2 show ip nhrp 172.20.0.3
```
172.20.0.3/32 via 172.20.0.3
   Tunnel0 created 00:00:07, expire 00:04:52
```

```
    Type: dynamic, Flags: router nhop bfd
    NBMA address: 100.64.0.3
```

R2 show crypto session brief

```
Status: A- Active, U - Up, D - Down, I - Idle, S - Standby, N - Negotiating
        K - No IKE
ivrf = (none)
Peer            I/F          Username        Group/Phase1_id          Uptime   Status
100.64.0.3      Tu0                          100.64.0.3               00:00:07 UA
100.64.0.1      Tu0                                                   00:00:00 DN
```

R2 ping 172.20.0.3 source Tunnel0 repeat 5

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.20.0.3, timeout is 2 seconds:
Packet sent with a source address of 172.20.0.2
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
```

R2 ping 10.20.3.1 source Loopback1 repeat 5

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.2.1
.....
Success rate is 0 percent (0/5)
```

R1 configure terminal

```
Enter configuration commands, one per line.  End with CNTL/Z.
```

R1 interface Tunnel0

```
${output} =
```

R1 no shutdown

```
${output} =
```

R1 end

```
${output} =
```

R1 show dmvpn

```
Legend: Attrb --> S - Static, D - Dynamic, I - Incomplete
        N - NATed, L - Local, X - No Socket
        T1 - Route Installed, T2 - Nexthop-override, B - BGP
        C - CTS Capable, I2 - Temporary
        # Ent --> Number of NHRP entries with same NBMA peer
        NHS Status: E --> Expecting Replies, R --> Responding, W --> Waiting
        UpDn Time --> Up or Down Time for a Tunnel
==========================================================================
```

R1 show dmvpn

```
Legend: Attrb --> S - Static, D - Dynamic, I - Incomplete
        N - NATed, L - Local, X - No Socket
        T1 - Route Installed, T2 - Nexthop-override, B - BGP
        C - CTS Capable, I2 - Temporary
        # Ent --> Number of NHRP entries with same NBMA peer
        NHS Status: E --> Expecting Replies, R --> Responding, W --> Waiting
        UpDn Time --> Up or Down Time for a Tunnel
==========================================================================

Interface: Tunnel0, IPv4 NHRP Details
Type:Unknown, NHRP Peers:1,

 # Ent  Peer NBMA Addr Peer Tunnel Add State  UpDn Tm Attrb
 ----- --------------- --------------- ----- -------- -----
... 1 more lines
```

R1 show dmvpn

```
Legend: Attrb --> S - Static, D - Dynamic, I - Incomplete
        N - NATed, L - Local, X - No Socket
        T1 - Route Installed, T2 - Nexthop-override, B - BGP
        C - CTS Capable, I2 - Temporary
        # Ent --> Number of NHRP entries with same NBMA peer
        NHS Status: E --> Expecting Replies, R --> Responding, W --> Waiting
        UpDn Time --> Up or Down Time for a Tunnel
==========================================================================

Interface: Tunnel0, IPv4 NHRP Details
Type:Unknown, NHRP Peers:1,

 # Ent  Peer NBMA Addr Peer Tunnel Add State  UpDn Tm Attrb
 ----- --------------- --------------- ----- -------- -----
... 2 more lines
```

R1 show dmvpn

```
Legend: Attrb --> S - Static, D - Dynamic, I - Incomplete
        N - NATed, L - Local, X - No Socket
        T1 - Route Installed, T2 - Nexthop-override, B - BGP
        C - CTS Capable, I2 - Temporary
        # Ent --> Number of NHRP entries with same NBMA peer
```

```
        NHS Status: E --> Expecting Replies, R --> Responding, W --> Waiting
        UpDn Time --> Up or Down Time for a Tunnel
==========================================================================

Interface: Tunnel0, IPv4 NHRP Details
Type:Hub, NHRP Peers:2,

 # Ent  Peer NBMA Addr Peer Tunnel Add State  UpDn Tm Attrb
 ----- --------------- --------------- ----- -------- -----
... 2 more lines
```

R2 show ip route 10.20.3.1

```
% Subnet not in table
```

R2 show ip route 10.20.3.1

```
% Subnet not in table
```

R2 show ip route 10.20.3.1

```
% Subnet not in table
```

R2 show ip route 10.20.3.1

```
% Subnet not in table
```

R2 show ip route 10.20.3.1

```
% Subnet not in table
```

R2 show ip route 10.20.3.1

```
% Subnet not in table
```

R2 show ip route 10.20.3.1

```
% Subnet not in table
```

R2 show ip route 10.20.3.1

```
Routing entry for 10.20.3.1/32
  Known via "bgp 65002", distance 20, metric 0
  Tag 65001, type external
  Last update from 172.20.0.3 00:00:03 ago
  Routing Descriptor Blocks:
  * 172.20.0.3, from 172.20.0.1, 00:00:03 ago
      opaque_ptr 0x798416842808
      Route metric is 0, traffic share count is 1
      AS Hops 2
      Route tag 65001
      MPLS label: none
```

R3 show ip route 10.20.2.1

```
Routing entry for 10.20.2.1/32
  Known via "bgp 65003", distance 20, metric 0
  Tag 65001, type external
  Last update from 172.20.0.2 00:00:31 ago
  Routing Descriptor Blocks:
  * 172.20.0.2, from 172.20.0.1, 00:00:31 ago
      opaque_ptr 0x7AED8E74CA10
      Route metric is 0, traffic share count is 1
      AS Hops 2
      Route tag 65001
      MPLS label: none
```

R2 ping 10.20.3.1 source Loopback1 repeat 5

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.2.1
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
```

Teardown (this test) Restore Hub Tunnel

23.02 PASS The Fabric Comes Back When The Hub Does (0.2s, 5 commands)

Recovery without operator action beyond restoring the hub: registration, BGP and the shortcut all return on their own.

Setup (suite) Open All Routers, Open All Hosts

Acceptance criteria

- output contains **${${r}_WAN_IP}** (checked 2×)
- output contains **Known via**
- output contains **${${peer}_WAN_IP}**
- output contains **${R3_DMVPN_IP}**
- output does not contain **${R1_DMVPN_IP}**

  the shortcut did not re-form after the hub returned

How it was checked

R1 show dmvpn

```
Legend: Attrb --> S - Static, D - Dynamic, I - Incomplete
        N - NATed, L - Local, X - No Socket
        T1 - Route Installed, T2 - Nexthop-override, B - BGP
        C - CTS Capable, I2 - Temporary
        # Ent --> Number of NHRP entries with same NBMA peer
        NHS Status: E --> Expecting Replies, R --> Responding, W --> Waiting
        UpDn Time --> Up or Down Time for a Tunnel
==========================================================================

Interface: Tunnel0, IPv4 NHRP Details
Type:Hub, NHRP Peers:2,

 # Ent  Peer NBMA Addr Peer Tunnel Add State  UpDn Tm Attrb
 ----- --------------- --------------- ----- -------- -----
... 2 more lines
```

R2 show ip route 10.20.3.1

```
Routing entry for 10.20.3.1/32
  Known via "bgp 65002", distance 20, metric 0
  Tag 65001, type external
  Last update from 172.20.0.3 00:00:03 ago
  Routing Descriptor Blocks:
  * 172.20.0.3, from 172.20.0.1, 00:00:03 ago
      opaque_ptr 0x798416842808
      Route metric is 0, traffic share count is 1
      AS Hops 2
      Route tag 65001
      MPLS label: none
```

R2 ping 10.20.3.1 source Loopback1 repeat 10

```
Type escape sequence to abort.
Sending 10, 100-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.2.1
!!!!!!!!!!
Success rate is 100 percent (10/10), round-trip min/avg/max = 1/1/1 ms
```

R2 show ip nhrp 172.20.0.3

```
172.20.0.3/32 via 172.20.0.3
   Tunnel0 created 00:01:07, expire 00:03:52
   Type: dynamic, Flags: router nhop bfd
   NBMA address: 100.64.0.3
```

R2 traceroute 10.20.3.1 source Loopback1 probe 1 timeout 1 numeric

```
Type escape sequence to abort.
Tracing the route to 10.20.3.1
VRF info: (vrf in name/id, vrf out name/id)
  1 172.20.0.3 2 msec
```

Teardown (suite) Restore Everything, Close All Connections

23.03 PASS Blocking The Spokes At The Underlay Falls Back Through The Hub (23.6s, 27 commands)

The realistic partial failure: two sites that can both reach the hub but not each other, which is what happens whenever a firewall between branches permits the head end and nothing else. Resolution cannot complete, so no shortcut forms -- and connectivity must survive anyway, through the hub. The suite's other fallback test only clears the cache, which is the easy version: nothing is stopping the shortcut from forming a moment later. Here it genuinely cannot.

Setup (suite) Open All Routers, Open All Hosts

Acceptance criteria

- **${rate}[0] > 0** holds
- output does not contain **${R3_WAN_IP}**
  a direct mapping formed even though the underlay between spokes is blocked
- output contains **${R1_DMVPN_IP}**
  traffic is not transiting the hub, so it is not using the fallback:\n${trace}
- output contains **${${r}_WAN_IP}** (checked 2×)
- output contains **Known via** (checked 2×)
- output contains **Success rate is 100 percent** (checked 2×)
  the spokes cannot reach each other end to end, so the fabric is not converged

How it was checked

```
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 ip access-list extended BLOCK-SPOKE
  ${output} =
R2 deny ip host 100.64.0.2 host 100.64.0.3
  ${output} =
R2 deny ip host 100.64.0.3 host 100.64.0.2
  ${output} =
R2 permit ip any any
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 interface GigabitEthernet2
  ${output} =
R2 ip access-group BLOCK-SPOKE in
  ${output} =
R2 end
  ${output} =
R2 clear ip nhrp
  ${output} =
R2 ping 10.20.3.1 source Loopback1 repeat 10
  Type escape sequence to abort.
  Sending 10, 100-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:
  Packet sent with a source address of 10.20.2.1
  ..!!!!!!!!
  Success rate is 80 percent (8/10), round-trip min/avg/max = 1/1/2 ms
R2 show ip nhrp 172.20.0.3
  172.20.0.3/32 via 172.20.0.3
     Tunnel0 created 00:00:04, expire 00:03:00
     Type: dynamic, Flags: temporary
     NBMA address: 100.64.0.1
R2 traceroute 10.20.3.1 source Loopback1 probe 1 timeout 1 numeric
  Type escape sequence to abort.
  Tracing the route to 10.20.3.1
  VRF info: (vrf in name/id, vrf out name/id)
    1 172.20.0.1 1 msec
    2 172.20.0.3 2 msec
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 interface GigabitEthernet2
  ${output} =
R2 no ip access-group BLOCK-SPOKE in
  ${output} =
R2 end
  ${output} =
R2 configure terminal
  Enter configuration commands, one per line.  End with CNTL/Z.
R2 no ip access-list extended BLOCK-SPOKE
  ${output} =
R2 end
  ${output} =
R2 clear ip nhrp
  ${output} =
R1 show dmvpn
  Legend: Attrb --> S - Static, D - Dynamic, I - Incomplete
          N - NATed, L - Local, X - No Socket
          T1 - Route Installed, T2 - Nexthop-override, B - BGP
          C - CTS Capable, I2 - Temporary
          # Ent --> Number of NHRP entries with same NBMA peer
          NHS Status: E --> Expecting Replies, R --> Responding, W --> Waiting
          UpDn Time --> Up or Down Time for a Tunnel
  ==========================================================================

  Interface: Tunnel0, IPv4 NHRP Details
  Type:Hub, NHRP Peers:2,

   # Ent  Peer NBMA Addr Peer Tunnel Add State  UpDn Tm Attrb
   ----- --------------- --------------- ----- -------- -----
```

```
... 2 more lines
```

R2 show ip route 10.20.3.1

```
Routing entry for 10.20.3.1/32
  Known via "bgp 65002", distance 20, metric 0
  Tag 65001, type external
  Last update from 172.20.0.3 00:00:20 ago
  Routing Descriptor Blocks:
  * 172.20.0.3, from 172.20.0.1, 00:00:20 ago
      opaque_ptr 0x798416842808
      Route metric is 0, traffic share count is 1
      AS Hops 2
      Route tag 65001
      MPLS label: none
```

R3 show ip route 10.20.2.1

```
Routing entry for 10.20.2.1/32
  Known via "bgp 65003", distance 20, metric 0
  Tag 65001, type external
  Last update from 172.20.0.2 00:00:48 ago
  Routing Descriptor Blocks:
  * 172.20.0.2, from 172.20.0.1, 00:00:48 ago
      opaque_ptr 0x7AED8E74CA10
      Route metric is 0, traffic share count is 1
      AS Hops 2
      Route tag 65001
      MPLS label: none
```

R2 ping 10.20.3.1 source Loopback1 repeat 5

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.2.1
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/1 ms
```

R2 ping 10.20.3.1 source Loopback1 repeat 5

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.2.1
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
```

Teardown (this test) Remove Spoke Block

23.04 PASS An NHRP Authentication Mismatch Is Refused (47.7s, 29 commands)

The authentication string must match across the cloud. A spoke that gets it wrong is refused registration and simply never appears at the hub. Both caches have to be cleared to see it. The hub holds its registration entry for the full holdtime, so clearing only the spoke leaves the hub answering from a stale cache -- and the test reads "still registered" and concludes, wrongly, that the mismatch had no effect.

Setup (suite) Open All Routers, Open All Hosts

Acceptance criteria

- output does not contain **${${peer}_WAN_IP}**
  the hub still holds a registration for ${peer}
- output contains **${${r}_WAN_IP}** (checked 8×)
- output contains **Known via** (checked 7×)
- output contains **Success rate is 100 percent** (checked 2×)
  the spokes cannot reach each other end to end, so the fabric is not converged

How it was checked

R3 configure terminal

```
Enter configuration commands, one per line.  End with CNTL/Z.
```

R3 interface Tunnel0

```
${output} =
```

R3 ip nhrp authentication WrongKey

```
${output} =
```

R3 end

```
${output} =
```

R1 clear ip nhrp

```
${output} =
```

R2 clear ip nhrp

```
${output} =
```

R3 clear ip nhrp
```
${output} =
```
R1 show dmvpn | include 100.64.0
```
1 100.64.0.2          172.20.0.2    UP 00:00:00     D
```
R3 configure terminal
```
Enter configuration commands, one per line.  End with CNTL/Z.
```
R3 interface Tunnel0
```
${output} =
```
R3 tunnel key 100
```
${output} =
```
R3 ip nhrp authentication LabNhrp
```
${output} =
```
R3 end
```
${output} =
```
R1 clear ip nhrp
```
${output} =
```
R2 clear ip nhrp
```
${output} =
```
R3 clear ip nhrp
```
${output} =
```
R1 show dmvpn
```
Legend: Attrb --> S - Static, D - Dynamic, I - Incomplete
        N - NATed, L - Local, X - No Socket
        T1 - Route Installed, T2 - Nexthop-override, B - BGP
        C - CTS Capable, I2 - Temporary
        # Ent --> Number of NHRP entries with same NBMA peer
        NHS Status: E --> Expecting Replies, R --> Responding, W --> Waiting
        UpDn Time --> Up or Down Time for a Tunnel
==========================================================================
```
R1 show dmvpn
```
Legend: Attrb --> S - Static, D - Dynamic, I - Incomplete
        N - NATed, L - Local, X - No Socket
        T1 - Route Installed, T2 - Nexthop-override, B - BGP
        C - CTS Capable, I2 - Temporary
        # Ent --> Number of NHRP entries with same NBMA peer
        NHS Status: E --> Expecting Replies, R --> Responding, W --> Waiting
        UpDn Time --> Up or Down Time for a Tunnel
==========================================================================

Interface: Tunnel0, IPv4 NHRP Details
Type:Hub, NHRP Peers:2,

 # Ent  Peer NBMA Addr Peer Tunnel Add State  UpDn Tm Attrb
 ----- --------------- --------------- ----- -------- -----
... 2 more lines
```
R1 show dmvpn
```
Legend: Attrb --> S - Static, D - Dynamic, I - Incomplete
        N - NATed, L - Local, X - No Socket
        T1 - Route Installed, T2 - Nexthop-override, B - BGP
        C - CTS Capable, I2 - Temporary
        # Ent --> Number of NHRP entries with same NBMA peer
        NHS Status: E --> Expecting Replies, R --> Responding, W --> Waiting
        UpDn Time --> Up or Down Time for a Tunnel
==========================================================================

Interface: Tunnel0, IPv4 NHRP Details
Type:Hub, NHRP Peers:2,

 # Ent  Peer NBMA Addr Peer Tunnel Add State  UpDn Tm Attrb
 ----- --------------- --------------- ----- -------- -----
... 2 more lines
```
R1 show dmvpn
```
Legend: Attrb --> S - Static, D - Dynamic, I - Incomplete
        N - NATed, L - Local, X - No Socket
        T1 - Route Installed, T2 - Nexthop-override, B - BGP
        C - CTS Capable, I2 - Temporary
        # Ent --> Number of NHRP entries with same NBMA peer
        NHS Status: E --> Expecting Replies, R --> Responding, W --> Waiting
        UpDn Time --> Up or Down Time for a Tunnel
==========================================================================

Interface: Tunnel0, IPv4 NHRP Details
Type:Hub, NHRP Peers:2,

 # Ent  Peer NBMA Addr Peer Tunnel Add State  UpDn Tm Attrb
```

```
    ----- --------------- -------------- ----- -------- -----
  ... 2 more lines
```

R2 show ip route 10.20.3.1

```
  Routing entry for 10.20.3.1/32
    Known via "bgp 65002", distance 20, metric 0
    Tag 65001, type external
    Last update from 172.20.0.3 00:00:05 ago
    Routing Descriptor Blocks:
    * 172.20.0.3, from 172.20.0.1, 00:00:05 ago
        opaque_ptr 0x798416842BC8
        Route metric is 0, traffic share count is 1
        AS Hops 2
        Route tag 65001
        MPLS label: none
```

R3 show ip route 10.20.2.1

```
  % Subnet not in table
```

R3 show ip route 10.20.2.1

```
  % Subnet not in table
```

R3 show ip route 10.20.2.1

```
  % Subnet not in table
```

R3 show ip route 10.20.2.1

```
  % Subnet not in table
```

R3 show ip route 10.20.2.1

```
  % Subnet not in table
```

R3 show ip route 10.20.2.1

```
  Routing entry for 10.20.2.1/32
    Known via "bgp 65003", distance 20, metric 0
    Tag 65001, type external
    Last update from 172.20.0.2 00:00:02 ago
    Routing Descriptor Blocks:
    * 172.20.0.2, from 172.20.0.1, 00:00:02 ago
        opaque_ptr 0x7AED8E74CA10
        Route metric is 0, traffic share count is 1
        AS Hops 2
        Route tag 65001
        MPLS label: none
```

R2 ping 10.20.3.1 source Loopback1 repeat 5

```
  Type escape sequence to abort.
  Sending 5, 100-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:
  Packet sent with a source address of 10.20.2.1
  .!!!!
  Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/1 ms
```

R2 ping 10.20.3.1 source Loopback1 repeat 5

```
  Type escape sequence to abort.
  Sending 5, 100-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:
  Packet sent with a source address of 10.20.2.1
  !!!!!
  Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/3 ms
```

Teardown (this test) Restore Spoke Tunnel Settings

23.05 PASS A Tunnel Key Mismatch Fails Silently Instead (12.5s, 23 commands)

The same outcome by a different route, and the reason both are worth having. A wrong authentication string is a refusal. A wrong tunnel key is not refused by anything -- the GRE headers simply do not match, the packets are discarded, and the tunnel interface goes on reporting itself up with nothing passing through it.

Setup (suite) Open All Routers, Open All Hosts

Acceptance criteria

- output does not contain **${${peer}_WAN_IP}**
  the hub still holds a registration for ${peer}
- output matches **up\s+up**
  the interface reported itself down, which would make this failure the
- output contains **${${r}_WAN_IP}** (checked 4×)
- output contains **Known via** (checked 2×)
- output contains **Success rate is 100 percent** (checked 2×)
  the spokes cannot reach each other end to end, so the fabric is not converged

How it was checked

R3 configure terminal

```
Enter configuration commands, one per line.  End with CNTL/Z.
```

R3 interface Tunnel0

```
${output} =
```

R3 tunnel key 999

```
${output} =
```

R3 end

```
${output} =
```

R1 clear ip nhrp

```
${output} =
```

R2 clear ip nhrp

```
${output} =
```

R3 clear ip nhrp

```
${output} =
```

R1 show dmvpn | include 100.64.0

```
${output} =
```

R3 show ip interface brief | include Tunnel0

```
Tunnel0                172.20.0.3      YES NVRAM  up                    up
```

R3 configure terminal

```
Enter configuration commands, one per line.  End with CNTL/Z.
```

R3 interface Tunnel0

```
${output} =
```

R3 tunnel key 100

```
${output} =
```

R3 ip nhrp authentication LabNhrp

```
${output} =
```

R3 end

```
${output} =
```

R1 clear ip nhrp

```
${output} =
```

R2 clear ip nhrp

```
${output} =
```

R3 clear ip nhrp

```
${output} =
```

R1 show dmvpn

```
Legend: Attrb --> S - Static, D - Dynamic, I - Incomplete
        N - NATed, L - Local, X - No Socket
        T1 - Route Installed, T2 - Nexthop-override, B - BGP
        C - CTS Capable, I2 - Temporary
        # Ent --> Number of NHRP entries with same NBMA peer
        NHS Status: E --> Expecting Replies, R --> Responding, W --> Waiting
        UpDn Time --> Up or Down Time for a Tunnel
==========================================================================

Interface: Tunnel0, IPv4 NHRP Details
Type:Hub, NHRP Peers:1,

 # Ent  Peer NBMA Addr Peer Tunnel Add State  UpDn Tm Attrb
 ----- --------------- --------------- ----- -------- -----
... 1 more lines
```

R1 show dmvpn

```
Legend: Attrb --> S - Static, D - Dynamic, I - Incomplete
        N - NATed, L - Local, X - No Socket
        T1 - Route Installed, T2 - Nexthop-override, B - BGP
        C - CTS Capable, I2 - Temporary
        # Ent --> Number of NHRP entries with same NBMA peer
        NHS Status: E --> Expecting Replies, R --> Responding, W --> Waiting
        UpDn Time --> Up or Down Time for a Tunnel
==========================================================================

Interface: Tunnel0, IPv4 NHRP Details
Type:Hub, NHRP Peers:2,

 # Ent  Peer NBMA Addr Peer Tunnel Add State  UpDn Tm Attrb
 ----- --------------- --------------- ----- -------- -----
... 2 more lines
```

R2 show ip route 10.20.3.1

```
Routing entry for 10.20.3.1/32
  Known via "bgp 65002", distance 20, metric 0
  Tag 65001, type external
  Last update from 172.20.0.3 00:00:42 ago
  Routing Descriptor Blocks:
```

```
    * 172.20.0.3, from 172.20.0.1, 00:00:42 ago
        opaque_ptr 0x798416842BC8
        Route metric is 0, traffic share count is 1
        AS Hops 2
        Route tag 65001
        MPLS label: none
```

R3 show ip route 10.20.2.1

```
  Routing entry for 10.20.2.1/32
    Known via "bgp 65003", distance 20, metric 0
    Tag 65001, type external
    Last update from 172.20.0.2 00:00:14 ago
    Routing Descriptor Blocks:
    * 172.20.0.2, from 172.20.0.1, 00:00:14 ago
        opaque_ptr 0x7AED8E74CA10
        Route metric is 0, traffic share count is 1
        AS Hops 2
        Route tag 65001
        MPLS label: none
```

R2 ping 10.20.3.1 source Loopback1 repeat 5

```
  Type escape sequence to abort.
  Sending 5, 100-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:
  Packet sent with a source address of 10.20.2.1
  .!!!!
  Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/2 ms
```

R2 ping 10.20.3.1 source Loopback1 repeat 5

```
  Type escape sequence to abort.
  Sending 5, 100-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:
  Packet sent with a source address of 10.20.2.1
  !!!!!
  Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
```

Teardown (this test) Restore Spoke Tunnel Settings

23.06 PASS Blocking The Too-Big Message Blackholes Large Traffic (7.3s, 18 commands)

The classic silent MTU failure. The router tells a host "fragmentation needed, MTU 1472" and the host adapts. Take that message away and nothing announces the fault: small packets keep working, ping keeps working, and anything bulk stalls with no diagnostic anywhere. The message is suppressed with "no ip unreachables" on the interface it would leave by -- R2's LAN side, facing the host. An outbound ACL was the obvious first choice and it does not work: an outbound ACL does not filter traffic the router originates, and this message is originated by R2. Applying it to Tunnel0 does nothing either, for the same reason -- the constraint is on the tunnel, but the message is emitted towards the host. Packet loss on its own proves nothing here: the oversized ping fails either way, because the host is forbidden to fragment it. What changes is whether anything says why, so the assertions are about the diagnostic and not the loss. This is what makes the boundary the MTU suite measures matter, and it is why that suite asserts the host learns the path MTU rather than merely that big packets fail.

Setup (this test) Wait For Converged Fabric

Acceptance criteria

- output contains **${${r}_WAN_IP}** (checked 2×)
- output contains **Known via** (checked 2×)
- output contains **Success rate is 100 percent**
  the spokes cannot reach each other end to end, so the fabric is not converged
- output contains **mtu = ${TUNNEL_MTU}**
  the host did not learn the path MTU even before anything was blocked
- output contains **100% packet loss**
  large traffic got through even with the too-big message suppressed
- output does not contain **mtu = ${TUNNEL_MTU}**
  the host still learned the MTU, so the message was not actually suppressed
- output does not contain **Frag needed**
  the host was still told fragmentation was needed, so the failure is not
- output contains **, 0% packet loss**
  small packets failed too, so this is an outage rather than a blackhole

How it was checked

R1 show dmvpn

```
  Legend: Attrb --> S - Static, D - Dynamic, I - Incomplete
          N - NATed, L - Local, X - No Socket
```

```
        T1 - Route Installed, T2 - Nexthop-override, B - BGP
        C - CTS Capable, I2 - Temporary
        # Ent --> Number of NHRP entries with same NBMA peer
        NHS Status: E --> Expecting Replies, R --> Responding, W --> Waiting
        UpDn Time --> Up or Down Time for a Tunnel
==========================================================================

Interface: Tunnel0, IPv4 NHRP Details
Type:Hub, NHRP Peers:2,

 # Ent  Peer NBMA Addr Peer Tunnel Add State  UpDn Tm Attrb
 ----- --------------- --------------- ----- -------- -----
... 2 more lines
```

R2 show ip route 10.20.3.1

```
Routing entry for 10.20.3.1/32
  Known via "bgp 65002", distance 20, metric 0
  Tag 65001, type external
  Last update from 172.20.0.3 00:00:49 ago
  Routing Descriptor Blocks:
  * 172.20.0.3, from 172.20.0.1, 00:00:49 ago
      opaque_ptr 0x798416842BC8
      Route metric is 0, traffic share count is 1
      AS Hops 2
      Route tag 65001
      MPLS label: none
```

R3 show ip route 10.20.2.1

```
Routing entry for 10.20.2.1/32
  Known via "bgp 65003", distance 20, metric 0
  Tag 65001, type external
  Last update from 172.20.0.2 00:00:21 ago
  Routing Descriptor Blocks:
  * 172.20.0.2, from 172.20.0.1, 00:00:21 ago
      opaque_ptr 0x7AED8E74CA10
      Route metric is 0, traffic share count is 1
      AS Hops 2
      Route tag 65001
      MPLS label: none
```

R2 ping 10.20.3.1 source Loopback1 repeat 5

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.2.1
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
```

H2 sudo ip route flush cache

```
${output} =
```

H2 ping -c 2 -W 2 -M do -s 1472 192.168.30.10

```
PING 192.168.30.10 (192.168.30.10) 1472(1500) bytes of data.
From 192.168.20.1 icmp_seq=1 Frag needed and DF set (mtu = 1472)
ping: local error: message too long, mtu=1472

--- 192.168.30.10 ping statistics ---
2 packets transmitted, 0 received, +2 errors, 100% packet loss, time 1007ms
```

R2 configure terminal

```
Enter configuration commands, one per line.  End with CNTL/Z.
```

R2 interface GigabitEthernet3

```
${output} =
```

R2 no ip unreachables

```
${output} =
```

R2 end

```
${output} =
```

H2 sudo ip route flush cache

```
${output} =
```

H2 ping -c 3 -W 2 -M do -s 1472 192.168.30.10

```
PING 192.168.30.10 (192.168.30.10) 1472(1500) bytes of data.

--- 192.168.30.10 ping statistics ---
3 packets transmitted, 0 received, 100% packet loss, time 2036ms
```

H2 ping -c 2 -W 2 192.168.30.10

```
PING 192.168.30.10 (192.168.30.10) 56(84) bytes of data.
64 bytes from 192.168.30.10: icmp_seq=1 ttl=62 time=1.24 ms
64 bytes from 192.168.30.10: icmp_seq=2 ttl=62 time=1.26 ms

--- 192.168.30.10 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1001ms
rtt min/avg/max/mdev = 1.240/1.249/1.259/0.009 ms
```

R2 configure terminal
```
Enter configuration commands, one per line.  End with CNTL/Z.
```
R2 interface GigabitEthernet3
```
${output} =
```
R2 ip unreachables
```
${output} =
```
R2 end
```
${output} =
```
H2 sudo ip route flush cache
```
${output} =
```
Teardown (this test) Remove Pmtu Block

23.07 PASS NHRP Mappings Are Soft State And Age Out (112.0s, 46 commands)

A shortcut is a cache entry, not configuration: it exists because traffic justified it and it should disappear when that stops being true. Proven with a shortened holdtime -- the configured 300s would add five minutes to every run for a property that is no truer at 300 than at 60. Expiry is not punctual. NHRP ages entries lazily, and a 60s holdtime was measured clearing at about 170s, so the wait here is generous rather than exact.

Setup (this test) Wait For Converged Fabric

Acceptance criteria

- output contains **${${r}_WAN_IP}** (checked 6×)
- output contains **Known via** (checked 4×)
- output contains **Success rate is 100 percent** (checked 3×)

  the spokes cannot reach each other end to end, so the fabric is not converged
- output contains **${${peer}_WAN_IP}** (checked 3×)
- output does not contain **${${peer}_WAN_IP}** (checked 9×)

How it was checked

R1 show dmvpn
```
Legend: Attrb --> S - Static, D - Dynamic, I - Incomplete
        N - NATed, L - Local, X - No Socket
        T1 - Route Installed, T2 - Nexthop-override, B - BGP
        C - CTS Capable, I2 - Temporary
        # Ent --> Number of NHRP entries with same NBMA peer
        NHS Status: E --> Expecting Replies, R --> Responding, W --> Waiting
        UpDn Time --> Up or Down Time for a Tunnel
==========================================================================

Interface: Tunnel0, IPv4 NHRP Details
Type:Hub, NHRP Peers:2,

 # Ent  Peer NBMA Addr Peer Tunnel Add State  UpDn Tm Attrb
 ----- --------------- --------------- ----- -------- -----
... 2 more lines
```
R2 show ip route 10.20.3.1
```
Routing entry for 10.20.3.1/32
  Known via "bgp 65002", distance 20, metric 0
  Tag 65001, type external
  Last update from 172.20.0.3 00:00:57 ago
  Routing Descriptor Blocks:
  * 172.20.0.3, from 172.20.0.1, 00:00:57 ago
      opaque_ptr 0x798416842BC8
      Route metric is 0, traffic share count is 1
      AS Hops 2
      Route tag 65001
      MPLS label: none
```
R3 show ip route 10.20.2.1
```
Routing entry for 10.20.2.1/32
  Known via "bgp 65003", distance 20, metric 0
  Tag 65001, type external
  Last update from 172.20.0.2 00:00:29 ago
  Routing Descriptor Blocks:
  * 172.20.0.2, from 172.20.0.1, 00:00:29 ago
      opaque_ptr 0x7AED8E74CA10
      Route metric is 0, traffic share count is 1
      AS Hops 2
      Route tag 65001
      MPLS label: none
```

R2 ping 10.20.3.1 source Loopback1 repeat 5

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.2.1
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
```

R2 configure terminal

```
Enter configuration commands, one per line.  End with CNTL/Z.
```

R2 interface Tunnel0

```
${output} =
```

R2 ip nhrp holdtime 60

```
${output} =
```

R2 end

```
${output} =
```

R3 configure terminal

```
Enter configuration commands, one per line.  End with CNTL/Z.
```

R3 interface Tunnel0

```
${output} =
```

R3 ip nhrp holdtime 60

```
${output} =
```

R3 end

```
${output} =
```

R1 clear ip nhrp

```
${output} =
```

R2 clear ip nhrp

```
${output} =
```

R3 clear ip nhrp

```
${output} =
```

R2 ping 10.20.3.1 source Loopback1 repeat 10

```
Type escape sequence to abort.
Sending 10, 100-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.2.1
.!!!!!!!!!
Success rate is 90 percent (9/10), round-trip min/avg/max = 1/1/2 ms
```

R2 show ip nhrp 172.20.0.3

```
172.20.0.3/32 via 172.20.0.3
   Tunnel0 created 00:00:02, expire 00:00:58
   Type: dynamic, Flags: router nhop bfd
   NBMA address: 100.64.0.3
```

R2 show ip nhrp 172.20.0.3

```
172.20.0.3/32 via 172.20.0.3
   Tunnel0 created 00:00:02, expire 00:00:57
   Type: dynamic, Flags: router nhop bfd
   NBMA address: 100.64.0.3
```

R2 show ip nhrp 172.20.0.3

```
172.20.0.3/32 via 172.20.0.3
   Tunnel0 created 00:00:12, expire 00:00:47
   Type: dynamic, Flags: router nhop bfd
   NBMA address: 100.64.0.3
```

R2 show ip nhrp 172.20.0.3

```
172.20.0.3/32 via 172.20.0.3
   Tunnel0 created 00:00:22, expire 00:00:52
   Type: dynamic, Flags: router nhop bfd
   NBMA address: 100.64.0.3
```

R2 show ip nhrp 172.20.0.3

```
172.20.0.3/32 via 172.20.0.3
   Tunnel0 created 00:00:32, expire 00:00:42
   Type: dynamic, Flags: router nhop bfd
   NBMA address: 100.64.0.3
```

R2 show ip nhrp 172.20.0.3

```
172.20.0.3/32 via 172.20.0.3
   Tunnel0 created 00:00:42, expire 00:00:32
   Type: dynamic, Flags: router nhop bfd
   NBMA address: 100.64.0.3
```

R2 show ip nhrp 172.20.0.3

```
172.20.0.3/32 via 172.20.0.3
   Tunnel0 created 00:00:52, expire 00:00:22
   Type: dynamic, Flags: router nhop bfd
   NBMA address: 100.64.0.3
```

R2 show ip nhrp 172.20.0.3

```
172.20.0.3/32 via 172.20.0.3
```

```
    Tunnel0 created 00:01:02, expire 00:00:12
    Type: dynamic, Flags: router nhop bfd
    NBMA address: 100.64.0.3
```

R2 show ip nhrp 172.20.0.3

```
172.20.0.3/32 via 172.20.0.3
   Tunnel0 created 00:01:12, expire 00:00:02
   Type: dynamic, Flags: router nhop bfd
   NBMA address: 100.64.0.3
```

R2 show ip nhrp 172.20.0.3

```
${output} =
```

R2 ping 10.20.3.1 source Loopback1 repeat 10

```
Type escape sequence to abort.
Sending 10, 100-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.2.1
.!!!!!!!!!
Success rate is 90 percent (9/10), round-trip min/avg/max = 1/1/2 ms
```

R2 show ip nhrp 172.20.0.3

```
172.20.0.3/32 via 172.20.0.3
   Tunnel0 created 00:00:02, expire 00:00:57
   Type: dynamic, Flags: router nhop bfd
   NBMA address: 100.64.0.3
```

R2 show ip nhrp 172.20.0.3

```
172.20.0.3/32 via 172.20.0.3
   Tunnel0 created 00:00:02, expire 00:00:57
   Type: dynamic, Flags: router nhop bfd
   NBMA address: 100.64.0.3
```

R2 configure terminal

```
Enter configuration commands, one per line.  End with CNTL/Z.
```

R2 interface Tunnel0

```
${output} =
```

R2 ip nhrp holdtime 300

```
${output} =
```

R2 end

```
${output} =
```

R3 configure terminal

```
Enter configuration commands, one per line.  End with CNTL/Z.
```

R3 interface Tunnel0

```
${output} =
```

R3 ip nhrp holdtime 300

```
${output} =
```

R3 end

```
${output} =
```

R1 clear ip nhrp

```
${output} =
```

R2 clear ip nhrp

```
${output} =
```

R3 clear ip nhrp

```
${output} =
```

R1 show dmvpn

```
Legend: Attrb --> S - Static, D - Dynamic, I - Incomplete
        N - NATed, L - Local, X - No Socket
        T1 - Route Installed, T2 - Nexthop-override, B - BGP
        C - CTS Capable, I2 - Temporary
        # Ent --> Number of NHRP entries with same NBMA peer
        NHS Status: E --> Expecting Replies, R --> Responding, W --> Waiting
        UpDn Time --> Up or Down Time for a Tunnel
==========================================================================
```

R1 show dmvpn

```
Legend: Attrb --> S - Static, D - Dynamic, I - Incomplete
        N - NATed, L - Local, X - No Socket
        T1 - Route Installed, T2 - Nexthop-override, B - BGP
        C - CTS Capable, I2 - Temporary
        # Ent --> Number of NHRP entries with same NBMA peer
        NHS Status: E --> Expecting Replies, R --> Responding, W --> Waiting
        UpDn Time --> Up or Down Time for a Tunnel
==========================================================================

Interface: Tunnel0, IPv4 NHRP Details
Type:Hub, NHRP Peers:2,

 # Ent  Peer NBMA Addr Peer Tunnel Add State  UpDn Tm Attrb
 ----- --------------- --------------- ----- -------- -----
```

```
... 2 more lines
```

R2 show ip route 10.20.3.1

```
Routing entry for 10.20.3.1/32
  Known via "bgp 65002", distance 20, metric 0
  Tag 65001, type external
  Last update from 172.20.0.3 00:02:42 ago
  Routing Descriptor Blocks:
  * 172.20.0.3, from 172.20.0.1, 00:02:42 ago
      opaque_ptr 0x798416842BC8
      Route metric is 0, traffic share count is 1
      AS Hops 2
      Route tag 65001
      MPLS label: none
```

R3 show ip route 10.20.2.1

```
Routing entry for 10.20.2.1/32
  Known via "bgp 65003", distance 20, metric 0
  Tag 65001, type external
  Last update from 172.20.0.2 00:02:14 ago
  Routing Descriptor Blocks:
  * 172.20.0.2, from 172.20.0.1, 00:02:14 ago
      opaque_ptr 0x7AED8E74CA10
      Route metric is 0, traffic share count is 1
      AS Hops 2
      Route tag 65001
      MPLS label: none
```

R2 ping 10.20.3.1 source Loopback1 repeat 5

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.2.1
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/2 ms
```

R2 ping 10.20.3.1 source Loopback1 repeat 5

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.20.3.1, timeout is 2 seconds:
Packet sent with a source address of 10.20.2.1
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
```

Teardown (this test) Restore Holdtime

24 Terraform

24.01 PASS The Terraform Configuration Is Valid And Canonically Formatted (0.2s, 0 commands)

The cheapest gate there is, and the one most worth having first: a syntax or type error here would otherwise surface as a confusing failure several tests later.

Setup (suite) Open All Routers, Generate Terraform Variables, Terraform Init, Terraform Destroy

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked no device commands of its own; the assertion read output captured earlier in the suite

Teardown (suite) Terraform Destroy, Close All Connections

24.02 PASS A Plan From An Empty State Proposes Every Managed Resource (0.3s, 0 commands)

Two resources on each of three routers. Asserting the count catches a provider that silently drops a device from the loop — a plan that proposes four resources instead of six still looks like success in the summary line.

Setup (suite) Open All Routers, Generate Terraform Variables, Terraform Init, Terraform Destroy

Acceptance criteria

- output contains **6 to add**

  Terraform did not propose one loopback and one access list per router:\n${out}

How it was checked no device commands of its own; the assertion read output captured earlier in the suite

Teardown (suite) Terraform Destroy, Close All Connections

24.03 PASS Apply Puts The Loopback On Every Router (1.1s, 3 commands)

The push itself, read back from the running configuration.

Setup (suite) Open All Routers, Generate Terraform Variables, Terraform Init, Terraform Destroy

Acceptance criteria

- output contains **description Managed by Terraform** (checked 3×)
- output contains **255.255.255.255** (checked 3×)

How it was checked

R1 show running-config interface Loopback99

```
Building configuration...

Current configuration : 100 bytes
!
interface Loopback99
 description Managed by Terraform
 ip address 10.99.1.1 255.255.255.255
end
```

R2 show running-config interface Loopback99

```
Building configuration...

Current configuration : 100 bytes
!
interface Loopback99
 description Managed by Terraform
 ip address 10.99.2.1 255.255.255.255
end
```

R3 show running-config interface Loopback99

```
Building configuration...

Current configuration : 100 bytes
!
interface Loopback99
 description Managed by Terraform
 ip address 10.99.3.1 255.255.255.255
end
```

Teardown (suite) Terraform Destroy, Close All Connections

24.04 PASS Each Router Receives Its Own Address, Not A Shared One (0.2s, 3 commands)

A per-device value has to survive the provider's device loop. If it does not, every router ends up with the first one's address — which still passes a test that only checks "an address is configured".

Setup (suite) Open All Routers, Generate Terraform Variables, Terraform Init, Terraform Destroy

Acceptance criteria

- output contains **ip address ${expected}[${r}][address]** (checked 3×)

How it was checked

R1 show running-config interface Loopback99
```
Building configuration...

Current configuration : 100 bytes
!
interface Loopback99
 description Managed by Terraform
 ip address 10.99.1.1 255.255.255.255
end
```
R2 show running-config interface Loopback99
```
Building configuration...

Current configuration : 100 bytes
!
interface Loopback99
 description Managed by Terraform
 ip address 10.99.2.1 255.255.255.255
end
```
R3 show running-config interface Loopback99
```
Building configuration...

Current configuration : 100 bytes
!
interface Loopback99
 description Managed by Terraform
 ip address 10.99.3.1 255.255.255.255
end
```
Teardown (suite) Terraform Destroy, Close All Connections

24.05 PASS The Access List Arrives With Its Entries In Order (0.4s, 3 commands)

An ordered list is where a provider and a device are most likely to disagree. Sequence numbers make the order explicit, so it can be asserted rather than assumed.

Setup (suite) Open All Routers, Generate Terraform Variables, Terraform Init, Terraform Destroy

Acceptance criteria

- output contains **10 remark Managed by Terraform** (checked 3×)
- output contains **20 permit 10.99.0.0 0.0.255.255** (checked 3×)
- output contains **30 deny any** (checked 3×)
- **${idx} >= 0** holds (checked 9×)
- **${ten} < ${twenty} < ${thirty}** holds (checked 3×)

How it was checked

R1 show running-config | section ip access-list standard TF-MANAGED
```
ip access-list standard TF-MANAGED
 10 remark Managed by Terraform
 20 permit 10.99.0.0 0.0.255.255
 30 deny any
```
R2 show running-config | section ip access-list standard TF-MANAGED
```
ip access-list standard TF-MANAGED
 10 remark Managed by Terraform
 20 permit 10.99.0.0 0.0.255.255
 30 deny any
```
R3 show running-config | section ip access-list standard TF-MANAGED
```
ip access-list standard TF-MANAGED
 10 remark Managed by Terraform
```

```
  20 permit 10.99.0.0 0.0.255.255
  30 deny any
```

Teardown (suite) Terraform Destroy, Close All Connections

24.06 PASS Applying The Same Configuration Twice Proposes Nothing (0.4s, 0 commands)

Idempotence, and the sharpest single check in this suite. A provider that cannot read back what it wrote reports a permanent difference here, which would mean every future plan is noisy and real drift becomes invisible in it.

Setup (suite) Open All Routers, Generate Terraform Variables, Terraform Init, Terraform Destroy

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked no device commands of its own; the assertion read output captured earlier in the suite

Teardown (suite) Terraform Destroy, Close All Connections

24.07 PASS A Change Made By Hand Is Reported As Drift (3.6s, 4 commands)

The reason to keep configuration in Terraform at all. The device is edited from the CLI behind Terraform's back, and the next plan has to notice. The wait is not padding. A change made at the CLI is not immediately visible over RESTCONF: measured on this platform, a plan run in the same second reports no drift, and the same plan two seconds later reports it. Polling until the plan sees it keeps the test honest about what is being asserted -- that the drift is detected at all, not that it is detected instantly. It is also worth knowing beyond this test: any tool that writes at the CLI and verifies over RESTCONF can read its own change back as absent.

Setup (suite) Open All Routers, Generate Terraform Variables, Terraform Init, Terraform Destroy

Acceptance criteria

- output contains **${HAND_EDIT}**
  the plan proposes changes but does not name the hand edit, so it may
- output contains **1 to change**
  drift on one attribute of one device did not produce exactly one change

How it was checked

R2 configure terminal

```
Enter configuration commands, one per line.  End with CNTL/Z.
```

R2 interface Loopback99

```
${output} =
```

R2 description Changed by hand

```
${output} =
```

R2 end

```
${output} =
```

Teardown (this test) Terraform Apply

24.08 PASS Applying Puts A Hand-Edited Device Back (0.4s, 1 commands)

Detection is only half of it; the correction is what makes the declared state authoritative.

Setup (suite) Open All Routers, Generate Terraform Variables, Terraform Init, Terraform Destroy

Acceptance criteria

- output contains **description Managed by Terraform**
  R2 was not returned to the state Terraform declares
- output does not contain **${HAND_EDIT}**
  the hand edit survived an apply

How it was checked

R2 show running-config interface Loopback99

```
Building configuration...

Current configuration : 100 bytes
!
interface Loopback99
 description Managed by Terraform
 ip address 10.99.2.1 255.255.255.255
```

```
end
```

Teardown (suite) Terraform Destroy, Close All Connections

24.09 PASS Destroy Removes Everything Terraform Created (3.2s, 6 commands)

Taking it back matters as much as putting it there: a lab that cannot be returned to a known state stops being a lab.

Setup (suite) Open All Routers, Generate Terraform Variables, Terraform Init, Terraform Destroy

Acceptance criteria

- output does not contain **Managed by Terraform** (checked 3×)
- output does not contain **${TF_ACL}** (checked 3×)
- **${count}** equals **0**

  Terraform still holds ${count} resources in state after destroy

How it was checked

R1 **`show running-config interface Loopback99`**

```
^
% Invalid input detected at '^' marker.
```

R1 **`show running-config | include ip access-list standard TF-MANAGED`**

```
${output} =
```

R2 **`show running-config interface Loopback99`**

```
^
% Invalid input detected at '^' marker.
```

R2 **`show running-config | include ip access-list standard TF-MANAGED`**

```
${output} =
```

R3 **`show running-config interface Loopback99`**

```
^
% Invalid input detected at '^' marker.
```

R3 **`show running-config | include ip access-list standard TF-MANAGED`**

```
${output} =
```

Teardown (suite) Terraform Destroy, Close All Connections

25 Terraform Nac

25.01 PASS The Module Resolves Against A NETCONF-Only Provider (0.7s, 0 commands)

Recording the constraint rather than discovering it again. The module requires iosxe >= 1.0.0; 1.0.0 is NETCONF-only. If a later provider restores RESTCONF, or the module relaxes its floor, this test is where that shows up -- and the device host in the model would need revisiting, because it points at the forwarded NETCONF port.

Setup (suite) Open All Routers, Render Nac Model, Nac Init, Nac Destroy

Acceptance criteria

- **${version}** starts with **1.**
  the locked provider is ${version}; this suite's NETCONF assumption was

How it was checked no device commands of its own; the assertion read output captured earlier in the suite

Teardown (suite) Nac Destroy, Close All Connections

25.02 PASS The Data Model Carries An Address For Every Router (0.0s, 0 commands)

The trap this module sets. Every optional key is read through try(), so a key at the wrong nesting depth is not an error -- it is dropped in silence. An address written at the top of an interface entry instead of under ipv4: yields a successful plan that creates a loopback with no address on it. So the model is checked for shape before anything is pushed, and the address is asserted on the device afterwards. Either check alone would have missed it.

Setup (suite) Open All Routers, Render Nac Model, Nac Init, Nac Destroy

Acceptance criteria

- **${devices}** has exactly **3** entries
  the model does not describe all three routers
- **${loopbacks}[0]** contains the key **ipv4** (checked 3×)
  would accept and quietly push without an address
- **${loopbacks}[0][ipv4]** contains the key **address** (checked 3×)

How it was checked no device commands of its own; the assertion read output captured earlier in the suite

Teardown (suite) Nac Destroy, Close All Connections

25.03 PASS Applying The Model Configures Every Router (2.4s, 3 commands)

The push itself, read back from running configuration.

Setup (suite) Open All Routers, Render Nac Model, Nac Init, Nac Destroy

Acceptance criteria

- output contains **description ${NAC_DESC}** (checked 3×)

How it was checked

R1 show running-config interface Loopback98
```
Building configuration...

Current configuration : 106 bytes
!
interface Loopback98
 description Managed by Network as Code
 ip address 10.98.1.1 255.255.255.255
end
```
R2 show running-config interface Loopback98
```
Building configuration...

Current configuration : 106 bytes
!
interface Loopback98
 description Managed by Network as Code
 ip address 10.98.2.1 255.255.255.255
end
```
R3 show running-config interface Loopback98
```
Building configuration...
```

```
Current configuration : 106 bytes
!
interface Loopback98
 description Managed by Network as Code
 ip address 10.98.3.1 255.255.255.255
end
```

Teardown (suite) Nac Destroy, Close All Connections

25.04 PASS The Address From The Model Reaches The Device (0.1s, 3 commands)

The half that catches a silently dropped key: not that a loopback exists, but that it carries the address the model declares for that specific router.

Setup (suite) Open All Routers, Render Nac Model, Nac Init, Nac Destroy

Acceptance criteria

- output contains **ip address ${d}[configuration][interfaces][loopbacks][0][ipv4][address]** (checked 3×)
  which is what a mis-nested key looks like from the device side

How it was checked

? show running-config interface Loopback98

```
Building configuration...

Current configuration : 106 bytes
!
interface Loopback98
 description Managed by Network as Code
 ip address 10.98.1.1 255.255.255.255
end
```

? show running-config interface Loopback98

```
Building configuration...

Current configuration : 106 bytes
!
interface Loopback98
 description Managed by Network as Code
 ip address 10.98.2.1 255.255.255.255
end
```

? show running-config interface Loopback98

```
Building configuration...

Current configuration : 106 bytes
!
interface Loopback98
 description Managed by Network as Code
 ip address 10.98.3.1 255.255.255.255
end
```

Teardown (suite) Nac Destroy, Close All Connections

25.05 PASS The Abstract Access List Model Renders To Ordered Entries (0.3s, 3 commands)

The model says action: permit with a prefix; the device gets "20 permit 10.98.0.0 0.0.255.255". This checks the translation, and that ordering survives it.

Setup (suite) Open All Routers, Render Nac Model, Nac Init, Nac Destroy

Acceptance criteria

- output contains **10 remark ${NAC_DESC}** (checked 3×)
- output contains **20 permit 10.98.0.0 0.0.255.255** (checked 3×)
- output contains **30 deny any** (checked 3×)

How it was checked

R1 show running-config | section ip access-list standard NAC-MANAGED

```
ip access-list standard NAC-MANAGED
 10 remark Managed by Network as Code
 20 permit 10.98.0.0 0.0.255.255
 30 deny any
```

R2 show running-config | section ip access-list standard NAC-MANAGED

```
ip access-list standard NAC-MANAGED
```

```
 10 remark Managed by Network as Code
 20 permit 10.98.0.0 0.0.255.255
 30 deny any
```

R3 **show running-config | section ip access-list standard NAC-MANAGED**

```
ip access-list standard NAC-MANAGED
 10 remark Managed by Network as Code
 20 permit 10.98.0.0 0.0.255.255
 30 deny any
```

Teardown (suite) Nac Destroy, Close All Connections

25.06 PASS Rendering The Same Model Twice Proposes Nothing (1.4s, 0 commands)

Idempotence across the whole rendering path: YAML to model to resources to device, and back again.

Setup (suite) Open All Routers, Render Nac Model, Nac Init, Nac Destroy

Acceptance criteria none asserted directly — this test captures state for the record, and is checked by the tests that read it

How it was checked no device commands of its own; the assertion read output captured earlier in the suite

Teardown (suite) Nac Destroy, Close All Connections

25.07 PASS A Change Made By Hand Is Reported As Drift (3.9s, 4 commands)

Worth comparing with the RESTCONF suite, which has to wait for this. Over RESTCONF a plan run in the same second as a CLI edit reports no drift and needs about two seconds to see it. Over NETCONF the same edit is visible immediately, measured at zero delay. That is a property of the transport, not of Terraform, and it is asserted here without a wait deliberately: if this test ever starts needing one, NETCONF has begun behaving like RESTCONF and that is worth being told about.

Setup (suite) Open All Routers, Render Nac Model, Nac Init, Nac Destroy

Acceptance criteria

- output contains **${HAND_EDIT}**
  the plan proposes changes but does not name the hand edit:\n${out}

How it was checked

R2 **configure terminal**

```
Enter configuration commands, one per line.  End with CNTL/Z.
```

R2 **interface Loopback98**

```
${output} =
```

R2 **description Changed by hand**

```
${output} =
```

R2 **end**

```
${output} =
```

Teardown (this test) Nac Apply

25.08 PASS Applying Puts A Hand-Edited Device Back (1.5s, 1 commands)

Setup (suite) Open All Routers, Render Nac Model, Nac Init, Nac Destroy

Acceptance criteria

- output contains **description ${NAC_DESC}**
  R2 was not returned to the state the model declares
- output does not contain **${HAND_EDIT}**

How it was checked

R2 **show running-config interface Loopback98**

```
Building configuration...

Current configuration : 106 bytes
!
interface Loopback98
 description Managed by Network as Code
 ip address 10.98.2.1 255.255.255.255
end
```

Teardown (suite) Nac Destroy, Close All Connections

25.09 PASS Destroy Removes Everything The Model Created (5.0s, 6 commands)

Setup (suite) Open All Routers, Render Nac Model, Nac Init, Nac Destroy

Acceptance criteria

- output does not contain **${NAC_DESC}** (checked 3×)
- output does not contain **${NAC_ACL}** (checked 3×)
- **${count}** equals **0**
  the module still holds ${count} resources in state after destroy

How it was checked

R1 `show running-config interface Loopback98`

```
^
% Invalid input detected at '^' marker.
```

R1 `show running-config | include ip access-list standard NAC-MANAGED`

```
${output} =
```

R2 `show running-config interface Loopback98`

```
^
% Invalid input detected at '^' marker.
```

R2 `show running-config | include ip access-list standard NAC-MANAGED`

```
${output} =
```

R3 `show running-config interface Loopback98`

```
^
% Invalid input detected at '^' marker.
```

R3 `show running-config | include ip access-list standard NAC-MANAGED`

```
${output} =
```

Teardown (suite) Nac Destroy, Close All Connections

25.10 PASS The Two Approaches Do Not Contend For The Same Objects (0.0s, 0 commands)

Both suites push a loopback and a standard access list to the same three routers. If they ever named the same object, each would revert the other and the symptom would be an intermittent failure in whichever ran second.

Setup (suite) Open All Routers, Render Nac Model, Nac Init, Nac Destroy

Acceptance criteria

- **${d}[configuration][interfaces][loopbacks][0][id]** equals **98** (checked 3×)
  the Network as Code model no longer owns Loopback98; check it has
- **${d}[configuration][access_lists][standard][0][name]** equals **${NAC_ACL}** (checked 3×)

How it was checked no device commands of its own; the assertion read output captured earlier in the suite

Teardown (suite) Nac Destroy, Close All Connections

Appendix — captured device state

Written by suite *05 Capture State* during the run. Routing tables and protocol state first, then what the NMS sees for each router -- polled over SNMPv3, and received as syslog and traps -- then the full running configuration of each router.

R1 — ip route

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
       n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       H - NHRP, G - NHRP registered, g - NHRP registration summary
       o - ODR, P - periodic downloaded static route, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR
       & - replicated local route overrides by connected

Gateway of last resort is 10.0.2.2 to network 0.0.0.0

S*    0.0.0.0/0 [254/0] via 10.0.2.2
      10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
C        10.0.2.0/24 is directly connected, GigabitEthernet1
L        10.0.2.15/32 is directly connected, GigabitEthernet1
C        10.20.1.1/32 is directly connected, Loopback1
B        10.20.2.1/32 [20/0] via 172.20.0.2, 09:05:56
B        10.20.3.1/32 [20/0] via 172.20.0.3, 09:06:58
S        10.30.1.0/24 is directly connected, Null0
B        10.30.2.0/24 [20/0] via 172.20.0.2, 09:05:56
B        10.30.3.0/24 [20/0] via 172.20.0.3, 09:06:58
      100.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        100.64.0.0/24 is directly connected, GigabitEthernet2
L        100.64.0.1/32 is directly connected, GigabitEthernet2
      172.16.0.0/32 is subnetted, 1 subnets
C        172.16.1.1 is directly connected, Loopback0
      172.20.0.0/16 is variably subnetted, 2 subnets, 2 masks
C        172.20.0.0/24 is directly connected, Tunnel0
L        172.20.0.1/32 is directly connected, Tunnel0
      192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.10.0/24 is directly connected, GigabitEthernet3
L        192.168.10.1/32 is directly connected, GigabitEthernet3
B     192.168.20.0/24 [20/0] via 172.20.0.2, 09:05:56
B     192.168.30.0/24 [20/0] via 172.20.0.3, 09:06:58
```

R2 — ip route

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
       n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       H - NHRP, G - NHRP registered, g - NHRP registration summary
       o - ODR, P - periodic downloaded static route, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR
       & - replicated local route overrides by connected

Gateway of last resort is 10.0.2.2 to network 0.0.0.0

S*    0.0.0.0/0 [254/0] via 10.0.2.2
      10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
C        10.0.2.0/24 is directly connected, GigabitEthernet1
L        10.0.2.15/32 is directly connected, GigabitEthernet1
B        10.20.1.1/32 [20/0] via 172.20.0.1, 09:05:56
C        10.20.2.1/32 is directly connected, Loopback1
B        10.20.3.1/32 [20/0] via 172.20.0.3, 09:05:56
B        10.30.1.0/24 [20/0] via 172.20.0.1, 09:05:56
S        10.30.2.0/24 is directly connected, Null0
B        10.30.3.0/24 [20/0] via 172.20.0.3, 09:05:56
      100.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        100.64.0.0/24 is directly connected, GigabitEthernet2
L        100.64.0.2/32 is directly connected, GigabitEthernet2
      172.16.0.0/32 is subnetted, 1 subnets
```

```
C        172.16.2.1 is directly connected, Loopback0
      172.20.0.0/16 is variably subnetted, 2 subnets, 2 masks
C        172.20.0.0/24 is directly connected, Tunnel0
L        172.20.0.2/32 is directly connected, Tunnel0
B     192.168.10.0/24 [20/0] via 172.20.0.1, 09:05:56
      192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.20.0/24 is directly connected, GigabitEthernet3
L        192.168.20.1/32 is directly connected, GigabitEthernet3
B     192.168.30.0/24 [20/0] via 172.20.0.3, 09:05:56
```

R3 — ip route

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
       n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       H - NHRP, G - NHRP registered, g - NHRP registration summary
       o - ODR, P - periodic downloaded static route, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR
       & - replicated local route overrides by connected

Gateway of last resort is 10.0.2.2 to network 0.0.0.0

S*    0.0.0.0/0 [254/0] via 10.0.2.2
      10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
C        10.0.2.0/24 is directly connected, GigabitEthernet1
L        10.0.2.15/32 is directly connected, GigabitEthernet1
B        10.20.1.1/32 [20/0] via 172.20.0.1, 09:06:58
B        10.20.2.1/32 [20/0] via 172.20.0.2, 09:05:27
C        10.20.3.1/32 is directly connected, Loopback1
B        10.30.1.0/24 [20/0] via 172.20.0.1, 09:06:58
B        10.30.2.0/24 [20/0] via 172.20.0.2, 09:05:27
S        10.30.3.0/24 is directly connected, Null0
      100.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        100.64.0.0/24 is directly connected, GigabitEthernet2
L        100.64.0.3/32 is directly connected, GigabitEthernet2
      172.16.0.0/32 is subnetted, 1 subnets
C        172.16.3.1 is directly connected, Loopback0
      172.20.0.0/16 is variably subnetted, 2 subnets, 2 masks
C        172.20.0.0/24 is directly connected, Tunnel0
L        172.20.0.3/32 is directly connected, Tunnel0
B     192.168.10.0/24 [20/0] via 172.20.0.1, 09:06:58
B     192.168.20.0/24 [20/0] via 172.20.0.2, 09:05:27
      192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.30.0/24 is directly connected, GigabitEthernet3
L        192.168.30.1/32 is directly connected, GigabitEthernet3
```

R1 — bgp summary

```
BGP router identifier 172.16.1.1, local AS number 65001
BGP table version is 34, main routing table version 34
9 network entries using 2232 bytes of memory
9 path entries using 1224 bytes of memory
7/7 BGP path/bestpath attribute entries using 2072 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
6 BGP community entries using 144 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 5720 total bytes of memory
BGP activity 303/294 prefixes, 324/315 paths, scan interval 60 secs
22 networks peaked at 02:27:59 Sep 8 2026 UTC (22:26:44.598 ago)

Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
172.20.0.2      4        65002    3445    3452       34    0    0 09:05:56        3
172.20.0.3      4        65003    3454    3458       34    0    0 09:06:58        3
```

R2 — bgp summary

```
BGP router identifier 172.16.2.1, local AS number 65002
BGP table version is 46, main routing table version 46
9 network entries using 2232 bytes of memory
9 path entries using 1224 bytes of memory
7/7 BGP path/bestpath attribute entries using 2072 bytes of memory
```

```
2 BGP AS-PATH entries using 64 bytes of memory
6 BGP community entries using 144 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 5736 total bytes of memory
BGP activity 333/324 prefixes, 348/339 paths, scan interval 60 secs
31 networks peaked at 02:28:40 Sep 8 2026 UTC (22:26:03.670 ago)

Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
172.20.0.1      4        65001    3452    3445       46    0    0 09:05:56        6
```

R3 — bgp summary

```
BGP router identifier 172.16.3.1, local AS number 65003
BGP table version is 490, main routing table version 490
9 network entries using 2232 bytes of memory
9 path entries using 1224 bytes of memory
7/7 BGP path/bestpath attribute entries using 2072 bytes of memory
2 BGP AS-PATH entries using 64 bytes of memory
6 BGP community entries using 144 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 5736 total bytes of memory
BGP activity 240/231 prefixes, 249/240 paths, scan interval 60 secs
9 networks peaked at 01:35:13 Sep 8 2026 UTC (23:19:30.752 ago)

Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
172.20.0.1      4        65001    3458    3454      490    0    0 09:06:58        6
```

R1 — bgp table

```
BGP table version is 34, local router ID is 172.16.1.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
              t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *>   10.20.1.1/32     0.0.0.0                  0         32768 i
 *>   10.20.2.1/32     172.20.0.2               0             0 65002 i
 *>   10.20.3.1/32     172.20.0.3               0             0 65003 i
 *>   10.30.1.0/24     0.0.0.0                  0         32768 i
 *>   10.30.2.0/24     172.20.0.2               0             0 65002 i
 *>   10.30.3.0/24     172.20.0.3               0             0 65003 i
 *>   192.168.10.0     0.0.0.0                  0         32768 i
 *>   192.168.20.0     172.20.0.2               0             0 65002 i
 *>   192.168.30.0     172.20.0.3               0             0 65003 i
```

R2 — bgp table

```
BGP table version is 46, local router ID is 172.16.2.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
              t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *>   10.20.1.1/32     172.20.0.1               0             0 65001 i
 *>   10.20.2.1/32     0.0.0.0                  0         32768 i
 *>   10.20.3.1/32     172.20.0.3                             0 65001 65003 i
 *>   10.30.1.0/24     172.20.0.1               0             0 65001 i
 *>   10.30.2.0/24     0.0.0.0                  0         32768 i
 *>   10.30.3.0/24     172.20.0.3                             0 65001 65003 i
 *>   192.168.10.0     172.20.0.1               0             0 65001 i
 *>   192.168.20.0     0.0.0.0                  0         32768 i
 *>   192.168.30.0     172.20.0.3                             0 65001 65003 i
```

R3 — bgp table

```
BGP table version is 490, local router ID is 172.16.3.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
```

```
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
              t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *>   10.20.1.1/32     172.20.0.1               0             0 65001 i
 *>   10.20.2.1/32     172.20.0.2                             0 65001 65002 i
 *>   10.20.3.1/32     0.0.0.0                  0         32768 i
 *>   10.30.1.0/24     172.20.0.1               0             0 65001 i
 *>   10.30.2.0/24     172.20.0.2                             0 65001 65002 i
 *>   10.30.3.0/24     0.0.0.0                  0         32768 i
 *>   192.168.10.0     172.20.0.1               0             0 65001 i
 *>   192.168.20.0     172.20.0.2                             0 65001 65002 i
 *>   192.168.30.0     0.0.0.0                  0         32768 i
```

R1 — crypto ikev2 sa

```
 IPv4 Crypto IKEv2  SA

Tunnel-id Local                 Remote                fvrf/ivrf            Status
1         100.64.0.1/500        100.64.0.3/500        none/none            READY
      Encr: AES-CBC, keysize: 256, PRF: SHA256, Hash: SHA256, DH Grp:14, Auth sign: PSK, Auth verify: PSK
      Life/Active Time: 86400/32545 sec
      CE id: 1154, Session-id: 149
      Local spi: 552EA031EC5AD516       Remote spi: DF541840554A4190

Tunnel-id Local                 Remote                fvrf/ivrf            Status
2         100.64.0.1/500        100.64.0.2/500        none/none            READY
      Encr: AES-CBC, keysize: 256, PRF: SHA256, Hash: SHA256, DH Grp:14, Auth sign: PSK, Auth verify: PSK
      Life/Active Time: 86400/32544 sec
      CE id: 1155, Session-id: 150
      Local spi: FA434BF0F1020BEE       Remote spi: E99229818F7EF5EB

 IPv6 Crypto IKEv2  SA
```

R2 — crypto ikev2 sa

```
 IPv4 Crypto IKEv2  SA

Tunnel-id Local                 Remote                fvrf/ivrf            Status
1         100.64.0.2/500        100.64.0.1/500        none/none            READY
      Encr: AES-CBC, keysize: 256, PRF: SHA256, Hash: SHA256, DH Grp:14, Auth sign: PSK, Auth verify: PSK
      Life/Active Time: 86400/32545 sec
      CE id: 1156, Session-id: 141
      Local spi: E99229818F7EF5EB       Remote spi: FA434BF0F1020BEE

 IPv6 Crypto IKEv2  SA
```

R3 — crypto ikev2 sa

```
 IPv4 Crypto IKEv2  SA

Tunnel-id Local                 Remote                fvrf/ivrf            Status
1         100.64.0.3/500        100.64.0.1/500        none/none            READY
      Encr: AES-CBC, keysize: 256, PRF: SHA256, Hash: SHA256, DH Grp:14, Auth sign: PSK, Auth verify: PSK
      Life/Active Time: 86400/32546 sec
      CE id: 1150, Session-id: 145
      Local spi: DF541840554A4190       Remote spi: 552EA031EC5AD516

 IPv6 Crypto IKEv2  SA
```

H1 — ip addr

```
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 52:54:00:aa:01:01 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
       valid_lft 45682sec preferred_lft 34882sec
```

```
    inet6 fec0::5054:ff:feaa:101/64 scope site dynamic noprefixroute flags 100
       valid_lft 86385sec preferred_lft 14385sec
    inet6 fe80::5054:ff:feaa:101/64 scope link
       valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 52:54:00:aa:01:02 brd ff:ff:ff:ff:ff:ff
    inet 192.168.10.10/24 scope global eth1
       valid_lft forever preferred_lft forever
    inet 169.254.229.55/16 brd 169.254.255.255 scope global noprefixroute eth1
       valid_lft forever preferred_lft forever
    inet 192.168.10.11/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.10.12/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.10.13/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.10.14/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.10.15/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.10.16/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.10.17/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.10.18/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.10.19/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.10.64/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.10.65/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.10.66/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.10.67/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.10.68/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.10.69/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.10.70/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.10.71/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.10.72/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.10.73/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.10.74/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.10.75/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.10.76/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.10.77/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.10.78/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.10.79/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.10.80/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.10.81/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.10.82/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.10.83/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.10.84/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.10.85/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.10.86/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.10.87/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.10.88/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.10.89/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.10.90/24 scope global secondary eth1
```

```
       valid_lft forever preferred_lft forever
    inet 192.168.10.91/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.10.92/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.10.93/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
```

H1 — ip route

```
default via 10.0.2.2 dev eth0  src 10.0.2.15  metric 1002
10.0.2.0/24 dev eth0 scope link  src 10.0.2.15  metric 1002
10.30.0.0/16 via 192.168.10.1 dev eth1
169.254.0.0/16 dev eth1 scope link  src 169.254.229.55  metric 1003
192.168.10.0/24 dev eth1 scope link  src 192.168.10.10
192.168.20.0/24 via 192.168.10.1 dev eth1
192.168.30.0/24 via 192.168.10.1 dev eth1
```

H2 — ip addr

```
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 52:54:00:aa:02:01 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
       valid_lft 45682sec preferred_lft 34882sec
    inet6 fec0::5054:ff:feaa:201/64 scope site dynamic noprefixroute flags 100
       valid_lft 86191sec preferred_lft 14191sec
    inet6 fe80::5054:ff:feaa:201/64 scope link
       valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 52:54:00:aa:02:02 brd ff:ff:ff:ff:ff:ff
    inet 192.168.20.10/24 scope global eth1
       valid_lft forever preferred_lft forever
    inet 169.254.199.41/16 brd 169.254.255.255 scope global noprefixroute eth1
       valid_lft forever preferred_lft forever
    inet 192.168.20.11/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.20.12/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.20.13/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.20.14/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.20.15/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.20.16/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.20.17/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.20.18/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.20.19/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.20.64/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.20.65/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.20.66/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.20.67/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.20.68/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.20.69/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.20.70/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.20.71/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.20.72/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.20.73/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
```

```
    inet 192.168.20.74/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.20.75/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.20.76/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.20.77/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.20.78/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.20.79/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.20.80/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.20.81/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.20.82/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.20.83/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.20.84/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.20.85/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.20.86/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.20.87/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.20.88/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.20.89/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.20.90/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.20.91/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.20.92/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.20.93/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
```

H2 — ip route

```
default via 10.0.2.2 dev eth0  src 10.0.2.15  metric 1002
10.0.2.0/24 dev eth0 scope link  src 10.0.2.15  metric 1002
10.30.0.0/16 via 192.168.20.1 dev eth1
169.254.0.0/16 dev eth1 scope link  src 169.254.199.41  metric 1003
192.168.10.0/24 via 192.168.20.1 dev eth1
192.168.20.0/24 dev eth1 scope link  src 192.168.20.10
192.168.30.0/24 via 192.168.20.1 dev eth1
```

H3 — ip addr

```
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 52:54:00:aa:03:01 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
       valid_lft 45683sec preferred_lft 34883sec
    inet6 fec0::5054:ff:feaa:301/64 scope site dynamic noprefixroute flags 100
       valid_lft 86305sec preferred_lft 14305sec
    inet6 fe80::5054:ff:feaa:301/64 scope link
       valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 52:54:00:aa:03:02 brd ff:ff:ff:ff:ff:ff
    inet 192.168.30.10/24 scope global eth1
       valid_lft forever preferred_lft forever
    inet 169.254.207.242/16 brd 169.254.255.255 scope global noprefixroute eth1
       valid_lft forever preferred_lft forever
    inet 192.168.30.11/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.30.12/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.30.13/24 scope global secondary eth1
```

```
       valid_lft forever preferred_lft forever
    inet 192.168.30.14/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.30.15/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.30.16/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.30.17/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.30.18/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.30.19/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.30.64/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.30.65/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.30.66/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.30.67/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.30.68/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.30.69/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.30.70/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.30.71/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.30.72/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.30.73/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.30.74/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.30.75/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.30.76/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.30.77/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.30.78/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.30.79/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.30.80/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.30.81/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.30.82/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.30.83/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.30.84/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.30.85/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.30.86/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.30.87/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.30.88/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.30.89/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.30.90/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.30.91/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.30.92/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
    inet 192.168.30.93/24 scope global secondary eth1
       valid_lft forever preferred_lft forever
```

H3 — ip route

```
default via 10.0.2.2 dev eth0  src 10.0.2.15  metric 1002
10.0.2.0/24 dev eth0 scope link  src 10.0.2.15  metric 1002
10.30.0.0/16 via 192.168.30.1 dev eth1
169.254.0.0/16 dev eth1 scope link  src 169.254.207.242  metric 1003
```

```
192.168.10.0/24 via 192.168.30.1 dev eth1
192.168.20.0/24 via 192.168.30.1 dev eth1
192.168.30.0/24 dev eth1 scope link  src 192.168.30.10
```

NMS — network

```
lo               UNKNOWN        127.0.0.1/8 ::1/128
ens3             UP             10.0.2.15/24 metric 100 fec0::5054:ff:febb:101/64 fe80::5054:ff:febb:101/64
ens4             UP             192.168.10.20/24 fe80::5054:ff:febb:102/64
ens5             UP             192.168.99.10/24 fe80::5054:ff:febb:103/64
default via 10.0.2.2 dev ens3 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev ens3 proto kernel scope link src 10.0.2.15 metric 100
10.0.2.2 dev ens3 proto dhcp scope link src 10.0.2.15 metric 100
10.0.2.3 dev ens3 proto dhcp scope link src 10.0.2.15 metric 100
10.30.0.0/16 via 192.168.10.1 dev ens4
192.168.10.0/24 dev ens4 proto kernel scope link src 192.168.10.20
192.168.20.0/24 via 192.168.10.1 dev ens4
192.168.30.0/24 via 192.168.10.1 dev ens4
192.168.99.0/24 dev ens5 proto kernel scope link src 192.168.99.10
```

NMS — snmp version

```
NET-SNMP version: 5.9.4.pre2
```

NMS — collectors

```
active
active
active
```

NMS — collector files

```
total 10600
-rw-r--r-- 1 syslog syslog  527673 Sep  9 00:54 192.168.99.1.log
-rw-r--r-- 1 syslog syslog  653218 Sep  9 00:54 192.168.99.2.log
-rw-r--r-- 1 syslog syslog  455348 Sep  9 00:54 192.168.99.3.log
-rw-r--r-- 1 root   root   2824382 Sep  9 00:54 snmptrapd.log
-rw-r--r-- 1 root   root   6362155 Sep  9 00:54 traps-raw.log
```

NMS — R1 syslog received

```
tail: cannot open '/var/log/lab/192.168.10.1.log' for reading: No such file or directory
```

NMS — R2 syslog received

```
tail: cannot open '/var/log/lab/192.168.20.1.log' for reading: No such file or directory
```

NMS — R3 syslog received

```
tail: cannot open '/var/log/lab/192.168.30.1.log' for reading: No such file or directory
```

NMS — snmptrapd

```
2026-09-09 00:54:16 <UNKNOWN> [UDP: [192.168.99.3]:50504->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:23:19:44.43       iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.171.2.0.7
iso.3.6.1.4.1.9.9.171.1.3.2.1.9.148 3600        iso.3.6.1.4.1.9.9.171.1.3.2.1.8.148 4608000
iso.3.6.1.4.1.9.9.171.1.3.3.1.3.148.1 1 iso.3.6.1.4.1.9.9.171.1.3.3.1.4.148.1 "64 40 00 03 "
iso.3.6.1.4.1.9.9.171.1.3.3.1.5.148.1 "64 40 00 03 "    iso.3.6.1.4.1.9.9.171.1.3.3.1.9.148.1 1
iso.3.6.1.4.1.9.9.171.1.3.3.1.10.148.1 "64 40 00 02 "   iso.3.6.1.4.1.9.9.171.1.3.3.1.11.148.1 "64 40 00 02
"   iso.3.6.1.4.1.9.9.171.1.3.3.1.6.148.1 47        iso.3.6.1.4.1.9.9.171.1.3.3.1.7.148.1 0
2026-09-09 00:54:16 <UNKNOWN> [UDP: [192.168.99.2]:51414->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:23:19:44.78       iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.171.2.0.7
iso.3.6.1.4.1.9.9.171.1.3.2.1.9.150 3600        iso.3.6.1.4.1.9.9.171.1.3.2.1.8.150 4608000
iso.3.6.1.4.1.9.9.171.1.3.3.1.3.150.1 1 iso.3.6.1.4.1.9.9.171.1.3.3.1.4.150.1 "64 40 00 02 "
iso.3.6.1.4.1.9.9.171.1.3.3.1.5.150.1 "64 40 00 02 "    iso.3.6.1.4.1.9.9.171.1.3.3.1.9.150.1 1
iso.3.6.1.4.1.9.9.171.1.3.3.1.10.150.1 "64 40 00 03 "   iso.3.6.1.4.1.9.9.171.1.3.3.1.11.150.1 "64 40 00 03
"   iso.3.6.1.4.1.9.9.171.1.3.3.1.6.150.1 47        iso.3.6.1.4.1.9.9.171.1.3.3.1.7.150.1 0
2026-09-09 00:54:16 <UNKNOWN> [UDP: [192.168.99.3]:50504->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:23:19:45.19       iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.10.137.0.1
```

```
iso.3.6.1.4.1.9.10.137.1.2.1.8.1708097 0        iso.3.6.1.4.1.9.10.137.1.2.1.8.1708097 0
iso.3.6.1.2.1.2.2.1.1.9 9
2026-09-09 00:54:16 <UNKNOWN> [UDP: [192.168.99.2]:51414->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:23:19:45.55       iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.10.137.0.1
iso.3.6.1.4.1.9.10.137.1.2.1.8.1708044 0        iso.3.6.1.4.1.9.10.137.1.2.1.8.1708044 0
iso.3.6.1.2.1.2.2.1.1.9 9
2026-09-09 00:54:17 <UNKNOWN> [UDP: [192.168.99.2]:51414->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:23:19:46.94       iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.10.62.2.0.4
iso.3.6.1.4.1.9.10.62.1.2.3.1.1.2.14.84.117.110.110.101.108.48.45.104.101.97.100.45.48 3
2026-09-09 00:54:17 <UNKNOWN> [UDP: [192.168.99.3]:50504->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:23:19:46.59       iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.171.2.0.8
iso.3.6.1.4.1.9.9.171.1.3.2.1.10.148 216        iso.3.6.1.4.1.9.9.171.1.4.3.1.1.12.147 216
iso.3.6.1.4.1.9.9.171.1.4.3.1.1.10.147 3600     iso.3.6.1.4.1.9.9.171.1.4.3.1.1.9.147 4608000
iso.3.6.1.4.1.9.9.171.1.4.3.1.1.2.147 1 iso.3.6.1.4.1.9.9.171.1.4.3.2.1.5.147 1
iso.3.6.1.4.1.9.9.171.1.4.3.2.1.11.147 1        iso.3.6.1.4.1.9.9.171.1.4.3.2.1.6.147 "64 40 00 03 "
iso.3.6.1.4.1.9.9.171.1.4.3.2.1.7.147 "64 40 00 03 "    iso.3.6.1.4.1.9.9.171.1.4.3.2.1.12.147 "64 40 00 02
"   iso.3.6.1.4.1.9.9.171.1.4.3.2.1.13.147 "64 40 00 02 "   iso.3.6.1.4.1.9.9.171.1.4.3.2.1.8.147 47
iso.3.6.1.4.1.9.9.171.1.4.3.2.1.9.147 0
2026-09-09 00:54:18 <UNKNOWN> [UDP: [192.168.99.2]:51414->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:23:19:46.94       iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.171.2.0.8
iso.3.6.1.4.1.9.9.171.1.3.2.1.10.150 215        iso.3.6.1.4.1.9.9.171.1.4.3.1.1.12.149 215
iso.3.6.1.4.1.9.9.171.1.4.3.1.1.10.149 3600     iso.3.6.1.4.1.9.9.171.1.4.3.1.1.9.149 4608000
iso.3.6.1.4.1.9.9.171.1.4.3.1.1.2.149 1 iso.3.6.1.4.1.9.9.171.1.4.3.2.1.5.149 1
iso.3.6.1.4.1.9.9.171.1.4.3.2.1.11.149 1        iso.3.6.1.4.1.9.9.171.1.4.3.2.1.6.149 "64 40 00 02 "
iso.3.6.1.4.1.9.9.171.1.4.3.2.1.7.149 "64 40 00 02 "    iso.3.6.1.4.1.9.9.171.1.4.3.2.1.12.149 "64 40 00 03
"   iso.3.6.1.4.1.9.9.171.1.4.3.2.1.13.149 "64 40 00 03 "   iso.3.6.1.4.1.9.9.171.1.4.3.2.1.8.149 47
iso.3.6.1.4.1.9.9.171.1.4.3.2.1.9.149 0
2026-09-09 00:54:18 <UNKNOWN> [UDP: [192.168.99.3]:50504->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:23:19:46.59       iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.10.62.2.0.4
iso.3.6.1.4.1.9.10.62.1.2.3.1.1.2.14.84.117.110.110.101.108.48.45.104.101.97.100.45.48 3
2026-09-09 00:54:18 <UNKNOWN> [UDP: [192.168.99.2]:51414->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:23:19:46.94       iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.10.137.0.2
iso.3.6.1.4.1.9.10.137.1.2.1.8.987148 3 iso.3.6.1.4.1.9.10.137.1.2.1.8.987148 3 iso.3.6.1.2.1.2.2.1.1.9 9
2026-09-09 00:54:18 <UNKNOWN> [UDP: [192.168.99.3]:50504->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:23:19:46.60       iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.171.2.0.2
iso.3.6.1.4.1.9.9.171.1.2.2.1.6.1.4.100.64.0.3.1.4.100.64.0.2.153 "64 40 00 03 "
iso.3.6.1.4.1.9.9.171.1.2.2.1.7.1.4.100.64.0.3.1.4.100.64.0.2.153 "64 40 00 02 "
iso.3.6.1.4.1.9.9.171.1.2.3.1.16.153 2  iso.3.6.1.4.1.9.9.171.1.4.2.1.1.2.153 4
2026-09-09 00:54:18 <UNKNOWN> [UDP: [192.168.99.2]:51414->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:23:19:46.95       iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.171.2.0.2
iso.3.6.1.4.1.9.9.171.1.2.2.1.6.1.4.100.64.0.2.1.4.100.64.0.3.155 "64 40 00 02 "
iso.3.6.1.4.1.9.9.171.1.2.2.1.7.1.4.100.64.0.2.1.4.100.64.0.3.155 "64 40 00 03 "
iso.3.6.1.4.1.9.9.171.1.2.3.1.16.155 2  iso.3.6.1.4.1.9.9.171.1.4.2.1.1.2.155 6
2026-09-09 00:54:18 <UNKNOWN> [UDP: [192.168.99.3]:50504->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:23:19:46.60       iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.10.137.0.2
iso.3.6.1.4.1.9.10.137.1.2.1.8.987201 3 iso.3.6.1.4.1.9.10.137.1.2.1.8.987201 3 iso.3.6.1.2.1.2.2.1.1.9 9
2026-09-09 00:54:18 <UNKNOWN> [UDP: [192.168.99.2]:51414->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:23:19:47.73       iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.41.2.0.1
iso.3.6.1.4.1.9.9.41.1.2.3.1.2.139 "DMVPN"      iso.3.6.1.4.1.9.9.41.1.2.3.1.3.139 4
iso.3.6.1.4.1.9.9.41.1.2.3.1.4.139 "DMVPN_NHRP_ERROR"   iso.3.6.1.4.1.9.9.41.1.2.3.1.5.139 " Tunnel0: NHRP
Encap Error for  Purge Reply , Reason:  protocol generic error (7) on (Tunnel: 172.20.0.2 NBMA: 100.64.0.2)"
iso.3.6.1.4.1.9.9.41.1.2.3.1.6.139 0:23:19:47.72
2026-09-09 00:54:18 <UNKNOWN> [UDP: [192.168.99.3]:50504->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:23:19:47.37       iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.41.2.0.1
iso.3.6.1.4.1.9.9.41.1.2.3.1.2.120 "DMVPN"      iso.3.6.1.4.1.9.9.41.1.2.3.1.3.120 4
iso.3.6.1.4.1.9.9.41.1.2.3.1.4.120 "DMVPN_NHRP_ERROR"   iso.3.6.1.4.1.9.9.41.1.2.3.1.5.120 " Tunnel0: NHRP
Encap Error for  Purge Request , Reason:  protocol generic error (7) on (Tunnel: 172.20.0.3 NBMA:
100.64.0.3)"       iso.3.6.1.4.1.9.9.41.1.2.3.1.6.120 0:23:19:47.37
2026-09-09 00:54:28 <UNKNOWN> [UDP: [192.168.99.2]:51414->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:23:19:57.05       iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.0.1
iso.3.6.1.4.1.9.2.9.3.1.1.435.1 6       iso.3.6.1.2.1.6.13.1.1.10.0.2.15.22.10.0.2.2.42402 8
iso.3.6.1.4.1.9.2.6.1.1.5.10.0.2.15.22.10.0.2.2.42402 3553
iso.3.6.1.4.1.9.2.6.1.1.1.10.0.2.15.22.10.0.2.2.42402 3504
iso.3.6.1.4.1.9.2.6.1.1.2.10.0.2.15.22.10.0.2.2.42402 52243     iso.3.6.1.4.1.9.2.9.2.1.18.435 "lab"
2026-09-09 00:54:28 <UNKNOWN> [UDP: [192.168.99.3]:50504->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:23:19:56.71       iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.0.1
iso.3.6.1.4.1.9.2.9.3.1.1.434.1 6       iso.3.6.1.2.1.6.13.1.1.10.0.2.15.22.10.0.2.2.35352 8
iso.3.6.1.4.1.9.2.6.1.1.5.10.0.2.15.22.10.0.2.2.35352 3541
iso.3.6.1.4.1.9.2.6.1.1.1.10.0.2.15.22.10.0.2.2.35352 2784
iso.3.6.1.4.1.9.2.6.1.1.2.10.0.2.15.22.10.0.2.2.35352 29635     iso.3.6.1.4.1.9.2.9.2.1.18.434 "lab"
2026-09-09 00:54:28 <UNKNOWN> [UDP: [192.168.99.1]:53525->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:23:19:56.81       iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.0.1
iso.3.6.1.4.1.9.2.9.3.1.1.435.1 6       iso.3.6.1.2.1.6.13.1.1.10.0.2.15.22.10.0.2.2.34690 8
iso.3.6.1.4.1.9.2.6.1.1.5.10.0.2.15.22.10.0.2.2.34690 3565
iso.3.6.1.4.1.9.2.6.1.1.1.10.0.2.15.22.10.0.2.2.34690 2688
iso.3.6.1.4.1.9.2.6.1.1.2.10.0.2.15.22.10.0.2.2.34690 28707     iso.3.6.1.4.1.9.2.9.2.1.18.435 "lab"
2026-09-09 00:54:31 <UNKNOWN> [UDP: [192.168.99.1]:53525->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:23:19:59.79       iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.43.2.0.1
iso.3.6.1.4.1.9.9.43.1.1.6.1.3.297 1    iso.3.6.1.4.1.9.9.43.1.1.6.1.4.297 3
iso.3.6.1.4.1.9.9.43.1.1.6.1.5.297 2    iso.3.6.1.4.1.9.9.43.1.1.6.1.8.297 "lab"
2026-09-09 00:54:35 <UNKNOWN> [UDP: [192.168.99.2]:51414->[192.168.99.10]:162]:
```

```
iso.3.6.1.2.1.1.3.0 0:23:20:04.63       iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.43.2.0.1
iso.3.6.1.4.1.9.9.43.1.1.6.1.3.728 1    iso.3.6.1.4.1.9.9.43.1.1.6.1.4.728 3
iso.3.6.1.4.1.9.9.43.1.1.6.1.5.728 2    iso.3.6.1.4.1.9.9.43.1.1.6.1.8.728 "lab"
2026-09-09 00:54:39 <UNKNOWN> [UDP: [192.168.99.3]:50504->[192.168.99.10]:162]:
iso.3.6.1.2.1.1.3.0 0:23:20:08.28       iso.3.6.1.6.3.1.1.4.1.0 iso.3.6.1.4.1.9.9.43.2.0.1
iso.3.6.1.4.1.9.9.43.1.1.6.1.3.276 1    iso.3.6.1.4.1.9.9.43.1.1.6.1.4.276 3
iso.3.6.1.4.1.9.9.43.1.1.6.1.5.276 2    iso.3.6.1.4.1.9.9.43.1.1.6.1.8.276 "lab"
```

NMS — R1 polled over SNMPv3: snmp system

```
.1.3.6.1.2.1.1.1.0 = STRING: "Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-
UNIVERSALK9-M), Version 17.15.6, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2026 by Cisco Systems, Inc.
Compiled Wed 22-Jul-26 03:39 b"
.1.3.6.1.2.1.1.2.0 = OID: .1.3.6.1.4.1.9.1.3004
.1.3.6.1.2.1.1.3.0 = Timeticks: (8401422) 23:20:14.22
.1.3.6.1.2.1.1.4.0 = ""
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r1.lab.local"
.1.3.6.1.2.1.1.6.0 = ""
.1.3.6.1.2.1.1.7.0 = INTEGER: 78
.1.3.6.1.2.1.1.8.0 = Timeticks: (0) 0:00:00.00
```

NMS — R1 polled over SNMPv3: snmp interfaces

```
.1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
.1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
.1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
.1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
.1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"
.1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"
.1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"
.1.3.6.1.2.1.2.2.1.2.8 = STRING: "Loopback1"
.1.3.6.1.2.1.2.2.1.2.9 = STRING: "Tunnel0"
```

NMS — R2 polled over SNMPv3: snmp system

```
.1.3.6.1.2.1.1.1.0 = STRING: "Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-
UNIVERSALK9-M), Version 17.15.6, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2026 by Cisco Systems, Inc.
Compiled Wed 22-Jul-26 03:39 b"
.1.3.6.1.2.1.1.2.0 = OID: .1.3.6.1.4.1.9.1.3004
.1.3.6.1.2.1.1.3.0 = Timeticks: (8401461) 23:20:14.61
.1.3.6.1.2.1.1.4.0 = ""
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r2.lab.local"
.1.3.6.1.2.1.1.6.0 = ""
.1.3.6.1.2.1.1.7.0 = INTEGER: 78
.1.3.6.1.2.1.1.8.0 = Timeticks: (0) 0:00:00.00
```

NMS — R2 polled over SNMPv3: snmp interfaces

```
.1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
.1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
.1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
.1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
.1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"
.1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"
.1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"
.1.3.6.1.2.1.2.2.1.2.8 = STRING: "Loopback1"
.1.3.6.1.2.1.2.2.1.2.9 = STRING: "Tunnel0"
```

NMS — R3 polled over SNMPv3: snmp system

```
.1.3.6.1.2.1.1.1.0 = STRING: "Cisco IOS Software [IOSXE], Virtual XE Software (X86_64_LINUX_IOSD-
UNIVERSALK9-M), Version 17.15.6, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2026 by Cisco Systems, Inc.
Compiled Wed 22-Jul-26 03:39 b"
.1.3.6.1.2.1.1.2.0 = OID: .1.3.6.1.4.1.9.1.3004
.1.3.6.1.2.1.1.3.0 = Timeticks: (8401439) 23:20:14.39
.1.3.6.1.2.1.1.4.0 = ""
.1.3.6.1.2.1.1.5.0 = STRING: "c8000v-r3.lab.local"
.1.3.6.1.2.1.1.6.0 = ""
```

```
.1.3.6.1.2.1.1.7.0 = INTEGER: 78
.1.3.6.1.2.1.1.8.0 = Timeticks: (0) 0:00:00.00
```

NMS — R3 polled over SNMPv3: snmp interfaces

```
.1.3.6.1.2.1.2.2.1.2.1 = STRING: "GigabitEthernet1"
.1.3.6.1.2.1.2.2.1.2.2 = STRING: "GigabitEthernet2"
.1.3.6.1.2.1.2.2.1.2.3 = STRING: "GigabitEthernet3"
.1.3.6.1.2.1.2.2.1.2.4 = STRING: "GigabitEthernet4"
.1.3.6.1.2.1.2.2.1.2.5 = STRING: "VoIP-Null0"
.1.3.6.1.2.1.2.2.1.2.6 = STRING: "Null0"
.1.3.6.1.2.1.2.2.1.2.7 = STRING: "Loopback0"
.1.3.6.1.2.1.2.2.1.2.8 = STRING: "Loopback1"
.1.3.6.1.2.1.2.2.1.2.9 = STRING: "Tunnel0"
```

R1 — running config

```
Building configuration...

Current configuration : 21292 bytes
!
! Last configuration change at 00:53:51 UTC Wed Sep 9 2026 by lab
!
version 17.15
service timestamps debug datetime msec
service timestamps log datetime msec show-timezone
service sequence-numbers
platform qfp utilization monitor load 80
platform sslvpn use-pd
platform console serial
!
hostname c8000v-r1
!
boot-start-marker
boot-end-marker
!
!
vrf definition MGMT
 description out-of-band management
 !
 address-family ipv4
 exit-address-family
!
aaa new-model
!
!
aaa group server tacacs+ LABTACGRP
 server name LAB-TAC
 ip vrf forwarding MGMT
 ip tacacs source-interface GigabitEthernet4
!
aaa authentication login default group LABTACGRP local
aaa authentication enable default group LABTACGRP enable
aaa authorization exec default group LABTACGRP local if-authenticated
aaa accounting exec default start-stop group LABTACGRP
aaa accounting commands 15 default start-stop group LABTACGRP
!
!
aaa session-id common
!
!
!
!
!
!
!
!
!
!
!
!
ip domain name lab.local
!
!
!
login on-success log
!
!
```

```
subscriber templating
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
crypto pki trustpoint TP-self-signed-2558801973
 enrollment selfsigned
 subject-name cn=IOS-Self-Signed-Certificate-2558801973
 revocation-check none
 rsakeypair TP-self-signed-2558801973
 hash sha512
!
crypto pki trustpoint SLA-TrustPoint
 enrollment pkcs12
 revocation-check crl
 hash sha512
!
!
crypto pki certificate chain TP-self-signed-2558801973
 certificate self-signed 01
  30820330 30820218 A0030201 02020101 300D0609 2A864886 F70D0101 0D050030
  31312F30 2D060355 04030C26 494F532D 53656C66 2D536967 6E65642D 43657274
  69666963 6174652D 32353538 38303139 3733301E 170D3236 30393037 32323134
  31335A17 0D333630 39303632 32313431 335A3031 312F302D 06035504 030C2649
  4F532D53 656C662D 5369676E 65642D43 65727469 66696361 74652D32 35353838
  30313937 33308201 22300D06 092A8648 86F70D01 01010500 0382010F 00308201
  0A028201 0100A3B0 B1A31C69 7CE0D771 3CC0FD10 E87466FF 8D56B97B 7AD85044
  91C4659E DCED2F64 193AAD2A 709A1A7F B0F9945F 797749AD F86A4510 7E1F80A1
  1FE80C60 D4E397B7 4846E65C C13B5B9C 05D21CDF EAB8EA3A 202D3F4F 693E67C9
  169D42DB F7000DF8 676793A5 8AEB92D4 877721B9 A083A1F4 72CFA262 004898FA
  CD75F76C 287F9957 6AAF716E 30D9DA53 CEBC2C14 5F41C80F 3247EDB8 C35AA5E6
  CC2628B6 F19CF7C9 2537F3FD 995067E0 20535AD0 B8B892C3 549EBA73 8A0E9DB3
  990300B6 76DD444A 8F5E5B8E A6E30FC6 ACF04379 0B7C0CBD 047905B0 EAB4E9F3
  DDB03158 10BCF670 6405B592 C4C94F7C 0A9A3A87 907BD41D 26C6734C F6CA0371
  9B89CA94 82710203 010001A3 53305130 1D060355 1D0E0416 0414DABC F43EC8F4
  32D3489E 11DFF4F8 E2C8D1CB D88B301F 0603551D 23041830 168014DA BCF43EC8
  F432D348 9E11DFF4 F8E2C8D1 CBD88B30 0F060355 1D130101 FF040530 030101FF
  300D0609 2A864886 F70D0101 0D050003 82010100 354BEB77 02B9B574 B6A2A7B3
  DB44EAF0 BD91B085 A514CC8F 9F3CBE65 28A9C7C1 17ECD656 73651630 FF4E1EDE
  0D09C01D F0C223D8 8B936734 EC239D25 4C9F5648 A5DA5CB6 D9144CA0 8CB781EA
  8E80B81E 07E2D33B 767E629F E52A6505 6C10A855 85E69C37 348B0823 7209371E
  CE045D6E FB230443 69EC02A4 830956CB 31AB98AF F9F426FD 52C3BC02 68B742EB
  5EE2700E EC289D1A 40499A5A 10AA5DEA 6C7B7346 45CB66A7 B91BA82E 8439642E
  687AE6AA FAA120C7 F7BE113A 20EF3FD2 BC31C20B 35F3EB98 4FA898A6 6D0434B6
  0D2B4A11 924B2538 0956F607 5BA8D57C 011E753D E62C7505 3E534885 97469D4E
  C86EDCB7 8895F4ED F4756850 60602A97 34796304
        quit
crypto pki certificate chain SLA-TrustPoint
 certificate ca 01
  30820321 30820209 A0030201 02020101 300D0609 2A864886 F70D0101 0B050030
  32310E30 0C060355 040A1305 43697363 6F312030 1E060355 04031317 43697363
  6F204C69 63656E73 696E6720 526F6F74 20434130 1E170D31 33303533 30313934
  3834375A 170D3338 30353330 31393438 34375A30 32310E30 0C060355 040A1305
  43697363 6F312030 1E060355 04031317 43697363 6F204C69 63656E73 696E6720
  526F6F74 20434130 82012230 0D06092A 864886F7 0D010101 05000382 010F0030
  82010A02 82010100 A6BCBD96 131E05F7 145EA72C 2CD686E6 17222EA1 F1EFF64D
  CBB4C798 212AA147 C655D8D7 9471380D 8711441E 1AAF071A 9CAE6388 8A38E520
  1C394D78 462EF239 C659F715 B98C0A59 5BBB5CBD 0CFEBEA3 700A8BF7 D8F256EE
  4AA4E80D DB6FD1C9 60B1FD18 FFC69C96 6FA68957 A2617DE7 104FDC5F EA2956AC
  7390A3EB 2B5436AD C847A2C5 DAB553EB 69A9A535 58E9F3E3 C0BD23CF 58BD7188
  68E69491 20F320E7 948E71D7 AE3BCC84 F10684C7 4BC8E00F 539BA42B 42C68BB7
  C7479096 B4CB2D62 EA2F505D C7B062A4 6811D95B E8250FC4 5D5D5FB8 8F27D191
  C55F0D76 61F9A4CD 3D992327 A8BB03BD 4E6D7069 7CBADF8B DF5F4368 95135E44
  DFC7C6CF 04DD7FD1 02030100 01A34230 40300E06 03551D0F 0101FF04 04030201
  06300F06 03551D13 0101FF04 05300301 01FF301D 0603551D 0E041604 1449DC85
  4B3D31E5 1B3E6A17 606AF333 3D3B4C73 E8300D06 092A8648 86F70D01 010B0500
  03820101 00507F24 D3932A66 86025D9F E838AE5C 6D4DF6B0 49631C78 240DA905
  604EDCDE FF4FED2B 77FC460E CD636FDB DD44681E 3A5673AB 9093D3B1 6C9E3D8B
  D98987BF E40CBD9E 1AECA0C2 2189BB5C 8FA85686 CD98B646 5575B146 8DFC66A8
  467A3DF4 4D565700 6ADF0F0D CF835015 3C04FF7C 21E878AC 11BA9CD2 55A9232C
```

```
  7CA7B7E6 C1AF74F6 152E99B7 B1FCF9BB E973DE7F 5BDDEB86 C71E3B49 1765308B
  5FB0DA06 B92AFE7F 494E8A9E 07B85737 F3A58BE1 1A48A229 C37C1E69 39F08678
  80DDCD16 D6BACECA EEBC7CF9 8428787B 35202CDC 60E4616A B623CDBD 230E3AFB
  418616A9 4093E049 4D10AB75 27E86F73 932E35B5 8862FDAE 0275156F 719BB2F0
  D697DF7F 28
  	quit
!
!
!
!
!
!
!
!
!
license udi pid C8000V sn 9VRWIY571LW
license boot level network-premier addon dna-premier
memory free low-watermark processor 31774
diagnostic bootup level minimal
!
!
spanning-tree extend system-id
!
!
enable secret 9 $9$uia9E1anibovxE$Oqyklcz.SLvh8epz2w2TjV6d9TggTO3p9KSxEGgTTD2
!
username lab privilege 15 secret 9 $9$ksvY7W/rHjFg0.$AdIeocLbIuwaSOCYSoWgOl/6LNagQ58URuynu0Dbtq6
!
redundancy
!
!
!
crypto ikev2 proposal LAB-PROP
 encryption aes-cbc-256
 integrity sha256
 group 14
!
crypto ikev2 policy LAB-POL
 proposal LAB-PROP
!
crypto ikev2 keyring LAB-KR
 peer ANY
  address 0.0.0.0 0.0.0.0
  pre-shared-key LabPreSharedKey123
 !
!
!
crypto ikev2 profile LAB-PROF
 match identity remote address 0.0.0.0
 authentication remote pre-share
 authentication local pre-share
 keyring local LAB-KR
!
!
!
!
!
!
!
!
!
!
!
!
!
crypto ipsec transform-set LAB-TS esp-aes 256 esp-sha256-hmac
 mode transport
!
crypto ipsec profile LAB-IPSEC
 set transform-set LAB-TS
 set ikev2-profile LAB-PROF
!
!
!
!
!
!
!
!
!
!
```

```
interface Loopback0
 description Protected endpoint
 ip address 172.16.1.1 255.255.255.255
!
interface Loopback1
 description BGP-advertised prefix
 ip address 10.20.1.1 255.255.255.255
!
interface Tunnel0
 description DMVPN hub
 ip address 172.20.0.1 255.255.255.0
 no ip redirects
 ip nat outside
 ip nhrp authentication LabNhrp
 ip nhrp network-id 100
 ip nhrp holdtime 300
 ip nhrp redirect
 ip tcp adjust-mss 1360
 bfd interval 500 min_rx 500 multiplier 3
 tunnel source GigabitEthernet2
 tunnel mode gre multipoint
 tunnel key 100
 tunnel protection ipsec profile LAB-IPSEC
!
interface GigabitEthernet1
 description MGMT - QEMU user-mode net (DHCP)
 ip address dhcp
 negotiation auto
!
interface GigabitEthernet2
 description WAN - shared provider segment, DMVPN underlay
 ip address 100.64.0.1 255.255.255.0
 negotiation auto
!
interface GigabitEthernet3
 description LAN toward the local host
 ip address 192.168.10.1 255.255.255.0
 ip nat inside
 negotiation auto
!
interface GigabitEthernet4
 description OOB management - NTP, syslog, SNMP
 vrf forwarding MGMT
 ip address 192.168.99.1 255.255.255.0
 negotiation auto
!
router bgp 65001
 bgp router-id 172.16.1.1
 bgp log-neighbor-changes
 neighbor 172.20.0.2 remote-as 65002
 neighbor 172.20.0.2 password LabBgpSecret
 neighbor 172.20.0.2 timers 10 30
 neighbor 172.20.0.2 fall-over bfd
 neighbor 172.20.0.3 remote-as 65003
 neighbor 172.20.0.3 password LabBgpSecret
 neighbor 172.20.0.3 timers 10 30
 neighbor 172.20.0.3 fall-over bfd
 !
 address-family ipv4
  network 10.20.1.1 mask 255.255.255.255
  network 10.30.1.0 mask 255.255.255.0
  network 192.168.10.0
  neighbor 172.20.0.2 activate
  neighbor 172.20.0.2 send-community both
  neighbor 172.20.0.2 route-map RM-IN-R2 in
  neighbor 172.20.0.2 route-map RM-OUT-R2 out
  neighbor 172.20.0.2 maximum-prefix 15 80
  neighbor 172.20.0.3 activate
  neighbor 172.20.0.3 send-community both
  neighbor 172.20.0.3 route-map RM-IN-R3 in
  neighbor 172.20.0.3 route-map RM-OUT-R3 out
  neighbor 172.20.0.3 maximum-prefix 15 80
 exit-address-family
!
ip forward-protocol nd
!
ip bgp-community new-format
ip community-list standard CL-VALID permit 65001:1
ip community-list standard CL-VALID permit 65001:2
ip community-list standard CL-VALID permit 65001:3
ip community-list standard CL-VALID permit 65002:1
```

```
ip community-list standard CL-VALID permit 65002:2
ip community-list standard CL-VALID permit 65002:3
ip community-list standard CL-VALID permit 65003:1
ip community-list standard CL-VALID permit 65003:2
ip community-list standard CL-VALID permit 65003:3
ip as-path access-list 10 permit ^65002$
ip as-path access-list 20 permit ^65003$
no ip http server
ip http secure-server
ip route 10.30.1.0 255.255.255.0 Null0
ip nat pool SITE-POOL 10.30.1.100 10.30.1.129 prefix-length 24
ip nat inside source static 192.168.10.10 10.30.1.10 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.11 10.30.1.11 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.12 10.30.1.12 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.13 10.30.1.13 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.14 10.30.1.14 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.15 10.30.1.15 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.16 10.30.1.16 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.17 10.30.1.17 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.18 10.30.1.18 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.10.19 10.30.1.19 route-map NAT-TO-DOMAIN
ip nat inside source list NAT-POOL-SRC pool SITE-POOL
ip ssh bulk-mode 131072
ip ssh server algorithm authentication password
!
ip prefix-list PL-FAR-R2 seq 40 permit 10.20.2.1/32
ip prefix-list PL-FAR-R2 seq 45 permit 10.30.2.0/24
ip prefix-list PL-FAR-R2 seq 50 permit 192.168.20.0/24
!
ip prefix-list PL-FAR-R3 seq 40 permit 10.20.3.1/32
ip prefix-list PL-FAR-R3 seq 45 permit 10.30.3.0/24
ip prefix-list PL-FAR-R3 seq 50 permit 192.168.30.0/24
!
ip prefix-list PL-OWN-LAN seq 5 permit 192.168.10.0/24
!
ip prefix-list PL-OWN-LO seq 5 permit 10.20.1.1/32
!
ip prefix-list PL-OWN-NAT seq 5 permit 10.30.1.0/24
!
ip access-list standard VTY-MGMT
 10 remark out-of-band management network
 10 permit 192.168.99.0 0.0.0.255
 20 remark hypervisor user-mode NAT: how the harness reaches these devices
 20 permit 10.0.2.0 0.0.0.255
 30 deny any log
!
ip access-list extended NAT-DOMAIN
 10 permit ip host 192.168.10.10 10.30.0.0 0.0.255.255
 20 permit ip host 192.168.10.11 10.30.0.0 0.0.255.255
 30 permit ip host 192.168.10.12 10.30.0.0 0.0.255.255
 40 permit ip host 192.168.10.13 10.30.0.0 0.0.255.255
 50 permit ip host 192.168.10.14 10.30.0.0 0.0.255.255
 60 permit ip host 192.168.10.15 10.30.0.0 0.0.255.255
 70 permit ip host 192.168.10.16 10.30.0.0 0.0.255.255
 80 permit ip host 192.168.10.17 10.30.0.0 0.0.255.255
 90 permit ip host 192.168.10.18 10.30.0.0 0.0.255.255
 100 permit ip host 192.168.10.19 10.30.0.0 0.0.255.255
ip access-list extended NAT-POOL-SRC
 10 permit ip 192.168.10.64 0.0.0.31 10.30.0.0 0.0.255.255
logging source-interface GigabitEthernet4 vrf MGMT
logging host 192.168.99.10 vrf MGMT
route-map RM-OUT-R2 permit 10
 description R1's own LO, tagged 65001:1
 match ip address prefix-list PL-OWN-LO
 set community 65001:1
!
route-map RM-OUT-R2 permit 20
 description R1's own NAT, tagged 65001:2
 match ip address prefix-list PL-OWN-NAT
 set community 65001:2
!
route-map RM-OUT-R2 permit 30
 description R1's own LAN, tagged 65001:3
 match ip address prefix-list PL-OWN-LAN
 set community 65001:3
!
route-map RM-OUT-R2 permit 55
 description re-advertise R3, preserving its communities
 match ip address prefix-list PL-FAR-R3
!
route-map RM-OUT-R3 permit 10
```

```
 description R1's own LO, tagged 65001:1
 match ip address prefix-list PL-OWN-LO
 set community 65001:1
!
route-map RM-OUT-R3 permit 20
 description R1's own NAT, tagged 65001:2
 match ip address prefix-list PL-OWN-NAT
 set community 65001:2
!
route-map RM-OUT-R3 permit 30
 description R1's own LAN, tagged 65001:3
 match ip address prefix-list PL-OWN-LAN
 set community 65001:3
!
route-map RM-OUT-R3 permit 55
 description re-advertise R2, preserving its communities
 match ip address prefix-list PL-FAR-R2
!
route-map NAT-TO-DOMAIN permit 10
 match ip address NAT-DOMAIN
!
route-map RM-IN-R2 permit 10
 description accept only known communities on an AS path R2 can produce
 match as-path 10
 match community CL-VALID
!
route-map RM-IN-R3 permit 10
 description accept only known communities on an AS path R3 can produce
 match as-path 20
 match community CL-VALID
!
snmp-server group LABGRP v3 priv read LABVIEW notify LABVIEW
snmp-server view LABVIEW iso included
snmp-server trap-source GigabitEthernet4
snmp-server enable traps snmp authentication linkdown linkup coldstart warmstart
snmp-server enable traps vrrp
snmp-server enable traps flowmon
snmp-server enable traps pfr
snmp-server enable traps ds1
snmp-server enable traps sdwan security policy system omp bfd
snmp-server enable traps entity-perf throughput-notif
snmp-server enable traps call-home message-send-fail server-fail
snmp-server enable traps tty
snmp-server enable traps ospf state-change
snmp-server enable traps ospf errors
snmp-server enable traps ospf retransmit
snmp-server enable traps ospf lsa
snmp-server enable traps ospf cisco-specific state-change nssa-trans-change
snmp-server enable traps ospf cisco-specific state-change shamlink interface
snmp-server enable traps ospf cisco-specific state-change shamlink neighbor
snmp-server enable traps ospf cisco-specific errors
snmp-server enable traps ospf cisco-specific retransmit
snmp-server enable traps ospf cisco-specific lsa
snmp-server enable traps eigrp
snmp-server enable traps casa
snmp-server enable traps xgcp
snmp-server enable traps cef resource-failure peer-state-change peer-fib-state-change inconsistency
snmp-server enable traps smart-license
snmp-server enable traps diameter
snmp-server enable traps entity-qfp mem-res-thresh throughput-notif
snmp-server enable traps entity-state
snmp-server enable traps ethernet cfm cc mep-up mep-down cross-connect loop config
snmp-server enable traps ethernet cfm crosscheck mep-missing mep-unknown service-up
snmp-server enable traps ether-oam
snmp-server enable traps ethernet evc status create delete
snmp-server enable traps aaa_server
snmp-server enable traps config-copy
snmp-server enable traps config
snmp-server enable traps config-ctid
snmp-server enable traps memory bufferpeak
snmp-server enable traps fru-ctrl
snmp-server enable traps entity-sensor
snmp-server enable traps resource-policy
snmp-server enable traps flash insertion removal lowspace
snmp-server enable traps hsrp
snmp-server enable traps entity
snmp-server enable traps ip local pool
snmp-server enable traps cpu threshold
snmp-server enable traps auth-framework sec-violation
snmp-server enable traps atm subif
snmp-server enable traps frame-relay
```

```
snmp-server enable traps frame-relay subif
snmp-server enable traps bgp state-changes all backward-trans limited updown-limited
snmp-server enable traps bgp threshold prefix
snmp-server enable traps bgp cbgp2 state-changes all backward-trans limited updown-limited
snmp-server enable traps bgp cbgp2 threshold prefix
snmp-server enable traps pim neighbor-change rp-mapping-change invalid-pim-message
snmp-server enable traps ipmulticast
snmp-server enable traps msdp
snmp-server enable traps mvpn
snmp-server enable traps rsvp
snmp-server enable traps isis
snmp-server enable traps dhcp
snmp-server enable traps nhrp nhs
snmp-server enable traps nhrp nhc
snmp-server enable traps nhrp nhp
snmp-server enable traps nhrp quota-exceeded
snmp-server enable traps ipsla
snmp-server enable traps bfd
snmp-server enable traps cnpd
snmp-server enable traps event-manager
snmp-server enable traps dlsw
snmp-server enable traps ospfv3 state-change
snmp-server enable traps ospfv3 errors
snmp-server enable traps pppoe
snmp-server enable traps l2tun session
snmp-server enable traps l2tun pseudowire status
snmp-server enable traps l2tun tunnel
snmp-server enable traps pki
snmp-server enable traps pimstdmib neighbor-loss invalid-register invalid-join-prune rp-mapping-change
interface-election
snmp-server enable traps adslline
snmp-server enable traps entity-diag boot-up-fail hm-test-recover hm-thresh-reached scheduled-test-fail
snmp-server enable traps vdsl2line
snmp-server enable traps vtp
snmp-server enable traps vlancreate
snmp-server enable traps vlandelete
snmp-server enable traps port-security
snmp-server enable traps isdn call-information
snmp-server enable traps isdn layer2
snmp-server enable traps isdn chan-not-avail
snmp-server enable traps isdn ietf
snmp-server enable traps c3g
snmp-server enable traps LTE
snmp-server enable traps lisp
snmp-server enable traps firewall serverstatus
snmp-server enable traps ike policy add
snmp-server enable traps ike policy delete
snmp-server enable traps ike tunnel start
snmp-server enable traps ike tunnel stop
snmp-server enable traps ipsec cryptomap add
snmp-server enable traps ipsec cryptomap delete
snmp-server enable traps ipsec cryptomap attach
snmp-server enable traps ipsec cryptomap detach
snmp-server enable traps ipsec tunnel start
snmp-server enable traps ipsec tunnel stop
snmp-server enable traps ipsec too-many-sas
snmp-server enable traps gdoi gm-start-registration
snmp-server enable traps gdoi gm-registration-complete
snmp-server enable traps gdoi gm-re-register
snmp-server enable traps gdoi gm-rekey-rcvd
snmp-server enable traps gdoi gm-rekey-fail
snmp-server enable traps gdoi ks-rekey-pushed
snmp-server enable traps gdoi gm-incomplete-cfg
snmp-server enable traps gdoi ks-no-rsa-keys
snmp-server enable traps gdoi ks-new-registration
snmp-server enable traps gdoi ks-reg-complete
snmp-server enable traps gdoi ks-role-change
snmp-server enable traps gdoi ks-gm-deleted
snmp-server enable traps gdoi ks-peer-reachable
snmp-server enable traps gdoi ks-peer-unreachable
snmp-server enable traps mpls rfc ldp
snmp-server enable traps mpls ldp
snmp-server enable traps mpls rfc traffic-eng
snmp-server enable traps mpls traffic-eng
snmp-server enable traps mpls fast-reroute protected
snmp-server enable traps otn
snmp-server enable traps pw vc
snmp-server enable traps frame-relay multilink bundle-mismatch
snmp-server enable traps dial
snmp-server enable traps dsp card-status
snmp-server enable traps dsp oper-state
```

```
snmp-server enable traps dsp video-usage
snmp-server enable traps dsp video-out-of-resource
snmp-server enable traps syslog
snmp-server enable traps sbc adj-status
snmp-server enable traps sbc blacklist
snmp-server enable traps sbc congestion-alarm
snmp-server enable traps sbc h248-ctrlr-status
snmp-server enable traps sbc media-source
snmp-server enable traps sbc radius-conn-status
snmp-server enable traps sbc sla-violation
snmp-server enable traps sbc sla-violation-rev1
snmp-server enable traps sbc svc-state
snmp-server enable traps sbc qos-statistics
snmp-server enable traps bulkstat collection transfer
snmp-server enable traps ethernet cfm alarm
snmp-server enable traps alarms informational
snmp-server enable traps transceiver all
snmp-server enable traps rf
snmp-server enable traps vrfmib vrf-up vrf-down vnet-trunk-up vnet-trunk-down
snmp-server enable traps ccme
snmp-server enable traps srst
snmp-server enable traps mpls vpn
snmp-server enable traps mpls rfc vpn
snmp-server enable traps voice
snmp-server host 192.168.99.10 vrf MGMT version 3 priv labmon udp-port 1162
snmp-server host 192.168.99.10 vrf MGMT version 3 priv labmon
tacacs server LAB-TAC
 address ipv4 192.168.99.10
 key LabTacacsKey123
!
!
!
!
!
control-plane
!
!
mgcp behavior rsip-range tgcp-only
mgcp behavior comedia-role none
mgcp behavior comedia-check-media-src disable
mgcp behavior comedia-sdp-force disable
!
mgcp profile default
!
!
!
!
!
banner login ^C
******************************************************************************
                          AUTHORISED ACCESS ONLY

  Access to this device is restricted to authorised personnel. If you are not
  an authorised user, disconnect now.

  All activity is monitored, logged and audited. Continued use of this system
  constitutes consent to that monitoring.

  Unauthorised access or use may result in disciplinary action and civil or
  criminal penalties.
******************************************************************************
^C
!
line con 0
 exec-timeout 0 0
 activation-character 13
 logging synchronous
 stopbits 1
line aux 0
 activation-character 13
line vty 0 4
 access-class VTY-MGMT in vrf-also
 exec-timeout 0 0
 activation-character 13
 transport input ssh
line vty 5 15
 access-class VTY-MGMT in vrf-also
 exec-timeout 0 0
 activation-character 13
 transport input ssh
!
```

```
ntp source GigabitEthernet4
ntp server vrf MGMT 192.168.99.10
!
!
!
!
!
!
netconf-yang
restconf
end
```

R2 — running config

```
Building configuration...

Current configuration : 19824 bytes
!
! Last configuration change at 00:53:31 UTC Wed Sep 9 2026 by lab
!
version 17.15
service timestamps debug datetime msec
service timestamps log datetime msec show-timezone
service sequence-numbers
platform qfp utilization monitor load 80
platform sslvpn use-pd
platform console serial
!
hostname c8000v-r2
!
boot-start-marker
boot-end-marker
!
!
vrf definition MGMT
 description out-of-band management
 !
 address-family ipv4
 exit-address-family
!
aaa new-model
!
!
aaa group server tacacs+ LABTACGRP
 server name LAB-TAC
 ip vrf forwarding MGMT
 ip tacacs source-interface GigabitEthernet4
!
aaa authentication login default group LABTACGRP local
aaa authentication enable default group LABTACGRP enable
aaa authorization exec default group LABTACGRP local if-authenticated
aaa accounting exec default start-stop group LABTACGRP
aaa accounting commands 15 default start-stop group LABTACGRP
!
!
aaa session-id common
!
!
!
!
!
!
!
!
!
!
!
!
ip domain name lab.local
!
!
!
login on-success log
!
!
subscriber templating
!
!
!
!
```

```
!
!
!
!
!
!
!
!
!
!
!
crypto pki trustpoint TP-self-signed-451475034
 enrollment selfsigned
 subject-name cn=IOS-Self-Signed-Certificate-451475034
 revocation-check none
 rsakeypair TP-self-signed-451475034
 hash sha512
!
crypto pki trustpoint SLA-TrustPoint
 enrollment pkcs12
 revocation-check crl
 hash sha512
!
!
crypto pki certificate chain TP-self-signed-451475034
 certificate self-signed 01
  3082032E 30820216 A0030201 02020101 300D0609 2A864886 F70D0101 0D050030
  30312E30 2C060355 04030C25 494F532D 53656C66 2D536967 6E65642D 43657274
  69666963 6174652D 34353134 37353033 34301E17 0D323630 39303732 32313431
  325A170D 33363039 30363232 31343132 5A303031 2E302C06 03550403 0C25494F
  532D5365 6C662D53 69676E65 642D4365 72746966 69636174 652D3435 31343735
  30333430 82012230 0D06092A 864886F7 0D010101 05000382 010F0030 82010A02
  82010100 C9CFA37B 39E41985 6120CCAF 0AE693B6 F671562D 0F4B676A 7D14741D
  929C02B1 B6E17AC0 E6F0C660 BA1A9C4E 45B077E6 7C7C3AF9 7F3BE083 A805381B
  84C48F6F 8806352B E69AA1FD 5D0D6D65 4B5276E3 6677EC55 851DBE2B 92AAB096
  8C5B49D6 05B47713 A8B77C13 1CD15A64 6F37D0A6 D35BBF4C A97471F7 07461355
  E9EEBD7E 16F1459D 22302CC8 BA056BD0 ADE0190A 00DBA2B8 8D6977E5 2CAACBE4
  FDBFD944 0D6771FD DC46CA79 327E1B61 5E2F4A7A 991E75E4 6723C838 249320FC
  597FACA0 77533D57 9E92FB49 873D1F4A EEC9C052 06CB9D3B 4A2360C5 5362DECA
  9271CF20 14DD10E5 BBB01C17 6556ADCF 533222FC 3869400A 2933F291 27F723E8
  BB49B7E3 02030100 01A35330 51301D06 03551D0E 04160414 F21BB9E2 A8780800
  8A7482B4 A872EB5E FE8688B9 301F0603 551D2304 18301680 14F21BB9 E2A87808
  008A7482 B4A872EB 5EFE8688 B9300F06 03551D13 0101FF04 05300301 01FF300D
  06092A86 4886F70D 01010D05 00038201 0100B90C 297103CA B34A6FA5 57C2310E
  D3C86AD7 863193A6 12D30D52 35310D37 3E948055 2B965E6B 505BFC95 A3107821
  FD7FB67F 7FE22E75 0512C3C1 0D13EE47 53B9D736 63B13871 324360F7 4B1D8067
  3236EF41 3A44DC6D 57540BE4 E2BED3A2 0E9A231C 9FC34FF9 03F30438 F71FA888
  2951138B 46D6C4F8 E94EC7F7 7FE4EB9C 987FDC76 7974B4F6 7834B88B 7A3A7766
  EEFDF646 437A7DBD 993C13D9 7D5CE081 55A05480 004CD0F5 7E81F48D 20AAD5C9
  E46B42EC 8C43C089 A3B8A082 D27D4DE8 621EFE73 6A7ECBB3 FDA089A7 1E1BCD34
  3210A8A8 3BE602C4 8F6104D2 AF27AF4C 76D7513E 0AE9CDDE 149BD06B D29D1593
  B6BECC04 8C99DE76 29F09F47 74FF35E2 0C74
  	quit
crypto pki certificate chain SLA-TrustPoint
 certificate ca 01
  30820321 30820209 A0030201 02020101 300D0609 2A864886 F70D0101 0B050030
  32310E30 0C060355 040A1305 43697363 6F312030 1E060355 04031317 43697363
  6F204C69 63656E73 696E6720 526F6F74 20434130 1E170D31 33303533 30313934
  3834375A 170D3338 30353330 31393438 34375A30 32310E30 0C060355 040A1305
  43697363 6F312030 1E060355 04031317 43697363 6F204C69 63656E73 696E6720
  526F6F74 20434130 82012230 0D06092A 864886F7 0D010101 05000382 010F0030
  82010A02 82010100 A6BCBD96 131E05F7 145EA72C 2CD686E6 17222EA1 F1EFF64D
  CBB4C798 212AA147 C655D8D7 9471380D 8711441E 1AAF071A 9CAE6388 8A38E520
  1C394D78 462EF239 C659F715 B98C0A59 5BBB5CBD 0CFEBEA3 700A8BF7 D8F256EE
  4AA4E80D DB6FD1C9 60B1FD18 FFC69C96 6FA68957 A2617DE7 104FDC5F EA2956AC
  7390A3EB 2B5436AD C847A2C5 DAB553EB 69A9A535 58E9F3E3 C0BD23CF 58BD7188
  68E69491 20F320E7 948E71D7 AE3BCC84 F10684C7 4BC8E00F 539BA42B 42C68BB7
  C7479096 B4CB2D62 EA2F505D C7B062A4 6811D95B E8250FC4 5D5D5FB8 8F27D191
  C55F0D76 61F9A4CD 3D992327 A8BB03BD 4E6D7069 7CBADF8B DF5F4368 95135E44
  DFC7C6CF 04DD7FD1 02030100 01A34230 40300E06 03551D0F 0101FF04 04030201
  06300F06 03551D13 0101FF04 05300301 01FF301D 0603551D 0E041604 1449DC85
  4B3D31E5 1B3E6A17 606AF333 3D3B4C73 E8300D06 092A8648 86F70D01 010B0500
  03820101 00507F24 D3932A66 86025D9F E838AE5C 6D4DF6B0 49631C78 240DA905
  604EDCDE FF4FED2B 77FC460E CD636FDB DD44681E 3A5673AB 9093D3B1 6C9E3D8B
  D98987BF E40CBD9E 1AECA0C2 2189BB5C 8FA85686 CD98B646 5575B146 8DFC66A8
  467A3DF4 4D565700 6ADF0F0D CF835015 3C04FF7C 21E878AC 11BA9CD2 55A9232C
  7CA7B7E6 C1AF74F6 152E99B7 B1FCF9BB E973DE7F 5BDDEB86 C71E3B49 1765308B
  5FB0DA06 B92AFE7F 494E8A9E 07B85737 F3A58BE1 1A48A229 C37C1E69 39F08678
  80DDCD16 D6BACECA EEBC7CF9 8428787B 35202CDC 60E4616A B623CDBD 230E3AFB
  418616A9 4093E049 4D10AB75 27E86F73 932E35B5 8862FDAE 0275156F 719BB2F0
  D697DF7F 28
```

```
        quit
!
!
!
!
!
!
!
!
!
license udi pid C8000V sn 99MT1OJLO06
license boot level network-premier addon dna-premier
memory free low-watermark processor 31774
diagnostic bootup level minimal
!
!
spanning-tree extend system-id
!
!
enable secret 9 $9$IzR6xMPRz4hVOU$O/V1R.nWlX98duswhDwnTGs67nYdgYArKX89ewvQ67s
!
username lab privilege 15 secret 9 $9$bqcYxwxYLmCe2E$3qzzD/xblonG/KBzgLi14CgQBnGkOqBWmswiHb1oxLU
!
redundancy
!
!
!
crypto ikev2 proposal LAB-PROP
 encryption aes-cbc-256
 integrity sha256
 group 14
!
crypto ikev2 policy LAB-POL
 proposal LAB-PROP
!
crypto ikev2 keyring LAB-KR
 peer ANY
  address 0.0.0.0 0.0.0.0
  pre-shared-key LabPreSharedKey123
 !
!
!
crypto ikev2 profile LAB-PROF
 match identity remote address 0.0.0.0
 authentication remote pre-share
 authentication local pre-share
 keyring local LAB-KR
!
!
!
!
!
!
!
!
!
!
!
!
!
crypto ipsec transform-set LAB-TS esp-aes 256 esp-sha256-hmac
 mode transport
!
crypto ipsec profile LAB-IPSEC
 set transform-set LAB-TS
 set ikev2-profile LAB-PROF
!
!
!
!
!
!
!
!
!
!
interface Loopback0
 description Protected endpoint
 ip address 172.16.2.1 255.255.255.255
!
interface Loopback1
```

```
 description BGP-advertised prefix
 ip address 10.20.2.1 255.255.255.255
!
interface Tunnel0
 description DMVPN spoke
 ip address 172.20.0.2 255.255.255.0
 no ip redirects
 ip nat outside
 ip nhrp authentication LabNhrp
 ip nhrp network-id 100
 ip nhrp holdtime 300
 ip nhrp nhs 172.20.0.1 nbma 100.64.0.1 multicast
 ip tcp adjust-mss 1360
 bfd interval 500 min_rx 500 multiplier 3
 tunnel source GigabitEthernet2
 tunnel mode gre multipoint
 tunnel key 100
 tunnel protection ipsec profile LAB-IPSEC
!
interface GigabitEthernet1
 description MGMT - QEMU user-mode net (DHCP)
 ip address dhcp
 negotiation auto
!
interface GigabitEthernet2
 description WAN - shared provider segment, DMVPN underlay
 ip address 100.64.0.2 255.255.255.0
 negotiation auto
!
interface GigabitEthernet3
 description LAN toward the local host
 ip address 192.168.20.1 255.255.255.0
 ip nat inside
 negotiation auto
!
interface GigabitEthernet4
 description OOB management - NTP, syslog, SNMP
 vrf forwarding MGMT
 ip address 192.168.99.2 255.255.255.0
 negotiation auto
!
router bgp 65002
 bgp router-id 172.16.2.1
 bgp log-neighbor-changes
 neighbor 172.20.0.1 remote-as 65001
 neighbor 172.20.0.1 password LabBgpSecret
 neighbor 172.20.0.1 timers 10 30
 neighbor 172.20.0.1 fall-over bfd
 !
 address-family ipv4
  network 10.20.2.1 mask 255.255.255.255
  network 10.30.2.0 mask 255.255.255.0
  network 192.168.20.0
  neighbor 172.20.0.1 activate
  neighbor 172.20.0.1 send-community both
  neighbor 172.20.0.1 route-map RM-IN-R1 in
  neighbor 172.20.0.1 route-map RM-OUT-R1 out
  neighbor 172.20.0.1 maximum-prefix 15 80
 exit-address-family
!
ip forward-protocol nd
!
ip bgp-community new-format
ip community-list standard CL-VALID permit 65001:1
ip community-list standard CL-VALID permit 65001:2
ip community-list standard CL-VALID permit 65001:3
ip community-list standard CL-VALID permit 65002:1
ip community-list standard CL-VALID permit 65002:2
ip community-list standard CL-VALID permit 65002:3
ip community-list standard CL-VALID permit 65003:1
ip community-list standard CL-VALID permit 65003:2
ip community-list standard CL-VALID permit 65003:3
ip as-path access-list 10 permit ^65001$
ip as-path access-list 10 permit ^65001_65003$
no ip http server
ip http secure-server
ip route 10.30.2.0 255.255.255.0 Null0
ip nat pool SITE-POOL 10.30.2.100 10.30.2.129 prefix-length 24
ip nat inside source static 192.168.20.10 10.30.2.10 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.11 10.30.2.11 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.12 10.30.2.12 route-map NAT-TO-DOMAIN
```

```
ip nat inside source static 192.168.20.13 10.30.2.13 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.14 10.30.2.14 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.15 10.30.2.15 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.16 10.30.2.16 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.17 10.30.2.17 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.18 10.30.2.18 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.20.19 10.30.2.19 route-map NAT-TO-DOMAIN
ip nat inside source list NAT-POOL-SRC pool SITE-POOL
ip ssh bulk-mode 131072
ip ssh server algorithm authentication password
!
ip prefix-list PL-OWN-LAN seq 5 permit 192.168.20.0/24
!
ip prefix-list PL-OWN-LO seq 5 permit 10.20.2.1/32
!
ip prefix-list PL-OWN-NAT seq 5 permit 10.30.2.0/24
!
ip access-list standard VTY-MGMT
 10 remark out-of-band management network
 10 permit 192.168.99.0 0.0.0.255
 20 remark hypervisor user-mode NAT: how the harness reaches these devices
 20 permit 10.0.2.0 0.0.0.255
 30 deny any log
!
ip access-list extended NAT-DOMAIN
 10 permit ip host 192.168.20.10 10.30.0.0 0.0.255.255
 20 permit ip host 192.168.20.11 10.30.0.0 0.0.255.255
 30 permit ip host 192.168.20.12 10.30.0.0 0.0.255.255
 40 permit ip host 192.168.20.13 10.30.0.0 0.0.255.255
 50 permit ip host 192.168.20.14 10.30.0.0 0.0.255.255
 60 permit ip host 192.168.20.15 10.30.0.0 0.0.255.255
 70 permit ip host 192.168.20.16 10.30.0.0 0.0.255.255
 80 permit ip host 192.168.20.17 10.30.0.0 0.0.255.255
 90 permit ip host 192.168.20.18 10.30.0.0 0.0.255.255
 100 permit ip host 192.168.20.19 10.30.0.0 0.0.255.255
ip access-list extended NAT-POOL-SRC
 10 permit ip 192.168.20.64 0.0.0.31 10.30.0.0 0.0.255.255
logging source-interface GigabitEthernet4 vrf MGMT
logging host 192.168.99.10 vrf MGMT
route-map RM-OUT-R1 permit 10
 description R2's own LO, tagged 65002:1
 match ip address prefix-list PL-OWN-LO
 set community 65002:1
!
route-map RM-OUT-R1 permit 20
 description R2's own NAT, tagged 65002:2
 match ip address prefix-list PL-OWN-NAT
 set community 65002:2
!
route-map RM-OUT-R1 permit 30
 description R2's own LAN, tagged 65002:3
 match ip address prefix-list PL-OWN-LAN
 set community 65002:3
!
route-map NAT-TO-DOMAIN permit 10
 match ip address NAT-DOMAIN
!
route-map RM-IN-R1 permit 10
 description accept only known communities on an AS path R1 can produce
 match as-path 10
 match community CL-VALID
!
snmp-server group LABGRP v3 priv read LABVIEW notify LABVIEW
snmp-server view LABVIEW iso included
snmp-server trap-source GigabitEthernet4
snmp-server enable traps snmp authentication linkdown linkup coldstart warmstart
snmp-server enable traps vrrp
snmp-server enable traps flowmon
snmp-server enable traps pfr
snmp-server enable traps ds1
snmp-server enable traps sdwan security policy system omp bfd
snmp-server enable traps entity-perf throughput-notif
snmp-server enable traps call-home message-send-fail server-fail
snmp-server enable traps tty
snmp-server enable traps ospf state-change
snmp-server enable traps ospf errors
snmp-server enable traps ospf retransmit
snmp-server enable traps ospf lsa
snmp-server enable traps ospf cisco-specific state-change nssa-trans-change
snmp-server enable traps ospf cisco-specific state-change shamlink interface
snmp-server enable traps ospf cisco-specific state-change shamlink neighbor
```

```
snmp-server enable traps ospf cisco-specific errors
snmp-server enable traps ospf cisco-specific retransmit
snmp-server enable traps ospf cisco-specific lsa
snmp-server enable traps eigrp
snmp-server enable traps casa
snmp-server enable traps xgcp
snmp-server enable traps cef resource-failure peer-state-change peer-fib-state-change inconsistency
snmp-server enable traps smart-license
snmp-server enable traps diameter
snmp-server enable traps entity-qfp mem-res-thresh throughput-notif
snmp-server enable traps entity-state
snmp-server enable traps ethernet cfm cc mep-up mep-down cross-connect loop config
snmp-server enable traps ethernet cfm crosscheck mep-missing mep-unknown service-up
snmp-server enable traps ether-oam
snmp-server enable traps ethernet evc status create delete
snmp-server enable traps aaa_server
snmp-server enable traps config-copy
snmp-server enable traps config
snmp-server enable traps config-ctid
snmp-server enable traps memory bufferpeak
snmp-server enable traps fru-ctrl
snmp-server enable traps entity-sensor
snmp-server enable traps resource-policy
snmp-server enable traps flash insertion removal lowspace
snmp-server enable traps hsrp
snmp-server enable traps entity
snmp-server enable traps ip local pool
snmp-server enable traps cpu threshold
snmp-server enable traps auth-framework sec-violation
snmp-server enable traps atm subif
snmp-server enable traps frame-relay
snmp-server enable traps frame-relay subif
snmp-server enable traps bgp state-changes all backward-trans limited updown-limited
snmp-server enable traps bgp threshold prefix
snmp-server enable traps bgp cbgp2 state-changes all backward-trans limited updown-limited
snmp-server enable traps bgp cbgp2 threshold prefix
snmp-server enable traps pim neighbor-change rp-mapping-change invalid-pim-message
snmp-server enable traps ipmulticast
snmp-server enable traps msdp
snmp-server enable traps mvpn
snmp-server enable traps rsvp
snmp-server enable traps isis
snmp-server enable traps dhcp
snmp-server enable traps nhrp nhs
snmp-server enable traps nhrp nhc
snmp-server enable traps nhrp nhp
snmp-server enable traps nhrp quota-exceeded
snmp-server enable traps ipsla
snmp-server enable traps bfd
snmp-server enable traps cnpd
snmp-server enable traps event-manager
snmp-server enable traps dlsw
snmp-server enable traps ospfv3 state-change
snmp-server enable traps ospfv3 errors
snmp-server enable traps pppoe
snmp-server enable traps l2tun session
snmp-server enable traps l2tun pseudowire status
snmp-server enable traps l2tun tunnel
snmp-server enable traps pki
snmp-server enable traps pimstdmib neighbor-loss invalid-register invalid-join-prune rp-mapping-change
interface-election
snmp-server enable traps adslline
snmp-server enable traps entity-diag boot-up-fail hm-test-recover hm-thresh-reached scheduled-test-fail
snmp-server enable traps vdsl2line
snmp-server enable traps vtp
snmp-server enable traps vlancreate
snmp-server enable traps vlandelete
snmp-server enable traps port-security
snmp-server enable traps isdn call-information
snmp-server enable traps isdn layer2
snmp-server enable traps isdn chan-not-avail
snmp-server enable traps isdn ietf
snmp-server enable traps c3g
snmp-server enable traps LTE
snmp-server enable traps lisp
snmp-server enable traps firewall serverstatus
snmp-server enable traps ike policy add
snmp-server enable traps ike policy delete
snmp-server enable traps ike tunnel start
snmp-server enable traps ike tunnel stop
snmp-server enable traps ipsec cryptomap add
```

```
snmp-server enable traps ipsec cryptomap delete
snmp-server enable traps ipsec cryptomap attach
snmp-server enable traps ipsec cryptomap detach
snmp-server enable traps ipsec tunnel start
snmp-server enable traps ipsec tunnel stop
snmp-server enable traps ipsec too-many-sas
snmp-server enable traps gdoi gm-start-registration
snmp-server enable traps gdoi gm-registration-complete
snmp-server enable traps gdoi gm-re-register
snmp-server enable traps gdoi gm-rekey-rcvd
snmp-server enable traps gdoi gm-rekey-fail
snmp-server enable traps gdoi ks-rekey-pushed
snmp-server enable traps gdoi gm-incomplete-cfg
snmp-server enable traps gdoi ks-no-rsa-keys
snmp-server enable traps gdoi ks-new-registration
snmp-server enable traps gdoi ks-reg-complete
snmp-server enable traps gdoi ks-role-change
snmp-server enable traps gdoi ks-gm-deleted
snmp-server enable traps gdoi ks-peer-reachable
snmp-server enable traps gdoi ks-peer-unreachable
snmp-server enable traps mpls rfc ldp
snmp-server enable traps mpls ldp
snmp-server enable traps mpls rfc traffic-eng
snmp-server enable traps mpls traffic-eng
snmp-server enable traps mpls fast-reroute protected
snmp-server enable traps otn
snmp-server enable traps pw vc
snmp-server enable traps frame-relay multilink bundle-mismatch
snmp-server enable traps dial
snmp-server enable traps dsp card-status
snmp-server enable traps dsp oper-state
snmp-server enable traps dsp video-usage
snmp-server enable traps dsp video-out-of-resource
snmp-server enable traps syslog
snmp-server enable traps sbc adj-status
snmp-server enable traps sbc blacklist
snmp-server enable traps sbc congestion-alarm
snmp-server enable traps sbc h248-ctrlr-status
snmp-server enable traps sbc media-source
snmp-server enable traps sbc radius-conn-status
snmp-server enable traps sbc sla-violation
snmp-server enable traps sbc sla-violation-rev1
snmp-server enable traps sbc svc-state
snmp-server enable traps sbc qos-statistics
snmp-server enable traps bulkstat collection transfer
snmp-server enable traps ethernet cfm alarm
snmp-server enable traps alarms informational
snmp-server enable traps transceiver all
snmp-server enable traps rf
snmp-server enable traps vrfmib vrf-up vrf-down vnet-trunk-up vnet-trunk-down
snmp-server enable traps ccme
snmp-server enable traps srst
snmp-server enable traps mpls vpn
snmp-server enable traps mpls rfc vpn
snmp-server enable traps voice
snmp-server host 192.168.99.10 vrf MGMT version 3 priv labmon udp-port 1162
snmp-server host 192.168.99.10 vrf MGMT version 3 priv labmon
tacacs server LAB-TAC
 address ipv4 192.168.99.10
 key LabTacacsKey123
!
!
!
!
!
control-plane
!
!
mgcp behavior rsip-range tgcp-only
mgcp behavior comedia-role none
mgcp behavior comedia-check-media-src disable
mgcp behavior comedia-sdp-force disable
!
mgcp profile default
!
!
!
!
!
banner login ^C
*******************************************************************************
```

```
                          AUTHORISED ACCESS ONLY

  Access to this device is restricted to authorised personnel. If you are not
  an authorised user, disconnect now.

  All activity is monitored, logged and audited. Continued use of this system
  constitutes consent to that monitoring.

  Unauthorised access or use may result in disciplinary action and civil or
  criminal penalties.
******************************************************************************
^C
!
line con 0
 exec-timeout 0 0
 activation-character 13
 logging synchronous
 stopbits 1
line aux 0
 activation-character 13
line vty 0 4
 access-class VTY-MGMT in vrf-also
 exec-timeout 0 0
 activation-character 13
 transport input ssh
line vty 5 15
 access-class VTY-MGMT in vrf-also
 exec-timeout 0 0
 activation-character 13
 transport input ssh
!
ntp source GigabitEthernet4
ntp server vrf MGMT 192.168.99.10
!
!
!
!
!
!
netconf-yang
restconf
end
```

R3 — running config

```
Building configuration...

Current configuration : 19832 bytes
!
! Last configuration change at 00:53:31 UTC Wed Sep 9 2026 by lab
!
version 17.15
service timestamps debug datetime msec
service timestamps log datetime msec show-timezone
service sequence-numbers
platform qfp utilization monitor load 80
platform sslvpn use-pd
platform console serial
!
hostname c8000v-r3
!
boot-start-marker
boot-end-marker
!
!
vrf definition MGMT
 description out-of-band management
 !
 address-family ipv4
 exit-address-family
!
aaa new-model
!
!
aaa group server tacacs+ LABTACGRP
 server name LAB-TAC
 ip vrf forwarding MGMT
 ip tacacs source-interface GigabitEthernet4
!
aaa authentication login default group LABTACGRP local
```

```
aaa authentication enable default group LABTACGRP enable
aaa authorization exec default group LABTACGRP local if-authenticated
aaa accounting exec default start-stop group LABTACGRP
aaa accounting commands 15 default start-stop group LABTACGRP
!
!
aaa session-id common
!
!
!
!
!
!
!
!
!
!
!
!
ip domain name lab.local
!
!
!
login on-success log
!
!
subscriber templating
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
crypto pki trustpoint TP-self-signed-1394225539
 enrollment selfsigned
 subject-name cn=IOS-Self-Signed-Certificate-1394225539
 revocation-check none
 rsakeypair TP-self-signed-1394225539
 hash sha512
!
crypto pki trustpoint SLA-TrustPoint
 enrollment pkcs12
 revocation-check crl
 hash sha512
!
!
crypto pki certificate chain TP-self-signed-1394225539
 certificate self-signed 01
  30820330 30820218 A0030201 02020101 300D0609 2A864886 F70D0101 0D050030
  31312F30 2D060355 04030C26 494F532D 53656C66 2D536967 6E65642D 43657274
  69666963 6174652D 31333934 32323535 3339301E 170D3236 30393037 32323134
  31345A17 0D333630 39303632 32313431 345A3031 312F302D 06035504 030C2649
  4F532D53 656C662D 5369676E 65642D43 65727469 66696361 74652D31 33393432
  32353533 39308201 22300D06 092A8648 86F70D01 01010500 0382010F 00308201
  0A028201 01009A72 1ABC8E0C 1072AA3A FC85A8A4 F55F1158 86CF558A 9797A864
  25ADAB27 09CEDE1F 69BA396E 076D1EB5 1FB18AD2 8BC8F432 D68FA7E1 D31CE5A2
  68609CD7 962FFD33 3079835F 1E692B04 7B476846 85B6AB56 C1149552 09ACBCC7
  F7CC6723 6A93AF71 12478A64 38AC2CFA 0E8B0E0A 23FCC8EA 7EDA498E 8AEB8561
  17118926 F930366A CAD27EC2 1C8F0312 4637F83B 1AC605CA 0C3CAC05 8553A775
  81A29BE3 4CC15CB9 CDE88881 B696A9A7 99596CB9 8C8E0E6B 0581BFCD 16AA1EDF
  C037C482 B7BE1108 AFDBFC42 5D101BA2 B3AF8A5C C8C5A51F 6AABF787 B9874536
  9C44B46F 2160F9EA 6E0B349C 4B3232EE 4109D8FC F15C76F1 C77EAFA8 DA61BA95
  30F50494 0D1F0203 010001A3 53305130 1D060355 1D0E0416 0414AE7B 0DEF0D53
  9A2DFBB8 2A7F4857 8DDF095E A4C3301F 0603551D 23041830 168014AE 7B0DEF0D
  539A2DFB B82A7F48 578DDF09 5EA4C330 0F060355 1D130101 FF040530 030101FF
  300D0609 2A864886 F70D0101 0D050003 82010100 65426ED5 210C76B1 913676FF
  D92847C4 DB87D6F8 C7D468A2 D1BB826A 822083B9 88DDD430 2DFA57EA FD22715D
  D73197D0 5A6C3582 5D12968B 56DE28B1 1D162D61 119BB282 E2239093 D96D95DD
  D1122D35 0BF5780D 0FD3EB53 EF40D3D1 61693953 E97694CD C6F369F8 6B010182
  D4978C4B 9316184D FDCDF993 82047F48 557B580A 8AFDB4D8 AB1CBD94 80E67D03
  EB61EACC 075A9CDF F272A80F 02544535 24886FBF 500A61CC D2F56993 9457B80D
  C3C63E80 359E03A4 1256F1C9 7F905BBA 20E341AD B99A8BCC 02C5B9FC 96C55D7B
```

```
  66EE9989 ABC1C984 101DE474 631A5848 E41FB71A 9654FF7A 0302C615 EF5E298B
  A2C2F138 17AE6A7D 9B630B20 26159088 5AF665AF
        quit
crypto pki certificate chain SLA-TrustPoint
 certificate ca 01
  30820321 30820209 A0030201 02020101 300D0609 2A864886 F70D0101 0B050030
  32310E30 0C060355 040A1305 43697363 6F312030 1E060355 04031317 43697363
  6F204C69 63656E73 696E6720 526F6F74 20434130 1E170D31 33303533 30313934
  3834375A 170D3338 30353330 31393438 34375A30 32310E30 0C060355 040A1305
  43697363 6F312030 1E060355 04031317 43697363 6F204C69 63656E73 696E6720
  526F6F74 20434130 82012230 0D06092A 864886F7 0D010101 05000382 010F0030
  82010A02 82010100 A6BCBD96 131E05F7 145EA72C 2CD686E6 17222EA1 F1EFF64D
  CBB4C798 212AA147 C655D8D7 9471380D 8711441E 1AAF071A 9CAE6388 8A38E520
  1C394D78 462EF239 C659F715 B98C0A59 5BBB5CBD 0CFEBEA3 700A8BF7 D8F256EE
  4AA4E80D DB6FD1C9 60B1FD18 FFC69C96 6FA68957 A2617DE7 104FDC5F EA2956AC
  7390A3EB 2B5436AD C847A2C5 DAB553EB 69A9A535 58E9F3E3 C0BD23CF 58BD7188
  68E69491 20F320E7 948E71D7 AE3BCC84 F10684C7 4BC8E00F 539BA42B 42C68BB7
  C7479096 B4CB2D62 EA2F505D C7B062A4 6811D95B E8250FC4 5D5D5FB8 8F27D191
  C55F0D76 61F9A4CD 3D992327 A8BB03BD 4E6D7069 7CBADF8B DF5F4368 95135E44
  DFC7C6CF 04DD7FD1 02030100 01A34230 40300E06 03551D0F 0101FF04 04030201
  06300F06 03551D13 0101FF04 05300301 01FF301D 0603551D 0E041604 1449DC85
  4B3D31E5 1B3E6A17 606AF333 3D3B4C73 E8300D06 092A8648 86F70D01 010B0500
  03820101 00507F24 D3932A66 86025D9F E838AE5C 6D4DF6B0 49631C78 240DA905
  604EDCDE FF4FED2B 77FC460E CD636FDB DD44681E 3A5673AB 9093D3B1 6C9E3D8B
  D98987BF E40CBD9E 1AECA0C2 2189BB5C 8FA85686 CD98B646 5575B146 8DFC66A8
  467A3DF4 4D565700 6ADF0F0D CF835015 3C04FF7C 21E878AC 11BA9CD2 55A9232C
  7CA7B7E6 C1AF74F6 152E99B7 B1FCF9BB E973DE7F 5BDDEB86 C71E3B49 1765308B
  5FB0DA06 B92AFE7F 494E8A9E 07B85737 F3A58BE1 1A48A229 C37C1E69 39F08678
  80DDCD16 D6BACECA EEBC7CF9 8428787B 35202CDC 60E4616A B623CDBD 230E3AFB
  418616A9 4093E049 4D10AB75 27E86F73 932E35B5 8862FDAE 0275156F 719BB2F0
  D697DF7F 28
        quit
!
!
!
!
!
!
!
!
!
license udi pid C8000V sn 98WSPSI3CQ2
license boot level network-premier addon dna-premier
memory free low-watermark processor 31774
diagnostic bootup level minimal
!
!
spanning-tree extend system-id
!
!
enable secret 9 $9$Bw47UP1ye/.yVU$K5CSpZUsnABYFZykCttzeySotiNrRBY/frbuYNtoXY2
!
username lab privilege 15 secret 9 $9$nTmPdPNMf5C6Nk$TZUyweyM6Pgquc3KoKrRbcfdLyeESXbTbt75p43gz7g
!
redundancy
!
!
!
crypto ikev2 proposal LAB-PROP
 encryption aes-cbc-256
 integrity sha256
 group 14
!
crypto ikev2 policy LAB-POL
 proposal LAB-PROP
!
crypto ikev2 keyring LAB-KR
 peer ANY
  address 0.0.0.0 0.0.0.0
  pre-shared-key LabPreSharedKey123
 !
!
!
crypto ikev2 profile LAB-PROF
 match identity remote address 0.0.0.0
 authentication remote pre-share
 authentication local pre-share
 keyring local LAB-KR
!
!
!
```

```
!
!
!
!
!
!
!
!
!
!
crypto ipsec transform-set LAB-TS esp-aes 256 esp-sha256-hmac
 mode transport
!
crypto ipsec profile LAB-IPSEC
 set transform-set LAB-TS
 set ikev2-profile LAB-PROF
!
!
!
!
!
!
!
!
!
!
interface Loopback0
 description Protected endpoint
 ip address 172.16.3.1 255.255.255.255
!
interface Loopback1
 description BGP-advertised prefix
 ip address 10.20.3.1 255.255.255.255
!
interface Tunnel0
 description DMVPN spoke
 ip address 172.20.0.3 255.255.255.0
 no ip redirects
 ip nat outside
 ip nhrp authentication LabNhrp
 ip nhrp network-id 100
 ip nhrp holdtime 300
 ip nhrp nhs 172.20.0.1 nbma 100.64.0.1 multicast
 ip tcp adjust-mss 1360
 bfd interval 500 min_rx 500 multiplier 3
 tunnel source GigabitEthernet2
 tunnel mode gre multipoint
 tunnel key 100
 tunnel protection ipsec profile LAB-IPSEC
!
interface GigabitEthernet1
 description MGMT - QEMU user-mode net (DHCP)
 ip address dhcp
 negotiation auto
!
interface GigabitEthernet2
 description WAN - shared provider segment, DMVPN underlay
 ip address 100.64.0.3 255.255.255.0
 negotiation auto
!
interface GigabitEthernet3
 description LAN toward the local host
 ip address 192.168.30.1 255.255.255.0
 ip nat inside
 negotiation auto
!
interface GigabitEthernet4
 description OOB management - NTP, syslog, SNMP
 vrf forwarding MGMT
 ip address 192.168.99.3 255.255.255.0
 negotiation auto
!
router bgp 65003
 bgp router-id 172.16.3.1
 bgp log-neighbor-changes
 neighbor 172.20.0.1 remote-as 65001
 neighbor 172.20.0.1 password LabBgpSecret
 neighbor 172.20.0.1 timers 10 30
 neighbor 172.20.0.1 fall-over bfd
 !
 address-family ipv4
```

```
  network 10.20.3.1 mask 255.255.255.255
  network 10.30.3.0 mask 255.255.255.0
  network 192.168.30.0
  neighbor 172.20.0.1 activate
  neighbor 172.20.0.1 send-community both
  neighbor 172.20.0.1 route-map RM-IN-R1 in
  neighbor 172.20.0.1 route-map RM-OUT-R1 out
  neighbor 172.20.0.1 maximum-prefix 15 80
 exit-address-family
!
ip forward-protocol nd
!
ip bgp-community new-format
ip community-list standard CL-VALID permit 65001:1
ip community-list standard CL-VALID permit 65001:2
ip community-list standard CL-VALID permit 65001:3
ip community-list standard CL-VALID permit 65002:1
ip community-list standard CL-VALID permit 65002:2
ip community-list standard CL-VALID permit 65002:3
ip community-list standard CL-VALID permit 65003:1
ip community-list standard CL-VALID permit 65003:2
ip community-list standard CL-VALID permit 65003:3
ip as-path access-list 10 permit ^65001$
ip as-path access-list 10 permit ^65001_65002$
no ip http server
ip http secure-server
ip route 10.30.3.0 255.255.255.0 Null0
ip nat pool SITE-POOL 10.30.3.100 10.30.3.129 prefix-length 24
ip nat inside source static 192.168.30.10 10.30.3.10 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.11 10.30.3.11 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.12 10.30.3.12 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.13 10.30.3.13 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.14 10.30.3.14 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.15 10.30.3.15 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.16 10.30.3.16 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.17 10.30.3.17 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.18 10.30.3.18 route-map NAT-TO-DOMAIN
ip nat inside source static 192.168.30.19 10.30.3.19 route-map NAT-TO-DOMAIN
ip nat inside source list NAT-POOL-SRC pool SITE-POOL
ip ssh bulk-mode 131072
ip ssh server algorithm authentication password
!
ip prefix-list PL-OWN-LAN seq 5 permit 192.168.30.0/24
!
ip prefix-list PL-OWN-LO seq 5 permit 10.20.3.1/32
!
ip prefix-list PL-OWN-NAT seq 5 permit 10.30.3.0/24
!
ip access-list standard VTY-MGMT
 10 remark out-of-band management network
 10 permit 192.168.99.0 0.0.0.255
 20 remark hypervisor user-mode NAT: how the harness reaches these devices
 20 permit 10.0.2.0 0.0.0.255
 30 deny any log
!
ip access-list extended NAT-DOMAIN
 10 permit ip host 192.168.30.10 10.30.0.0 0.0.255.255
 20 permit ip host 192.168.30.11 10.30.0.0 0.0.255.255
 30 permit ip host 192.168.30.12 10.30.0.0 0.0.255.255
 40 permit ip host 192.168.30.13 10.30.0.0 0.0.255.255
 50 permit ip host 192.168.30.14 10.30.0.0 0.0.255.255
 60 permit ip host 192.168.30.15 10.30.0.0 0.0.255.255
 70 permit ip host 192.168.30.16 10.30.0.0 0.0.255.255
 80 permit ip host 192.168.30.17 10.30.0.0 0.0.255.255
 90 permit ip host 192.168.30.18 10.30.0.0 0.0.255.255
 100 permit ip host 192.168.30.19 10.30.0.0 0.0.255.255
ip access-list extended NAT-POOL-SRC
 10 permit ip 192.168.30.64 0.0.0.31 10.30.0.0 0.0.255.255
logging source-interface GigabitEthernet4 vrf MGMT
logging host 192.168.99.10 vrf MGMT
route-map RM-OUT-R1 permit 10
 description R3's own LO, tagged 65003:1
 match ip address prefix-list PL-OWN-LO
 set community 65003:1
!
route-map RM-OUT-R1 permit 20
 description R3's own NAT, tagged 65003:2
 match ip address prefix-list PL-OWN-NAT
 set community 65003:2
!
route-map RM-OUT-R1 permit 30
```

```
 description R3's own LAN, tagged 65003:3
 match ip address prefix-list PL-OWN-LAN
 set community 65003:3
!
route-map NAT-TO-DOMAIN permit 10
 match ip address NAT-DOMAIN
!
route-map RM-IN-R1 permit 10
 description accept only known communities on an AS path R1 can produce
 match as-path 10
 match community CL-VALID
!
snmp-server group LABGRP v3 priv read LABVIEW notify LABVIEW
snmp-server view LABVIEW iso included
snmp-server trap-source GigabitEthernet4
snmp-server enable traps snmp authentication linkdown linkup coldstart warmstart
snmp-server enable traps vrrp
snmp-server enable traps flowmon
snmp-server enable traps pfr
snmp-server enable traps ds1
snmp-server enable traps sdwan security policy system omp bfd
snmp-server enable traps entity-perf throughput-notif
snmp-server enable traps call-home message-send-fail server-fail
snmp-server enable traps tty
snmp-server enable traps ospf state-change
snmp-server enable traps ospf errors
snmp-server enable traps ospf retransmit
snmp-server enable traps ospf lsa
snmp-server enable traps ospf cisco-specific state-change nssa-trans-change
snmp-server enable traps ospf cisco-specific state-change shamlink interface
snmp-server enable traps ospf cisco-specific state-change shamlink neighbor
snmp-server enable traps ospf cisco-specific errors
snmp-server enable traps ospf cisco-specific retransmit
snmp-server enable traps ospf cisco-specific lsa
snmp-server enable traps eigrp
snmp-server enable traps casa
snmp-server enable traps xgcp
snmp-server enable traps cef resource-failure peer-state-change peer-fib-state-change inconsistency
snmp-server enable traps smart-license
snmp-server enable traps diameter
snmp-server enable traps entity-qfp mem-res-thresh throughput-notif
snmp-server enable traps entity-state
snmp-server enable traps ethernet cfm cc mep-up mep-down cross-connect loop config
snmp-server enable traps ethernet cfm crosscheck mep-missing mep-unknown service-up
snmp-server enable traps ether-oam
snmp-server enable traps ethernet evc status create delete
snmp-server enable traps aaa_server
snmp-server enable traps config-copy
snmp-server enable traps config
snmp-server enable traps config-ctid
snmp-server enable traps memory bufferpeak
snmp-server enable traps fru-ctrl
snmp-server enable traps entity-sensor
snmp-server enable traps resource-policy
snmp-server enable traps flash insertion removal lowspace
snmp-server enable traps hsrp
snmp-server enable traps entity
snmp-server enable traps ip local pool
snmp-server enable traps cpu threshold
snmp-server enable traps auth-framework sec-violation
snmp-server enable traps atm subif
snmp-server enable traps frame-relay
snmp-server enable traps frame-relay subif
snmp-server enable traps bgp state-changes all backward-trans limited updown-limited
snmp-server enable traps bgp threshold prefix
snmp-server enable traps bgp cbgp2 state-changes all backward-trans limited updown-limited
snmp-server enable traps bgp cbgp2 threshold prefix
snmp-server enable traps pim neighbor-change rp-mapping-change invalid-pim-message
snmp-server enable traps ipmulticast
snmp-server enable traps msdp
snmp-server enable traps mvpn
snmp-server enable traps rsvp
snmp-server enable traps isis
snmp-server enable traps dhcp
snmp-server enable traps nhrp nhs
snmp-server enable traps nhrp nhc
snmp-server enable traps nhrp nhp
snmp-server enable traps nhrp quota-exceeded
snmp-server enable traps ipsla
snmp-server enable traps bfd
snmp-server enable traps cnpd
```

```
snmp-server enable traps event-manager
snmp-server enable traps dlsw
snmp-server enable traps ospfv3 state-change
snmp-server enable traps ospfv3 errors
snmp-server enable traps pppoe
snmp-server enable traps l2tun session
snmp-server enable traps l2tun pseudowire status
snmp-server enable traps l2tun tunnel
snmp-server enable traps pki
snmp-server enable traps pimstdmib neighbor-loss invalid-register invalid-join-prune rp-mapping-change
interface-election
snmp-server enable traps adslline
snmp-server enable traps entity-diag boot-up-fail hm-test-recover hm-thresh-reached scheduled-test-fail
snmp-server enable traps vdsl2line
snmp-server enable traps vtp
snmp-server enable traps vlancreate
snmp-server enable traps vlandelete
snmp-server enable traps port-security
snmp-server enable traps isdn call-information
snmp-server enable traps isdn layer2
snmp-server enable traps isdn chan-not-avail
snmp-server enable traps isdn ietf
snmp-server enable traps c3g
snmp-server enable traps LTE
snmp-server enable traps lisp
snmp-server enable traps firewall serverstatus
snmp-server enable traps ike policy add
snmp-server enable traps ike policy delete
snmp-server enable traps ike tunnel start
snmp-server enable traps ike tunnel stop
snmp-server enable traps ipsec cryptomap add
snmp-server enable traps ipsec cryptomap delete
snmp-server enable traps ipsec cryptomap attach
snmp-server enable traps ipsec cryptomap detach
snmp-server enable traps ipsec tunnel start
snmp-server enable traps ipsec tunnel stop
snmp-server enable traps ipsec too-many-sas
snmp-server enable traps gdoi gm-start-registration
snmp-server enable traps gdoi gm-registration-complete
snmp-server enable traps gdoi gm-re-register
snmp-server enable traps gdoi gm-rekey-rcvd
snmp-server enable traps gdoi gm-rekey-fail
snmp-server enable traps gdoi ks-rekey-pushed
snmp-server enable traps gdoi gm-incomplete-cfg
snmp-server enable traps gdoi ks-no-rsa-keys
snmp-server enable traps gdoi ks-new-registration
snmp-server enable traps gdoi ks-reg-complete
snmp-server enable traps gdoi ks-role-change
snmp-server enable traps gdoi ks-gm-deleted
snmp-server enable traps gdoi ks-peer-reachable
snmp-server enable traps gdoi ks-peer-unreachable
snmp-server enable traps mpls rfc ldp
snmp-server enable traps mpls ldp
snmp-server enable traps mpls rfc traffic-eng
snmp-server enable traps mpls traffic-eng
snmp-server enable traps mpls fast-reroute protected
snmp-server enable traps otn
snmp-server enable traps pw vc
snmp-server enable traps frame-relay multilink bundle-mismatch
snmp-server enable traps dial
snmp-server enable traps dsp card-status
snmp-server enable traps dsp oper-state
snmp-server enable traps dsp video-usage
snmp-server enable traps dsp video-out-of-resource
snmp-server enable traps syslog
snmp-server enable traps sbc adj-status
snmp-server enable traps sbc blacklist
snmp-server enable traps sbc congestion-alarm
snmp-server enable traps sbc h248-ctrlr-status
snmp-server enable traps sbc media-source
snmp-server enable traps sbc radius-conn-status
snmp-server enable traps sbc sla-violation
snmp-server enable traps sbc sla-violation-rev1
snmp-server enable traps sbc svc-state
snmp-server enable traps sbc qos-statistics
snmp-server enable traps bulkstat collection transfer
snmp-server enable traps ethernet cfm alarm
snmp-server enable traps alarms informational
snmp-server enable traps transceiver all
snmp-server enable traps rf
snmp-server enable traps vrfmib vrf-up vrf-down vnet-trunk-up vnet-trunk-down
```

```
snmp-server enable traps ccme
snmp-server enable traps srst
snmp-server enable traps mpls vpn
snmp-server enable traps mpls rfc vpn
snmp-server enable traps voice
snmp-server host 192.168.99.10 vrf MGMT version 3 priv labmon udp-port 1162
snmp-server host 192.168.99.10 vrf MGMT version 3 priv labmon
tacacs server LAB-TAC
 address ipv4 192.168.99.10
 key LabTacacsKey123
!
!
!
!
!
control-plane
!
!
mgcp behavior rsip-range tgcp-only
mgcp behavior comedia-role none
mgcp behavior comedia-check-media-src disable
mgcp behavior comedia-sdp-force disable
!
mgcp profile default
!
!
!
!
!
banner login ^C
******************************************************************************
                          AUTHORISED ACCESS ONLY

  Access to this device is restricted to authorised personnel. If you are not
  an authorised user, disconnect now.

  All activity is monitored, logged and audited. Continued use of this system
  constitutes consent to that monitoring.

  Unauthorised access or use may result in disciplinary action and civil or
  criminal penalties.
******************************************************************************
^C
!
line con 0
 exec-timeout 0 0
 activation-character 13
 logging synchronous
 stopbits 1
line aux 0
 activation-character 13
line vty 0 4
 access-class VTY-MGMT in vrf-also
 exec-timeout 0 0
 activation-character 13
 transport input ssh
line vty 5 15
 access-class VTY-MGMT in vrf-also
 exec-timeout 0 0
 activation-character 13
 transport input ssh
!
ntp source GigabitEthernet4
ntp server vrf MGMT 192.168.99.10
!
!
!
!
!
!
netconf-yang
restconf
end
```